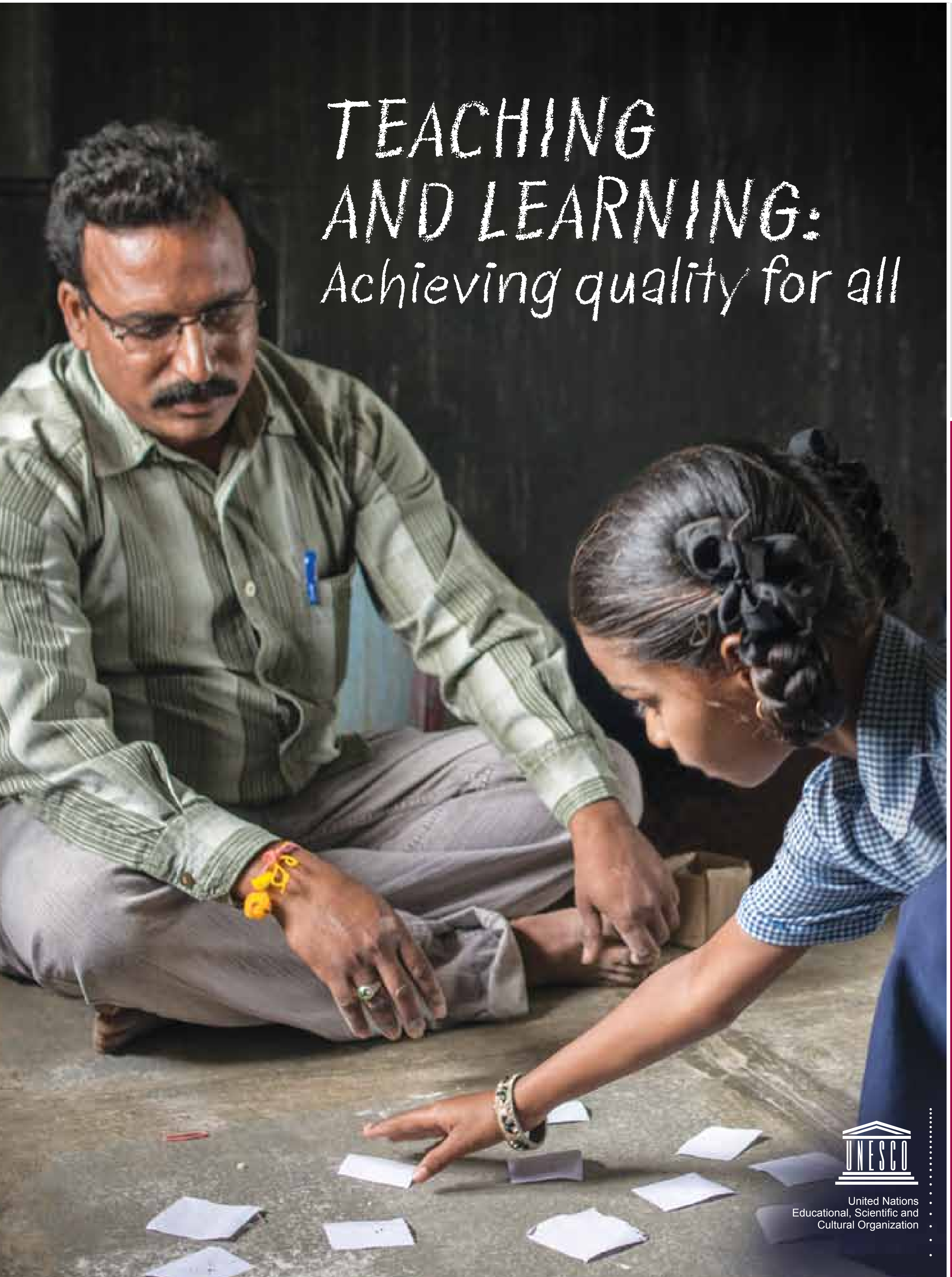
EFA Global Monitoring Report

2013/4

# TEACHING AND LEARNING:

## Achieving quality for all

Education for All



United Nations
Educational, Scientific and
Cultural Organization

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# Foreword

This 11th *EFA Global Monitoring Report* provides a timely update on progress that countries are making towards the global education goals that were agreed in 2000. It also makes a powerful case for placing education at the heart of the global development agenda after 2015. In 2008, the *EFA Global Monitoring Report* asked – 'will we make it?' With less than two years left before 2015, this Report makes it clear that we will not.

Fifty-seven million children are still failing to learn, simply because they are not in school. Access is not the only crisis – poor quality is holding back learning even for those who make it to school. One third of primary school age children are not learning the basics, whether they have been to school or not. To reach our goals, this Report calls on Governments to redouble efforts to provide learning to all who face disadvantages – whether from poverty, gender, where they live or other factors.

An education system is only as good as its teachers. Unlocking their potential is essential to enhancing the quality of learning. Evidence shows that education quality improves when teachers are supported – it deteriorates if they are not, contributing to the shocking levels of youth illiteracy captured in this Report.

Governments must step up efforts to recruit an additional 1.6 million teachers to achieve universal primary education by 2015. This Report identifies four strategies to provide the best teachers to reach all children with a good quality education. First, the right teachers must be selected to reflect the diversity of the children they will be teaching. Second, teachers must be trained to support the weakest learners, starting from the early grades. A third strategy aims to overcome inequalities in learning by allocating the best teachers to the most challenging parts of a country. Lastly, governments must provide teachers with the right mix of incentives to encourage them to remain in the profession and to make sure all children are learning, regardless of their circumstances.

But teachers cannot shoulder the responsibility alone. The Report shows also that teachers can only shine in the right context, with well-designed curricula and assessment strategies to improve teaching and learning.

These policy changes have a cost. This is why we need to see a dramatic shift in funding. Basic education is currently underfunded by US$26 billion a year, while aid is continuing to decline. At this stage, governments simply cannot afford to reduce investment in education – nor should donors step back from their funding promises. This calls for exploring new ways to fund urgent needs.

We must learn from the evidence as we shape a new global sustainable development agenda after 2015. As this Report shows, equality in access and learning must stand at the heart of future education goals. We must ensure that all children and young people are learning the basics and that they have the opportunity to acquire the transferable skills needed to become global citizens. We must also set goals that are clear and measurable, to allow for the tracking and monitoring that is so essential for governments and donors alike, and to bridge the gaps that remain.

As we advance towards 2015 and set a new agenda to follow, all governments must invest in education as an accelerator of inclusive development. This Report's evidence clearly shows that education provides sustainability to progress against all development goals. Educate mothers, and you empower women and save children's lives. Educate communities, and you transform societies and grow economies. This is the message of this *EFA Global Monitoring Report*.

Irina Bokova
Director-General of UNESCO

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2007. Strong foundations — Early childhood care and education
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# Overview

With the deadline for the Education for All goals less than two years away, it is clear that, despite advances over the past decade, not a single goal will be achieved globally by 2015. This year's *EFA Global Monitoring Report* vividly underlines the fact that people in the most marginalized groups have continued to be denied opportunities for education over the decade. It is not too late, however, to accelerate progress in the final stages. And it is vital to put in place a robust global post-2015 education framework to tackle unfinished business while addressing new challenges. Post-2015 education goals will only be achieved if they are accompanied by clear, measurable targets with indicators tracking that no one is left behind, and if specific education financing targets for governments and aid donors are set.

The *2013/4 EFA Global Monitoring Report* is divided into three parts. Part 1 provides an update of progress towards the six EFA goals. The second part presents clear evidence that progress in education is vital for achieving development goals after 2015. Part 3 puts the spotlight on the importance of implementing strong policies to unlock the potential of teachers so as to support them in overcoming the global learning crisis.

## Monitoring the Education for All goals

Since the Education for All framework was established in 2000, countries have made progress towards the goals. However, too many will still be far from the target in 2015.

### Goal 1: Early childhood care and education

The foundations set in the first thousand days of a child's life, from conception to the second birthday, are critical for future well-being. It is therefore vital that families have access to adequate health care, along with support to make the right choices for mothers and babies. In addition, access to good nutrition holds the key to developing children's immune systems and the cognitive abilities they need in order to learn.

Despite improvements, an unacceptably high number of children suffer from ill health: under-5 mortality fell by 48% from 1990 to 2012, yet 6.6 million children still died before their fifth birthday in 2012.Progress has been slow. In 43 countries, more than one in ten children died before age 5 in 2000. If the annual rate of reduction for child mortality in these 43 countries between 2000 and 2011 is projected to 2015, only eight countries will reach the target of reducing child deaths by two-thirds from their

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1990 levels. Some poorer countries that invested in early childhood interventions, including Bangladesh and Timor-Leste, reduced child mortality by at least two-thirds in advance of the target date.

Progress in improving child nutrition has been considerable. Yet, as of 2012, some 162 million children under 5 were still malnourished; three-quarters of them live in sub-Saharan Africa and South and West Asia. While the share of children under 5 who were stunted – a robust indicator of long-term malnutrition – was 25%, down from 40% in 1990, the annual rate of reduction needs to almost double if global targets are to be achieved by 2025.

The links between early childhood care and education are strong and mutually reinforcing. Early childhood care and education services help build skills at a time when children's brains are developing, with long-term benefits for children from disadvantaged backgrounds. In Jamaica, for example, infants who were stunted and from disadvantaged backgrounds receiving weekly psychosocial stimulation were earning 42% more than their peers by their early 20s.

Since 2000, pre-primary education has expanded considerably. The global pre-primary education gross enrolment ratio increased from 33% in 1999 to 50% in 2011, although it reached only 18% in sub-Saharan Africa. The number of children enrolled in pre-primary schools grew by almost 60 million over the period.

In many parts of the world, however, there is a wide gap in enrolment between the richest and poorest. Part of the reason is that governments have yet to assume sufficient responsibility for pre-primary education: as of 2011, private providers were catering for 33% of all enrolled children, rising to 71% in the Arab States. The cost of private provision is one of the factors that contribute to inequity in access at this level.

No target was set at Dakar in 2000 to guide assessment of success in early childhood education. To gauge progress, this Report has set a pre-primary education gross enrolment ratio of 80% as an indicative target for 2015. Of the 141 countries with data, 21% had reached the target in 1999. By 2011, the number had risen to 37%. Looking ahead to 2015, it is projected that 48% of countries will reach the target.

An 80% target is modest, leaving many young children, often the most vulnerable, out of pre-school. Any post-2015 goal must provide a clear target to make sure all young children have access to pre-primary education, and a way to track the progress of disadvantaged groups to be sure they do not miss out.

## Goal 2: Universal primary education

With just two years until the 2015 deadline for the Education for All goals, the goal of universal primary education (UPE) is likely to be missed by a wide margin. By 2011, 57 million children were still out of school.

There is some good news: between 1999 and 2011, the number of children out of school fell almost by half. Following a period of stagnation, there was a small improvement between 2010 and 2011. But that reduction of 1.9 million is scarcely more than a quarter of the average between 1999 and 2004. Had the rate of decline between 1999 and 2008 been maintained, UPE could almost have been achieved by 2015.

Sub-Saharan Africa is the region that is lagging most behind, with 22% of the region's primary school age population still not in school in 2011. By contrast, South and West Asia experienced the fastest decline, contributing more than half the total reduction in numbers out of school.

Girls make up 54% of the global population of children out of school. In the Arab States, the share is 60%, unchanged since 2000. In South and West Asia, by contrast, the percentage of girls in the out-of-school population fell steadily, from 64% in 1999 to 57% in 2011. Almost half the children out of school globally are expected never to make it to school, and the same is true for almost two of three girls in the Arab States and sub-Saharan Africa.

The top three performers in the last five years have been the Lao People's Democratic Republic, Rwanda and Viet Nam, which reduced their out-of-school populations by at least 85%. There has been little change in the list of countries with the highest numbers of children out of school. The top 10 was unchanged over the period with one exception: Ghana was replaced by Yemen.

Some countries that might have been in the list with the largest out-of-school populations are not there simply because they have no recent reliable data. Using household surveys, this Report estimates that 14 countries had more than 1 million children out of school in 2011, including Afghanistan, China, the Democratic Republic of the Congo, Somalia, Sudan (pre-secession) and the United Republic of Tanzania.

Around half the world's out-of-school population lives in conflict-affected countries, up from 42% in 2008. Of the 28.5 million primary school age children out of school in conflict-affected countries, 95% live in low and lower middle income countries. Girls, who make up 55% of the total, are the worst affected.

Often children do not make it to school because of disadvantages they are born with. One of the most neglected disadvantages is disability. New analysis from four countries shows that children at higher risk of disability are far more likely to be denied a chance to go to school, with differences widening depending on the type of disability. In Iraq, for instance, 10% of 6- to 9-year-olds with no risk of disability had never been to school in 2006, but 19% of those with a risk of hearing impairment and 51% of those who were at higher risk of mental disability had never been to school.

Children are more likely to complete primary schooling if they enter at the right age. However, the net intake rate for the first year of primary school increased only slightly between 1999 and 2011, from 81% to 86% – and it rose by less than one percentage point over the last four years of the period. Some countries have made great progress in getting children into school on time, however, including Ethiopia, which increased its rate from 23% in 1999 to 94% in 2011.

Dropout before completing a full primary cycle has hardly changed since 1999. In 2010, around 75% of those who started primary school reached the last grade. In sub-Saharan Africa, the proportion of those starting school who reached the last grade worsened from 58% in 1999 to 56% in 2010; by contrast, in the Arab States this proportion improved from 79% in 1999 to 87% in 2010.

Universal participation in primary school is likely to remain elusive in many countries by 2015. Of 122 countries, the proportion reaching universal primary enrolment rose from 30% in 1999 to 50% in 2011. Looking ahead to 2015, it is projected that 56% of countries will reach the target. In 2015, 12% of countries will still have fewer than 8 in 10 enrolled, including two-thirds of countries in sub-Saharan Africa.

Assessing whether UPE has been achieved should be judged not by participation alone, but also by whether children complete primary education. Among the 90 countries with data, it is expected that at least 97% of children will reach the last grade of primary school by 2015 in just 13 countries, 10 of which are OECD or EU member states.

## Goal 3: Youth and adult skills

The third EFA goal has been one of the most neglected, in part because no targets or indicators were set to monitor its progress. The 2012 Report proposed a framework for various pathways to skills – including foundation, transferable, and technical and vocational skills – as a way of improving monitoring efforts, but the international community is still a long way from measuring the acquisition of skills systematically.

The most effective route to acquiring foundation skills is through lower secondary schooling. The lower secondary gross enrolment ratio increased from 72% to 82% over 1999–2011. The fastest growth was in sub-Saharan Africa, where enrolment more than doubled, albeit from a low base, reaching 49% in 2011.

Children need to complete lower secondary education to acquire foundation skills. Analysis using household surveys shows that completion rates had only reached 37% in low income countries by around 2010. There are wide inequalities in completion, with rates reaching 61% for the richest households but 14% for the poorest.

**In low income countries, only 14% of the poorest complete lower secondary school**

The number of out-of-school adolescents has fallen since 1999 by 31%, to 69 million. However it has all but stagnated since 2007, leaving many young people needing access to second-chance programmes to acquire foundation skills. Slow progress towards reducing the number of adolescents out of school in South and West Asia resulted in the region's share of the total number increasing from 39% in 1999 to 45% in 2011. In sub-Saharan Africa, the number of adolescents out of school remained at 22 million between 1999 and 2011 as population growth cancelled out enrolment growth.

**The number of illiterate adults fell by just 1% since 2000**

Given that universal lower secondary education is expected to become an explicit goal after 2015, it is vital to assess where the world is likely to stand in 2015. An assessment of progress based on 82 countries finds that only 26% achieved universal lower secondary education in 1999. By 2011, 32% of countries had reached that level. By 2015, the proportion of countries reaching that level is expected to grow to 46%.

This assessment is based on information from only 40% of all countries. It includes two-thirds of the countries in North America and Western Europe but only a quarter of sub-Saharan African countries, so it is not representative. Taking into account the countries that have not yet achieved universal primary enrolment, the percentage of countries that could achieve lower secondary enrolment by 2015 would be much lower.


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## Goal 4: Adult literacy

Universal literacy is fundamental to social and economic progress. Literacy skills are best developed in childhood through good quality education. Few countries offer genuine second chances to illiterate adults. As a result, countries with a legacy of low access to school have been unable to eradicate adult illiteracy.

The number of illiterate adults remains stubbornly high at 774 million, a fall of 12% since 1990 but just 1% since 2000. It is projected only to fall to 743 million by 2015. Ten countries are responsible for almost three-quarters of the world's illiterate adults. Women make up almost two-thirds of the total, and there has been no progress in reducing this share since 1990. Of the 61 countries with data, around half are expected to achieve gender parity in adult literacy by 2015, and 10 will be very close.

Since 1990, adult literacy rates have risen fastest in the Arab States. Nevertheless, population growth has meant that the number of illiterate adults has only fallen from 52 million to 48 million. Similarly, the region with the second fastest increase in adult literacy rates, South and West Asia, has seen its population of illiterate adults remain stable at just over 400 million. In sub-Saharan Africa, the number of illiterate adults has increased by 37% since 1990, mainly due to population growth, reaching 182 million in 2011. By 2015, it is projected that 26% of all illiterate adults will live in sub-Saharan Africa, up from 15% in 1990.

Slow progress means that there has been little change in the number of countries achieving universal adult literacy. Of 87 countries, 21% had reached universal adult literacy in 2000. Between 2000 and 2011, the number of countries that had reached this level increased to 26%. By contrast, 26% of countries were very far from this level in 2011. In 2015, 29% of countries are expected to achieve universal adult literacy, while 37% will still be very far.

The 15 countries of West Africa are among those with the worst adult literacy rates globally, and include the five countries with the world's lowest literacy rates, below 35%. Those five countries also have female literacy rates below 25%, compared with an average for sub-Saharan

Africa of 50%. These trends are not likely to improve soon. In 12 out of the 15 countries, fewer than half of young women are literate.

## Goal 5: Gender parity and equality

Gender parity – ensuring an equal enrolment ratio of girls and boys – is the first step towards the fifth EFA goal. The full goal – gender equality – also demands appropriate schooling environments, practices free of discrimination, and equal opportunities for boys and girls to realize their potential.

Gender disparity patterns vary between countries in different income groups. Among low income countries, disparities are commonly at the expense of girls: 20% achieve gender parity in primary education, 10% in lower secondary education and 8% in upper secondary education. Among middle and high income countries, where more countries achieve parity at any level, the disparities are increasingly at the expense of boys as one moves up to the lower and upper secondary levels. For example, 2% of upper middle income countries have disparity at the expense of boys in primary school, 23% in lower secondary school and 62% in upper secondary school.

Reaching parity at both the primary and secondary levels was singled out to be achieved by 2005, earlier than the other goals. Yet, even by 2011, many countries had not achieved this goal. At the primary level, for 161 countries, 57% had achieved gender parity in 1999. Between 1999 and 2011, the proportion of countries that had reached the target increased to 63%. The number of countries furthest from the target, with fewer than 90 girls for every 100 boys enrolled, fell from 19% in 1999 to 9% in 2011.

Looking ahead, it is projected that by 2015, 70% of countries will have reached the goal and 9% of countries will be close. By contrast, 14% of countries will still be far from the target, and 7% will be very far, of which three-quarters are in sub-Saharan Africa.

A move towards gender parity does not always mean more children in school. Burkina Faso, for example, is projected to achieve parity in primary school by 2015, but it still has the seventh-lowest gross enrolment ratio in the world. And in Senegal, where progress has been made in narrowing the gender gap, this is due to improvement in female enrolment while the male enrolment ratio has not increased since 2004.

At the lower secondary level, of 150 countries, 43% had achieved gender parity in 1999. It is projected that by 2015, 56% of countries will have achieved the target. At the other extreme, 33% of countries were far from the target in 1999, of which three-quarters had disparity at the expense of girls. It is expected that by 2015, 21% of countries will still experience gender disparity in lower secondary school, in 70% of which the disparity will be at girls' expense.

Fast progress is feasible, as the example of Turkey shows: it has almost achieved parity at both lower and upper secondary levels, even though the gender parity index in 1999 was 0.74 in lower secondary and 0.62 in upper secondary. But there is no room for complacency. Traditional perceptions of gender roles that permeate society filter down to schools.

## Goal 6: Quality of education

Improving quality and learning is likely to be more central to the post-2015 global development framework. Such a shift is vital to improve education opportunities for the 250 million children who are unable to read, write, or do basic mathematics, 130 million of whom are in school.

The pupil/teacher ratio is one measure for assessing progress towards goal 6. Globally, average pupil/teacher ratios have barely changed at the pre-primary, primary and secondary levels. In sub-Saharan Africa, with teacher recruitment lagging behind growth in enrolment, ratios stagnated and are now the highest in the world at the pre-primary and primary levels. Of the 162 countries with data in 2011, 26 had a pupil/teacher ratio in primary education exceeding 40:1, 23 of which are in sub-Saharan Africa.

**By 2015, 70% of countries are expected to reach gender parity in primary enrolment**

Between 1999 and 2011, the pupil/teacher ratio in primary education increased by at least 20% in nine countries. By contrast, it fell by at least 20% in 60 countries. Congo, Ethiopia and Mali more than doubled primary school enrolment and yet decreased their pupil/teacher ratios by more than 10 pupils per teacher.

**In a third of countries, less than 75% of primary school teachers are trained**

However, many countries have expanded teacher numbers rapidly by hiring people to teach without training. This may serve to get more children into school, but jeopardizes education quality. In a third of countries with data, less than 75% of teachers are trained according to national standards. The pupil/trained teacher ratio exceeds the pupil/teacher ratio by 10 pupils in 29 of the 98 countries, of which two-thirds are in sub-Saharan Africa.

At the secondary level, in 14 of 130 countries with data the pupil/teacher ratio exceeds 30:1. Although the vast majority of the countries facing the largest challenges are in sub-Saharan Africa, the region nevertheless managed to double the number of secondary teachers over 1999–2011. Of the 60 countries with data on the share of trained secondary teachers, half had less than 75% of the teaching force trained to the national standard, while 11 had less than 50% trained.

The proportion of teachers trained to national standards is particularly low in pre-primary education. Although the number of teachers at this level has increased by 53% since 2000, in 40 of the 75 countries with data, less than 75% of teachers are trained to the national standard.

In some contexts, the presence of female teachers is crucial to attract girls to school and improve their learning outcomes. Yet women teachers are lacking in some countries with high gender disparity in enrolment, such as Djibouti and Eritrea.

Teachers need good quality learning materials to be effective but many do not have access to textbooks. In the United Republic of Tanzania, only 3.5% of all grade 6 pupils had sole use of a reading textbook. Poor physical infrastructure is another problem for students in many poor countries. Children are often squeezed into overcrowded classrooms, with those in early grades particularly disadvantaged. In Malawi, there are 130 children per class in grade 1, on average, compared with 64 in the last grade. In Chad, just one in four schools has a toilet, and only one in three of those toilets is reserved for girls' use.

With 250 million children not learning the basics, it is vital for a global post-2015 goal to be set which will monitor whether, by 2030, all children and youth, regardless of their circumstances, acquire foundation skills in reading, writing and mathematics. Meeting this need requires countries to strengthen their national assessment systems and ensure that they are used to inform policy. Many national assessment systems are lacking in this respect. Governments often consider their public examination system as equivalent to a national assessment system, even though it is mainly used to promote students between levels of education. National assessments should be a diagnostic tool that can establish whether students achieve the learning standards expected by a particular age or grade, and how this achievement changes over time for subgroups of the population.

Regional and international assessments are critical for monitoring a global learning goal after 2015. Just as improved global monitoring of access has helped maintain pressure on governments to ensure that each child completes primary school, better global monitoring of learning can push governments to make certain that all children not only go to school but are achieving the basics.

For these assessments to facilitate monitoring post-2015 global learning goals, three key principles need to be adhered to. First, all children and young people need to be taken into account when interpreting results, not just those who were in school and took part in the assessment. Disadvantaged children may already be out of the school system, and therefore unlikely to have reached minimum learning standards, by the time the assessment is administered. Not counting them means that the scale of the problem is understated. Second, better information on background characteristics of students is needed to identify which groups of students are not learning. Third, information on the quality of education systems should also be included as part of the assessments.

## Leaving no one behind — how long will it take?

After 2015, unfinished business will remain across the six EFA goals, while new priorities are likely to emerge. Analysis of the time it will take to achieve universal primary and lower secondary school completion and youth literacy paints a worrying picture.

While rich boys are expected to reach universal primary completion by 2030 in 56 of 74 low and middle income countries, poor girls will reach the goal by that date in only 7 countries, just one of which is low income. Even by 2060, poor girls will not have reached the goal in 24 of the 28 low income countries in the sample. In sub-Saharan Africa, if recent trends continue, the richest boys will achieve universal primary completion in 2021, but the poorest girls will not catch up until 2086.

Spending time in primary school is no guarantee that a child will be able to read and write. Among 68 countries with data, the poorest young women are projected to achieve universal literacy only in 2072.

The situation is even more dire for lower secondary school completion. In 44 of the 74 countries analysed, there is at least a 50-year gap between when all the richest boys complete lower secondary school and when all the poorest girls do so. And, if recent trends continue, girls from the poorest families in sub-Saharan Africa will only achieve this target in 2111, 64 years later than the boys from the richest families.

While these projections are extremely disconcerting, they are based on recent trends that can be changed if governments, aid donors and the international education community take concerted action to make education available to all, including the marginalized. The projections also show how vital it will be to track progress towards education goals for the most disadvantaged groups after 2015 and to put policies in place that maintain and accelerate progress by redressing imbalances.

## Monitoring global education targets after 2015

Since the six Education for All goals were adopted in Dakar, Senegal, in 2000, a lack of precise targets and indicators has prevented some education priorities from receiving the attention they deserve.

The shape of new goals after 2015 should be guided by the principles of upholding education as a right, making sure all children have an equal chance of education and recognizing the learning stages at each phase of a person's life. There should be one core set of goals aligned with the global development agenda, accompanied by a more detailed set of targets that make up a post-2015 education framework. Each goal must be clear and measurable, with the aim of ensuring that no one is left behind. To achieve this, progress should be tracked by the achievements of the lowest performing groups, making sure the gap between them and the better-off is narrowing.

The number of years young people spend in school is one measure of overall progress in access to education. To achieve a goal of universal lower secondary completion by 2030, young people will need to stay in school about nine years. By 2010, the richest urban young men had already spent more than 9.5 years in school in low income countries, on average, and more than 12 years in lower middle income countries. But the poorest young women in rural areas had spent less than 3 years in school in both low and lower middle income countries, leaving them well below the 6-year target that is associated with universal primary completion, which is supposed to be achieved by 2015. In sub-Saharan Africa, the gap between the time the poorest rural females and the richest urban males spent in school actually widened between 2000 and 2010, from 6.9 years to 8.3 years.

Over the past decade there has been more progress in getting children into primary school than in ensuring that children complete primary or lower secondary education. And extreme inequality persists and in some cases has widened. In sub-Saharan Africa, for example, almost all the richest boys in urban areas were entering school in 2000. By the end of the decade, their primary school completion rate

**In sub-Saharan Africa, the poorest girls will not achieve universal primary completion until 2086**

**Only 8 of 53 countries plan to monitor inequality in learning**

had reached 87%, and their lower secondary school completion rate 70%. By contrast, among the poorest girls in rural areas, 49% were entering school at the beginning of the decade and 61% by the end of the decade. Only 25% were completing primary education and 11% lower secondary education in 2000; furthermore, these rates had fallen by the end of the decade to 23% and 9%, respectively. Inequality in South and West Asia is also wide and largely unchanged: by the end of the decade, while 89% of the richest urban males completed lower secondary school, only 13% of the poorest rural girls did so.

A goal on learning is an indispensable part of a future global education monitoring framework, but focusing only on learning assessments can be misleading if large numbers of children never make it to the grade where skills are tested. In the United Republic of Tanzania, for example, the proportion of children in grade 6 who achieved a minimum standard in reading in 2007 ranged from 80% of the poorest rural girls to 97% of the richest urban boys. However, while 92% of the richest urban boys of grade 6 age had reached that grade, only 40% of the poorest rural girls had done so. If it is assumed that children who did not reach grade 6 could not have achieved the minimum standard, then the proportions among children of that cohort who learned the basics were 90% for the richest urban boys and 32% for the poorest rural girls.

To make sure inequality is overcome by 2030, country plans need to include specific targets so that education participation and learning can be monitored for individual population groups. Very few currently do so. Gender is the disadvantage most frequently covered in education plans, yet only 24 of 53 country plans reviewed for this report included gender equality targets in primary and lower secondary education.

Four plans included indicators on participation of particular ethnic groups. Only three specifically targeted disparity in access between rural and urban areas in primary and lower secondary education. Moreover, only three plans had an enrolment indicator that differentiated between poorer and richer children. Bangladesh's plan included a monitoring framework for tracking progress in enrolment ratios across wealth quintiles, and Namibia's included a target of making sure 80% of orphans and other vulnerable children in each region were enrolled in primary and secondary education by the final year of the plan.

Even fewer countries plan to monitor inequality in learning outcomes. Only 8 of the 53 countries did so at the primary level and 8 at the lower secondary level, and in most cases monitoring was restricted to gender inequality. Sri Lanka is one exception, with targets for achievement scores in mathematics and native language in the lowest performing regions.

The failure over the past decade to assess progress in education goals by various population subgroups has concealed wide inequality. Its invisibility is further reflected in country plans' lack of national targets for assessing progress in narrowing gaps in access or learning. Post-2015 goals need to include a commitment to making sure the most disadvantaged groups achieve benchmarks set for goals. Failure to do so could mean that measurement of progress continues to mask the fact that the most advantaged benefit the most.

## Monitoring progress on financing Education for All

Insufficient financing is one of the main obstacles to achieving Education for All. The finance gap to achieve good quality basic education for all by 2015 has reached US$26 billion, putting the goal of getting every child into school far out of reach. Unfortunately, donors seem more likely to reduce their aid than increase it in coming years. Unless urgent action is taken to change aid patterns, the goal of ensuring that every child is in school and learning by 2015 will be seriously jeopardized.

With little time left before 2015, closing the financing gap might seem impossible. But analysis in this Report shows that the gap could be filled by raising more domestic revenue, devoting an adequate share of existing and projected government resources to education and sharpening the focus of external assistance.

If, as expected, new education goals after 2015 extend to lower secondary education, the finance gap will rise to US$38 billion. The post-2015 framework must include explicit financing targets, demanding full transparency, so that all donors are accountable for their commitments, and finance gaps do not thwart our promises to children.

## Many countries far from EFA need to spend more on education

Domestic spending on education has increased in recent years, particularly in low and lower middle income countries, partly because of improvements in economic growth. Government spending on education increased from 4.6% to 5.1% of gross national product (GNP) between 1999 and 2011, on average. In low and middle income countries it rose faster: 30 of these countries increased their spending on education by one percentage point of GNP or more between 1999 and 2011.

The Dakar Framework for Action did not establish how much countries should commit to education. The failure to set a common financing target for the EFA goals should be addressed after 2015, with a specific goal set: that countries should allocate at least 6% of GNP to education. Of the 150 countries with data, only 41 spent 6% or more of GNP on education in 2011, and 25 countries dedicated less than 3%.

It is widely accepted that countries should allocate at least 20% of their budget to education. Yet the global average in 2011 was only 15%, a proportion that has hardly changed since 1999. Of the 138 countries with data, only 25 spent more than 20% in 2011, while at least 6 low and middle income countries decreased their education expenditure as a share of total government expenditure by 5 percentage points or more between 1999 and 2011.

This situation is not expected to improve in coming years. Of 49 countries with data in 2012, 25 planned to shrink their education budget between 2011 and 2012. Of these, 16 were in sub-Saharan Africa. However, some countries, including Afghanistan, Benin and Ethiopia, are resisting this negative trend and are expected to increase their education budgets.

To tap into the potential for economic growth in many of the world's poorest countries, governments need to expand their tax base and devote a fifth of their budget to education. If governments in 67 low and middle income countries did this, they could raise an additional US$153 billion for education in 2015. That would increase the average share of GDP spent on education from 3% to 6% by 2015.

Few poor countries manage to raise 20% of their GDP in taxes, as needed to achieve the Millennium Development Goals. Only 7 of the 67 countries with data both generate 20% of GDP in taxes and allocate the recommended 20% of the revenue to education. In Pakistan, tax revenue is just 10% of GDP and education receives only around 10% of government expenditure. If the government increased its tax revenue to 14% of GDP by 2015 and allocated one-fifth of this to education, it could raise sufficient funds to get all of Pakistan's children and adolescents into school.

Ethiopia is one of 11 among the 67 countries that have been successful in prioritizing education in the government budget but could do far more to maximize revenue from taxation. In 2011, the government received 12% of GDP from taxes, on average. If the proportion were to increase to 16% by 2015, the sector would receive 18% more resources – enabling US$19 more to be spent per primary school age child.

**67 countries could increase education resources by US$153 billion through reforms to expand the tax base**

Tax revenue as a share of GDP is, however, growing far too slowly in poorer countries. At present rates, only 4 of the 48 countries currently raising less than 20% of GDP in tax would reach the 20% threshold by 2015.

A well-functioning taxation system enables governments to support their education system with domestic finance. Some middle income countries, such as Egypt, India and the Philippines, have far greater potential to mobilize domestic resources for education through improved taxes. Higher levels of tax revenue in Brazil help explain how it spends ten times as much as India per primary school child.

Some countries in South Asia grant large tax exemptions to strong domestic interest groups, resulting in some of the lowest tax-to-GDP ratios in the world. In Pakistan, the tax/GDP ratio of 10% can be partly explained by the political influence of the agricultural lobby. While the agricultural sector makes up 22.5% of Pakistan's GDP, its share in tax revenue is just 1.2%. In

**Reforms to expand domestic resources could meet 56% of the annual financing gap in basic education**

India, the majority of tax revenue forgone is due to exemptions from custom and excise duties. The revenue lost to exemptions came to the equivalent of 5.7% of GDP in 2012/13. If 20% of this had been earmarked for education, the sector would have received an additional US$22.5 billion in 2013, increasing funding by almost 40% compared with the current education budget.

Some governments sell concessions to exploit natural resources for less than their true value. The Democratic Republic of the Congo lost US$1.36 billion from deals with mining companies over three years in 2010 to 2012, equal to the amount allocated to education over two years in 2010 and 2011.

For many of the world's poorest countries, tax evasion results in the elite building personal fortunes, rather than strong education systems for the majority. If the trillions of dollars estimated to be hidden away in tax havens were subject to capital gains tax, and 20% of the resulting income was allocated to education, it would add between US$38 billion and US$56 billion to funding for the sector.

Illegal tax practices cost African governments an estimated US$63 billion a year. If these practices were halted and 20% of the resulting income spent on education, it would raise an additional US$13 billion for the sector each year.

While governments must lead the drive to reform taxation, donors can play an important complementary role. Just US$1 of donor aid to strengthen tax regimes, for example, can generate up to US$350 in tax revenue. Yet less than 0.1% of total aid was spent supporting tax programmes between 2002 and 2011. In addition, donor country governments should demand transparency from corporations registered in their countries.

Increasing tax revenue and allocating an adequate share to education could raise considerable extra resources for the sector in a short time. The EFA Global Monitoring Report team estimates that 67 low and middle income countries could increase education resources by US$153 billion, or 72%, by 2015 through reforms to raise tax/GDP ratios and public expenditure on education.

On average across the 67 countries, spending per primary school age child would increase from US$209 to US$466 in 2015. In the low income countries among the 67, the average amount spent per primary school age child would increase from US$102 to US$158.

Fourteen of these 67 countries have already reached the proposed target of spending at least 6% of GDP on education. Of the 53 yet to reach the target, 19 could achieve it if they expanded and diversified the tax base and prioritized education spending by 2015.

These additions to domestic resources could meet 56% of the US$26 billion average annual financing gap in basic education for 46 low and lower middle income countries, or 54% of the US$38 billion gap in basic and lower secondary education.

Such reforms are not without precedent. Ecuador renegotiated contracts with oil companies, widened its tax base and made education a higher priority, tripling its education expenditure between 2003 and 2010.

To achieve Education for All, it is necessary not only to increase domestic resources for education but also to redistribute these resources so that a fair share reaches those most in need. More often, however, resources are skewed towards the most privileged. To shift education spending in favour of the marginalized, many governments have introduced funding mechanisms that allocate more resources to parts of the country or groups of schools that need greater support to overcome educational deprivation and inequality. Countries have different methods of redistributing resources. Brazil, for example, guarantees a certain minimum spending level per pupil, giving priority to schools in rural areas, with greater weight on highly marginalized indigenous groups. The reforms have led to improvement in enrolment and learning in the disadvantaged north of the country.

Other redistribution programmes have been less successful, however. One reason is that allocations per child still do not adequately reflect the costs of delivering quality education to the marginalized. In one of India's wealthier states, Kerala, education spending per pupil was about US$685. By contrast, in the poorer state of Bihar it was just US$100.

Poorer countries can find it difficult to identify and target the groups most in need. As a result, many base allocations on enrolment figures, to the detriment of areas where large numbers of children are out of school. In Kenya, for example, the capitation grant is distributed on the basis of number of students enrolled, a disadvantage for the 12 counties in the arid and semi-arid areas that are home to 46% of the out-of-school population.

To realize the full potential of redistributive measures, governments need to ensure that such resources cover the entire cost of a quality education for the most vulnerable and that far-reaching reforms strengthen education systems' capacity to implement such measures.

## Trends in aid to education

With improvement in the numbers of children out of school stagnating, a final push is needed to ensure that all children are in school by 2015. Even before the economic downturn, donors were off track to fulfil their education finance promises. A more recent decline in aid to basic education increases the difficulty of this task.

While aid to education rose steadily after 2002, it peaked in 2010 and is now falling: total aid to all levels of education declined by 7% between 2010 and 2011. Aid to basic education fell for the first time since 2002, by 6%: from US$6.2 billion in 2010 to US$5.8 billion in 2011. Aid to secondary education declined by 11% between 2010 and 2011 from an already low level. This puts at risk the chance of meeting Education for All goals and any hope of more ambitious goals to include universal lower secondary education after 2015.

Low income countries, which only receive around one-third of aid to basic education, witnessed a larger decrease in aid to basic education than middle income countries. Aid fell by 9% in low income countries between 2010 and 2011, from US$2.05 billion to US$1.86 billion. As a result, the resources available per child dropped from US$18 in 2010 to US$16 in 2011.

In sub-Saharan Africa, home to over half the world's out-of-school population, aid to basic education declined by 7% between 2010 and 2011. The US$134 million reduction in basic education aid to the region would have been enough to fund good quality school places for over 1 million children.

**In 2011, aid to basic education fell for the first time since 2002**

In some countries, aid has been falling for more than a year. Aid has played a key part in supporting efforts to get more children into school in the United Republic of Tanzania, but it fell by 12% between 2009 and 2010 and by a further 57% in 2011.

The world's poorest children, who are the most likely to be out of school, live not only in low income countries but also in some lower middle income countries. Since 2000, 25 countries have graduated to this group, which now comprises 54 countries, while 36 are classified as low income. In 1999, 84% of the world's out-of-school children lived in low income countries and 12% in lower middle income countries; by 2011, 37% lived in low income countries and 49% in lower middle income countries. This shift was largely due to the graduation to lower middle income status of some large-population countries, such as India, Nigeria and Pakistan. While these countries could do far more to raise their own resources for education and distribute them to those most in need, such reforms will take time. In the meantime, donors should direct aid to the areas in lower middle income countries where poverty is concentrated to make sure another generation of children in these countries is not denied its right to education.

Direct aid to education fell slightly more than overall aid to other sectors between 2010 and 2011, and thus the share of aid to education declined from 12% to 11%. The fall in aid to education reflects the changing spending patterns of a large number of donors. Canada, France, the Netherlands and the United States, in particular, cut spending on education more than they reduced overall aid. Between 2010 and 2011, 21 bilateral and multilateral donors reduced their aid disbursements to basic education. The largest decreases in volume terms were by Canada, the European Union, France, Japan, the Netherlands, Spain and the United States, which together accounted for 90% of the reduction in aid to basic education.

The United States moved from being the largest bilateral donor to basic education in 2010 to second place in 2011. The United Kingdom is now the largest donor to education. The Netherlands' aid to basic education fell by over a third between 2010 and 2011; it had been the largest donor to basic education in 2007, but by 2011 was in 11th place.

Australia, the IMF and the World Bank increased their overall aid to basic education between 2010 and 2011 but reduced their spending in low income countries. World Bank aid to basic education increased by 13% overall, but fell by 23% in low income countries. The United Republic of Tanzania saw World Bank disbursements fall from US$88 million in 2002 to less than US$0.3 million in 2011.

The Global Partnership for Education (GPE) is an important source of financing for some low income countries. In 2011, the GPE disbursed a record US$385 million to basic education, making it the fourth largest donor to low and lower middle income countries that year. For the 31 countries with a programme implementation grant in 2011, 24% of basic education aid was disbursed by the GPE. It is unlikely, however, to have made up for the reduction in World Bank spending. The United Republic of Tanzania became a GPE partner in 2013 with a US$5.2 million grant for its education plan. All this, however, is allocated to Zanzibar, and the amount is small compared with what the country received from the World Bank early in the 2000s. To improve monitoring of its contributions, the GPE needs to report its aid flows to the OECD-DAC, just as global health funds such as the GAVI Alliance and the Global Fund to Fight AIDS, Tuberculosis and Malaria do.

**Education receives just 1.4% of humanitarian aid**

There is no sign that overall aid will stop declining before the 2015 deadline. From 2011 to 2012, total aid decreased by 4%, with 16 DAC donors decreasing their aid: 13 DAC donors made aid a lower priority by decreasing aid as a proportion of gross national income (GNI). Less developed countries are expected to bear the brunt of these reductions, with cuts to their bilateral aid of 12.8% from 2011 to 2012. In 2013, aid is expected to fall in 31 of 36 low income countries, the majority of which are in sub-Saharan Africa.

In addition, only five of the fifteen members of the European Union that agreed to increase their aid to 0.7% of GNI by 2015 are expected to meet their commitment. If these countries met their promises, they would raise US$9 billion more for education in 2015.

Education in conflict-affected countries should be a priority for donors. These countries house half of the world's out-of-school children. Currently education receives just 1.4% of humanitarian aid, far from the 4% called for by the UN Secretary-General's Global Education First Initiative. In plans for 2013, education's share of overall humanitarian aid is likely to reach no more than 2%.

In Mali, where most schools in the north remained closed due to conflict, education made up 5% of appeal requests for 2013, but just 15% of the requested funds had been pledged by September 2013. Similarly, 36% of the resources requested in 2013 for education in the Syrian Arab Republic had actually been pledged. While these countries might receive more of the requested funds later in 2013, they would come too late for the millions of children who have had to drop out of school due to conflict.

It is not only the amount of aid that counts but also whether it is used to target the most disadvantaged. These children do not receive all the aid available, however: a quarter of direct aid to education is spent on students studying in universities in rich countries. Even though scholarships and imputed student costs may be vital to strengthen human resource capacity in low income countries, most of this funding actually goes to upper middle income countries, with China the largest recipient, receiving 21% of the total.

On average over 2010–2011, donors – primarily Germany and Japan – disbursed US$656 million per year to China for scholarships and student imputed costs, which was 77 times the amount of aid disbursed to Chad for basic education over the same period. The total funding in the form of imputed student costs and scholarships received annually by Algeria, China, Morocco, Tunisia and Turkey was equivalent to the total amount of direct aid to basic education for all 36 low income countries in 2010–2011, on average.

Aid can also be delivered on unfavourable terms for poorer countries: 15% of aid is in the form of loans that countries have to pay back at concessional interest rates. Without funding from bilateral donors, poorer countries risk becoming dependent on these loans, leading to debt that could restrict their ability to fund education from their own resources.

Removing imputed student costs, scholarships and loans, Germany would drop from being the largest donor in direct aid to education to being the fifth largest; the World Bank would fall from third to fourteenth place. The United Kingdom and United States, by contrast, would jump from sixth and seventh position to first and second.

Information on the whole spectrum of education financing – including aid, domestic resources and household spending – is often insufficient and fragmented, resulting in only partial analysis and diagnosis of how much money is needed and where. New analysis from seven countries shows that households bear up to 37% of education expenditure in primary education and up to 58% in secondary education, which places a particular burden on the poorest households. In addition, it shows how vital aid is to education financing in some of the poorest countries: it accounts for almost a quarter of education spending in Malawi and Rwanda. These findings highlight the importance of building a comprehensive national education accounts system, which could be modelled on the experience in health.

To avoid failing another generation of children due to lack of resources after 2015, national governments aid donors need to be held to account for their commitments to provide the resources necessary to reach education goals. Drawing on analysis included in *EFA Global Monitoring Reports* over the years, the Report team proposes that a target should be set for national governments to allocate at least 6% of their GNP to education. Targets for governments and aid donors should also include commitments for them to spend at least 20% of their budgets on education. Setting these targets, and making sure governments and aid donors keep to them, will be an important contribution to the education opportunities of children and young people in the future.

## Education transforms lives

Treaties and laws worldwide recognize that education is a fundamental human right. In addition, education imparts knowledge and skills that enable people to realize their full potential, and so it becomes a catalyst for the achievement of other development goals. Education reduces poverty, boosts job opportunities and fosters economic prosperity. It also increases people's chances of leading a healthy life, deepens the foundations of democracy, and changes attitudes to protect the environment and empower women.

Educating girls and women, in particular, has unmatched transformative power. As well as boosting their own chances of getting jobs, staying healthy and participating fully in society, educating girls and young women has a marked impact on the health of their children and accelerates their countries' transition to stable population growth.

To unlock the wider benefits of education and achieve development goals after 2015, it needs to be equitable and to extend at least to lower secondary school. And the schooling that children receive needs to be of good quality so that they actually learn the basics.

### Education reduces poverty and boosts jobs and growth

Education is a key way of helping individuals escape poverty and of preventing poverty from being passed down through the generations. It enables those in paid formal employment to earn higher wages and offers better livelihoods for those who work in agriculture and the urban informal sector.

**If all students left school with basic reading skills, 171 million people could be lifted out of poverty**

EFA Global Monitoring Report team calculations show that if all students in low income countries left school with basic reading skills, 171 million people could be lifted out of poverty, which would be equivalent to a 12% cut in world poverty. An important way education reduces poverty is by increasing people's income. Globally, one year of school increases earnings by 10%, on average.

Education can help people escape from working poverty. In the United Republic of Tanzania, 82% of workers who had less than primary education were below the poverty line. But working adults with primary education were 20% less likely to be poor, while secondary education reduced the chances of being poor by almost 60%. It is not just time in school, but skills acquired that count. In Pakistan, working women with good literacy skills earned 95% more than women with weak literacy skills.

In the formal sector, higher wages reflect the higher productivity of more educated workers. But many of the poorest are involved in informal sector work, running small businesses. Educated people are more likely to start a business, and their businesses are likely to be more profitable. In Uganda, owners of household enterprises with a primary education earned 36% more than those with no education; those with a lower secondary education earned 56% more.

In rural areas, farmers with good literacy and numeracy skills can interpret and respond to new information, making better use of modern inputs and technologies to increase the productivity of traditional crops and diversify into higher value crops. In Mozambique, literate farmers were 26 percentage points more likely than non-literate ones to cultivate cash crops.

Education also helps people in rural areas diversify their income by involvement in off-farm work. In rural Indonesia, 15% of men and 17% of women with no education were employed in non-farm work, compared with 61% of men and 72% of women with secondary education.

**For each additional year of mothers' education, a child spends an extra 0.32 years in school**

One of the benefits of increased education is that educated parents are likely to have more educated children. Analysis of household surveys from 56 countries finds that, for each additional year of the mother's education, the average child attains an extra 0.32 years, and for girls the benefit is slightly larger.

By benefiting women in particular, education can help narrow gender gaps in work opportunities and pay. In Argentina and Jordan, for instance, among people with primary education, women earned around half the average wage of men, while among those with secondary education, women earned around two-thirds as much as men.

Education helps protect working adults from exploitation by increasing their opportunities to obtain secure contracts. In urban El Salvador, only 7% of working adults with less than primary education had an employment contract, compared with 49% of those with secondary education.

Education not only facilitates individuals' escape from poverty, but also generates productivity that fuels economic growth. A one-year increase in the average educational attainment of a country's population increases annual per capita GDP growth from 2% to 2.5%.

Education can help explain differences in regional growth trajectories. In 1965, adults in East Asia and the Pacific had spent 2.7 more years in school than those in sub-Saharan Africa. Over the next 45 years, the average growth rate was more than four times faster in East Asia and the Pacific.

Comparing experiences within regions further illustrates the importance of education. In Guatemala, adults had 3.6 years of schooling, on average, in 2005 and the average level increased by only 2.3 years from 1965 to 2005. If the country had matched the average for Latin America and the Caribbean, where the average number of years that adults spent at school rose from 3.6 in 1965 to 7.5 in 2005, it could have more than doubled its average annual growth rate between 2005 and 2010, equivalent to an additional US$500 per person.

Only by investing in equitable education – making sure that the poorest complete more years in school – can countries achieve the kind of growth that banishes poverty. Equality in education can be measured by the Gini coefficient, which ranges from zero, indicating perfect equality, to one, indicating maximum inequality. An improvement in the Gini coefficient by 0.1 accelerates growth by half a percentage point, increasing income per capita by 23% over 40 years. If sub-Saharan Africa's education Gini coefficient of 0.49 had been halved to the level in Latin America and the Caribbean, the annual growth rate in GDP per capita over 2005–2010 could have risen by 47% (from 2.4% to 3.5%) and income could have grown by US$82 per capita over the period.

Comparing Pakistan and Viet Nam illustrates starkly the importance of equitable education. In 2005, the average number of years adults had spent at school was similar: 4.5 in Pakistan and 4.9 in Viet Nam. Education levels, however, were very unequally distributed in Pakistan: Pakistan's Gini coefficient for education inequality was more than double the level in Viet Nam. The difference in education inequality between the countries accounts for 60% of the difference in their per capita growth between 2005 and 2010. Viet Nam's per capita income, which was around 40% below Pakistan's in the 1990s, not only caught up with Pakistan's but was 20% higher by 2010.

## Education improves people's chances of a healthier life

Education is one of the most powerful ways of improving people's health. It saves the lives of millions of mothers and children, helps prevent and contain disease, and is an essential element of efforts to reduce malnutrition. Educated people are better informed about diseases, take preventative measures, recognize signs of illness early and tend to use health care services more often. Despite its benefits, education is often neglected as a vital health intervention in itself and as a means of making other health interventions more effective.

There are few more dramatic illustrations of the power of education than the estimate that the lives of 2.1 million children under 5 were saved between 1990 and 2009 because of improvements in the education of women of reproductive age. However, the challenge remaining is enormous. In 2012, 6.6 million children under 5 died, most of them in low and lower middle income countries. If all women completed primary education in these countries, the under-5 mortality rate would fall by 15%. If all women completed secondary education, it would fall by 49%, equal to around 2.8 million lives a year.

Around 40% of all under-5 deaths occur within the first 28 days of life, the majority being due to complications during delivery. Yet the most recent estimates suggest there were no skilled birth attendants present in over half the births in sub-Saharan Africa and South Asia. Across 57 low and middle income countries, a literate mother was 23% more likely to have a skilled attendant at birth than an illiterate mother. In Mali, maternal literacy more than tripled this likelihood.

Educated mothers are better informed about specific diseases, so they can take measures to prevent them. Pneumonia is the largest cause of child deaths, accounting for 17% of the total worldwide. One additional year of maternal education can lead to a 14% decrease in the pneumonia death rate – equivalent to 160,000 child lives saved every year. Diarrhoea is the fourth biggest child killer, accounting for 9% of child deaths. If all women completed primary education, the incidence of diarrhoea would fall by 8% in low and lower middle income countries; with secondary education, it would fall by 30%. The probability of a child being immunized against diphtheria, tetanus and whooping cough would increase by 10% if all women in low and lower middle income countries completed primary education, and by 43% if they completed secondary education.

**If all women completed primary education, there would be 66% fewer maternal deaths**

A mother's education is just as crucial for her own health as it is for her offspring's. Every day, almost 800 women die from preventable causes related to pregnancy and childbirth. If all women completed primary education, there would be 66% fewer maternal deaths, saving 189,000 lives per year. In sub-Saharan Africa alone, if all women completed primary education, there would be 70% fewer maternal deaths, saving 113,400 women's lives.

Some countries have seen considerable gains. Thanks to education reforms in the 1970s, the average amount of schooling young women received increased by 2.2 years in Nigeria. This accounted for a 29% reduction in the maternal mortality rate.

Improving education is a powerful way to help reduce the incidence of infectious diseases such as HIV/AIDS. Education helps increase awareness about HIV prevention, for example. In South and West Asia and sub-Saharan Africa, literate women were as much as 30 percentage points more likely than those who were not literate to be aware that they had the right to refuse sex or request condom use if they knew that their partner had a sexually transmitted disease. Knowing where to get tested for HIV is a first step to receiving treatment if needed. But only 52% of women who were not literate in sub-Saharan Africa knew where to get tested for HIV, compared with 85% of the literate.

Malaria is one of the world's deadliest diseases, killing one child every minute in Africa. Improved access to education is crucial in ensuring the effectiveness of preventative measures, such as the use of drugs or bed nets treated with insecticide. In the Democratic Republic of the Congo, where a fifth of the world's malaria-related deaths occur, the education of the household head or the mother increased the probability that the family slept under a bed net. Such changes result in fewer infections, especially in areas of high transmission risk. In these areas, the odds of children having malaria parasites were 22% lower if their mothers had primary education than if they had no education, or 36% less if they had secondary education.

Education – especially education that empowers women – is key to tackling malnutrition, the underlying cause of more than 45% of child deaths. Educated mothers are more likely to know about appropriate health and hygiene practices at home, and have more power to ensure that household resources are allocated so as to meet children's nutrition needs. In low and lower middle income countries, providing all women with a primary education would reduce stunting – a robust indicator of malnutrition – by 4%, or 1.7 million children; providing a secondary education would reduce stunting by 26%, or 11.9 million children.

By age 1, adverse effects of malnutrition on life prospects are likely to be irreversible. Infants in Peru whose mothers had reached lower secondary education were 60% less likely to be stunted than children whose mothers had no education.

## Education promotes healthy societies

Education helps people understand democracy, promotes the tolerance and trust that underpin it, and motivates people to participate in politics. Education also has a vital role in preventing environmental degradation and limiting the causes and effects of climate change. And it can empower women to overcome discrimination and assert their rights.

Education improves people's understanding of politics and how to participate in it. Across 12 sub-Saharan African countries, 63% of individuals without formal schooling had an understanding of democracy, compared with 71% of those with primary education and 85% of those with secondary. People with higher levels of education are more interested in politics and so more likely to seek information. In Turkey, for example, the share of those who said they were interested in politics rose from 40% among those

Credit: Nguyen Thanh Tuan/UNESCO

with primary education to 52% among those with secondary education.

Education increases people's support for democracy, particularly where there have been recent democratic transitions. Across 18 sub-Saharan African countries, those of voting age with primary education were 1.5 times more likely to express support for democracy than those with no education, and twice as likely if they had completed secondary education.

Educated people are more likely to vote. In 14 Latin American countries, turnout was five percentage points higher for those with primary education and nine percentage points higher for individuals with secondary education compared to those with no education. The effect was larger in countries where average levels of education were lower, such as El Salvador, Guatemala and Paraguay. Education also encourages other forms of political participation. In Argentina, China and Turkey, citizens were twice as likely to sign a petition or boycott products if they had secondary education than if they just had primary education.

Education has an indispensable role in strengthening the bonds that hold communities and societies together. In Latin America, people with secondary education were 47% less likely than those with primary education to express intolerance for people of a different race. In the Arab States, people with secondary education were 14% less likely than those with only primary education to express intolerance towards people of a different religion. In sub-Saharan Africa, people with primary education were 10% less likely to express intolerance towards people with HIV and 23% less likely if they had a secondary education. In Central and Eastern Europe, those with secondary education were 16% less likely than those who had not completed secondary education to express intolerance towards immigrants.

Education also helps overcome gender biases in political behaviour to deepen democracy. In India, reducing the gender literacy gap by 40% increased the probability of women standing for state assembly election by 16% and the share of votes that they received by 13%.

Education lowers tolerance towards corruption and helps build accountability. In 31 countries, those with secondary education were one-sixth more likely than average to complain about deficient government services.

Increasing access to school for all generally reduces feelings of injustice in society that have fuelled many conflicts. But it needs to increase equally for all population groups; otherwise, perceived unfairness can reinforce disillusionment. In 55 low and middle income countries where the level of educational inequality doubled, the probability of conflict more than doubled, from 3.8% to 9.5%.

By improving knowledge, instilling values, fostering beliefs and shifting attitudes, education has considerable potential to change environmentally harmful lifestyles and behaviour. A key way in which education can increase environmental awareness and concern is by improving understanding of the science behind climate change and other environmental issues. Students with higher science scores across 57 countries reported being more aware of complex environmental issues. Similarly, across 29 mostly high income countries, 25% of people with less than secondary education expressed concern for the environment, compared with 37% of people with secondary education and 46% of people with tertiary education.

**In Argentina, China and Turkey, citizens are twice as likely to sign a petition if they have secondary education**

Education is also critical for helping people adapt to the consequences of climate change, especially in poorer countries, where threats to livelihoods are being felt most strongly by farmers dependent on rain-fed agriculture. In Ethiopia, six years of education increased by 20% the chance that a farmer would adapt to climate change through techniques such as practising soil conservation, varying planting dates and changing crop varieties.

Education can empower women to claim their rights and overcome barriers that prevent them from getting a fair share of the fruits of overall progress. Having the freedom to choose one's spouse is one such right. Women in India with at least secondary education were 30 percentage points more likely to have a say over their choice of spouse than their less educated peers.

Likewise, ensuring that girls stay in school is one of the most effective ways to prevent child marriage. If all girls completed primary school in sub-Saharan Africa and South and West Asia, the number of girls getting married by age 15 would fall by 14%; with secondary education, 64% fewer girls would get married.

Staying in school longer also gives girls more confidence to make choices that avert the health risks of early births and births in quick succession. Currently one in seven girls have children before age 17 in sub-Saharan Africa and South and West Asia. In these regions, 10% fewer girls would become pregnant if they all had primary education, and 59% fewer would if they all had secondary education. This would result in around 2 million fewer early births.

Women with more education tend to have fewer children, which benefits them, their families and society more generally. One reason for this is because education allows women to have a greater influence on family size. In Pakistan, only 30% of women with no education believe they have a say over how many children they have, compared with 52% of women with primary education and 63% of those with lower secondary education.

In some regions, education has been a key factor in bringing forward the demographic transition. Other parts of the world are lagging, however, particularly sub-Saharan Africa, where women average 5.4 live births. In the region, women with no education have 6.7 births. The figure falls to 5.8 for those with primary education and to 3.9 for those with secondary education.

**Education's power to transform lives should secure it a central place in the post-2015 framework**

### Conclusion

The striking evidence laid out in this chapter demonstrates not only education's capacity to accelerate progress towards other development goals, but also how best to tap that potential, most of all by making sure that access to good quality education is available to all, regardless of their circumstances. Education's unique power should secure it a central place in the post-2015 development framework, and in the plans of policy-makers in poor and rich countries alike.

## Supporting teachers to end the learning crisis

A lack of attention to education quality and a failure to reach the marginalized have contributed to a learning crisis that needs urgent attention. Worldwide, 250 million children – many of them from disadvantaged backgrounds – are not learning even basic literacy and numeracy skills, let alone the further skills they need to get decent work and lead fulfilling lives.

To solve the learning crisis, all children must have teachers who are trained, motivated and enjoy teaching, who can identify and support weak learners, and who are backed by well-managed education systems.

As this Report shows, governments can increase access while also making sure that learning improves for all. Adequately funded national education plans that aim explicitly to meet the needs of the disadvantaged and that ensure equitable access to well-trained teachers must be a policy priority. Attracting and retaining the best teachers as a means to end the learning crisis requires a delicate juggling act on the part of policy-makers.

To ensure that all children are learning, teachers also need the support of an appropriate curriculum and assessment system that pays particular attention to the needs of children in early grades, when the most vulnerable are in danger of dropping out. Beyond teaching the basics, teachers must help children gain important transferable skills to help them become responsible global citizens.

### The learning crisis hits the disadvantaged hardest

Despite impressive gains in access to education over the past decade, improvements in quality have not always kept pace. Many countries are not ensuring that their children achieve even the most basic skills in reading and mathematics. The disadvantaged are the most likely to suffer because of insufficient numbers of trained teachers, overstretched infrastructure and inadequate materials. Yet it is possible for countries to extend access to school while also improving equitable learning.

*The global learning crisis: action is urgent*

Of the world's 650 million primary school age children, at least 250 million are not learning the basics in reading and mathematics. Of these, almost 120 million have little or no experience of primary school, having not even reached grade 4. The remaining 130 million are in primary school but have not achieved the minimum benchmarks for learning. Often unable to understand a simple sentence, these children are ill equipped to make the transition to secondary education.

There is a vast divide between regions in learning achievement. In North America and Western Europe, 96% of children stay in school until grade 4 and achieve the minimum reading standards, compared with only one-third of children in South and West Asia and two-fifths in sub-Saharan Africa. These two regions account for more than three-quarters of those not crossing the minimum learning threshold.

The learning crisis is extensive. New analysis shows that less than half of children are learning the basics in 21 out of the 85 countries with full data available. Of these, 17 are in sub-Saharan Africa; the others are India, Mauritania, Morocco and Pakistan.

This learning crisis has costs not only for the future ambitions of children, but also for the current finances of governments. The cost of 250 million children not learning the basics is equivalent to US$129 billion, or 10% of global spending on primary education.

*Global disparities mask huge inequalities within countries*

While average figures on learning achievement provide an overall picture of the scale of the learning crisis, they can conceal large disparities within countries. Poverty, gender, location, language, ethnicity, disability and other factors mean some children are likely to get less support from schools to improve their learning.

How much a child learns is strongly influenced by family wealth. Analysis of 20 African countries for this year's Report shows that children from richer households are more likely not only to complete school, but also to achieve a minimum level of learning. By contrast, in 15 of the countries, no more than one in five poor children reach the last grade and learn the basics.

In Latin America, where performance is higher in general, children from disadvantaged backgrounds also lag far behind their wealthier peers. In El Salvador, 42% of children from the poorest households complete primary education and master the basics, compared with 84% of those from richest households.

Girls in poor households face some of the worst disadvantages, indicating that there is an urgent need to tackle gender gaps through education policies. In Benin, for example, around 60% of rich boys stay in school and attain basic numeracy skills, compared with 6% of poor girls.

Living in a disadvantaged area – especially rural ones, which often lack teachers and teaching resources – is a huge barrier to learning. In the United Republic of Tanzania, only 25% of poor children in rural areas learn the basics, compared with 63% of rich children in urban areas. In some Latin American countries, including El Salvador, Guatemala, Panama and Peru, achievement gaps in mathematics and reading between rural and urban students exceed 15 percentage points.

**The cost of 250 million children not learning the basics is equivalent to US$129 billion**

Location-related disadvantages begin in the early grades and widen. In Ghana, urban students were twice as likely as rural students to reach minimum levels of English in 2011 in grade 3, and more than three times as likely by grade 6.

Geographical disadvantage is often aggravated by poverty and gender. In Balochistan province, Pakistan, only 45% of children of grade 5 age could solve a two-digit subtraction, compared with 73% in wealthier Punjab province. Only around one-quarter of girls from poor households in Balochistan achieved basic numeracy skills, while boys from rich households in the province fared much better, approaching the average in Punjab.

The discrimination some indigenous or ethnic groups face is reinforced by the fact that the language used in the classroom may not be one that they speak. In Peru in 2011, Spanish speakers were more than seven times as likely as indigenous language speakers to reach a satisfactory standard in reading.

**Less than 60% of immigrants in France passed the minimum learning benchmark**

Well-designed bilingual programmes taught by qualified teachers can help children overcome this challenge.

Children who learn less are more likely to leave school early. In Ethiopia, India, Peru and Viet Nam, children who achieved lower scores in mathematics at age 12 were more likely than others to drop out by age 15. In Viet Nam, for instance, almost half the poorer performers at age 12 had dropped out by age 15, compared with around one in five of the stronger performers.

The disadvantages children face in gaining access to school and remaining there will stay with them into secondary school. In South Africa, for example, there is a vast gap in learning between rich and poor, with only 14% of poor adolescents achieving the minimum standard in mathematics, comparable to the performance of poor students in Ghana, a country that has less than one-fifth South Africa's wealth. Such gaps are not inevitable. Botswana has achieved much higher levels of learning thanks largely to its much narrower gap between rich and poor.

In some countries, the gap between rich and poor becomes more apparent in later grades. In Chile, for example, while the gap is narrow at grade 4, 77% of rich students achieve the minimum standards by grade 8, compared with 44% of poor students.

### *Rich countries are also failing to ensure that the marginalized can learn*

While rich countries' achievement levels are generally higher, their education systems also fail significant minorities. For example, over 10% of grade 8 students in Norway and England performed below minimum learning levels in mathematics in 2011.

While East Asian countries, including Japan, the Republic of Korea and Singapore, have shown it is possible to overcome the disadvantages those living in poverty face, the same cannot be said for some OECD countries and for wealthy countries in the Arab States. The chance of a poor student in Oman achieving minimum learning standards, for example, is similar to that of a student in less wealthy countries, such as Ghana. In New Zealand, only two-thirds of poor students achieved the minimum standards, compared with 97% of rich students.

Immigrant students face a high risk of marginalization in education, resulting in lower levels of learning achievement. In France, Germany and the United Kingdom, over 80% of 15-year-old students achieve minimum benchmarks in reading. But immigrants perform far worse: in the United Kingdom, the proportion of immigrants making it above the minimum benchmark is no better than the average for Turkey, while Germany's immigrants are on a par with students in Chile. Immigrants in France face particular problems, with less than 60% passing the minimum benchmark – equivalent to the average for students in Mexico.

Indigenous children in high income countries often face disadvantage, and the gap in learning outcomes with the rest of the population has been persistent. In Australia, around two-thirds of indigenous students achieved the minimum benchmark in grade 8 between 1994/95 and 2011, compared with almost 90% of their non-indigenous peers.

### *Improving learning while expanding access*

It is often claimed that expanding access to primary school in poorer countries means lowering the quality of education. Yet, although vast numbers of children are not learning the basics, some countries have been able to get more children into school while ensuring that they learn once there. This balance is particularly impressive given that new entrants are more likely to come from marginalized households. Even so, far more needs to be done to bridge the learning gap more quickly, even in richer countries.

The United Republic of Tanzania made great strides in the numbers of students reaching the end of primary school, partly because primary school fees were abolished in 2001. Between 2000 and 2007, the proportion of children who completed primary school rose from half to around two-thirds, while the proportion learning the basics in mathematics increased from 19% to 36%. This is equivalent to around 1.5 million additional children learning the basics. While it is unacceptable that 27% were in school but not learning the basics, the fact that the problems with quality were already apparent in 2000 suggests that they were more inherent in the education system than directly associated with education expansion.

In Malawi and Uganda, access and quality did not improve significantly between 2000 and 2007, and the learning gap between rich and poor widened. These countries face a triple challenge, needing to strengthen access, quality and equity.

At secondary level, too, efforts to increase access have not always succeeded in increasing learning and reaching the disadvantaged. In Mexico, as access increased, the share of students performing above minimum benchmarks also increased, from one-third in 2003 to one-half in 2009. Targeted social protection programmes aimed at disadvantaged families helped improve learning outcomes for the rich and poor alike. In Ghana, by contrast, while secondary enrolment increased from 35% in 2003 to 46% in 2009, and performance in numeracy also increased by 10 percentage points, gender gaps in learning more than doubled and the poorest barely benefited at all.

Malaysia experienced a particularly worrying trend of worsening learning outcomes coupled with widening inequality and an increasing number of adolescents out of school. In 2003, the vast majority of adolescents passed the minimum benchmark, whether rich or poor. However, only around half the poorest boys reached the minimum benchmark in 2011, compared with over 90% in 2003. Poor boys in Malaysia went from being similar to average performers in the United States to being similar to those in Botswana.

## *Poor quality education leaves a legacy of illiteracy*

The quality of education during childhood has a marked bearing on youth literacy. New analysis for this Report, based on direct assessments of literacy in household surveys, shows that youth illiteracy is more widespread than previously believed: around 175 million young people in low and lower middle income countries – equivalent to around one-quarter of the youth population – cannot read all or part of a sentence. In sub-Saharan Africa, 40% of young people are not able to do so.

Young women are the worst affected of all, making up 61% of youth who are not literate. In South and West Asia, two out of three of young people who cannot read are young women.Comparisons among countries expose the widespread problems of illiteracy. In 9 of the 41 low and lower middle income countries in the analysis, more than half of 15- to 24-year-olds are not literate. All these countries are in sub-Saharan Africa.

**In low and lower middle income countries, one-quarter of youth cannot read a sentence**

The analysis confirms the assumption that children need to spend at least four years in school to become literate: among those who have spent four years or less in school, around 77% are not able to read all or part of a sentence. In 9 of the 41 countries analysed, more than half of young people have spent no more than four years in school, and almost none of them are literate.

Spending five or six years in school, equivalent in some systems to completing a full cycle of primary schooling, does not guarantee literacy, however. In the 41 countries in the analysis, around 20 million young people still cannot read all or part of a sentence – equivalent to one in three of those who leave school after grade 5 or 6.

Young people from poorer households are far less likely to be literate. Among the countries analysed, more than 80% of those from rich households can read a sentence in 32 countries, but 80% of the poor can do so in only 4 countries. At the other end of the scale, less than half of poor youth can read a sentence in 22 countries, while the rich fall below this threshold only in Niger. In several countries, including Cameroon, Ghana, Nigeria and Sierra Leone, the difference in youth literacy rates between rich and poor is more than 50 percentage points.

Disadvantages in acquiring basic skills are further compounded by a combination of poverty, gender, location and ethnicity. In Senegal, only 20% of rural young women could read in everyday situations in 2010, compared with 65% of urban young men. In Indonesia, almost all rich young women in Bali province have literacy skills, while just 60% of poor women in Papua province are literate.

These outcomes may reflect the combined effects of poverty, isolation, discrimination and cultural practices. However, they also echo failures of education policy to provide learning opportunities for the most disadvantaged populations, and they indicate an urgent need to provide these people with a second chance.

**Ethiopia's youth literacy rate increased from 34% in 2000 to 52% in 2011**

There are some signs of improvement in youth literacy that offer hope. Thanks to the expansion of primary schooling over the past decade, the youth literacy rate increased in Ethiopia from 34% in 2000 to 52% in 2011. Youth literacy has also improved in Nepal, especially among the most disadvantaged, who started with very low levels of literacy. Literacy among poor young women increased from 20% in 2001 to 55% in 2011.

Information on how much children with disabilities are learning is so scarce that analysis is difficult. Uganda provides a rare example where information is sufficient to compare literacy rates of young people according to types of impairment. In 2011, around 60% of young people with no identified impairment were literate, compared with 47% of those with physical or hearing impairments and 38% of those with mental impairments.

Striving for equal learning, including for children and youth with disabilities, requires identifying the particular difficulties children and young people with various types of disadvantage face, and implementing policies to tackle them.

## Making teaching quality a national priority

Strong national policies that give a high priority to improving learning and teaching are essential to ensure that all children in school obtain the skills and knowledge they are meant to acquire. Education plans should describe goals and establish benchmarks against which governments can be held to account, as well as ways to achieve the goals. Improving learning, especially among the most disadvantaged children, needs to be made a strategic objective. Plans should include a range of approaches to improve teacher quality, devised in consultation with teachers and teacher unions. They also need to guarantee that strategies will be backed by sufficient resources.

### *Quality must be made a strategic objective in education plans*

The global learning crisis cannot be overcome unless policies aim to improve learning for the disadvantaged. Of 40 national education plans reviewed for this Report, 26 list improved learning outcomes as a strategic objective. While the plans of all 40 countries address the needs of disadvantaged groups to some extent, learning is often only addressed as a by-product of increased access.

To improve learning for all, national education plans must improve teacher management and quality. Only 17 of the 40 plans include strategies for improving teacher education programmes, and only 16 envisage further training of teacher educators.

It is even less common for plans to recognize explicitly that improving teaching quality can enhance learning outcomes. In Kenya, in-service training is aimed at substantially boosting the learning of primary school leavers in poorly performing districts. South Africa and Sri Lanka link recruitment of teachers with improvements in quality and learning.

Governments need to get incentives right to attract and retain the best teachers. Of the 40 plans reviewed, 10 include reforms to improve teacher pay and 18 emphasize better career paths and promotion prospects.

Only some of the plans target teacher reforms at improving learning for disadvantaged students, mainly by getting teachers into disadvantaged areas. Among the 28 plans that aim to send teachers to disadvantaged areas, 22 aim to provide incentives, such as housing benefits and salary supplements. In 14 countries, education plans include incentives to promote deployment to rural areas, while 8, including Afghanistan, actively encourage female teachers. Cambodia's plan is notable for strategies to recruit teachers from target areas and ethnic groups and deploy them where they are most needed. In remote areas, where student numbers are often small, teachers may have to teach more than one age group at the same time. In Cambodia, Kenya and Papua New Guinea, there are plans to provide training in multigrade teaching.

Few plans highlight the need for support to students who are falling behind. Guyana's is one exception; it gives a high priority to building teachers' capacity to deliver targeted programmes.

For plans to be successfully implemented, they need to be backed by sufficient resources, but only 16 of the 40 policy documents reviewed include a budget breakdown. Countries assign different levels of importance in their budgets to policies aimed at improving education quality: they amount to over a fifth of the budget in Papua New Guinea, but to 5% or less in Palestine, for example. Few plans earmark expenditure for the disadvantaged, however.

Policies can only be effective if those responsible for implementing them are involved in shaping them. However, a survey in 10 countries showed that only 23% of teachers thought they had influence over policy and practice. Given their reach, teacher unions are key partners for governments. In some countries, engaging teacher unions has improved policies aimed at helping disadvantaged groups. In the Plurinational State of Bolivia, for example, teacher unions campaigned to ensure that indigenous rights were enshrined in the constitution.

Overall, teachers and their unions can help make sure policies are effective. Thus it is important to include them from the early stages in designing strategies aimed at tackling learning deficits.

### *Getting enough teachers into classrooms*

The quality of education is held back in many of the poorest countries by a lack of teachers, which often results in large class sizes in early grades and in the poorest areas. Future teacher recruitment needs are determined by current deficits, demographics, enrolment trends and numbers of children out of school. Analysis by the UNESCO Institute for Statistics shows that, between 2011 and 2015, 5.2 million teachers – including replacement and additional teachers – need to be recruited to ensure that there are sufficient teachers to achieve universal primary education. This amounts to over 1 million teachers per year, equivalent to about 5% of the current primary school teaching force.

Most initial teacher education programmes last at least two years. With 57 million children still out of school, it is unlikely that countries with a lack of teachers will be able to meet the 2015 Education for All deadline for achieving universal primary education. Nevertheless, countries must start planning now to make up the shortfall. If the deadline were extended to 2020, accounting for projected increases in enrolment, the number of teachers required would rise to 13.1 million over 9 years. If it were extended to 2030, 20.6 million teachers would be needed over 19 years.

Of the teachers required between 2011 and 2015, 3.7 million are needed to replace teachers who retire, change occupations or leave due to illness or death. The remaining 1.6 million are the additional teachers needed to make up the shortfall, address expanding enrolment and underwrite quality by ensuring that there are no more than 40 students for every teacher. Thus, around 400,000 additional teachers need to be recruited each year if there are to be sufficient teachers by 2015.

Sub-Saharan Africa accounts for 58% of the additional primary teachers needed, requiring approximately 225,000 per year between 2011 and 2015. However, over the past decade the average annual increase in the region has been only 102,000.

Nigeria has by far the largest gap to fill. Between 2011 and 2015, it needs 212,000 primary school teachers, 13% of the global total. Of the 10 countries needing the most additional primary teachers, all but one are in sub-Saharan Africa, the exception being Pakistan.

The challenge of recruiting teachers becomes even greater when the needs of lower secondary education are taken into account. To achieve universal lower secondary education by 2030 with 32 students per teacher, an additional 5.1 million would be needed, or 268,000 per year. Sub-Saharan Africa accounts for half of the additional lower secondary school teachers needed over this period.

It is unlikely that the countries with the most severe teacher gaps can recruit the numbers needed by 2015. Of the 93 countries that need to find additional primary school teachers by 2015, only 37 will be able to bridge the gap; 29 will not

**In Bolivia, teacher unions campaigned to ensure that indigenous rights were enshrined in the constitution**

even be able to fill the gap by 2030. Meanwhile, 148 countries need more teachers for lower secondary schools by 2015; 29 countries will not have filled this gap by 2030.

**29 countries will not be able to fill the primary school teacher gap by 2030**

To fill the teacher gap by 2015, some countries need to speed up expansion of their teacher force. Rwanda and Uganda would need to expand recruitment by 6%, on average, compared with a current average increase of 3% per year. In Malawi, the teaching force is growing by just 1% per year, which is far from sufficient to reduce the pupil/teacher ratio from 76:1 to 40:1. For Malawi to meet the universal primary education goal by 2015, it would need to increase its teaching force by 15% annually between 2011 and 2015.

Many poor countries will not be able to fill their teacher gap simply because they do not have enough upper secondary school graduates – the minimum qualification for primary teacher trainees. In 8 out of 14 countries in sub-Saharan Africa, at least 5% of all upper secondary school graduates in 2020 would need to be drawn into teaching to fill the teacher gap, rising to almost 25% in Niger. By comparison, just over 3% of those in the labour force with at least secondary education are primary school teachers in middle income countries.

Teachers need not only to be recruited, but also to be trained. Many countries, especially in sub-Saharan Africa, also need to train existing teachers. Mali, for example, recruited teachers at a rate of 9% per year over the past decade, which helped lower the number of pupils per teacher from 62 in 1999 to 48 in 2011. However, many of these teachers are untrained. The result is that Mali's ratio of pupils per trained teacher, 92:1, is one of the world's highest. On its past trend of trained teacher recruitment, Mali would not achieve a ratio of 40 pupils per trained teacher until 2030.

Countries that have many untrained teachers need to find ways to train them. In 10 out of 27 countries with available data, this challenge is greater than that of recruiting and training new teachers. In Benin, 47% of teachers were trained in 2011. The country needs to expand teacher recruitment by just 1.4% per year to achieve UPE by 2020, but the number of existing teachers who need to be trained would have to grow by almost 9% per year, well above Benin's 6% average annual growth rate since 1999 for trained teachers.

The shortage of trained teachers is likely to affect disadvantaged areas in particular. In the northern state of Kano, one of the poorer parts of Nigeria, the pupil/trained teacher ratio exceeded 100 in 2009/10, with at least 150 pupils per trained teacher in the most disadvantaged 25% of schools.

Children in the early grades who live in remote areas often face a double disadvantage. In Ethiopia, for example, where 48% of teachers are trained, only around 20% of teachers were trained in grades 1 to 4 in 2010, compared with 83% in grades 5 to 8.

Countries that require additional teachers will have to increase their overall budgets for teacher salaries. New analysis by the UNESCO Institute for Statistics for this Report finds that US$4 billion annually is needed in sub-Saharan Africa to pay the salaries of the additional primary school teachers required to achieve UPE by 2020, after taking into account projected economic growth. This is equivalent to 19% of the region's total education budget in 2011. Nigeria alone accounts for two-fifths of the gap.

While the required increases may seem vast, most countries should be able to meet them if their economies grow as projected and if they dedicate a larger share of their GDP to education while staying within the benchmark of 3% allocated to primary education. On average, sub-Saharan African countries would have to increase the share of the budget they allocate to education from 12% to 14% in 2011 to close the teacher gap by 2020.

The financing challenge is inevitably greater for lower secondary school. In sub-Saharan Africa, recruiting enough teachers to achieve universal lower secondary education by 2030 would add US$9.5 billion to the annual education budget.

While many countries should be able to meet the cost of recruiting and paying the required additional primary teachers from their national budgets, they will also need to pay for teacher training, as well as school construction and learning materials, to ensure that children receive an education of a good quality. Expanding the lower secondary teacher workforce will place a further burden on national budgets. Some of the poorest countries are therefore likely to face a substantial financing gap and will require the support of aid donors. This need is likely to be even greater when the cost of expanding teacher education programmes is taken into account.

However, between 2008 and 2011, donors spent only US$189 million per year, on average, on pre-service and in-service teacher education programmes, equivalent to 2% of the education aid budget. While the countries most in need are in sub-Saharan Africa, the largest country recipients of this aid included richer middle income countries such as Brazil, China and Indonesia.

## Four strategies to provide the best teachers

Policy-makers need to give teachers every chance to put their motivation, energy, knowledge and skills to work in improving learning for all. This Report describes the four strategies that governments need to adopt to attract and retain the best teachers, improve teacher education, allocate teachers more fairly and provide incentives in the form of appropriate salaries and attractive career paths. It then highlights the areas of teacher governance that need to be strengthened to ensure that the benefits of these four strategies are realized.

### *Strategy 1: Attract the best teachers*

*I chose to be a teacher because I believe that education has the power to transform the society we live in. What motivates me to be a good teacher is to be an active agent in this change that is so necessary for my country to fight against discrimination, injustice, racism, corruption, poverty.*

– Ana, teacher, Lima, Peru

The first step to getting good teachers is to attract the best and most motivated candidates into the profession. Many people who decide to become teachers are driven by the satisfaction of helping students learn, fulfil their potential and develop into confident, responsible citizens.

It is not enough just to want to teach. People should enter the profession having received a good education themselves. They need to have at least completed secondary schooling of appropriate quality and relevance, so that they have a sound knowledge of the subjects they will be teaching and the ability to acquire the skills needed to teach.

Teaching does not always draw the best candidates, however. In some countries, teaching is seen as a second-class job for those who do not do well enough academically to enter more prestigious careers, such as medicine or engineering. The level of qualification required to enter teaching is a signal of the field's professional status. To elevate the status of teaching and attract talented applicants, for example, Egypt has introduced more stringent entry requirements, requiring candidates to have strong performance in secondary school as well as a favourable interview assessment. Once selected, candidates also have to pass an entrance examination to establish whether they match the profile of a good teacher.

**Donors spend only US$189 million per year on teacher education**

Making sure there are enough female teachers and recruiting teachers from a wide range of backgrounds are important factors in providing an inclusive, good quality education. Flexible policies for entry qualifications may be required to improve diversity of the teaching force. In South Sudan, women make up about 65% of the post-war population, yet less than 10% of all teachers are women. To increase the number of female teachers, financial and material incentives have been given to over 4,500 girls to complete secondary school and to women trainees to enter the teaching profession.

Recruiting teachers from under-represented groups to work in their own communities guarantees that children have teachers familiar with their culture and language. Flexible policies on entry requirements can help increase the number of candidates recruited from ethnic minority groups. In Cambodia, where teacher trainees normally have to have completed grade 12, this requirement is waived for remote areas where upper secondary education is unavailable, increasing the pool of teachers from ethnic minorities.

### *Strategy 2: Improve teacher education so all children can learn*

Initial teacher education should impart the skills needed to teach – especially for teaching the disadvantaged and those in early grades – and lay the foundation for ongoing training. But initial teacher education is not always effective in preparing teachers to deliver good quality, equitable education.

Trainees need a good understanding of the subjects they will be teaching. In low income countries, however, teachers often enter the profession lacking core subject knowledge because their own education has been poor. In a 2010 survey of primary schools in Kenya, grade 6 teachers scored only 60% on tests designed for their students. In such circumstances, teacher education programmes need to start by ensuring that all trainees acquire a good understanding of the subjects they will be teaching.

Teacher education institutions often do not have time to upgrade weak subject knowledge, partly because of competing curriculum demands. In Kenya, teacher trainees are required to take up to 10 subjects and participate in teaching practice in the first year. This leaves little time to fill gaps in subject knowledge. Ghana has addressed this problem by making trainees pass an examination on subject knowledge in their first year.

**In Turkey, a pre-service course on gender equity had a significant impact on female teachers' gender attitudes**

Teachers need not only sound subject knowledge but also training in how to teach, particularly in the early grades. However, teachers are seldom trained in these skills. In Mali, few teachers were able to teach their pupils how to read. Teachers had been inadequately prepared to apply the required teaching methods and did not give sufficient attention to supporting pupils' individual reading. This is no doubt an important reason why nearly half the pupils in Mali could not read a word in their own language at the end of grade 2. Teachers are also rarely prepared for the reality of multilingual classrooms. In Senegal, for example, only 8% of trainees expressed any confidence about teaching reading in local languages.

As a result of inadequate training, including overemphasis on theory rather than practice, many newly qualified teachers are not confident that they have the skills necessary to support children with more challenging learning needs, including those with severe physical or intellectual disabilities, in mainstream classrooms. To address this, teachers in Viet Nam create individual education plans for all learners, designing and adapting activities for children with different learning needs, and assessing learning outcomes of children with special needs.

Teacher education should also prepare teachers for remote or under-resourced schools, where some teachers need to teach multiple grades, ages and abilities in one classroom. In some countries in sub-Saharan Africa, including Burkina Faso, Mali, Niger, Senegal and Togo, at least 10% of students study in such classrooms. A small project in Sri Lanka trained teachers to develop lesson plans and grade-appropriate tasks for classes combining grades 4 and 5. Results indicated that such methods had a positive impact on pupils' achievement in mathematics.

Countries with high student learning outcomes require trainees to receive practical training in classrooms before teaching. This is especially important for teachers who teach in under-resourced and diverse classrooms, but is rarely provided. In Pakistan, trainee teachers only spend 10% of their training time in classrooms. To address this need, an NGO teacher education programme in Malawi includes a full year of practice teaching in rural districts. Some 72% of the participants said the school practice component was the area of study that most prepared them for teaching in rural areas. In addition, 80% of them gained experience in providing remedial support to students, compared with 14% in government colleges.

Teachers also need training in how attitudes to gender can affect learning outcomes. In Turkey, a one-term pre-service course on gender equity had a significant impact on female teachers' gender attitudes and awareness.

All teachers require continuing support once they reach the classroom to enable them to reflect on teaching practices, to foster motivation and to help them adapt to change, such as using a new curriculum or language of instruction.

Ongoing training can also provide teachers with new ideas about how to support weak learners. Teachers who have received some in-service training are generally found to teach better than those who have not, although it depends on the purpose and quality of the training.

Ongoing training is even more important for teachers who enter classrooms with little or no pre-service teacher education, or whose training has not sufficiently exposed them to the reality of the classroom. In Benin, many teachers have been hired as community or contract teachers with no pre-service teacher education. A programme designed in 2007 offers them three years of training to give them qualifications equivalent to those of civil service teachers.

Teachers in conflict zones are among those most in need of a coherent strategy to upgrade their skills. In the Dadaab refugee camps in northern Kenya, 90% of teachers are hired from the refugee community, only 2% of whom are qualified. Refugee teachers are ineligible for admission to higher education institutions in Kenya, and so require alternative qualification options. A teacher management and development strategy for 2013–2015 aims to provide them with training, including school-based practice. The strategy also recommends qualification and certification options for teachers who meet minimum higher education admission requirements, as well as options for the majority who do not meet the requirements.

Ongoing training can bridge gaps in the quality and relevance of pre-service teacher education, but often fails to foster the skills teachers need to respond to particular learning needs, especially in the early grades. An Early Grade Reading Assessment in Liberia found that around one-third of grade 2 students were unable to read a word. As a result, in 2008, the Ministry of Education launched a new programme consisting of teacher education and support, structured lesson plans, teaching resource materials, and books for children to take home. Teachers participated in an intensive one-week course in early grade reading instruction and how to use formative and diagnostic assessment to identify and support weak learners. This was followed up with classroom-based support from trained mentors over two years. Pupils in this programme increased their reading comprehension scores by 130%, compared with 33% for non-participants.

In many low income countries, teaching relies on traditional approaches such as lecturing, rote learning and repetition, rather than fostering transferable skills such as critical thinking. A school-based teacher development programme in Kenya has shown that training can be effective in helping teachers adopt learner-centred methods. The programme, involving self-study using distance-learning materials and meetings with tutors at cluster resource centres, led to teaching becoming more interactive, with improved use of lesson plans and resources.

The key role that teacher educators play in shaping teachers' skills is often the most neglected aspect of teacher preparation systems, particularly in developing countries. Many teacher educators seldom set foot in local schools to learn about the challenges prospective teachers face. Analysis of six sub-Saharan African countries found that teacher educators helping train teachers how to teach reading skills were rarely experts in approaches used in the field.

Reforms aimed at helping disadvantaged students need to ensure that teacher educators are trained to give teachers appropriate support. In Viet Nam, many teacher educators had limited awareness of how to deal with diversity until training was provided for teacher educators from universities and colleges to act as experts on inclusive education in pre-service programmes.

Not only is the quality of teacher education often insufficient, but many teacher education institutions also lack the capacity for the huge numbers of people needing to be trained, and expanding capacity is costly. Using technology to provide training from a distance is one way to reach larger numbers of trainees. Distance education programmes must be of adequate quality, and should be complemented by mentoring and face-to-face support at key stages.

The extent to which information and communication technology (ICT) is used in distance learning for teacher education is dictated by ICT infrastructure and resources, and the needs of target audiences. In South Africa, where surveys revealed that only 1% of teachers had regular Internet access but the vast majority had access to mobile phones, a teacher education programme supplements paper-based distance learning with text messaging. In Malawi, battery-powered DVD players and interactive instructional DVDs are used to assist with training.

Distance teacher education programmes could reach more future teachers at lower cost than programmes in teacher education institutions. Costs per student graduating from distance programmes have been estimated at between one-third and two-thirds of conventional programmes.

### *Strategy 3: Get teachers where they are most needed*

Teachers are understandably reluctant to work in deprived areas, which lack basic facilities such as electricity, good housing and health care. If the best teachers seldom work in remote, rural, poor or dangerous areas, however, the learning opportunities of children who are already disadvantaged suffer further as a result of larger class sizes, high rates of teacher turnover and a scarcity of trained teachers.

Governments need to devise strategies to ensure that teachers are equally allocated, but they rarely do so. In Yemen, schools with 500 students were found to have between 4 and 27 teachers. In South Sudan, average pupil/teacher ratios varied from 51:1 in Central Equatoria to 145:1 in Jonglei.

**In South Sudan, pupil/teacher ratios reached 145:1 in Jonglei**

Unequal distribution of teachers is one reason some children leave school before learning the basics. In Bangladesh, only 60% of students reach the last grade of primary school in subdistricts where there are 75 students per teacher, compared with three-quarters where there are 30.

The unequal allocation of teachers is affected by four main factors:

*Urban bias:* Weak infrastructure in rural areas means teachers are less keen to teach there. In Swaziland, for example, remote rural schools are mostly staffed with newly recruited, inexperienced teachers and teachers with low qualifications.

*Ethnicity and language:* Because education levels of ethnic minorities are often lower, fewer can apply to be teachers. In India, states cannot fill their caste-based quotas for recruitment of teachers unless teachers with lower levels of qualifications are hired.

*Gender:* Women are less likely than men to work in disadvantaged areas. In Rwanda, only 10% of primary school teachers were female in Burera district, compared with 67% in wealthier Gisagara district.

*Subjects:* In secondary schools, in particular, there are often teacher shortages in specific subjects. In Indonesia, for example, at junior secondary level there is a surplus of teachers in religion, but a shortage in computer science.

To achieve a balance of teachers across the country, some governments post teachers to disadvantaged areas. One reason for the Republic of Korea's strong and more equitable learning outcomes is that disadvantaged groups have better access to more qualified and experienced teachers. Over three-quarters of teachers in villages have at least a bachelor's degree, compared with 32% in large cities, and 45% have more than 20 years of experience, compared with 30% in large cities. Teachers working in disadvantaged schools benefit from incentives such as an additional stipend, smaller class sizes, less teaching time, the chance to choose their next school after teaching in a difficult area and greater promotion opportunities.

Providing incentives is a way to encourage teachers to accept difficult postings. Safe housing is particularly important in encouraging women to teach in rural areas, as in Bangladesh. The Gambia introduced an allowance of 30% to 40% of their base salary for positions in remote regions. By 2007, 24% of teachers had requested a transfer to hardship schools.

Alternatively, countries can recruit teachers from within their own communities. In Lesotho, a system of local recruitment allows school management committees to hire teachers, who apply directly to the schools for vacant posts. As a result, there is relatively little difference in pupil/teacher ratios between rural and urban areas.

Some countries are opening alternative pathways into teaching to attract highly qualified professionals with strong subject knowledge. One approach is exemplified by the Teach for All programmes in a range of countries, which recruit graduates with strong subject-level degree qualifications to teach in schools that predominantly serve disadvantaged students. Evidence from evaluations of Teach for America suggests that, once they have gained some experience, these teachers help improve students' learning, provided they receive some training.

### *Strategy 4: Provide the right incentives to retain the best teachers*

Salaries are just one of many factors that motivate teachers, but they are a key consideration in attracting the best candidates and retaining the best teachers. Low salaries are likely to damage morale and can lead teachers to switch to other careers. At the same time, teacher salaries make up the largest share of most education budgets, so they need to be set at a realistic level to ensure that enough teachers can be recruited.

The level of teacher salaries influences education quality. In 39 countries, a 15% rise in pay increased student performance by 6% to 8%. However, teachers in some countries do not even earn enough to lift their households above the poverty line. A teacher who is the main breadwinner, and has at least four family members to support, needs to earn at least US$10 per day to keep the family above the poverty line of US$2 per day per person. However, average teacher salaries are below this level in eight countries. In the Central African Republic, Guinea-Bissau and Liberia, teachers are paid no more than US$5, on average. Teacher salaries are similarly low in the Democratic Republic of the Congo, where communities often have to supplement their pay. Communities that are too poor to do so suffer from further disadvantage, losing good teachers.

In some countries, few teachers can afford basic necessities without taking a second job. In Cambodia, where a teacher salary did not cover the cost of basic food items in 2008, over two-thirds of teachers had a second job. National data on average teacher pay disguise variations in pay among different types of teachers: salaries are often considerably less than average for teachers at the beginning of their career, unqualified teachers and those on temporary contracts. In Malawi, those entering the profession, or lacking the academic qualifications needed for promotion, earn less than one-third of teachers in the highest pay category. Their salary was equivalent to just US$4 per day in 2007/08.

**Average teacher salaries are below US$10 per day in 8 countries**

When teachers are paid less than people in comparable fields, the best students are less likely to become teachers, and teachers are more likely to lose motivation or leave the profession. In Latin America, teachers are generally paid above the poverty threshold, but their salaries do not compare favourably with those working in professions requiring similar qualifications. In 2007, professionals and technicians with similar characteristics earned 43% more than pre-school and primary school teachers in Brazil, and 50% more in Peru.

In sub-Saharan Africa and South and West Asia, policy-makers have responded to the need to expand education systems rapidly by recruiting teachers on temporary contracts with little formal training. Contract teachers are usually paid considerably less than civil service teachers; some are hired directly by the community or by schools.

In West Africa, contract teachers made up half the teaching force by the mid-2000s. At the end of the decade, there were far more teachers on temporary contracts than on civil service contracts in some of these countries: the proportion reached almost 80% in Mali and Niger and over 60% in Benin and Cameroon. In Niger, contract teachers earn half as much as civil service teachers.

In some countries, governments eventually hire contract teachers as civil service teachers. In Benin, for example, contract teachers, with the support of teacher unions, campaigned to obtain more stable employment conditions and better pay. In 2007, the government issued a decree absorbing into the civil service contract all teachers who had achieved the required qualifications. Thus, despite the share of contract teachers having increased dramatically, the average teacher salary in Benin rose by

45% between 2006 and 2010 as the salaries of contract and civil service teachers converged.

In Indonesia, where contract teachers made up over a third of the primary school teaching force in 2010, regular teachers earned up to 40 times their salary. The government ensured that contract teachers would eventually attain civil service status, with implications for the education budget: giving all contract teachers permanent status would increase the salary bill for basic education by 35%, to about US$9 billion.

Where contract teachers are paid by the community, sustaining their services depends on parents' ability to mobilize funding, putting considerable financial pressure on poorer communities. In some cases this can lead to the government taking over some of the responsibility, ultimately adding to the budget. In Madagascar, community teachers, who made up around half of all teachers in 2005/06, are hired directly by parent-teacher associations and generally receive less than half of regular teachers' salaries. Since 2006, the government has increasingly taken on the responsibility for paying community teachers.

**In the Republic of Korea, an experienced teacher can earn more than twice as much as a new teacher**

While hiring contract teachers to alleviate teacher shortages can help in the short term, it is unlikely to meet the long-term need to extend quality education. Countries that rely heavily on contract teachers, notably in West Africa, rank at or near the bottom for education access and learning.

Teachers' salaries – and the rates at which they increase – are conventionally determined by formal qualifications, the amount of training and years of experience. But pay structures based on these criteria do not necessarily lead to better learning outcomes. Relating teachers' pay to the performance of their students is an alternative approach that has intuitive appeal. This appeal is supported by PISA data from 28 OECD countries: the countries where teachers' salaries are adjusted for student performance have higher scores in reading, mathematics and science. However, a closer look at the evidence on performance-related pay from around the world does not show clear-cut benefits.

It is difficult to find reliable ways to evaluate which teachers are the best and add the most value, as experience from the United States shows. Performance-related pay can also have unintended side effects on teaching and learning. In Portugal, it has led to competition between teachers in ways that can be harmful for the weakest students. In Mexico, many teachers are excluded from participating in such programmes, with those teaching in schools with low achievement at a disadvantage. Experience in Brazil suggests that rewarding schools with collective bonuses may be a more effective way to improve learning outcomes.

In poorer countries, performance-related pay has rarely been tried on a large scale, but experience suggests there is a risk of it encouraging teachers to teach to the test, rather than promoting wider learning. In an experiment in Kenyan primary schools, teachers were rewarded for good student test scores and penalized if students did not take end-of-year examinations. Test scores and examination attendance increased, but test scores did not go up in subject areas that were not taken into account in the teacher pay formula.

A more appropriate way of motivating teachers is to offer an attractive career path. In some OECD countries, the difference in pay between a more experienced teacher and a new teacher is small and there is little scope for promotion. In England, for example, a beginning teacher earns US$32,000 while the most experienced teacher can receive, at most, US$15,000 more. By contrast, the Republic of Korea has a considerably steeper pay structure: a new teacher earns a similar salary to new teachers in England, but an experienced teacher can earn more than twice that. In France, insufficient career management and other inadequate teacher policies are contributing to poor learning.

In many developing countries, teachers' career structures are not sufficiently linked to prospects of promotion that recognize and reward teacher effectiveness. In 2010, Ghana began reviewing its teacher management and development policy to address such concerns.

### *Strengthening teacher governance*

Better teacher governance is vital to reduce disadvantage in learning. If days are lost because teachers are absent or devote more attention to private tuition than classroom teaching, for example, the learning of the poorest children can be harmed. Understanding the reasons behind these problems is crucial for the design of effective strategies to solve them. Strong school leadership is required to ensure that teachers show up on time, work a full week and provide equal support to all. Gender-based violence, which is sometimes perpetrated by teachers, damages girls' chances of learning. Strategies to prevent and respond to teacher misconduct, and take action against perpetrators, require advocacy and support from head teachers, teachers and their unions, as well as communities, if girls are to be protected.

The scale of absenteeism is evident from surveys carried out in a range of poor countries over the past decade: in the mid-2000s, teacher absenteeism ranged from 11% in Peru to 27% in Uganda. Absenteeism exacerbates the problem of teacher shortages. In Kenya, where the typical primary school faces, on average, a shortage of four teachers, 13% of teachers were absent during school visits. Absenteeism can also affect disadvantaged students in particular. Across India, absenteeism varied from 15% in Maharashtra and 17% in Gujarat – two richer states – to 38% in Bihar and 42% in Jharkhand, two of the poorest states.

Teacher absenteeism harms learning. In Indonesia, a 10% increase in teacher absenteeism was estimated to lead to a 7% decrease in mathematics scores, on average, and absenteeism was most likely to harm weaker students: the teacher absence rate was 19% for the quarter of students with the highest mathematics scores, and 22% for the quarter with the lowest scores.

Head teachers themselves are sometimes absent, impeding effective monitoring of teacher attendance and demonstrating inadequate leadership regarding the problem. A 2011 survey of schools in Uganda found that, on average, 21% of head teachers were absent on the day the schools were visited.

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Policy-makers need to understand why teachers miss school. In some countries, teachers are absent because their pay is extremely low, in others because working conditions are poor. In Malawi, where teachers' pay is low and payment often erratic, 1 in 10 teachers stated that they were frequently absent from school in connection with financial concerns, such as travelling to collect salaries or dealing with loan payments. High rates of HIV/AIDS can take their toll on teacher attendance. Zambia has introduced strategies to improve living conditions for HIV-positive teachers, including greater access to treatment, provision of nutritional supplements and loans.

Gender-based violence in schools is a major barrier to quality and equality in education. A survey in Malawi found that around one-fifth of teachers said they were aware of teachers coercing or forcing girls into sexual relationships.

Programmes and policies addressing gender discrimination and gender-based violence need to protect and empower girls, challenge entrenched practices, bring perpetrators to light and take action against them. Legal and policy frameworks that provide general protection for children need to be strengthened and publicized, and teachers need to be made aware of their own roles and responsibilities. In Kenya, for example, a range of penalties is available to discipline teachers in breach of professional conduct, including suspension and interdiction; new regulations state that a teacher convicted of a sexual offence against a pupil is to be deregistered.

Advocacy and lobbying constitute an important first step in seeing that policies tackling gender-based violence are in place and enforced. In Malawi, a project lobbied successfully for revisions to codes of conduct and stronger reporting mechanisms. When it ran an awareness campaign, the number of teachers who said they knew how to report a code violation rose by over one-third.

**In Egypt, rich students are almost twice as likely as poorer students to receive private tuition**

Working directly with teacher unions is a way to build support for taking action against teachers who violate codes of conduct. In Kenya, the National Union of Teachers collaborated with the Teachers' Service Commission, Ministry of Education and Children's Department to help draft a parliamentary bill that would reinforce procedures for reporting abuse or violence by teachers and prevent convicted teachers from simply being transferred to other schools.

Private tuition is another outcome of poor teacher governance. If unchecked or uncontrolled, it can be a detriment to learning outcomes, especially for the poorest students who are unable to afford it. Private tutoring by teachers is often a symptom of badly functioning school systems and low pay that forces teachers to supplement their income. In Cambodia, teacher salaries are small and often paid late. One consequence is that 13% of primary school teachers and 87% of secondary teachers provide private tuition. This reinforces disparities between those who can afford fees and those who cannot. In urban areas, grade 9 students scored 8.3 points out of 10 in Khmer with tutoring and 3.8 points without.

In Egypt, the situation has become extreme, partly due to a decline in the quality of education and partly because teachers need to supplement their low income. The amount spent annually on private tutoring is reported to be US$2.4 billion, equivalent to 27% of government spending on education in 2011. Private tuition is a significant part of household education spending, averaging 47% in rural areas and 40% in urban areas. Children from rich households are almost twice as likely as poorer students to receive private tuition. Teachers may be their own students' private tutors, and thus responsible for their grades. Students complain that teachers do not cover the curriculum during the school day, forcing them to take private tuition to cover the syllabus to enable them to pass exams. Strategies should at least be in place to prevent tutoring of pupils by teachers who are responsible for teaching them in their daily classes. This would ensure that full curriculum coverage is available to all students, even those unable to afford tutoring.

Private schools that charge low fees are seen by some as a way of expanding access to better quality education for disadvantaged children where government schools are failing. In Pakistan, a child in a low fee private school performs better than the average child in the top one-third of children in government schools. However, even in private schools, many pupils barely reach expected competency levels. According to analysis by the Annual State of Education Report team in Pakistan, 36% of grade 5 students in private schools could not read a sentence in English, which they should have been able to do by grade 2.

Learning outcomes may be better in low fee private schools in part because lower salaries enable these schools to hire more teachers and keep pupil/teacher ratios low. In private schools in parts of Nairobi, there are 15 students per teacher, compared with 80 in government schools. Small class sizes also enable teachers at private schools to interact more with their students. In Andhra Pradesh, India, 82% of teachers regularly corrected exercises given to children, compared with only 40% in government schools.

Private school teachers are generally thought to work under conditions of greater accountability. In India, only one head teacher in 3,000 government schools reported dismissing a teacher for repeated absence. By contrast, 35 private school head teachers, out of 600 surveyed, reported having dismissed teachers for this reason.

The benefits of low fee private schools do not mean they are better per se; often their students face far fewer disadvantages than students in government schools. In Andhra Pradesh, over 70% of students attending government schools belong to the poorest 40% of households, compared with 26% in private schools. Around one-third of teachers in government schools are teaching students of different ages in multigrade classrooms, compared with 3% in private schools.

There are no excuses for students not having the right conditions to learn: ultimately, it is vital that all children, regardless of their background and the type of school they attend, have the best teachers to offer them this opportunity.

## Curriculum and assessment strategies that improve learning

To improve learning for all children, teachers need the support of curriculum and assessment strategies that can reduce disparities in school achievement and offer all children and young people the opportunity to acquire vital transferable skills. Such strategies need to build strong foundation skills by starting early, moving at the right pace, enabling disadvantaged pupils to catch up, meeting the language needs of ethnic minorities and building a culture of reading.

### *Ensuring all children acquire foundation skills*

The key to ensuring that children succeed at school is to enable them to attain critical foundation skills, such as reading and basic mathematics. Without these basic skills, many children will struggle to keep up with the prescribed curriculum, and learning disparities will widen for disadvantaged children.

The quality of pre-school education makes a crucial difference to children's learning in early primary grades. In Bangladesh, primary school children who had attended pre-school performed better than children without any pre-school experience in skills relating to reading, writing and oral mathematics.

It is crucial that primary school pupils master the foundation skills of basic numeracy and literacy in the early grades so they can understand what is taught in later grades, but they sometimes fail to do so because curricula are too ambitious. Viet Nam's curriculum focuses on foundation skills, is closely matched to what children are able to learn and pays particular attention to disadvantaged learners. By contrast, India's curriculum, which outpaces what pupils can realistically learn and achieve in the time given, is a factor in widening learning gaps. In Viet Nam, 86% of 8-year-olds answered grade-specific test items correctly. Similarly, 90% of children aged 8 in India did so. However, when 14- to 15-year-olds were asked a two-stage word problem involving multiplication and addition, 71% of children in Viet Nam answered correctly, while in India the percentage was 33%.

For children from ethnic and linguistic minorities to acquire strong foundation skills, schools need to teach the curriculum in a language children understand. A bilingual approach that combines continued teaching in a child's mother tongue with the introduction of a second language can improve performance in the second language as well as in other subjects. To reduce learning disparities in the long term, bilingual programmes should be sustained over several years. In Cameroon, children taught in their local language, Kom, showed a marked advantage in achievement in reading and comprehension compared with children taught only in English. Kom-educated children also scored twice as high on mathematics tests at the end of grade 3. However, these learning gains were not sustained when the students switched to English-only instruction in grade 4. By contrast, in Ethiopia, children in regions where local language instruction extends through to upper primary school performed better in grade 8 subjects than pupils taught only in English.

Language policies may be difficult to implement, particularly where there is more than one language group in the same classroom and teachers are not proficient in the local language. For bilingual education to be effective, governments need to recruit and deploy teachers from minority language groups. Initial and ongoing programmes are also needed to train teachers to teach in two languages and to understand the needs of second-language learners.

For early grade literacy and bilingual education to be successful, pupils need access to inclusive learning materials that are relevant to their situation and in a language they are familiar with. Open licensing and new technology can make learning materials more widely available, including in local languages. In South Africa, open source educational materials are being developed and made available in several African languages. Digital distribution is increasing the number of districts, schools and teachers with access to curricular resources.

Providing appropriate reading materials may not be enough on its own, however, to improve children's learning; children and families must also be encouraged to use them. In poor or remote communities where there is little access to print media, providing reading materials and supporting activities to practice reading can improve children's learning. Save the Children's Literacy Boost programme aims to improve early grade reading skills in government schools through interventions such as training teachers to teach core reading skills and monitor pupils' mastery of them. In addition, communities are encouraged to support children's reading. Evaluations in Malawi, Mozambique, Nepal and Pakistan all showed greater learning gains by children in Literacy Boost schools than by their peers, including a reduction in the number of children whose scores were zero, suggesting that the programme benefited low achievers.

**In India, schools with trained female community volunteers helped increase children's learning**

Support outside school hours is one reason for such success. In Pakistan, children who had attended after-school reading camps coordinated by community volunteers showed greater learning gains in reading fluency and accuracy in both Pashto and Urdu than classmates in the same schools. In Malawi, pupils whose parents had received training to support their children's reading made greater vocabulary gains than those whose parents had not.

Curricula need to address issues of inclusion to enhance the chances of students from marginalized backgrounds to learn effectively. Where gender-responsive curricula have been developed, as in projects in Mumbai, India, and in Honduras, test scores measuring attitudes on several gender-related issues improved. In Honduras, adolescents who participated in the project also demonstrated better problem-solving skills and higher test scores.

More needs to be done to design curricula that pay attention to the needs of disabled learners. In Canberra, Australia, curriculum reform aims to help teachers improve student attitudes regarding students with disabilities, improve the quality of interactions between students with and without disabilities, and enhance the well-being and academic achievement of students with disabilities.

Classroom-based assessment tools can help teachers identify, monitor and support learners at risk of low achievement. In Liberia, the EGRA Plus project, which trained teachers in the use of classroom-based assessment tools and provided reading resources and scripted lesson plans to guide instruction, raised previously low levels of reading achievement among grade 2 and 3 pupils.

Assessments need to be aligned with the curriculum so that they do not add significantly to teachers' workloads. In South Africa, well-designed assessments with clear guidelines on how to interpret results helped teachers with little training who were working in difficult conditions: 80% of teachers were able to use them in class.

Students can make considerable gains if they are offered opportunities to monitor their own learning. In the Indian state of Tamil Nadu, primary students learn at their own pace, using self-evaluation cards that can be administered alone or with the help of another child; teachers strategically pair more advanced learners with less advanced ones for certain exercises. Overall, children's self-confidence has grown as a result of the approach, and learning achievement in the state is high.

Targeted additional support for students via trained teaching assistants is another way of improving learning for students at risk of falling behind. An initial early reading intervention delivered by teaching assistants in London schools in the United Kingdom was found to improve reading skills and have longer-term positive effects for children with poor literacy skills. In India, schools with trained female community volunteers helped increase the proportion of children able to do two-digit addition. While only 5% of pupils were able to carry out simple subtraction at the start of the study, 52% could by the end of the year, compared with 39% in other classes.

Interactive radio instruction can lead to improvement in learning outcomes for disadvantaged groups by addressing barriers such as distance and poor access to resources and quality teachers, as identified in a review of 15 projects. The use of interactive radio can be particularly beneficial in conflict contexts. Between 2006 and 2011, the South Sudan Interactive Radio Instruction project enrolled over 473,000 pupils, providing half-hour lessons linked to the national curriculum and including instruction in English, local language literacy, mathematics and life skills elements such as HIV/AIDS and land mine risk awareness. In locations that were out of range of any radio signal, the project distributed digital MP3 players to be used by trained teachers.

Digital classrooms can complement classes given by less qualified teachers. In India, the Digital Study Hall project provides digital video recordings of live classes taught by expert teachers, which are shown by DVD in rural and slum schools. An evaluation of four schools in Uttar Pradesh found that, after eight months, 72% of pupils had improved test scores.

Innovation in the use of technology can help improve learning by enriching teachers' curriculum delivery and encouraging flexibility in pupil learning. Greater access to computers in schools can also help reduce the digital divide between low and high income groups. However, new technology is not a substitute for good teaching.

Teachers' ability to use ICT as an educational resource plays a critical role in improving learning. A study in Brazil found that the introduction of computer laboratories in schools had a negative impact on student performance, but that teachers' use of the Internet as a pedagogical resource supported innovative classroom teaching and learning, resulting in improved test scores.

Children from low income groups are less likely to have experience of ICT outside school and may thus take longer to adapt to it and need additional support. In Rwanda, 79% of students who used computers in secondary school had previously used ICT and the Internet outside school (primarily in Internet cafés). However, girls and rural children were at a disadvantage because they were less likely to have access to Internet cafés or other ICT resources in their communities.

One promising way of increasing the accessibility of ICT for teaching and learning is 'mobile learning' – the use of mobile phones and other portable electronic devices, such as MP3 players. In rural India, an after-school programme for children from low income families used mobile phone games to help them learn English. This resulted in significant learning gains in tests of the spelling of common English nouns, particularly for children in higher grades who had stronger foundation skills.

Where children are learning little and dropping out early, second-chance programmes can teach foundation skills through a shorter cycle of learning, which is one way of accelerating children's progress and raising achievement for disadvantaged groups. Several such accelerated learning programmes raise achievement for disadvantaged groups in less time than formal government schools, allowing them the opportunity to catch up and to re-enter formal schools. They usually benefit from small classes and teachers speaking the local language recruited from surrounding communities. In northern Ghana, for example, 46% of those who had attended an accelerated learning programme and re-entered primary school attained grade-appropriate levels in grade 4, compared with 34% of other students.

Formal primary schools can also use accelerated learning programmes in situations where large proportions of students are over-age for their grade. In Brazil, over-age students in grades 5 to 8 were taught a substantially modified curriculum, covering more than one grade in a year. Overall, schools' share of students with a two-year age grade gap was reduced from 46% in 1998 to 30% in 2003. Once the students were restored to the right grade for their age, they were able to maintain their performance and their promotion rates in secondary school were comparable with those of other students.

*Beyond the basics: transferable skills for global citizenship*

Curricula need to ensure that all children and young people learn not just foundation skills, but also transferable skills, such as critical thinking, problem-solving, advocacy and conflict-resolution, to help them become responsible global citizens. An interdisciplinary approach involving hands-on, locally relevant educational activities can also develop students' understanding of the environment and build skills to promote sustainable development.

**The Philippines has integrated disaster risk reduction into education**

Between 1999 and 2004, Germany introduced an interdisciplinary programme that fostered participatory learning and provided opportunities for students to work together on innovative projects for sustainable living. An evaluation found that participants had a greater understanding of sustainable development than their peers, and up to 80% of the students said they had gained transferable skills. In South Africa, an initiative links the curriculum to practical actions such as adopting recycling systems and water harvesting in schools, using alternative energy sources for cooking, cleaning up public spaces, creating indigenous gardens and planting trees. Participating schools have reported increased environmental awareness and improved sustainability practices at school and in homes.

Empowering children through communication and advocacy can help them reduce their vulnerability to environmental risk. In the Philippines, which is prone to environmental disasters, a strong commitment to integrating disaster risk reduction into education has led children to take an active role in making their communities safer.

Programmes that emphasize inclusion and conflict resolution can also help bolster individuals' rights and build peace. In Burundi in 2009, secondary schools taught communication and conflict mediation skills to help returning refugees. After two years, trained teachers had abandoned corporal punishment, issues of sexual abuse and corruption were more easily debated, relationships had improved among pupils, their peers and their teachers, and pupils were acting as mediators in the resolution of minor conflicts at school and in the community.


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## Unlocking teachers' potential to solve the learning crisis

This Report identifies the 10 most important teaching reforms that policy-makers should adopt to achieve equitable learning for all.

### 1 *Fill teacher gaps*

On current trends, some countries will not be able to meet their primary school teacher needs by 2030. The challenge is even greater for other levels of education. Thus, countries need to activate policies that begin to address the vast shortfall.

### 2 *Attract the best candidates to teaching*

It is important for all children to have teachers with at least a good secondary-level qualification. Therefore, governments should invest in improving access to quality secondary education to enlarge the pool of good teacher candidates. This reform is particularly important if the pool of better-educated female teachers is to increase in disadvantaged areas. In some countries, this will mean introducing affirmative measures to attract more women into teaching.

Policy-makers also need to focus their attention on hiring and training teachers from under-represented groups, such as ethnic minorities, to serve in their own communities. Such teachers, familiar with the cultural context and local language, can improve learning opportunities for disadvantaged children.

### 3 *Train teachers to meet the needs of all children*

All teachers need to receive training to enable them to meet the learning needs of all children. Before teachers enter the classroom, they should undergo good quality pre-service teacher education programmes that provide a balance between knowledge of the subjects to be taught and knowledge of teaching methods.

Pre-service teacher education should also make adequate classroom teaching experience an essential part of training to become a qualified teacher. It should equip teachers with practical skills to teach children to read and to understand basic mathematics. In ethnically diverse societies, teachers should learn to teach in more than one language. Teacher education programmes should also prepare teachers to teach multiple grades and ages in one classroom, and to understand how teachers' attitudes to gender differences can affect learning outcomes.

Ongoing training is vital for every teacher to develop and strengthen teaching skills. It can also provide teachers with new ideas to support weak learners, especially in the early grades, and help teachers adapt to changes such as a new curriculum.

Innovative approaches such as distance teacher education, combined with face-to-face training and mentoring, should also be encouraged so as to extend both pre-service and ongoing teacher education to greater numbers of teachers.

### 4 *Prepare teacher educators and mentors to support teachers*

To ensure that teachers have the best training to improve learning for all children, it is important for those who train teachers to have knowledge and experience of real classroom teaching challenges and how to tackle them. Policy-makers should thus make sure teacher educators are trained and have adequate exposure to the classroom learning requirements facing those teaching in difficult circumstances.

To enable newly qualified teachers to translate teaching knowledge into activities that improve learning for all children, policy-makers should provide for trained mentors to help them achieve this transition.

### 5 *Get teachers to where they are needed most*

Governments need to ensure that the best teachers are not only recruited and trained, but also deployed to the areas where they are most needed. Adequate compensation, bonus pay, good housing and support in the form of professional development opportunities should be used to encourage trained teachers to accept positions in rural or disadvantaged areas. In addition, governments should recruit teachers locally and provide them with ongoing training so that all children, irrespective of their location, have teachers who understand their language and culture and thus can improve their learning.

### 6 *Use a competitive career and pay structure to retain the best teachers*

Governments should ensure that teachers earn at least enough to lift their families above the poverty line and make their pay competitive with comparable professions. Performance-related pay has intuitive appeal as a way to motivate teachers to improve learning. However, it can be a disincentive to teach students who achieve less well, have learning difficulties or live in poor communities. Instead, an attractive career and pay structure should be used as an incentive for all teachers to improve their performance. It can also be used to recognize and reward teachers in remote areas and those who support the learning of disadvantaged children.

### 7 *Improve teacher governance to maximize impact*

Governments should improve governance policies to address the problems of teacher misconduct such as absenteeism, tutoring their students privately and gender-based violence in schools. Governments can also do more to address teacher absenteeism by improving teachers' working conditions, making sure they are not overburdened with non-teaching duties and offering them access to good health care. Strong school leadership is required to ensure that teachers show up on time, work a full week and provide equal support to all. School leaders also need training in offering professional support to teachers.

Governments need to work closely with teacher unions and teachers to formulate policies and adopt codes of conduct to tackle unprofessional behaviour such as gender-based violence. Codes of practice should refer clearly to violence and abuse, making penalties consistent with legal frameworks for child rights and protection.

Where private tutoring by teachers is prevalent, explicit guidelines, backed up with legislation, so that teachers do not sacrifice classroom time to teach the school curriculum privately are needed.

### 8 *Equip teachers with innovative curricula to improve learning*

Teachers need the support of inclusive and flexible curriculum strategies designed to meet the learning needs of children from disadvantaged groups. Equipped with the appropriate curriculum content and delivery methods, teachers can reduce learning disparities, allowing low achievers to catch up.

Policy-makers should ensure that early grade curricula focus on securing strong foundation skills for all and are delivered in a language children understand. It is important for curriculum expectations to match learners' abilities, as overambitious curricula limit what teachers can achieve in helping children progress.

Getting out-of-school children back into school and learning is vital. Governments and donor agencies should support second-chance accelerated learning programmes to achieve this goal.

In many countries, radio, television, computers and mobile technologies are being used to supplement and improve children's learning. Teachers in both formal and non-formal settings need to be given skills to maximize the benefits of technology in ways that help narrow the digital divide.

It is not sufficient for children to learn foundation skills in school. A curriculum that promotes interdisciplinary and participatory learning, and fosters skills for global citizenship, is vital for teachers to help children develop transferable skills.

### 9 *Develop classroom assessments to help teachers identify and support students at risk of not learning*

Classroom-based assessments are vital tools to identify and help learners who are struggling. Teachers need to be trained to use them so that they can detect learning difficulties early and use appropriate strategies to tackle these difficulties.

Providing children with learning materials to evaluate their own progress, and training teachers to support their use, can help children make great strides in learning. Targeted additional support via trained teaching assistants or community volunteers is another key way of improving learning for students at risk of falling behind.

### 10 *Provide better data on trained teachers*

Countries should invest in collecting and analysing annual data on the number of trained teachers available throughout the country, including characteristics such as gender, ethnicity and disability, at all levels of education. These data should be complemented by information on the capacity of teacher education programmes, with an assessment of the competencies teachers are expected to acquire. Internationally agreed standards need to be established for teacher education programmes so that their comparability is ensured.

More and better data on teacher salaries in low and middle income countries are also needed to enable national governments and the international community to monitor how well teachers are paid and to raise global awareness of the need to pay them well.

## Conclusion

To end the learning crisis, all countries, rich and poor, have to ensure that every child has access to a well-trained and motivated teacher. The 10 strategies outlined here are based on the evidence of successful policies, programmes, strategies from a wide range of countries and educational environments. By implementing these reforms, countries can ensure that all children and young people, especially the disadvantaged, receive the good quality education they need to realize their potential and lead fulfilling lives.

# Part 1 Monitoring progress

With the deadline for the Education for All goals less than two years away, it is clear that despite advances over the decade, not a single goal will be achieved globally by 2015. This means millions of children, young people and adults will have been let down by the signatories of the Dakar Framework for Action. It is not too late, however, to accelerate progress in the final stages. It is also vital to put in place a robust global post-2015 education framework to tackle unfinished business while addressing new challenges.

One of the starkest reminders that EFA is not attainable is the fact that 57 million children were still out of school in 2011. If progress continues to be as slow as in recent years, 53 million children will remain out of school in 2015. Maintaining the speed of progress between 1999 and 2008 would have left just 23 million children out of school.

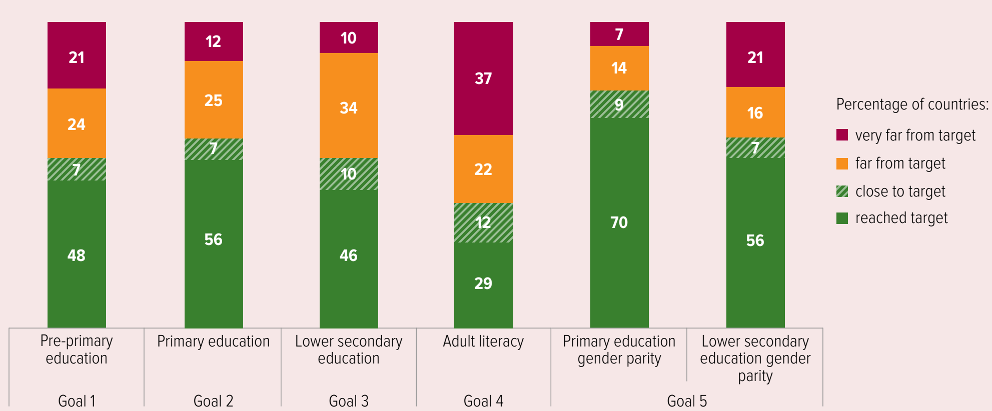
This Report projects that universal primary enrolment (goal 2) will be reached by just over half the world's countries by 2015 (Figure I.1). In one out of eight countries, less than 80% of primary school age children will be enrolled. For primary school completion, less than one in seven countries has achieved this target.

The world will be much closer to ensuring that equal numbers of boys and girls are enrolled in primary education (goal 5). By 2015, 7 in 10 countries will have reached the target. Gender parity in lower secondary education, however, will have been achieved in fewer than 6 in 10 countries. And gender parity should have been achieved by 2005. Moreover, parity is only the first step towards full gender equality.

Some countries have made fast progress towards improving adult literacy (goal 4), reaping the benefits of having expanded their basic education systems. In some regions, however, the rate of improvement has not kept pace with population growth. As a result, the number of illiterate adults worldwide remains stubbornly high at 774 million, a fall of just 1% since 2000. The number is projected to fall only to 743 million by 2015, when the adult literacy rate will still be below 80% in more than one in three countries.

Other goals established in 2000 have been difficult to monitor because clear targets were not set. For the first EFA goal, early childhood care and education, we assume for this Report that at least 80% of young children should be enrolled in pre-primary education programmes by 2015,

**Figure I.1: By 2015, many countries will still not have reached the EFA goals**
Percentage of countries projected to achieve a benchmark for five EFA goals by 2015




*Notes:* Countries are assessed as having reached the target if they have achieved a pre-primary education gross enrolment ratio of 80% (goal 1); a primary education adjusted net enrolment ratio of 97% (goal 2); a lower secondary education adjusted net enrolment ratio of 97% (goal 3); an adult literacy rate of 97% (goal 4); a gender parity index between 0.97 and 1.03 at primary and lower secondary education respectively (goal 5). The analysis was conducted on the subset of countries for which a projection was possible.
*Source:* Bruneforth (2013).

# towards the EFA goals

but only 5 in 10 countries are likely to reach this target. For the third EFA goal, skills for young people and adults, universal lower secondary education can be taken as one measure of progress towards ensuring that all young people have foundation skills, but fewer than 5 in 10 countries will have reached this level by 2015.

It is even more difficult to estimate the number of countries still to achieve the sixth EFA goal, improving the quality of education to ensure that all are learning. To help give education quality the greater focus that it deserves after 2015, this Report proposes ways to strengthen international and regional assessments so that progress towards a global learning goal can be measured.

New analysis for this Report vividly underlines the fact that the most marginalized have continued to be denied opportunities for education over the past decade. In 62 low and middle income countries, the richest urban young men had spent more than 9.5 years in school by 2010 in low income countries and more than 12 years in lower middle income countries. But the poorest rural young women had spent fewer than 3 years in school in both low and lower middle income countries, leaving them well below the 6-year target that is associated with universal primary completion, which is supposed to be achieved by 2015.

Although the percentage of girls from the poorest rural households entering school in sub-Saharan Africa grew over the decade, their primary school completion rate fell from 25% in the early 2000s to 23% in the late 2000s. Similarly, the rate of lower secondary school completion fell from 11% to 9% for this disadvantaged group.

Unless special efforts are urgently taken to extend educational opportunities to the marginalized, the poorest countries may take several generations to achieve universal completion of primary and lower secondary education as well as universal youth literacy, according to new analysis for this Report. In sub-Saharan Africa, if recent trends continue, the richest boys will achieve universal primary completion in 2021, but the poorest girls will not catch up until 2086 – and will only achieve lower secondary school completion in 2111.

The gap between the EFA goals and what has actually been achieved shows that additional resources must be better targeted at those most in need. Many of the success stories of the past few years can be traced back to a strong commitment by some of the poorest governments to invest in education. The global share of public expenditure devoted to education increased from 4.6% to 5.1% of gross national product (GNP) between 1999 and 2011, with the largest increases in low income countries.

Many low and middle income countries have the opportunity to expand their education spending still further. This Report estimates that a modest increase in tax-raising efforts, combined with growth in the share of government budgets allocated to education, could help raise education spending by US$153 billion by 2015 in 67 countries, a 72% increase from 2011 levels.

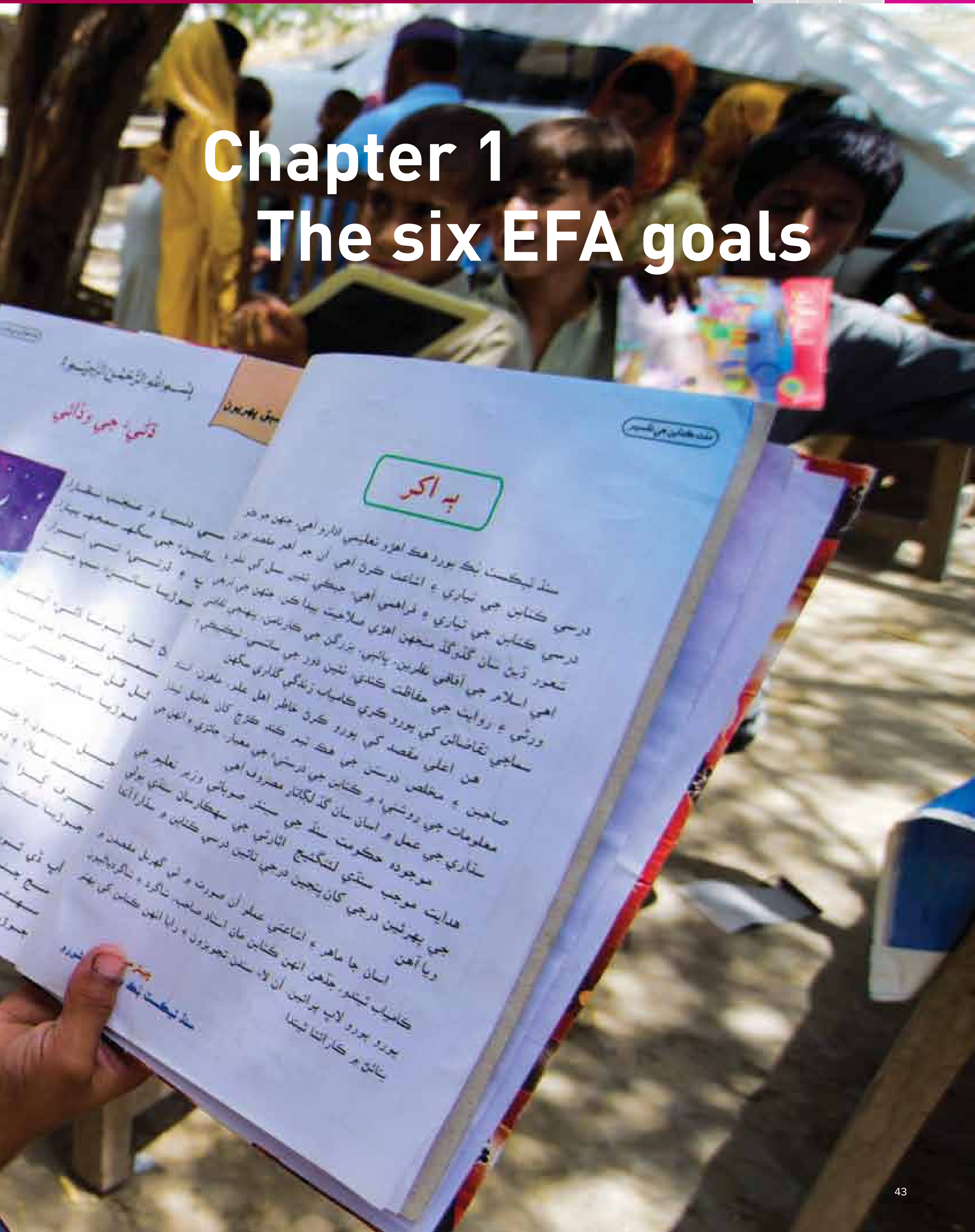
The Dakar Framework for Action included a commitment that no country should be left behind due to lack of resources. A failure by donors to keep this promise has left an education financing gap of US$26 billion per year in some of the world's poorest countries. There are signs that this gap will widen further. Between 2010 and 2011, aid to basic education declined for the first time, by 6%, with low income countries bearing the brunt. It is forecast that many low income countries will see their levels of aid cut further by 2015.

Experience over the past decade of monitoring progress towards the EFA goals offers vital lessons for designing a post-2015 education framework. Goals must be accompanied by clear, measurable targets and indicators. The progress of subgroups must be measured to make sure the most disadvantaged not only advance, but also narrow the gap between them and the more privileged. Post-2015 education goals will only be achieved if governments and aid donors also set specific targets for education financing, with a focus on the poorest, and demonstrate the leadership and political will necessary to keep their commitments.

Tough conditions: At a primary school in Baqir Shah, a village in Sindh, Pakistan, classes take place outside because the buildings collapsed years ago.

Credit: Amina Sayeed/UNESCO

# Chapter 1
# The six EFA goals





More children than ever are going to school, but it is now a certainty that the EFA goals will not be met by the 2015 deadline, in large part because the disadvantaged have been left behind. This chapter, which presents evidence-based analysis of progress, should serve as an indispensable guide for the shape of post-2015 education goals. It shows that, unless special efforts are urgently taken to reach the marginalized, the poorest countries may take several generations to achieve universal completion of primary and lower secondary education, as well as universal youth literacy.

# Goal 1 Early childhood care and education

*Expanding and improving comprehensive early childhood care and education, especially for the most vulnerable and disadvantaged children.*

## Highlights

- Despite major improvements, an unacceptably high number of children suffer from ill health: under-5 mortality was 48 deaths per 1,000 live births in 2012, equivalent to 6.6 million deaths. Of the 43 countries where child mortality was greater than 100 deaths per 1,000 live births in 2000, 35 are not expected to reach the target of reducing child deaths by two-thirds from the 1990 level by 2015.
- Progress in improving child nutrition has been considerable. Yet, as of 2012, the share of children under 5 who were short for their age – a sign of chronic deficiency in essential nutrients – was 25%. Sub-Saharan Africa and South and West Asia account for three-quarters of the world's malnourished children.
- Between 1999 and 2011, the pre-primary education gross enrolment ratio increased from 33% to 50%, although it reached only 18% in sub-Saharan Africa. It is projected that by 2015, only 68 out of 141 countries will have a pre-primary education gross enrolment ratio above 80%.
- In many parts of the world, governments have yet to assume responsibility for pre-primary education: as of 2011, private providers were catering for 33% of all enrolled children. The cost of this provision is one of the factors that contribute most to inequity in access.

**Table 1.1.1: Key indicators for goal 1**

| | Care | | | Education | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Under-5 mortality rate | | Moderate or severe stunting (children under age 5) | Total enrolment | | Gross enrolment ratio (GER) | | Gender parity index of GER | |
| | 2000 (‰) | 2012 (‰) | 2012 (%) | 2011 (000) | Change since 1999 (%) | 1999 (%) | 2011 (%) | 1999 (F/M) | 2011 (F/M) |
| **World** | **75** | **48** | **25** | **170 008** | **52** | **33** | **50** | **0.97** | **1.00** |
| Low income countries | 134 | 82 | 37 | 10 743 | 88 | 11 | 17 | 0.98 | 0.99 |
| Lower middle income countries | 93 | 61 | 35 | 65 195 | 112 | 22 | 46 | 0.93 | 1.01 |
| Upper middle income countries | 38 | 20 | 8 | 64 164 | 29 | 43 | 67 | 1.00 | 1.02 |
| High income countries | 10 | 6 | 3 | 29 906 | 17 | 72 | 82 | 0.99 | 1.01 |
| Sub-Saharan Africa | 156 | 97 | … | 12 222 | 126 | 10 | 18 | 0.95 | 1.00 |
| Arab States | 53 | 33 | … | 4 142 | 72 | 15 | 23 | 0.77 | 0.94 |
| Central Asia | 62 | 36 | … | 1 713 | 35 | 19 | 32 | 0.96 | 1.00 |
| East Asia and the Pacific | 39 | 20 | … | 47 603 | 29 | 39 | 62 | 1.00 | 1.01 |
| South and West Asia | 92 | 58 | … | 49 539 | 130 | 22 | 50 | 0.93 | 1.02 |
| Latin America and the Caribbean | 32 | 19 | … | 20 999 | 31 | 54 | 73 | 1.02 | 1.01 |
| North America and Western Europe | 7 | 6 | … | 22 341 | 17 | 76 | 85 | 0.98 | 1.01 |
| Central and Eastern Europe | 24 | 11 | … | 11 448 | 21 | 51 | 72 | 0.96 | 0.98 |

*Note:* Gender parity is reached when the gender parity index is between 0.97 and 1.03.
*Sources:* Annex, Statistical Tables 3A and 3B (print) and Statistical Table 3A (website); UIS database; Inter-agency Group for Child Mortality Estimation (2013); UNICEF et al. (2013).

**The total number of children dying before their fifth birthday fell by 48% between 1990 and 2012**

The foundations set in the first thousand days of a child's life, from conception to the second birthday, are critical for the child's future well-being. Hence it is vital for women of reproductive age to have access to adequate health care so that they are well prepared for the risks of pregnancy and the postnatal and infanthood periods. Families need support to make the right choices for mothers and babies. And access to good nutrition holds the key to ensuring that children develop strong immune systems and the cognitive abilities they need in order to learn.

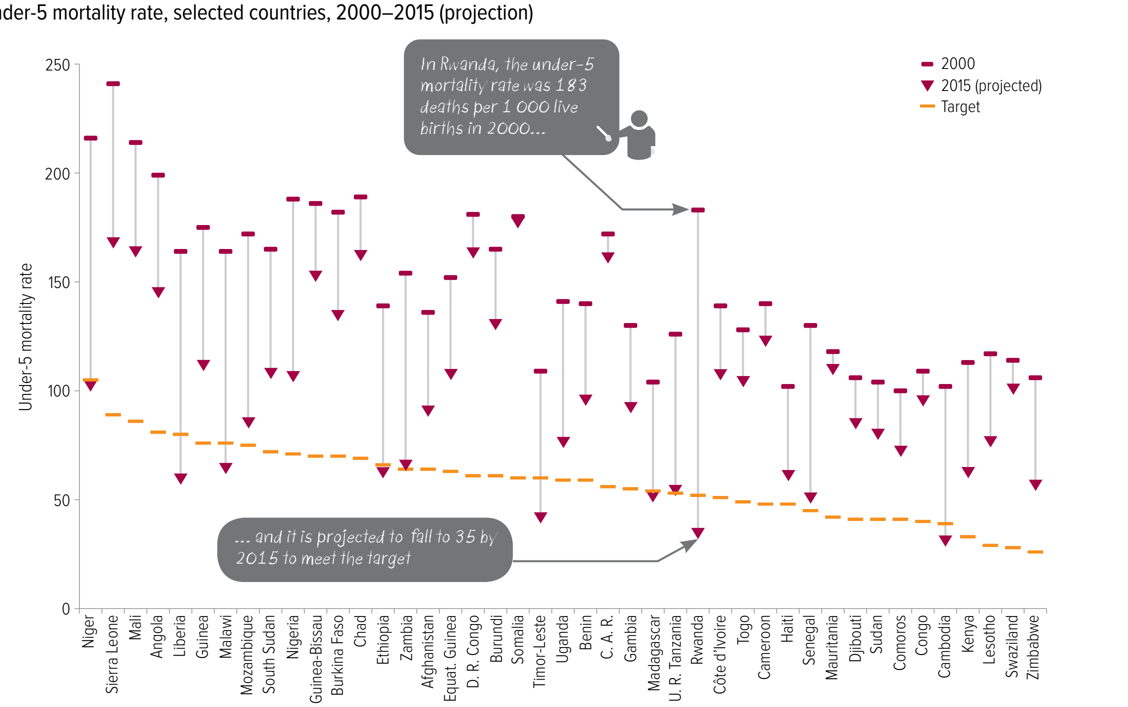
Realization of the significance of early childhood has led to a stronger monitoring system and a better understanding of the scale of the problems in this area, as well as the progress made, which has been considerable.

A decline in the child mortality rate is a key indicator of child health. Between 1990 and 2000, global child mortality fell from 90 deaths per 1,000 live births to 75, and it fell further to 48 in 2012. However, it is still well above the 2015 target of reducing child mortality by two-thirds from the 1990 level to 30 deaths per 1,000 live births (Inter-agency Group for Child Mortality Estimation, 2013).

The total number of children dying before their fifth birthday fell by 48% from 12.6 million in 1990 to 6.6 million in 2012. However, it fell by only 14% in sub-Saharan Africa, the region with the highest mortality rate. On the positive side, the rate of progress in sub-Saharan Africa reached 3.8% per year between 2000 and 2012, compared with 1.4% in the 1990s.

Out of the 43 countries where the child mortality rate was in excess of 100 deaths per 1,000 live births in 2000, all but seven were in sub-Saharan Africa. Two countries had achieved the target by 2011: Liberia and Timor-Leste. If the annual rate of reduction between 2000 and 2011 is projected to 2011–2015, only six more countries reach the target: Cambodia, Ethiopia, Madagascar, Malawi, Niger and Rwanda. By contrast, child mortality has fallen by less than 1% per year in seven countries, including Cameroon, the Democratic Republic of the Congo and Somalia (Figure 1.1.1).

**Figure 1.1.1: Despite progress, few countries are expected to meet the target for child survival**
Under-5 mortality rate, selected countries, 2000–2015 (projection)




*Notes:* The countries shown are those where the child mortality rate exceeded 100 in 2000. The projection from 2011 to 2015 is based on the average rate of reduction between 2000 and 2011.
*Source:* EFA Global Monitoring Report team calculations (2013), based on the Inter-agency Group for Child Mortality Estimation (2012).

Among the 41 countries that had reached the target by 2011, most were in North America and Western Europe or Central and Eastern Europe, or were high income countries in other regions. Nevertheless, some of the world's most populous developing countries, which invested in early childhood interventions, also reduced child mortality by at least two-thirds well in advance of the target date, including Bangladesh, Brazil, China, Egypt, Mexico, and Turkey. Other countries reduced child mortality by more than 70% in the short space of two decades, including El Salvador, the Lao People's Democratic Republic, Maldives and Mongolia.

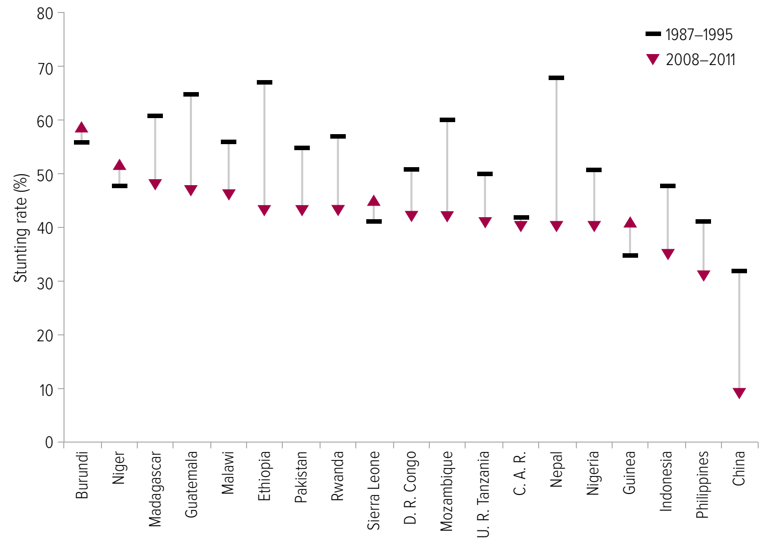
Nutrition has improved in recent years, yet, in 2012, it is estimated that 162 million children under 5 were moderately or severely stunted – that is, short for their age, a robust indicator of long-term malnutrition. The proportion of children who were malnourished reduced from 40% in 1990 to 25% in 2012 (UNICEF et al., 2013). This means the annual rate of reduction needs to increase from 2% to 3.6% if the global target adopted by the World Health Assembly in 2012, to reduce the number of stunted children to 100 million (equivalent to a stunting rate of 15%), is to be achieved by 2025. Achieving the target could, if anything, become more difficult in coming years as a result of factors such as the impact of climate change on agriculture and the increasing volatility of food prices.

The global burden of malnutrition is unequally distributed, with 38% of children in sub-Saharan Africa and in South Asia suffering from malnutrition in 2012. These two regions account for three-quarters of the total population of malnourished children.

Progress has so far also been unequal. China reduced its stunting rate by two-thirds. As a result, upper middle income countries reduced stunting at twice the rate of low and lower middle income countries. Ethiopia and Nepal reduced stunting by more than one-third over little more than 15 years, albeit from excessively high levels. By contrast, in Burundi and Sierra Leone, the proportion of malnourished children slightly increased (Figure 1.1.2).

A lesson from the countries that have made the most progress is that their policies saw

**Figure 1.1.2: Despite improvements, over 40% of young children are malnourished in many countries**
Moderate or severe stunting rate, selected countries, 1987–1995 and 2008–2011




*Source:* UNICEF (2013a), based on Demographic and Health Surveys, Multiple Indicator Cluster Surveys or other national surveys.

malnutrition as a social problem that required an integrated approach, rather than as an isolated challenge to be addressed through technical solutions. In Guatemala, despite some progress over the last 20 years, almost half of all children are still stunted. The signing of the Pacto Hambre Cero (Zero Hunger Pact) in 2012 exemplifies the government's resolve to accelerate progress, especially for indigenous communities. The government's aim is to reduce chronic malnutrition among children under 3 by 10% by 2016 as a first step towards a reduction of 25% by 2022. The implementation plan, which is part of the National Food Security and Nutrition System Law, envisages measures to promote breastfeeding and folic acid and micronutrient supplements to combat chronic malnutrition, along with support for subsistence agriculture and temporary employment programmes to deal with seasonal malnutrition (Guatemala Government, 2012).

**Nepal reduced stunting by more than one-third over 15 years**

The links between early childhood health care and education are strong and mutually reinforcing. Educated mothers adopt better practices that lead to better health outcomes for their children (see Chapter 3). Healthy and well-nourished children, in turn, are more likely to spend more years at school. In one of the most comprehensive studies of the long-term impact of a healthy start to life, groups of children in Brazil and India were followed from birth to adulthood. Boys and girls who were born 500 grams heavier attained 0.3 to 0.4 more years of schooling than their peers. Likewise, boys and girls who had grown faster than expected by the age of 2 attained 0.4 to 0.6 more years of school and were more likely to complete secondary school (Adair et al., 2013).

Early childhood care and education services help build cognitive and non-cognitive skills at a time when children's brains are developing, with long-term benefits for children from disadvantaged backgrounds. In Uganda, a cash transfer programme conditional on pre-school attendance, funded by UNICEF and the World Food Programme, led to an improvement in visual reception skills, fine motor skills and receptive or expressive language (Gilligan and Roy, 2013). In Jamaica, a survey tracked young adults from disadvantaged backgrounds who, as infants, had received psychosocial stimulation through weekly visits by community health personnel. By their early 20s, individuals who had taken part in the programme were earning 42% more than other disadvantaged youth (Gertler et al., 2013).

**In Chad, hardly any poor children attended early childhood education in 2010**

Since 2000, early childhood education services have expanded considerably. The pre-primary education gross enrolment ratio increased from 33% in 1999 to 50% in 2011. Almost 60 million more children have been enrolled in pre-primary schools over the period. The enrolment ratio more than doubled in South and West Asia, from 22% in 1999 to 50% in 2011. Low income and sub-Saharan African countries lag behind, however, with gross enrolment ratios of 17% and 18%, respectively.

Governments face significant challenges in expanding pre-primary education. As a result, the proportion of enrolment in private sector institutions increased from 28% in 1999 to 33% in 2011. In the Arab States, it was as high as 71% (Box 1.1.1). Constraints in collecting data from unregulated non-state providers mean these figures are likely to be an underestimate in some countries.

As private providers tend to serve households who can afford fees, generally those who are better off and concentrated in urban areas, the network of public nurseries and pre-schools needs to expand if more disadvantaged groups are also to be reached. The absence of such provision is a key reason why poorer children are unable to benefit from early childhood education services. In Chad and the Central African Republic, for example, hardly any poor boy or girl attended any form of early childhood education in 2010, but one in six richer children did so. Even in middle income countries with better coverage on average, such as Ghana, Mongolia and Serbia, there is a wide gap in access between the richest and poorest boys and girls (Figure 1.1.3).

**Box 1.1.1: In the Arab States, the private sector is the main provider of pre-primary education**

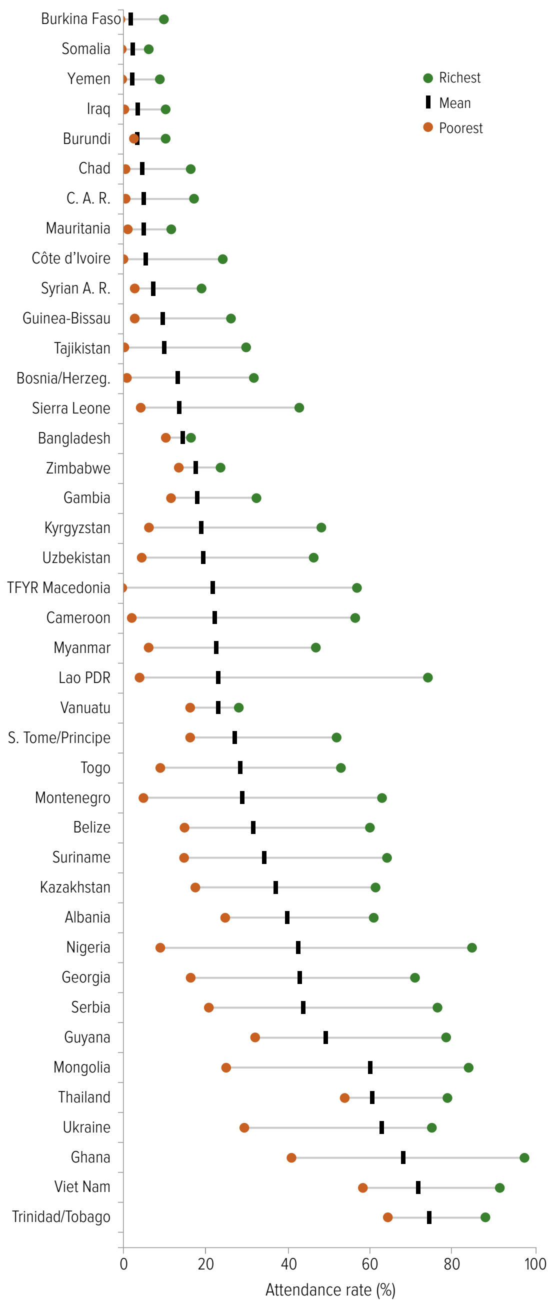
The pre-primary gross enrolment ratio in the Arab States increased from 15% in 1999 to 23% in 2011, but remains the second lowest among the regions. Moreover, this region has the highest share of private provision, with more than two-thirds of total enrolment in private pre-schools and nurseries.

Algeria stands out as the country with the highest share of government provision in the region, at 86%. In addition, it achieved the largest expansion in pre-primary education, from just 2% in 1999 to 75% in 2011. This was the result of a reform that introduced a pre-primary curriculum in 2004 and aimed to increase the gross enrolment ratio to 80% by 2010. At the same time Algeria undertook this rapid expansion of the public system, it encouraged private provision in urban areas. An inspection system monitors implementation of the curriculum in both types of institutions.

In Jordan, the construction of almost 400 kindergartens under a World Bank loan resulted in the share of enrolment in government institutions increasing from zero in 1999 to 17% by 2010. However, enrolment levels remained almost constant over the period, reaching 38% in 2012 despite a target of 50% for that year. It is likely that the continuing predominance of private sector provision is an obstacle to universal access. The share of expenditure on pre-primary education in the education budget was just 0.3% in 2011. As a result, many commendable initiatives taken as part of the national early childhood development strategy – which are unique for the region, including programmes aimed at improving teacher qualifications, setting and monitoring standards, and raising parental participation – have yet to benefit the majority of the population.

*Sources:* Benamar (2010); Jordan Ministry of Education (2013); UNICEF (2009).

**Figure 1.1.3: Few poor 4-year-olds receive pre-primary education**
Percentage of children aged 36 to 59 months who attended some form of organized early childhood education programme, by wealth, selected countries, 2005–2012




*Source:* World Inequality Database on Education, www.education-inequalities.org.

A stronger role for the state calls for clear plans on the objectives of the system, including paying attention to the preparation of teachers and caregivers to address the particular learning needs of young children.

## How many countries are likely to reach the pre-primary education goal by 2015?

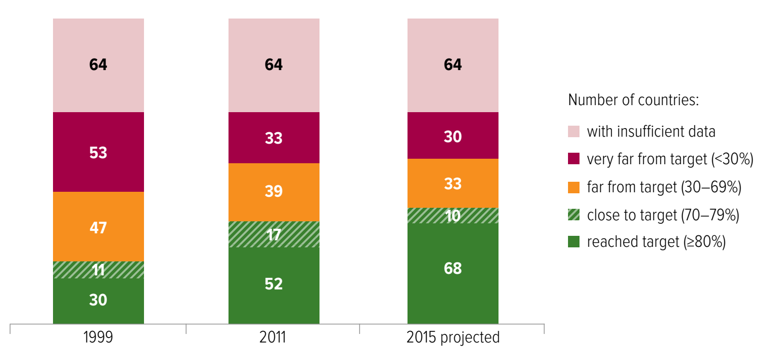
While the Millennium Development Goals were a clear guide towards the early childhood health components of the first Education for All goal, no target was set at the World Education Forum in Dakar in 2000 to guide assessment of success in promoting early childhood education. To gauge comparative progress across countries over the decade, this Report has set an indicative pre-primary education gross enrolment ratio of 80% as a target, which should be realistic for countries to reach by 2015. The analysis, which looks at how close countries are to achieving the target, not only takes stock of the situation as of 2011, but also examines whether countries have a realistic chance of reaching the target by 2015.

**By 2015, 68 countries are expected to have reached pre-primary enrolment of 80%**

Out of the 141 countries for which there are data, in 1999 just 30 had a gross enrolment ratio above 80%, while 11 were close. Between 1999 and 2011, the number of countries that reached the target increased to 52, and 17 had come close (Figure 1.1.4). Looking ahead to 2015, it is projected that 68 countries will reach the target, including Costa Rica, Lithuania and South Africa, and 10 will come close (Table 1.1.2).

The number of countries with a gross enrolment ratio of less than 30% fell from 53 in 1999 to 33 in 2011. Of the 20 countries that moved out of this group, Angola and Mongolia have since

**Figure 1.1.4: Goal 1 – pre-primary education will need to grow faster after 2015**
Number of countries by level of pre-primary gross enrolment ratio, 1999, 2011 and 2015 (projected)




*Note:* The analysis was conducted on the subset of countries for which a projection was possible; therefore, it covers fewer countries than those for which information is available for either 1999 or 2011.
*Source:* Bruneforth (2013).

**Table 1.1.2: Likelihood of countries achieving a pre-primary enrolment target of at least 70% by 2015**

| Level expected by 2015 | | Change between 1999 and 2011: Strong relative progress 41 | Change between 1999 and 2011: Slow progress or moving away from target 22 |
|---|---|---|---|
| Target reached or close (≥70%) | 78 | Algeria, Angola, Antigua and Barbuda, Argentina, Aruba, Australia, Austria, Barbados, Belarus, Belgium, Brunei Darussalam, Bulgaria, Canada, Cape Verde, Chile, Cook Islands, Costa Rica, Cuba, Cyprus, Czech Republic, Denmark, Dominica, Ecuador, El Salvador, Equatorial Guinea, Estonia, Finland, France, Germany, Ghana, Greece, Grenada, Guatemala, Guyana, Hungary, Iceland, India, Israel, Italy, Jamaica, Japan, Latvia, Lebanon, Lithuania, Luxembourg, Malaysia, Maldives, Malta, Mexico, Mongolia, Nepal, Netherlands, New Zealand, Nicaragua, Norway, Panama, Peru, Poland, Portugal, Qatar, Republic of Moldova, Romania, Russian Federation, Saint Kitts and Nevis, Seychelles, Slovakia, Slovenia, South Africa, Spain, Suriname, Sweden, Switzerland, Thailand, Ukraine, United Kingdom, Uruguay, Venezuela (Bolivarian Republic of), Viet Nam | |
| Far from target (30–69%) | 33 | Armenia, Belize, Cameroon, China, Croatia, Egypt, Honduras, Indonesia, Iran (Islamic Republic of), Kazakhstan, Lesotho, Montenegro, Philippines, Sao Tome and Principe, Solomon Islands, Turkey | Albania, Bermuda, Bolivia (Plurinational State of), Cayman Islands, Colombia, Dominican Republic, Jordan, Kenya, Marshall Islands, Morocco, Palestine, Paraguay, Saint Lucia, Samoa, Serbia, United States, Vanuatu |
| Very far from target (<30%) | 30 | Azerbaijan, Bangladesh, Benin, Bhutan, Burkina Faso, Burundi, Cambodia, Congo, Côte d'Ivoire, Democratic Republic of the Congo, Djibouti, Eritrea, Ethiopia, Guinea-Bissau, Kyrgyzstan, Lao People's Democratic Republic, Madagascar, Mali, Myanmar, Niger, Nigeria, Rwanda, Senegal, Togo, Yemen | Fiji, Syrian Arab Republic, Tajikistan, The former Yugoslav Republic of Macedonia, Uzbekistan |

| | | |
|---|---|---|
| Countries not included in analysis because of insufficient data | 64 | Afghanistan, Andorra, Anguilla, Bahamas, Bahrain, Bosnia and Herzegovina, Botswana, Brazil, British Virgin Islands, Central African Republic, Chad, Comoros, Democratic People's Republic of Korea, Gabon, Gambia, Georgia, Guinea, Haiti, Iraq, Ireland, Kiribati, Kuwait, Liberia, Libya, Macao (China), Malawi, Mauritania, Mauritius, Micronesia (Federated States of), Monaco, Montserrat, Mozambique, Namibia, Nauru, Netherlands Antilles, Niue, Oman, Pakistan, Palau, Papua New Guinea, Republic of Korea, Saint Vincent and the Grenadines, San Marino, Saudi Arabia, Sierra Leone, Singapore, Somalia, South Sudan, Sri Lanka, Sudan, Swaziland, Timor-Leste, Tokelau, Tonga, Trinidad and Tobago, Tunisia, Turkmenistan, Turks and Caicos Islands, Tuvalu, Uganda, United Arab Emirates, United Republic of Tanzania, Zambia, Zimbabwe |

*Source:* Bruneforth (2013).

achieved the target, while Algeria and Equatorial Guinea have come close. It is projected that, by 2015, 30 countries will still be very far from the target, of which 16 are in sub-Saharan Africa, including Côte d'Ivoire, Ethiopia and Mali.

In assessing performance, it is important to recognize how quickly some countries have moved towards the target even if they have not yet made it. Of the 72 countries that were far or very far from the target in 2011, 50 had nevertheless made strong progress since 1999, increasing their gross enrolment ratio by at least 33%. Eight countries expanded access by more than 25 percentage points, including India (from 19% to 55%), the Islamic Republic of Iran (from 14% to 43%), Nicaragua (from 28% to 55%) and South Africa (from 21% to 65%).[1]

By contrast, enrolment ratios stagnated in some countries, including the Plurinational State of Bolivia, which remained at 46%, Morocco (58%) and Serbia (53%). Of the 53 countries that were furthest from the target in 1999, only 5 did not rapidly increase their gross enrolment ratio, remaining below 30% by 2011: Fiji, the Syrian Arab Republic, Tajikistan, the Former Yugoslav Republic of Macedonia and Uzbekistan. Countries in Central and Eastern Europe have experienced different trajectories. Their speed of progress towards achieving universal pre-primary education depends on their political commitment towards equality (Box 1.1.2).

Not only is it a cause for concern that 63 countries are likely to be far or very far from the 80% target by 2015, but this excludes the 64 countries without data. Moreover, 80% is a modest target, leaving some young children without access to pre-primary education. Given that those excluded are often the most vulnerable, it is vital for any post-2015 goal to provide a clear target to ensure that all young children have access to pre-primary education.

1. In such comparisons it is important to recognize that differences between countries are sometimes due to variation in age groups: the chances of a country with a one-year target age group (e.g. 5-year-olds in Angola) are higher than those of a country with a three-year target age group (e.g. 3- to 5-year-olds in India).

### Box 1.1.2: In Central and Eastern Europe, the quest for universal access to pre-primary education continues

Access to pre-primary education increased rapidly in Central and Eastern Europe between 1999 and 2011, from 51% to 72%. This region, together with Central Asia, has the highest share of government provision at more than 97%. However, countries in Central and Eastern Europe have progressed at different paces, reflecting the varying degree of attention that governments have paid to this level of education.

In the Republic of Moldova, where there has been a concerted effort to expand pre-primary education, the gross enrolment ratio of children aged 3 to 6 increased from 43% in 2000 to 77% in 2011. Strong political commitment led to pre-primary education's share of the education budget reaching as much as 20% in 2011. Education already receives a high share of the government budget, at 22%, or 7.9% of GNP. The national strategy aims to achieve enrolment ratios of 78% by 2015 for children aged 3 to 6 and 98% for those aged 6 and 7 while reducing inequality between urban and rural areas by 5%. The strategy focuses on both access – targeting lagging areas and disadvantaged groups – and quality, expanding the training systems of educators and nurses.

The Former Yugoslav Republic of Macedonia appears to have a very low enrolment ratio, at 25% in 2010 – placing it in the group of countries most off track – but this figure is partly the result of a reform in 2005/06 that converted the final year of pre-primary education into a compulsory preparatory first year of primary education. In addition, early childhood education enrolment ratios are calculated for all children aged 6 and under, a much wider age group than other countries in the region.

Evidence from household surveys suggests that there has been progress in recent years but that inequality is high and increasing. Between 2005 and 2011, the percentage of 3- and 4-year-olds who attended some form of organized early childhood education programme doubled from 11% to 22%. However, the beneficiaries were almost exclusively children from the wealthiest 40% of families.

To improve access, the cost to households needs to be reduced. Families have to contribute 30% of the cost, restricting the access of the poorest. Children from the country's minorities also benefited far less, with attendance ratios for members of the Albanian and Roma communities at less than 4%. In the case of the Roma, this is despite initiatives to promote their inclusion in pre-primary schools.

*Sources:* Open Society Foundations et al. (2011); Republic of Moldova Ministry of Education (2010); TFYR Macedonia Ministry of Health et al. (2011); TFYR Macedonia Ministry of Labour and Social Policy (2010); UNESCO IBE (2011).

# Goal 2 Universal primary education

*Ensuring that by 2015 all children, particularly girls, children in difficult circumstances and those belonging to ethnic minorities, have access to and complete, free and compulsory primary education of good quality.*

## Highlights

- The goal of universal primary education is likely to be missed by a wide margin, as 57 million children were still out of school in 2011. Half of these children live in conflict-affected countries.
- If the global rate of reduction of the out-of-school population observed between 1999 and 2008 had been maintained, the target could have been achieved by 2015.
- The region lagging the most is sub-Saharan Africa, where no progress has been made since 2007, leaving 22% of primary school age children out of school in 2011. No progress was made in the region on preventing children from dropping out of school over the decade: of those who entered school, the proportion who reached the last grade fell from 58% in 1999 to 56% in 2010.
- Girls make up about 54% of the global population of children out of school. The proportion rises to 60% in the Arab States, a share that has remained unchanged since 1999.
- Of the 57 million children out of school, nearly half are expected never to make it to school. In the Arab States and sub-Saharan Africa, almost two-thirds of girls who are out of school are expected never to go to school.
- It is projected that by 2015, only 68 out of 122 countries will achieve universal primary enrolment. In 15 countries, on current trends, the ratio will still be below 80%.
- The record is likely to be far worse for primary school completion: it is expected that only 13 out of 90 countries will achieve universal primary school completion.

**Table 1.2.1: Key indicators for goal 2**

| | Total primary enrolment | | Primary adjusted net enrolment ratio | | Out-of-school children | | | Survival rate to last grade of primary education | |
|---|---|---|---|---|---|---|---|---|---|
| | 2011 (000) | Change since 1999 (%) | 1999 (%) | 2011 (%) | 2011 (000) | Change since 1999 (%) | Female (%) | 1999 (%) | 2010 (%) |
| **World** | **698 693** | **7** | **84** | **91** | **57 186** | **-47** | **54** | **74** | **75** |
| Low income countries | 126 870 | 70 | 59 | 82 | 21 370 | -46 | 54 | 55 | 59 |
| Lower middle income countries | 293 937 | 20 | 79 | 90 | 27 826 | -49 | 55 | 68 | 69 |
| Upper middle income countries | 204 934 | -20 | 95 | 97 | 6 337 | -44 | 49 | 85 | 90 |
| High income countries | 72 951 | -4 | 97 | 98 | 1 653 | -24 | 46 | 92 | 94 |
| Sub-Saharan Africa | 136 423 | 66 | 59 | 78 | 29 798 | -29 | 54 | 58 | 56 |
| Arab States | 42 771 | 22 | 79 | 89 | 4 823 | -42 | 60 | 79 | 87 |
| Central Asia | 5 468 | -20 | 94 | 95 | 290 | -34 | 55 | 97 | 98 |
| East Asia and the Pacific | 184 257 | -18 | 95 | 97 | 5 118 | -50 | 44 | 84 | 89 |
| South and West Asia | 192 850 | 24 | 77 | 93 | 12 450 | -69 | 57 | 62 | 64 |
| Latin America and the Caribbean | 65 686 | -6 | 94 | 95 | 2 726 | -24 | 45 | 77 | 84 |
| North America and Western Europe | 51 686 | -2 | 98 | 98 | 1 249 | 39 | 45 | 92 | 94 |
| Central and Eastern Europe | 19 552 | -21 | 93 | 96 | 732 | -57 | 49 | 96 | 98 |

*Sources:* Annex, Statistical Tables 5 and 6; UIS database.

With just two years until the 2015 deadline for the Education for All (EFA) goals, the world is unlikely to fulfil one of the most modest promises: to get every child into primary school. More than 57 million children continue to be denied the right to education, and almost half of them will probably never enter a classroom.

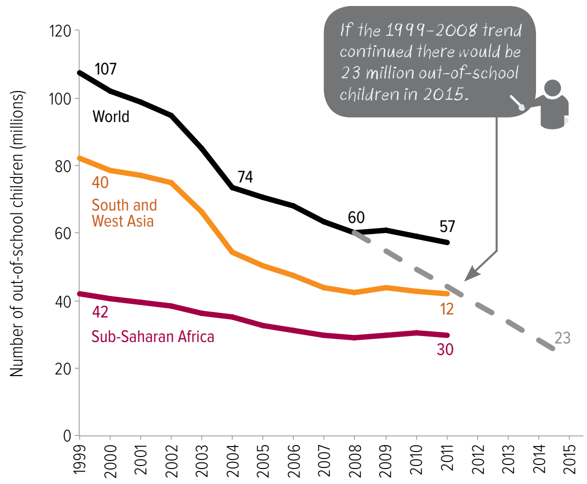
There is some good news: between 1999 and 2011, the number of children out of school fell almost by half, from 107 million to 57 million. Following a period of stagnation, the number of children out of school began to fall again between 2010 and 2011. Yet that reduction, of 1.9 million, is little more than half the average annual level of decline between 2004 and 2008 of 3.4 million and scarcely more than a quarter of the average annual level of decline between 1999 and 2004 of 6.8 million. Had the rate of decline between 1999 and 2008 been maintained, the number of children out of school would be 23 million by 2015, just below the EFA target of a 97% net enrolment ratio (Figure 1.2.1). But since 2008 the rate of decline has been so slow that 53 million children are likely to be out of school in 2015.

There are wide regional variations in the fall in out-of-school numbers since 1999. South and West Asia and sub-Saharan Africa have accounted for three-quarters of the world's out-of-school population throughout the Dakar period, but have seen very different trends. South and West Asia experienced the fastest decline of all regions, contributing more than half the total reduction in the number out of school. In India and the Islamic Republic of Iran, the out-of-school population fell by more than 90%. By contrast, in sub-Saharan Africa, after three years of stagnation, there was a small reduction in the number out of school between 2010 and 2011, of 0.8 million – leaving it at the same level as in 2007. In 2011, 22% of the region's primary school age population was still not in school.

Girls make up about 54% of the global population of children out of school. In the Arab States the share is 60%, unchanged since 1999. In South and West Asia, by contrast, the share of girls in the out-of-school population fell steadily from 64% in 1999 to 57% in 2011.

Over the past five years, there has been little change in the list of countries with the highest numbers of children out of school. Among the countries with data for both 2006 and 2011,

**Figure 1.2.1: Millions of children remain out of school in 2011**
Number of primary school age children out of school, by region, 1999–2011




*Note:* The dotted line from 2008 to 2015 is based on the average annual absolute reduction in the number of out-of-school children between 1999 and 2008.
*Sources:* UIS database; EFA Global Monitoring Report team calculations (2013).

the 10 worst performers for both periods are identical, with one exception: Ghana was replaced by Yemen, whose out-of-school numbers increased (Table 1.2.2A).

The top three performers since 2006 are the Lao People's Democratic Republic, Rwanda and Viet Nam, which reduced their out-of-school populations by at least 85% (Table 1.2.2B). Ethiopia and India are among the 10 countries that reduced their out-of-school populations the most in relative terms, and they also contributed significantly to the overall reduction in out-of-school numbers – by 2.2 million and 4.5 million, respectively.

**Ethiopia and India have contributed significantly to the overall reduction in out-of-school numbers**

Five of the 10 countries with the largest relative increases in out-of-school populations since 2004–2006 are in sub-Saharan Africa (Table 1.2.2C). Nigeria's out-of-school population not only grew the most in absolute terms, by 3.4 million, but also had the fourth highest rate of growth. In some countries, including Colombia, Paraguay and Thailand, the out-of-school population rose even though they are in regions that have been performing better on average.

Some countries that might have been on these lists are not there simply because no data are publicly available (Box 1.2.1). Half of the world's out-of-school children live in conflicted-affected countries, many of which lack data (Box 1.2.2).

**Table 1.2.2: Changes in the out-of-school population, 2006–2011**

*A. The ten countries with the highest out-of-school populations*

| | Out-of-school population 2011 (000) |
|---|---|
| Nigeria | 10 542 |
| Pakistan | 5 436 |
| Ethiopia | 1 703 |
| India | 1 674 |
| Philippines | 1 460 |
| Côte d'Ivoire | 1 161 |
| Burkina Faso | 1 015 |
| Kenya | 1 010 |
| Niger | 957 |
| Yemen | 949 |

*B. The ten countries with the highest relative decreases in out-of-school populations*

| | Out-of-school population | | |
|---|---|---|---|
| | 2006 (000) | 2011 (000) | Change (%) |
| Rwanda | 273 | 20 | -93 |
| Viet Nam | 436 | 39 | -91 |
| Lao PDR | 123 | 19 | -85 |
| Congo | 239 | 47 | -81 |
| Lebanon | 58 | 12 | -78 |
| India | 6 184 | 1 674 | -73 |
| Timor-Leste | 61 | 18 | -71 |
| Morocco | 419 | 134 | -68 |
| Cambodia | 91 | 31 | -66 |
| Ethiopia | 3 947 | 1 703 | -57 |

*C. The ten countries with the highest relative increases in out-of-school populations*

| | Out-of-school population | | |
|---|---|---|---|
| | 2006 (000) | 2011 (000) | Change (%) |
| Paraguay | 57 | 136 | 139 |
| Colombia | 206 | 435 | 112 |
| Thailand | 387 | 611 | 58 |
| Nigeria | 7 150 | 10 542 | 47 |
| Eritrea | 295 | 422 | 43 |
| Gambia | 65 | 86 | 33 |
| South Africa | 519 | 679 | 31 |
| Liberia | 325 | 386 | 19 |
| Mauritania | 113 | 131 | 16 |
| Yemen | 853 | 949 | 11 |

*Note*: The countries considered were those that had data for both 2006 and 2011 and an out-of-school population in the first period that exceeded 50,000.
*Source*: UIS database.

## Box 1.2.1: Which countries have more than 1 million children out of school?

Despite improvements in the availability of education data over the past decade, it is regrettable that recent data on out-of-school numbers, one of the simplest education indicators, are not publicly available for 57 countries. Focusing only on countries with publishable data is likely to be misleading for global policy debates, particularly as many of the countries without data are likely to be the furthest from achieving UPE. Recent improvements in the availability of household survey data help better understand the global picture. However, this picture will only be complete once all countries have publishable data.

The global figure for out-of-school children includes publicly available data from 147 out of 204 countries, and unpublished estimates by the UNESCO Institute for Statistics (UIS) for the remaining countries. Countries for which the UIS publishes no data either do not have enrolment by age or population data, or these data are not considered sufficiently reliable.

The publicly available data for individual countries account for only 68% of the 57 million children out of school. For example, published data account for 95% of out-of-school children in North America and Western Europe, but only 38% in the Arab States.

The largest gaps are in sub-Saharan Africa, where 14 countries lack data. These include countries affected by conflict, such as the Democratic Republic of the Congo and Somalia, where the chances of going to school remain slim, but also countries such as Benin, Sierra Leone, Togo and the United Republic of Tanzania. While most countries in Latin America have published data, Brazil, for example, does not, due to lack of agreed population figures.

For this Report, an attempt was made to identify which countries lacking data are likely to have more than 1 million children out of school, using estimates of the primary net attendance rate from household surveys carried out between 2008 and 2011.[1]

These calculations add six countries to the eight known to have more than 1 million children out of school (Table 1.2.3). Together the

**Table 1.2.3: Fourteen countries are likely to have more than 1 million children out of school**

| | |
|---|---|
| **Afghanistan** | Kenya |
| Burkina Faso | Nigeria |
| **China** | Pakistan |
| Côte d'Ivoire | Philippines |
| **D. R. Congo** | **Somalia** |
| Ethiopia | **Sudan (pre-secession)** |
| India | **U. R. Tanzania** |

*Notes:* Countries in bold are estimated to have more than 1 million children out of school, according to EFA Global Monitoring Team calculations using household survey data. During the period in question, Sudan is still included in the area that is now South Sudan.
*Source:* EFA Global Monitoring Report team calculations, based on the UIS database, the 2010 World Population Prospects and data from Demographic and Health Surveys and Multiple Indicator Cluster Surveys.

14 countries would account for around two-thirds of the global out-of-school population.

It is probably not surprising that Afghanistan, the Democratic Republic of the Congo, Somalia and pre-secession Sudan have out-of-school populations of over 1 million. China's estimated out-of-school population of more than 1 million is consistent with its having achieved UPE – defined as a net enrolment ratio of at least 97% – because of its large population.

Among the countries added to the list, the most unexpected is the United Republic of Tanzania. The most recent publishable UIS estimate for the United Republic of Tanzania put the out-of-school population at 137,000 in 2008. According to the 2010 Demographic and Health Survey, however, the primary net attendance rate was 80%, well below the 98% net enrolment ratio estimate provided by the UIS for 2008. One reason for this discrepancy is likely to be differences in the way information is collected on child age.[2] Moreover, primary completion rates estimated by both the Demographic and Health Survey and the UIS indicate that only around 7 out of 10 children finish primary school. This supports the possibility that the out-of-school population is indeed over 1 million.

1. Somalia is an exception, with data from the 2006 Multiple Indicator Cluster Survey. Continuing conflict in the country makes it unlikely that enrolment improved significantly over the following five years.

2. See Box 2.5 in the 2010 *EFA Global Monitoring Report* for an explanation of the difference between administrative and household data (UNESCO, 2010).

## Box 1.2.2: The continuing hidden crisis for children in conflict-affected areas

In many of the world's poorest countries, armed conflict continues to destroy the education chances of a whole generation of children. This Report draws on the international reporting systems used for the 2011 *EFA Global Monitoring Report* to construct an updated list of conflict-affected countries. Thirty-two countries were identified as affected by armed conflict in the period from 2002 to 2011, three fewer than in 1999–2008 (Table 1.2.4). New countries have joined the list, however, including Libya, Mali and the Syrian Arab Republic.

The analysis shows that, while the global number of children out of school fell slightly from 60 million in 2008 to 57 million in 2011, the benefits of this progress have not reached children in conflict-affected countries. These children make up 22% of the world's primary school age population, but 50% of its out-of-school children, and the proportion has increased from 42% in 2008.

Of the 28.5 million primary school age children out of school in conflict-affected countries, 12.6 million live in sub-Saharan Africa, 5.3 million in South and West Asia, and 4 million in the Arab States. The vast majority – 95% – live in low and lower middle income countries. Girls, who make up 55% of the total, are the worst affected.

**Table 1.2.4: Conflict-affected countries, 1999–2008 and 2002–2011**

| | | | |
|---|---|---|---|
| Afghanistan | Ethiopia | Myanmar | Sierra Leone |
| Algeria | Georgia | Nepal | Somalia |
| Angola | Guinea | **Niger** | Sri Lanka |
| Burundi | India | Nigeria | Sudan (pre-secession) |
| Central African Republic | Indonesia | Pakistan | **Syrian Arab Republic** |
| Chad | **Iran, Isl. Rep.** | Palestine | Thailand |
| Colombia | Iraq | Philippines | Timor-Leste |
| Côte d'Ivoire | Liberia | Russian Federation | Turkey |
| D. R. Congo | **Libya** | Rwanda | Uganda |
| Eritrea | **Mali** | Serbia | Yemen |

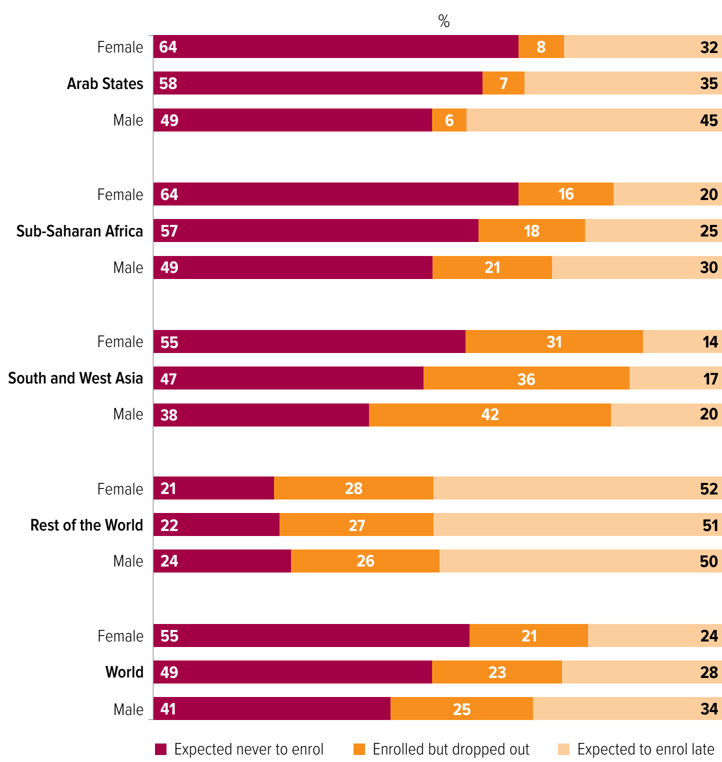
*Notes:* The 2002–2011 list comprises countries with 1,000 or more battle-related deaths over the period, plus those with more than 200 battle-related deaths in any one year between 2009 and 2011. Data are compiled using the Peace Research Institute Oslo and Uppsala Conflict Data Program data sets on armed conflict and battle deaths. See Box 3.1 in the 2011 *EFA Global Monitoring Report* for further information.
Countries in light blue were on the list in 2011 but are no longer identified as conflict-affected in 2013.
Countries in red joined the list in 2013.
During the period in question, Sudan still included the area that is now South Sudan.

One of the biggest disappointments since the EFA goals were established in 2000 is that nearly half of the 57 million children currently out of school are expected never even to make it to school. The percentage is considerably higher in the Arab States and sub-Saharan Africa, where girls are also much more affected: in these regions almost two in three out-of-school girls are expected never to go to school. The remaining half of the out-of-school population is split almost equally between children who enrolled but dropped out and those who are expected to enter school but will be older than the official primary school age, and so are more likely to eventually drop out (Figure 1.2.2).

The reasons for children not making it into school vary, but are usually associated with disadvantages children are born with – poverty, gender, ethnicity, or living in a rural area or a slum, for example. One of the most neglected of such disadvantages is disability (Box 1.2.3).

Children are more likely to complete primary schooling if they enter at the right age. The global adjusted net intake rate for the first year of primary school increased between 1999 and 2011, but only slightly, from 81% to 86% – and rose by less than

**Figure 1.2.2: Almost half of children out of school are expected never to enrol**
Distribution of out-of-school children by school exposure, gender and region, 2011




*Source:* UIS database.

## Box 1.2.3: Children with disabilities are often overlooked

Children with disabilities are often denied their right to education. However, little is known about their school attendance patterns. The collection of data on children with disabilities is not straightforward, but they are vital to ensure that policies are in place to address the constraints they face. Statistics on the education experience of children with disabilities are rare in part because household surveys, which tend to be the best source of information on access to school by different population groups, do not have sufficient information on the degree or type of disability, or their sample size is too small to make it possible to draw accurate conclusions.

By one estimate, 93 million children under age 14, or 5.1% of the world's children, were living with a 'moderate or severe disability' in 2004. Of these, 13 million, or 0.7% of the world's children, experience severe disabilities.

According to the World Health Survey, in 14 of 15 low and middle income countries, people of working age with disabilities were about one-third less likely to have completed primary school. For example, in Bangladesh, 30% of people with disabilities had completed primary school, compared with 48% of those with no disabilities. The corresponding shares were 43% and 57% in Zambia; 56% and 72% in Paraguay.

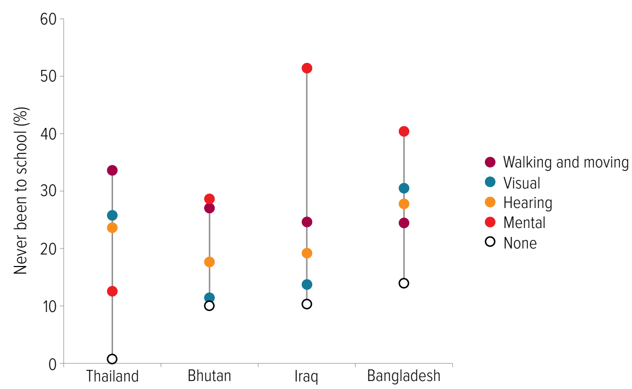
The obstacles that children face depend on the type of disability they experience. Since 2005, Multiple Indicator Cluster Surveys have used a tool with 10 questions to screen children aged 2 to 9 for the risk of various types of impairment. This information points to a risk of disability ranging from 3% in Uzbekistan to 49% in the Central African Republic. While data on the risk of disability may overestimate the number of children actually living with a disability, they throw some light on the barriers that children at risk of disability face in getting to school.

Analysis for this Report of Multiple Indicator Cluster Surveys from four countries shows that children at higher risk of disability are far more likely to be denied a chance to go to school. In Bangladesh, Bhutan and Iraq, children with mental impairments were most likely to be denied this right. In Iraq, for instance, 10% of 6- to 9-year-olds with no risk of disability had never been to school in 2006, but 19% of those at risk of having a hearing impairment and 51% of those who were at higher risk of mental disability had never been to school. In Thailand, almost all 6- to 9-year-olds who had no disability had been to school in 2005/06, and yet 34% of those with walking or moving impairments had never been to school (Figure 1.2.3).

*Sources:* Gottlieb et al. (2009); Mitra et al. (2011); WHO and World Bank (2011).

**Figure 1.2.3: Children at risk of disability face major barriers in gaining access to school**

Percentage of children aged 6 to 9 who have never been to school, by type of impairment, selected countries, 2005–2007




*Source:* EFA Global Monitoring Report team calculations (2013), based on Multiple Indicator Cluster Surveys.

one percentage point over the last four years of the period. Some countries nevertheless have made great progress in getting children into school by the official school starting age, including Ethiopia (where the rate rose from 23% to 94%), the Islamic Republic of Iran (from 46% to 99%) and Mozambique (from 24% to 71%).

Despite improvements in getting children into school, dropout before the last grade remains a serious problem in many low and middle income countries. The chances of children completing the primary cycle have hardly changed since 1999. In 2010, around 75% of those who started primary school reached the last grade. In sub-Saharan Africa, the proportion making it to the last grade even fell slightly, from 58% in 1999 to 56% in 2010. In South and West Asia, fewer than two out of three children entering school manage to reach the last grade. By contrast, in the Arab States there has been progress from 79% of students reaching the last grade in 1999 to 87% doing so in 2010.

## How many countries are likely to reach UPE by 2015?

While the UPE goal has clear indicators, it is not always easy to assess overall progress towards it. The commonly used benchmark of whether at least 97% of children of primary school age are in school does not show whether children complete primary school. Nor does it account for the varied experiences of children from different backgrounds. Thus it is likely to provide an overly optimistic picture of progress.

New analysis for this Report uses data on net enrolment as well as completion to assess progress towards UPE. It also looks at the speed at which countries have moved towards the target, given their starting points. From this information, the analysis projects whether countries have a realistic chance of reaching UPE between 2011 (the latest year for which data are available) and 2015.[1]

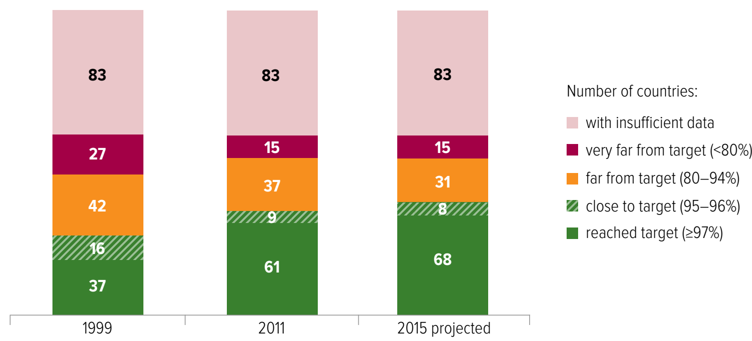
### *Universal access to primary schooling is likely to remain elusive in many countries*

In 1999, 37 of the 122 countries with available data had already reached the target of a 97% primary net enrolment ratio (Figure 1.2.4). By 2011, the number of countries that had reached the target had increased to 61. Looking ahead to 2015, it is projected that seven more countries will reach the target: Croatia, El Salvador, Latvia, Nepal, Nicaragua, Qatar and the United Arab Emirates. This leaves 54 countries that are not expected to meet this target by 2015, only eight of which are even close to it.

Of the countries furthest from UPE, 27 had enrolment ratios below 80% in 1999. In 2015, 15 countries are still expected to be in this situation, including 10 in sub-Saharan Africa: Burkina Faso, Côte d'Ivoire, Equatorial Guinea, Eritrea, the Gambia, Lesotho, Liberia, Mali, Niger and Nigeria. Djibouti and Pakistan are among the remaining countries in this group (Table 1.2.5). Of the 12 countries that moved out of this group, three have since reached the target (the Lao People's Democratic Republic, Rwanda and Zambia), and two are expected to come very close (Morocco and Mozambique). In addition,

1. For this analysis, the countries used are those with at least three data points since 1999, the most recent dating from 2009 or after. See Bruneforth (2013) for further information.

**Figure 1.2.4: Goal 2 – progress towards achieving universal primary education by 2015 is less than usually assumed**
Number of countries by level of adjusted net enrolment ratio, 1999, 2011 and 2015 (projected)




*Note:* The analysis was conducted on the subset of countries for which a projection was possible; therefore, it covers fewer countries than those for which information is available for either 1999 or 2011.
*Source:* Bruneforth (2013).

Mauritania, Senegal and Yemen will move out of the group of countries that were furthest from the target in 2011, as their net enrolment ratio is projected to exceed 80% by 2015.

While some countries are still off track, others have made considerable progress. Of the 46 countries with data that are expected to be far from UPE by 2015, 15 will have made strong progress, increasing their net enrolment ratio by at least 15% since 1999. Of the 27 countries that were furthest from the target in 1999, 20 have made strong progress; Botswana, Côte d'Ivoire and Eritrea have made weak progress, with increases between 5% and 15%, while Equatorial Guinea, the Gambia, Liberia and Nigeria have made even less progress than that, or have regressed.

**By 2015, 54 countries are not expected to have achieved universal primary enrolment**

### *Progress towards primary school completion is even more disappointing*

Enrolment is a partial measure of whether UPE has been achieved, given that success should be judged with respect to whether all children 'have access to and complete' primary education. Hence this Report considers an indicator related to completion: the expected cohort completion rate (Box 1.2.4). One disadvantage of this new indicator is that fewer data are available and no trend can be established to monitor progress since 1999 or make projections to 2015.

In three-quarters of the 90 countries with data since 2008, at least 95% of children were

### Box 1.2.4: Measuring primary school completion

A common measure of whether children are completing primary school is the gross intake rate to the last grade.[1] While this is relatively straightforward to calculate from existing data, over-age enrolment and insufficient data on repetition mean it exceeded 100% in 44 of the 153 countries with data for 2011, so it does not provide an accurate picture. This indicator has been used more broadly in recent years to make the case that the MDG on education has been almost achieved. A preferable measure of completion is one that avoids mixing children from different age groups, following instead a group (cohort) of children of primary school entrance age.

The expected cohort completion rate is one such measure. This new indicator measures the percentage of children of primary school entrance age expected to enter and complete primary education. It is estimated by multiplying two indicators:

- The *expected net intake rate* to the first grade of primary education, showing the probability that a child of official primary school entrance age will ever enrol in the first grade. In Senegal, for example, the expected net intake rate was 83% in 2010.
- The *survival rate to the last grade* of primary school, which shows the probability that a child who has entered primary school will ever reach the last grade. Reaching the last grade is taken as a proxy for completion because actual graduation records from primary school are not collected in many countries. In Senegal, 60% of those who enrolled in school reached the last grade. Taking account of the expected net intake rate, 49% of current 7-year-olds are expected to complete primary school.

Gross intake to the last grade in Senegal was higher, at 63% in 2011, showing that it can overestimate progress towards completion. An indicator in the spirit of the expected cohort completion rate provides a more accurate picture of completion, and so is more appropriate for measuring progress towards post-2015 goals.

1. This indicator is equal to the number of students of any age in the last grade of primary school (excluding repeaters), divided by the number of children of official graduating age.

expected to enter primary school in 2011. Yet, just four years before 2015, there were still eight countries – all in sub-Saharan Africa, and notably including Nigeria – where at least one-fifth of the children were not expected to enrol in primary school (Figure 1.2.5).

Progression remains an even greater challenge. In 26 of the 90 countries, almost three-quarters of them in sub-Saharan Africa, at least one–fifth of the children who entered school were not expected to reach the last grade. In Uganda, for example, fewer than one in three children were expected to reach the last grade.

Combining the two indicators to measure whether a given group of children will both enter primary school and reach the last grade conveys a sobering message. In only 13 out of the 90 countries with data is it expected that at least 97% of children will both enter primary school and reach the last grade. Of these 13 countries, 10 are OECD or EU members. The other three are Belarus, Kazakhstan and Tajikistan.

These findings raise the question whether assessing progress towards UPE solely on the basis of the net enrolment ratio is sufficient. Among the 70 countries that, in net enrolment ratio terms, were considered to have achieved the target in 2011 or to be close to achieving it, data on the expected cohort completion rate are available for 42. Only 15 of these would be considered to have achieved the target in 2011 or to be close to achieving it if the expected cohort completion rate to the last grade were the criterion.

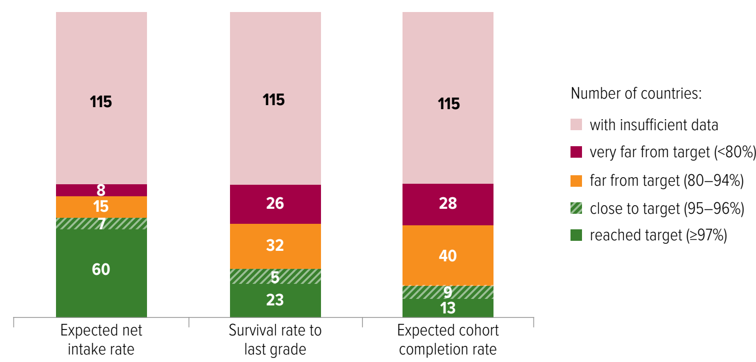
For some countries, the discrepancy between the two measures is particularly large. For example, Rwanda, with a net enrolment ratio of 99% in 2010, had an expected cohort completion rate of 38% in 2009. Guatemala, with a net enrolment ratio of 98% in 2010, had an expected cohort completion rate of 79% in 2009.

Comparing the experiences of the two sub-Saharan African countries that had the largest out-of-school populations in 1999 shows what can be achieved, with strong political commitment, in improving primary school completion. Ethiopia has made good progress while narrowing inequalities, and though it remains far from the target, it is advancing towards the goal. Nigeria, by contrast, has the world's largest number of children out of school. Its out-of-school population grew by 42% between 1999 and 2010 and it is among the 15 countries that are likely to be off track in 2015. In addition the high level of inequality has remained unchanged (Box 1.2.5).

The contrasting records of two countries in East Asia further show how variable degrees of political commitment to EFA have resulted in different outcomes since 2000: while Indonesia championed access to school and is among the countries that have reached the net enrolment ratio target, the Philippines is far from the target and in danger of going backwards (Box 1.2.6).

As the 2015 deadline for the EFA goals approaches, it is disconcerting to note not only that too many countries are far from the goal, but also that data remain insufficient to enable a full global monitoring of progress towards whether children are accessing, and also completing, primary school. Nor do the available data allow assessment of progress for various population groups. Promising developments in the availability of comparable household surveys offer encouraging signs that this situation could change in the post-2015 era.

**Figure 1.2.5: Goal 2 – progress towards universal primary completion is disappointing**
Number of countries by level of expected net intake rate, survival rate to the last grade and expected cohort completion rate, 2011 or latest available year




*Source:* UIS database.

**Table 1.2.5: Likelihood of countries achieving a primary enrolment target of at least 95% by 2015**

| Level expected by 2015 | | | |
|---|---|---|---|
| Target reached or close (≥95%) | 76 | Algeria, Aruba, Australia, Bahamas, Belgium, Belize, Bulgaria, Cambodia, Croatia, Cuba, Cyprus, Dominica, Ecuador, Egypt, El Salvador, Estonia, Fiji, Finland, France, Germany, Greece, Grenada, Guatemala, Honduras, Hungary, Iceland, India, Indonesia, Iran (Islamic Republic of), Ireland, Israel, Italy, Japan, Kazakhstan, Kyrgyzstan, Lao People's Democratic Republic, Latvia, Lebanon, Luxembourg, Malawi, Maldives, Mexico, Mongolia, Morocco, Mozambique, Nepal, Netherlands, New Zealand, Nicaragua, Norway, Oman, Panama, Peru, Portugal, Qatar, Republic of Korea, Rwanda, Saint Vincent and the Grenadines, Samoa, Sao Tome and Principe, Slovenia, Spain, Sweden, Switzerland, Syrian Arab Republic, Tajikistan, The former Yugoslav Republic of Macedonia, Trinidad and Tobago, Tunisia, Turkey, United Arab Emirates, United Kingdom, United States, Venezuela (Bolivarian Republic of), Viet Nam, Zambia | |
| Far from target (80-94%) | 31 | Bhutan, Ethiopia, Ghana, Guinea, Kenya, Mauritania, Senegal, Yemen | Azerbaijan, Belarus, Bolivia (Plurinational State of), Botswana, British Virgin Islands, Cape Verde, Colombia, Denmark, Dominican Republic, Jordan, Lithuania, Malta, Namibia, Palestine, Paraguay, Philippines, Poland, Republic of Moldova, Romania, Saint Lucia, South Africa, Sri Lanka, Suriname |
| Very far from target (<80%) | 15 | Burkina Faso, Djibouti, Eritrea, Lesotho, Mali, Niger, Pakistan | Cayman Islands, Côte d'Ivoire, Equatorial Guinea, Gambia, Jamaica, Liberia, Nigeria, Saint Kitts and Nevis |
| | | Strong relative progress 15 | Slow progress or moving away from target 31 |
| | | **Change between 1999 and 2011** | |

| | | |
|---|---|---|
| Countries not included in analysis because of insufficient data | 83 | Afghanistan, Albania, Andorra, Angola, Anguilla, Antigua and Barbuda, Argentina, Armenia, Austria, Bahrain, Bangladesh, Barbados, Benin, Bermuda, Bosnia and Herzegovina, Brazil, Brunei Darussalam, Burundi, Cameroon, Canada, Central African Republic, Chad, Chile, China, Comoros, Congo, Cook Islands, Costa Rica, Czech Republic, Democratic People's Republic of Korea, Democratic Republic of the Congo, Gabon, Georgia, Guinea-Bissau, Guyana, Haiti, Iraq, Kiribati, Kuwait, Libya, Macao (China), Madagascar, Malaysia, Marshall Islands, Mauritius, Micronesia (Federated States of), Monaco, Montenegro, Montserrat, Myanmar, Nauru, Netherlands Antilles, Niue, Palau, Papua New Guinea, Russian Federation, San Marino, Saudi Arabia, Serbia, Seychelles, Sierra Leone, Singapore, Slovakia, Solomon Islands, Somalia, South Sudan, Sudan, Swaziland, Thailand, Timor-Leste, Togo, Tokelau, Tonga, Turkmenistan, Turks and Caicos Islands, Tuvalu, Uganda, Ukraine, United Republic of Tanzania, Uruguay, Uzbekistan, Vanuatu, Zimbabwe |

*Source:* Bruneforth (2013).

## Box 1.2.5: The contrasting education fortunes of Ethiopia and Nigeria

The education experience of the 2000s for the two most populous countries of sub-Saharan Africa could not have been more different. It was a lost decade for Nigeria in terms of educational development. Despite the optimism generated by the restoration of democracy in 1999 and the introduction of the Universal Basic Education Law in 2004, the percentage of children out of school showed no improvement. By contrast, Ethiopia made major advances, with the number of children out of school falling by three-quarters between 1999 and 2011. While major challenges remain, the reforms that Ethiopia has put in place suggest that continued progress should be possible.

Household survey data confirm this divergent picture. Between 1998 and 2008, the share of children who had not completed primary school remained at about 29% in Nigeria (Figure 1.2.6). The gap between the country's two most populous areas hardly changed, with the rate of non-completers six times higher in the North-west than in the South-west. The gender gap within the two regions did not change either. As a result, 70% of young women in the North-west have not completed primary school.

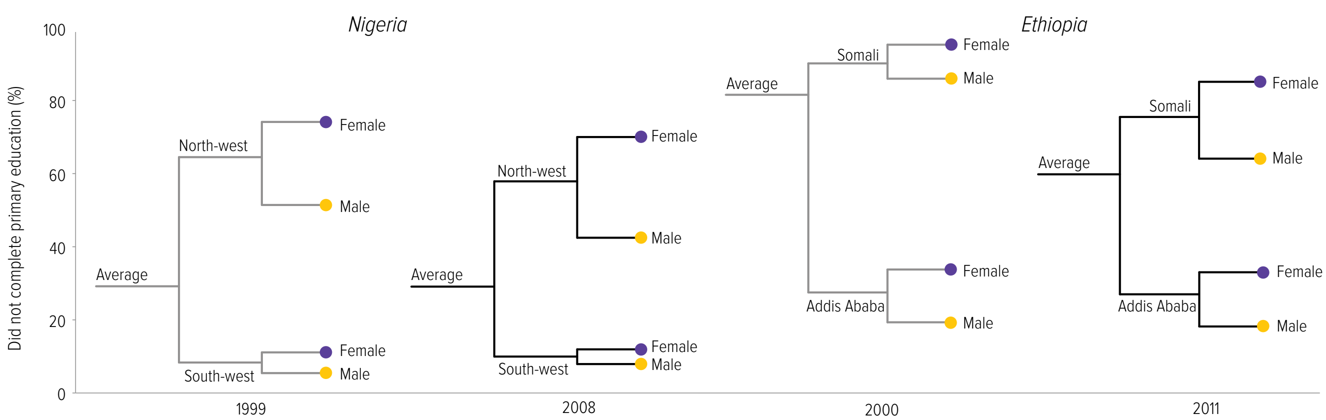
By contrast, Ethiopia witnessed improvement, albeit from a low base. In 2000, 82% of children had not completed primary school, falling to 60% by 2011. While there are still fewer benefiting from education than in Nigeria, on average, it is notable that inequalities have narrowed while progress has been made overall. The gap in primary school completion between Addis Ababa and the Somali region fell from 63 to 49 percentage points. Given that there has been a large reduction in the percentage of children who have never been to school in the Somali region (from 84% in 2000 to 37% in 2011), this gap is expected to fall even further in coming years.

Since Ethiopia became a federal republic in 1994, its central government has been effective both in exercising its coordinating function and in promoting policy decisions. It more than doubled the share of the budget allocated to education between 2000 and 2010, to 25%. These resources were used to fund rapid classroom construction and teacher recruitment. At the same time, the government ambitiously devolved power to the regions and districts, while closely monitoring results in the delivery of education and other social services. An analysis of almost 200 urban and rural districts in the Oromiya region and the Southern Nations, Nationalities and People's Region showed that the introduction of formula-based funding led to declines in inequality between districts not only in terms of funding per student but also in terms of enrolment outcomes.

In Nigeria, the federal government has less control over the states. The absence of data on the share of the budget spent on education since 1999 is one sign of poor accountability. An attempt to track whether schools in Kaduna and Enugu states received allocated resources revealed that, for most basic inputs, including maintenance, textbooks and in-service training, there were not even any norms as to what each school should receive.

*Sources:* Garcia and Rajkumar (2008); Nigeria National Bureau of Statistics et al. (2013); World Bank (2008b).

**Figure 1.2.6: Ethiopia has made progress in primary education while Nigeria has stagnated**
Percentage of children who have not completed primary school, by gender and selected regions, Ethiopia 2000–2011 and Nigeria 1999–2008




*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Survey data.

## Box 1.2.6: The cost of delaying action for disadvantaged children – the diverging paths of Indonesia and the Philippines

The Philippines is one of 14 countries estimated to have more than 1 million out-of-school children and also one where there has been lack of progress towards UPE. By contrast, Indonesia managed to reduce its out-of-school population by 84% between 2000 and 2011.

Evidence from household surveys shows, moreover, that inequality gaps in the Philippines have remained wide. The percentage of young people who had not completed primary education slightly increased in the Philippines over 1998–2008, from 11% to 13%, with inequalities between the poorest and richest and between young men and young women remaining entrenched. In Indonesia, the percentage of youth who had not completed primary education almost halved over a similar period, 1997–2007, from 14% to 8%, while gender inequalities remained narrow (Figure 1.2.7).

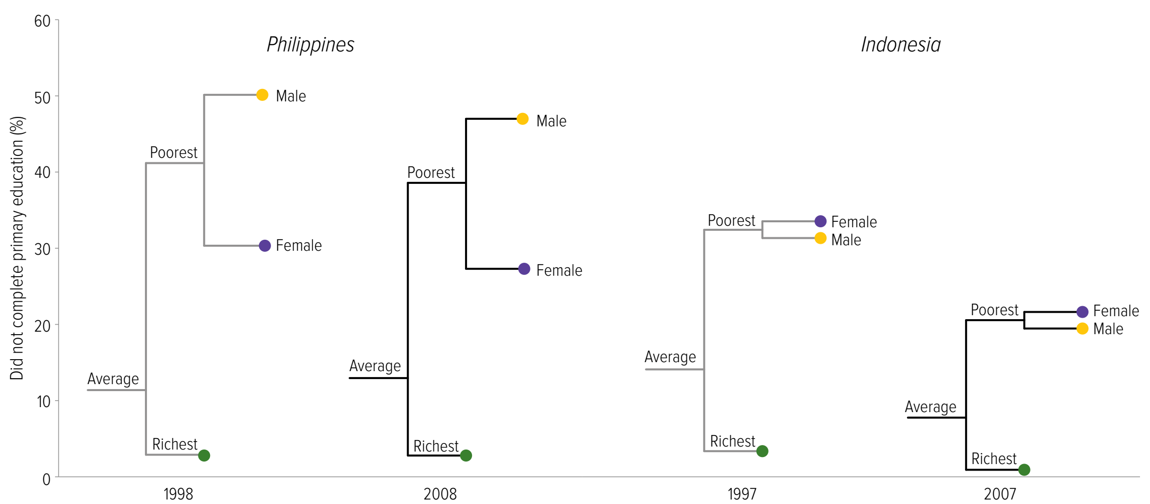
Whether children in the Philippines have a chance of attending school depends strongly on where they live. In the Autonomous Region of Muslim Mindanao, which continues to suffer from conflict, 21% of primary school age children were not in school in 2008 – more than twice the national average of 9%. The share of national income invested in education, which equalled the subregional average in 1999, had fallen behind by 2009 at 2.7% of GNP, compared with an average of 3.2% for East Asia.

Indonesia, for its part, showed strong political will towards education reform. After the 1997/98 financial crisis, the country embarked on an ambitious set of reforms regarding decentralization, social protection and education, which all contributed towards the improvement of educational outcomes. Since 2005, the School Operational Assistance programme has provided grants to schools to cover their running costs so they do not have to charge fees.

In 2009, the central government fulfilled a constitutional commitment to allocate 20% of its budget to education. The increase in resources has resulted in an overhaul of the scholarship system for poor students, which was insufficient, poorly targeted and not timed well enough to prevent dropout in the last grade of primary school. The reform will improve the targeting mechanism, increase the scholarship amount and modernize the administration to ensure that households are paid in time.

*Sources:* Albert et al. (2012); Chaudhury et al. (2013); Indonesia National Development Planning Agency et al. (2012); Satriawan (2013); World Bank (2013e).

**Figure 1.2.7: Indonesia has moved much faster than the Philippines towards universal primary education**
Percentage of young people who have not completed primary school, by gender and wealth, Indonesia 1997–2007 and the Philippines 1998–2008




*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Survey data.

# Goal 3 Youth and adult skills

*Ensuring that the learning needs of all young people and adults are met through equitable access to appropriate learning and life skills programmes.*

## Highlights

- Participation in lower secondary education increased from 72% in 1999 to 82% in 2011. The fastest growth was in sub-Saharan Africa, where enrolment more than doubled, albeit from a low base, reaching 49% in 2011.
- Progress in completing lower secondary school – a prerequisite for acquiring the foundation skills necessary for decent jobs – has been more modest, with little improvement in low income countries, where just 37% of adolescents complete that level. Among adolescents from poor families in these countries, completion rates are as low as 14%.
- The number of adolescents out of school stood at 69 million in 2011. While this constituted a reduction of 31% since 1999, most of the improvement had been achieved by 2004. In sub-Saharan Africa, the number of adolescents out of school remained at 22 million between 1999 and 2011, as population growth cancelled out any improvement in enrolment growth.
- Of the 82 countries with data, 38 are expected to achieve universal lower secondary enrolment by 2015. But three-quarters of the countries in sub-Saharan Africa are not included among these 82 countries. Given most of these countries have not yet achieved universal primary completion, it is extremely unlikely that they will achieve universal lower secondary education by 2015.
- Across 19 high income countries around half of young people have poor problem-solving skills in part due to too low upper-secondary completion.

**Table 1.3.1: Key indicators for goal 3**

| | Total secondary enrolment | | Lower secondary gross enrolment ratio | | Upper secondary gross enrolment ratio | | Technical and vocational education as a share of secondary enrolment | | Out-of-school adolescents of lower secondary school age | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 (000) | Change since 1999 (%) | 1999 (%) | 2011 (%) | 1999 (%) | 2011 (%) | 1999 (%) | 2011 (%) | 2011 (000) | Change since 1999 (%) | Female (%) |
| **World** | **543 226** | **25** | **72** | **82** | **45** | **59** | **11** | **11** | **69 413** | **-31** | **49** |
| Low income countries | 49 393 | 83 | 36 | 54 | 23 | 31 | 5 | 5 | 18 435 | -13 | 51 |
| Lower middle income countries | 203 179 | 48 | 61 | 77 | 31 | 48 | 5 | 5 | 42 359 | -25 | 53 |
| Upper middle income countries | 205 015 | 11 | 88 | 96 | 52 | 75 | 13 | 16 | 7 810 | -65 | 30 |
| High income countries | 85 640 | -2 | 102 | 105 | 97 | 100 | 17 | 14 | 809 | -52 | 39 |
| Sub-Saharan Africa | 46 282 | 114 | 29 | 49 | 21 | 32 | 8 | 8 | 21 832 | -1 | 55 |
| Arab States | 30 726 | 37 | 73 | 88 | 44 | 52 | 14 | 9 | 3 757 | -31 | 56 |
| Central Asia | 10 288 | 12 | 85 | 97 | 81 | 102 | 6 | 13 | 397 | -56 | 57 |
| East Asia and the Pacific | 159 783 | 22 | 78 | 90 | 44 | 70 | 15 | 17 | 8 944 | -64 | 33 |
| South and West Asia | 144 402 | 48 | 61 | 76 | 30 | 47 | 1 | 2 | 31 277 | -21 | 50 |
| Latin America and the Caribbean | 60 525 | 15 | 95 | 102 | 62 | 77 | 10 | 10 | 1 494 | -55 | 49 |
| North America and Western Europe | 61 433 | 1 | 102 | 106 | 97 | 99 | 14 | 14 | 583 | -50 | 35 |
| Central and Eastern Europe | 29 787 | -27 | 92 | 95 | 82 | 83 | 18 | 21 | 1 129 | -68 | 48 |

*Sources:* Annex, Statistical Table 7 (print) and Statistical Table 8 (website); UIS database.

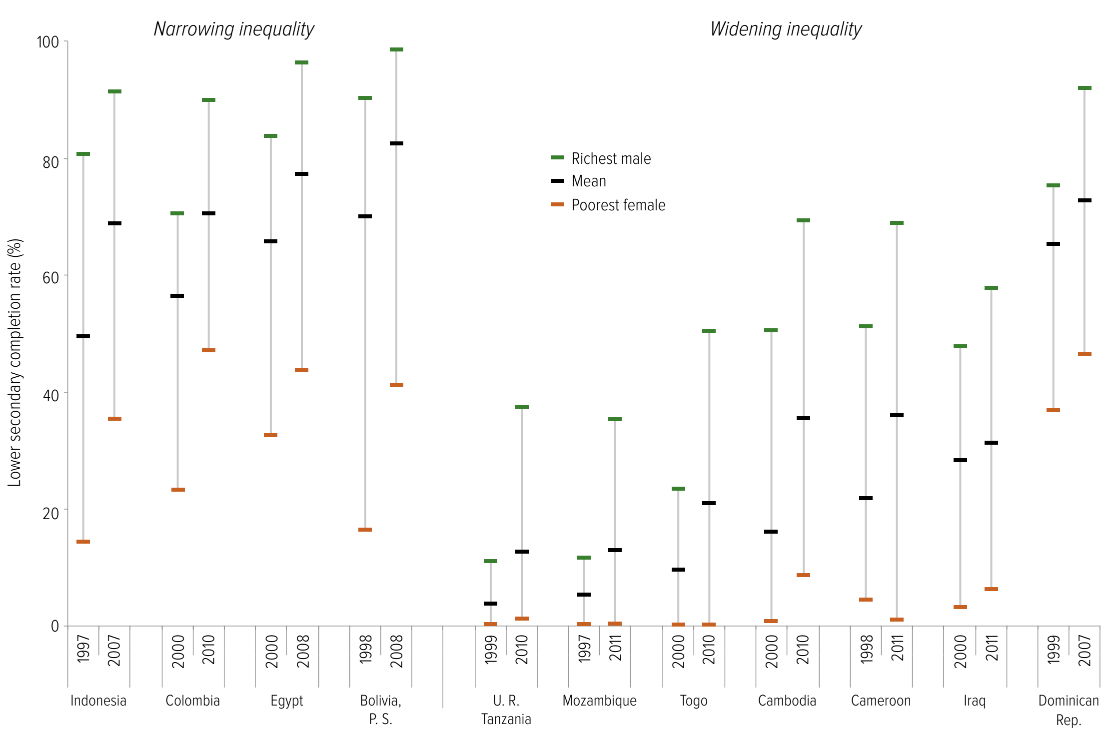
The 2012 *EFA Global Monitoring Report* put a spotlight on goal 3, which has been one of the most neglected EFA goals in part because no targets or indicators were set for its monitoring. To address this shortcoming, the 2012 Report proposed a framework for the various types of skills as a way of improving monitoring efforts. While highlighting some promising initiatives, the 2012 Report recognized that the international community was still a long way from measuring systematically the provision of skills-oriented programmes and the acquisition of skills, so monitoring progress on goal 3 is likely to remain challenging.

The framework emphasizes the crucial importance of foundation skills, including literacy and numeracy, which are essential for meeting daily needs, succeeding in the world of work and acquiring transferable skills and technical and vocational skills. While there are other pathways young people can take to acquire foundation skills, the most effective is lower secondary schooling, hence the calls for universal completion of lower secondary school to be a goal in the post-2015 framework. The global lower secondary gross enrolment ratio increased from 72% in 1999 to 82% in 2011. The largest increase was in sub-Saharan Africa, where the number of students more than doubled, although the share reaching this level was still only 49%.

Analysis by the EFA Global Monitoring Report team of household surveys shows that completion rates are significantly lower than these enrolment ratios suggest. In low income countries, completion rates increased from 27% in the early 2000s to 37% in the late 2000s. Moreover, almost all the increase was recorded among the richest fifth of the population, where the rates rose from 53% to 61%, while completion rates among the poorest increased from 11% to 14% (Figure 1.3.1). Similar patterns of stagnation for the poorest groups are apparent in countries in sub-Saharan Africa (Box 1.3.1).

**In low income countries, completion rates among the poorest reached just 14% in 2010**

**Figure 1.3.1: Inequality in lower secondary completion is growing in poorer countries**
Lower secondary school completion rate by wealth and gender, selected countries, 1997–2000 and 2007–2011




*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Surveys and Multiple Indicator Cluster Surveys.

## Box 1.3.1: In sub-Saharan Africa, far too few young people complete lower secondary school

Many countries in sub-Saharan Africa have expanded access to lower secondary school, but it will take more time and effort to translate these gains into higher completion rates. The poorest, in particular, appear not to be benefiting from current policies. In cash-strapped secondary education systems, unplanned growth in private schooling appears to be excluding many of the continent's more disadvantaged adolescents.

Enrolment data from the education ministries in Rwanda and Malawi show the divergent paths countries can take. Rwanda has carried out a major expansion of lower secondary schooling, with the gross enrolment ratio quadrupling in about 10 years to reach 47% in 2011, overtaking Malawi, where the ratio hovered around 40% for most of the decade (Figure 1.3.2).

Information collected from households, however, shows that completion rates remain low in both countries. Between 2000 and 2010, the lower secondary school completion rate increased from 9% to 15% in Rwanda and from 16% to 25% in Malawi (Figure 1.3.2). It is of particular concern that in both countries completion is extremely inequitable, with fewer than 5% of poor, rural girls completing lower secondary school. This is partly because poorer households are obliged to spend a high share of their income to educate their children (see Chapter 2).

The approaches the two countries have taken towards expanding access to lower secondary education differ, however, and Rwanda's could improve completion rates in coming years. In 2006, Rwanda extended basic education from six to nine years, through to lower secondary level, and abolished fees for the entire cycle. In 2009, it also removed the obstacle of the primary completion examination. The share of private enrolment in lower secondary education decreased from 43% in 2001/02 to 27% in 2008.

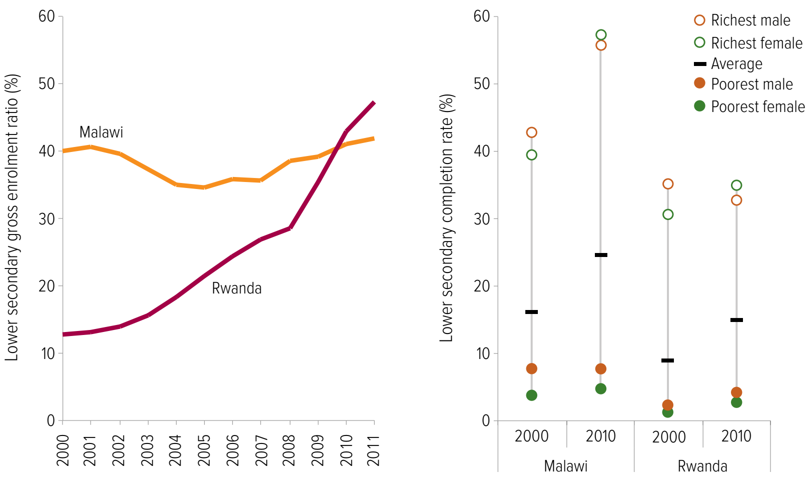
In Malawi, by contrast, expansion of lower secondary schooling has remained limited, even though a universal primary education programme in the 1990s increased demand for secondary education. Enrolment has increased only in private schools, whose share of total secondary enrolment rose from 13% in 2001 to 23% in 2007.

Because supply has not kept pace with demand, a highly selective system has developed. Only the students who perform best in an examination at the end of primary school enter higher quality government secondary schools. Those who do not pass the exam, but can afford the cost of private school, take that option. The vast majority, however, have to settle for community day secondary schools, which are widely considered to be of poor quality. As a result, adolescents in these schools are likely to drop out or have poor learning outcomes.

*Sources:* Chiche (2010); de Hoop (2011); World Bank (2010, 2011).

**Figure 1.3.2: Completion of lower secondary school remains elusive for many adolescents in Malawi and Rwanda**

Lower secondary gross enrolment ratios and completion rates, Malawi and Rwanda, 2000–2010




*Note:* The completion rate has been calculated for people aged 18 to 22.
*Sources:* Gross enrolment ratio: UIS database; Completion rate: EFA Global Monitoring Report team calculations (2013), based on Demographic and Health Survey data.

There is wide inequality between population groups in lower secondary completion. In poorer countries, only the most advantaged have benefited from progress. In Mozambique and the United Republic of Tanzania, for example, almost no young women from the poorest families completed lower secondary school in 2010/2011. By contrast, young men from the richest families more than tripled their completion rates, to over 35%, between the late 1990s and 2010/2011. This pattern is also apparent in middle income countries, such as the Dominican Republic and Iraq. In some countries, while poorer adolescents have benefited from progress, wide gaps remain: in Indonesia, lower secondary school completion increased from 81% to 92% for the richest males and from 15% to 36% for the poorest females (Figure 1.3.1).

Another measure of progress towards goal 3, the number of adolescents of lower secondary school age who are out of school, also shows that advances have been limited. While the number has fallen since 1999 by 31% to 69 million in 2011, progress has all but stagnated since 2007. Moreover, the reduction has been much more modest than for primary school age children (Figure 1.3.3). As a result, many young people have been denied a lower secondary education and so need access to second-chance programmes to acquire foundation skills.

South and West Asia has made great strides in reducing the number of primary school age children out of school, but progress has been slower for adolescents with progress achieved only up to 2004. The region's share of the global number of out-of-school adolescents increased from 39% in 1999 to 45% in 2011. In sub-Saharan Africa, 22 million adolescents remained out of school over the entire period. Any progress in enrolment has been cancelled by the 33% increase in the population of this age group since 1999.

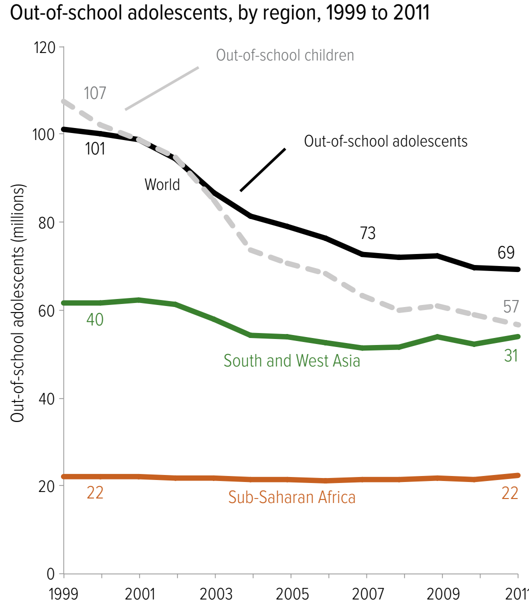
Transferable skills are acquired at all education levels. But upper secondary education has a distinct role in instilling such skills (Box 1.3.2). The global upper secondary gross enrolment ratio increased from 45% in 1999 to 59% in 2011. The largest increase was in East Asia and the Pacific, where the ratio increased from 44% to 70%. In absolute terms, the largest expansion took place in sub-Saharan Africa, where the number of students doubled over the period, although the region's enrolment ratio had only reached 32% by 2011.

Data sources are fragmented for technical and vocational skills. While this kind of knowledge is often best acquired on the job, for instance in apprenticeship programmes, no systematic information on such programmes currently exists. The information that is available indicates that the share of technical and vocational education in total secondary education enrolment has remained constant at 11% since 1999, with relatively small variations in regional trends, such as a decline in the proportion of enrolment in technical and vocational education in the Arab States. Governments around the world need to pay more attention to improving the quality of these programmes and ensuring their relevance to the world of work.

**In sub-Saharan Africa, 22 million adolescents remained out of school between 1999 and 2011**

**Figure 1.3.3: The number of adolescents out of school has hardly fallen since 2007**

Out-of-school adolescents, by region, 1999 to 2011




*Source:* UIS database.

## Box 1.3.2: Upper secondary education is key to acquiring transferable skills

Transferable skills, such as problem-solving, are vital for adapting knowledge to different work contexts, but there have been few attempts to measure how many people have acquired such skills.

One important step towards filling this gap is the OECD's Programme for the International Assessment of Adult Competencies (PIAAC). The results of the 2011 PIAAC survey confirm that upper secondary education is a vital way of improving transferable skills such as analysing information. Another key finding of the survey is that low literacy, associated with lower educational attainment, is as much of a barrier to effective engagement in the online world as lack of basic information and communication (ICT) skills. The digital divide may therefore also be a literacy divide.

Twenty-four countries participated in the 2011 PIAAC survey, which assessed the literacy, numeracy and problem-solving skills of 16- to 65-year-olds. It also collected information on the acquisition of problem-solving skills in technology-rich environments.

Across the 19 high income countries and regions where problem-solving skills in technology-rich environments were surveyed, 51% of those aged 16 to 24 were proficient at the two highest levels (2 and 3). At these levels of proficiency, individuals are typically able to solve problems involving several steps, evaluate the relevance of information and use several computer applications.

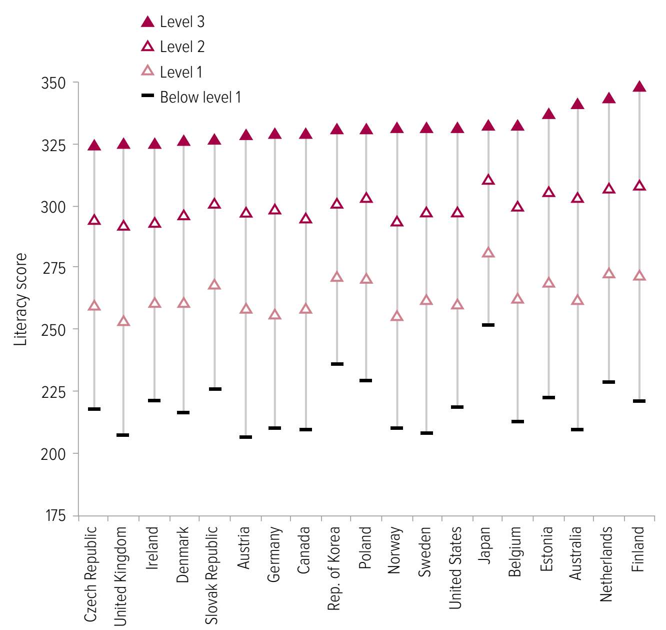
This leaves around half of young adults with low levels of such skills. These young adults are at most able to use familiar applications to solve problems that involve few steps and explicit criteria, such as sorting e-mails into pre-existing folders.

Proficiency in problem-solving in technology-rich environments among young adults is strongly associated with literacy skills. On average, moving from level 1 to level 2 in terms of performance in problem-solving skills is associated with an increase in the PIAAC literacy score from 264 to 300 points. This is equivalent to the difference in literacy skills between a person who can only draw lower level inferences from a text compared with a person who can interpret information from different sources (Figure 1.3.4).

Proficiency in problem-solving in technology-rich environments is also associated with the level of education completed. Nearly 47% of 16- to 24-year-olds who had not completed upper secondary education scored at level 1 or below, compared with 39% of those who had completed upper secondary education. In England and Northern Ireland (United Kingdom) 88% of those who had not completed upper secondary education scored at or below this minimum benchmark, compared with 54% of those who had completed it.

The PIAAC results show that countries need a more effective mix of policies and practices aimed at developing the skills necessary to manage information in digital environments. Policies should ensure that young people have solid foundation skills in literacy and numeracy, and that access to upper secondary education is expanded. In economies and societies in which access to resources and information is increasingly governed by digital ability, poor problem-solving skills increase chances of exclusion in the job market and other areas of life.

**Figure 1.3.4: Literacy is crucial to developing ICT skills**
Mean scores of 16- to 24-year-olds on the literacy scale, by level of proficiency in problem-solving in technology-rich environments, PIAAC, 2011




*Notes:* A literacy score between 225 and 275 points means respondents could paraphrase or draw low level inferences from the text. A score between 275 and 325 points means respondents could identify, interpret, or evaluate one or more pieces of information that required varying levels of inference and where competing information was often present.
The UK data refer to England and Northern Ireland. The Belgian data refer to Flanders.
*Source:* OECD (2013c).

## How many countries are likely to achieve universal lower secondary education by 2015?

No clear target was set in 2000 to guide an assessment of global success in promoting skills development and, at that time, universal lower secondary education was not an explicit goal. Since it is likely to become one after 2015, it is vital to assess where the world may stand in 2015.

The lower secondary school completion rate would be an appropriate indicator to measure progress, but data are lacking. The adjusted net enrolment ratio for lower secondary school age adolescents is a second-best indicator which has information on trends for just 82 countries.

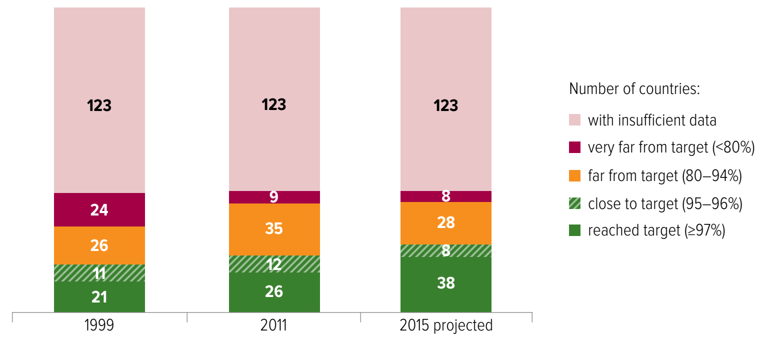
Of the 82 countries, 21 had an adjusted net enrolment ratio above 97% in 1999, and 11 were close. By 2011, 26 countries had reached the target and 12 were close. By 2015, the number of countries meeting the target is expected to grow to 38, with 8 countries close (Figure 1.3.5).

The number of countries furthest from the target, with a ratio below 80%, fell from 24 in 1999 to 9 in 2011. Of the countries that moved out of this group, Oman and Turkey have achieved the target of 97% enrolment, while Tajikistan has come close. By 2015, 8 countries are likely to have net enrolment below 80% – 7 in sub-Saharan Africa plus the Lao People's Democratic Republic – and 28 countries are expected to remain far from the target with net enrolment between 80% and 95% (Table 1.3.2).

While many countries will not have met the target of universal lower secondary net enrolment by 2015, assessing how quickly countries have moved towards the target indicates how soon they may achieve it. Of the 56 countries that had not reached the target by 2011, 14 made strong progress, increasing

**Figure 1.3.5: Goal 3 – of 82 countries with data, less than half will achieve universal lower secondary education by 2015**

Number of countries by level of adjusted net enrolment ratio for lower secondary education, 1999, 2011 and 2015 (projected)




*Note:* The analysis was conducted on the subset of countries for which a projection was possible; therefore, it covers fewer countries than those for which information is available for either 1999 or 2011.
*Source:* Bruneforth (2013).

their net enrolment ratio by at least 15% since 1999. Eleven countries expanded access by more than 20 percentage points, including Ecuador (from 72% to 93%), Ghana (from 62% to 83%) and Indonesia (from 65% to 89%). The net enrolment ratio increased for all countries with data in Central America, including El Salvador, Guatemala, Nicaragua and Panama (Box 1.3.3).

This assessment is based on information from only 40% of all countries. It includes two-thirds of the countries in North America and Western Europe but only a quarter of sub-Saharan African countries, some of which are unlikely to have achieved universal lower primary education so will not be in a position to achieve universal secondary schooling.

After 2015, it will be vital to set a target for lower secondary education. In order to monitor progress towards such a goal, more comprehensive data will be needed on access to and completion of lower secondary education.

**Between 1999 and 2011, Ecuador, Ghana and Indonesia rapidly expanded access to lower secondary school**

## Box 1.3.3: Latin America has made good progress in secondary education, but wide inequality remains

Secondary education has expanded in Latin America over the past decade, although at only half the speed of the previous decade. By 2011, the net enrolment ratio had reached 77%. In general, the countries that lagged behind were those that expanded secondary education fastest over the decade. Among the countries that had the lowest enrolment ratios in 1999, Ecuador and Guatemala's ratios improved faster, while Paraguay's increased more slowly.

Regardless of the speed of progress, household surveys show that the education systems of the three countries are still characterized by high levels of inequality. Ecuador appears to have made greater strides towards narrowing inequality gaps. Its gap in the secondary net attendance ratio between urban and rural males fell from 32 percentage points in 2001 to 13 percentage points in 2011. The gap between urban and rural youth fell more slowly in Paraguay, while it hardly changed in Guatemala, with the result that inequality there remained very high: only 28% of rural females attended secondary school compared to 62% of urban females in 2011 (Figure 1.3.6).

What explains this divergent performance in expanding secondary education? One factor is differences in spending priorities. Public education expenditure as a share of GNP almost tripled in Ecuador, from 2% in 1999 to 5.3% in 2011, while it fell in Paraguay from 5.1% in 1999 to 4.2% in 2010, the largest absolute decline in the subregion.

Another reason for the divergent performance is that the three countries have differed in the success of their policy approaches. In Ecuador, the Bono de Desarrollo Humano cash transfer programme, introduced in 2003, was conditional on school attendance. An evaluation at an early stage of implementation showed that the programme was having a positive effect on school attendance among children aged 6 to 17. The level of the transfer more than doubled in the second half of the decade. By 2010, it provided 44% of the beneficiaries' income. While the effects of this change have not yet been evaluated, the acceleration in the growth of enrolment after 2006 is likely to be related to the increasing availability of income to poor households.

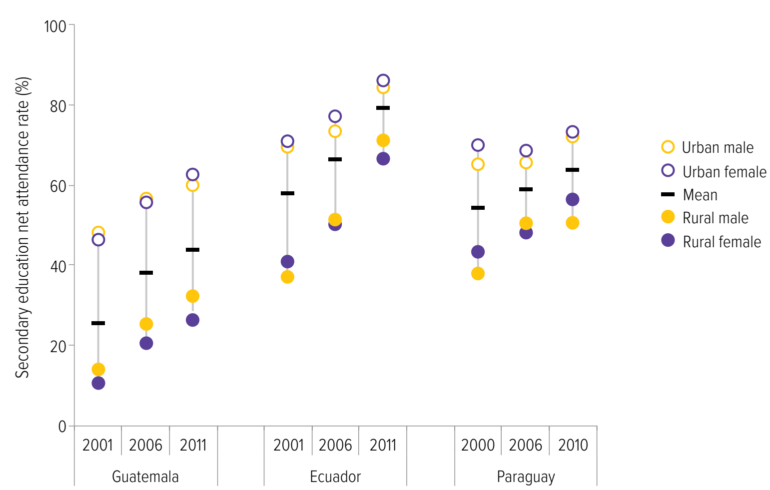
In Guatemala, part of the increase in secondary enrolment is explained by the introduction of distance education programmes previously implemented on a large scale in Mexico. The country is however a latecomer to the social policy revolution that has taken place in many countries in the region. A major conditional cash transfer programme, Mi Familia Progresa, was introduced in 2008 and expanded rapidly to cover 23% of the population within two years. However, in 2011 it provided only 9% of beneficiaries' income, much lower than the average in the region.

In Paraguay, the main government intervention at the lower secondary level is the second phase of the Escuela Viva programme. As a means of increasing retention and learning outcomes, it aims to improve participation in school management by training stakeholders to identify school needs and then develop, implement and monitor a school improvement plan. In addition, it provides transport subsidies, scholarships and boarding school arrangements. However, the expected results are relatively modest: from a baseline of 15% in 2005, the target is to raise the lower secondary completion rate in a thousand targeted rural schools to 25% by 2014.

Since 2005, Paraguay has also had a cash transfer programme, Tekoporã, which is targeted at the poorest households in the poorest districts and is conditional on children attending school. An evaluation at the pilot stage found that it had a positive effect on school attendance. However, the level of the transfer was low, and coverage was limited until it increased in 2009. Overall, the social policies of the government, including education transfers in kind, achieved one of the lowest income redistribution rates in the sub-region in 2010, reducing inequality by only 3.5%, compared with 15% in Mexico and 19% in Brazil. Social protection policies need to be enhanced if they are to produce a stronger effect on enrolment in secondary education.

*Sources:* Higgins et al. (2013); Inter-American Development Bank (2007, 2013); Ponce (2010, 2011); PREAL and Instituto Desarollo (2013); SITEAL (2013a, 2013b); Stampini and Tornarolli (2012); Teixeira et al. (2011).

**Figure 1.3.6: In Latin America, inequalities in secondary education remain**
Secondary education net attendance rate, by gender and location, Ecuador, Guatemala and Paraguay, 2000–2011




*Source:* SITEAL (2013b).

**Table 1.3.2: Likelihood of countries achieving a lower secondary education net enrolment target of at least 95% by 2015**

| Level expected by 2015 | | Strong relative progress 14 | Slow progress or moving away from target 22 |
|---|---|---|---|
| | Target reached or close (≥95%) 46 | Bahamas, Belarus, Botswana, Colombia, Croatia, Cyprus, Denmark, Dominica, Estonia, Fiji, Finland, France, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Kazakhstan, Kenya, Luxembourg, Mexico, Mongolia, Netherlands, Norway, Oman, Peru, Philippines, Qatar, Republic of Korea, Romania, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Samoa, Seychelles, Slovenia, Spain, Switzerland, Syrian Arab Republic, Tajikistan, Turkey, United Kingdom, United States, Venezuela (Bolivarian Republic of) | |
| | Far from target (80-94%) 28 | Bhutan, Dominican Republic, Ecuador, El Salvador, Ghana, Guatemala, Indonesia, Nicaragua, Panama | Aruba, Azerbaijan, Barbados, Bolivia (Plurinational State of), Bulgaria, Cayman Islands, Cook Islands, Cuba, Jamaica, Jordan, Lithuania, Malawi, Malaysia, Palestine, Paraguay, Poland, Republic of Moldova, Sweden, Trinidad and Tobago |
| | Very far from target (<80%) 8 | Burkina Faso, Ethiopia, Guinea, Mozambique, Niger | Eritrea, Lao People's Democratic Republic, Lesotho |
| | | **Change between 1999 and 2011** | |

| | |
|---|---|
| Countries not included in analysis because of insufficient data 123 | Afghanistan, Albania, Algeria, Andorra, Angola, Anguilla, Antigua and Barbuda, Argentina, Armenia, Australia, Austria, Bahrain, Bangladesh, Belgium, Belize, Benin, Bermuda, Bosnia and Herzegovina, Brazil, British Virgin Islands, Brunei Darussalam, Burundi, Cambodia, Cameroon, Canada, Cape Verde, Central African Republic, Chad, Chile, China, Comoros, Congo, Costa Rica, Côte d'Ivoire, Czech Republic, Democratic People's Republic of Korea, Democratic Republic of the Congo, Djibouti, Egypt, Equatorial Guinea, Gabon, Gambia, Georgia, Germany, Grenada, Guinea-Bissau, Guyana, Haiti, Honduras, India, Iran (Islamic Republic of), Iraq, Kiribati, Kuwait, Kyrgyzstan, Latvia, Lebanon, Liberia, Libya, Macao (China), Madagascar, Maldives, Mali, Malta, Marshall Islands, Mauritania, Mauritius, Micronesia (Federated States of), Monaco, Montenegro, Montserrat, Morocco, Myanmar, Namibia, Nauru, Nepal, Netherlands Antilles, New Zealand, Nigeria, Niue, Pakistan, Palau, Papua New Guinea, Portugal, Russian Federation, Rwanda, San Marino, Sao Tome and Principe, Saudi Arabia, Senegal, Serbia, Sierra Leone, Singapore, Slovakia, Solomon Islands, Somalia, South Africa, South Sudan, Sri Lanka, Sudan, Suriname, Swaziland, Thailand, The former Yugoslav Republic of Macedonia, Timor-Leste, Togo, Tokelau, Tonga, Tunisia, Turkmenistan, Turks and Caicos Islands, Tuvalu, Uganda, Ukraine, United Arab Emirates, United Republic of Tanzania, Uruguay, Uzbekistan, Vanuatu, Viet Nam, Yemen, Zambia, Zimbabwe |

*Source:* Bruneforth (2013).

# Goal 4 Adult literacy

*Achieving a 50% improvement in levels of adult literacy by 2015, especially for women, and equitable access to basic and continuing education for all adults.*

## Highlights

- The adult illiteracy rate fell from 24% in 1990 to 18% in 2000 and 16% in 2011. However, the number of illiterate adults remains stubbornly high at 774 million, a fall of 12% since 1990 but just 1% since 2000.
- In sub-Saharan Africa, the number of illiterate adults has increased by 37% since 1990, mainly as a result of population growth. This region, together with South and West Asia, accounts for three-quarters of the global population of illiterate adults.
- The number of illiterate adults is projected only to fall to 743 million by 2015. In 32 out of 89 countries, the adult literacy rate will still be below 80%.
- Almost two-thirds of illiterate adults are women. Only half of the 61 countries with data for the beginning and end of the decade are expected to achieve gender parity in adult literacy by 2015.
- West African countries account for 44% of illiterate young people in sub-Saharan Africa. Household survey data suggest that in 12 of 15 West African countries fewer than half of young women are literate, which means the quest for universal youth literacy will continue for at least another generation.

**Table 1.4.1: Key indicators for goal 4**

| | Illiterate adults | | | | Adult literacy rates | | | | Youth literacy rates | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Total | | Women | | Total | | Gender parity index | | Total | | Gender parity index | |
| | 2005–2011 (000) | Change since 1985–1994 (%) | 1985–1994 (%) | 2005–2011 (%) | 1985–1994 (%) | 2005–2011 (%) | 1985–1994 (F/M) | 2005–2011 (F/M) | 1985–1994 (%) | 2005–2011 (%) | 1985–1994 (F/M) | 2005–2011 (F/M) |
| **World** | **773 549** | **-12** | **63** | **64** | **76** | **84** | **0.85** | **0.90** | **83** | **89** | **0.90** | **0.94** |
| Low income countries | 183 552 | 23 | 60 | 60 | 51 | 61 | 0.69 | 0.79 | 60 | 73 | 0.79 | 0.90 |
| Lower middle income countries | 470 164 | 2 | 61 | 65 | 59 | 71 | 0.71 | 0.78 | 71 | 84 | 0.80 | 0.88 |
| Upper middle income countries | 112 671 | -57 | 67 | 67 | 82 | 94 | 0.86 | 0.96 | 94 | 99 | 0.96 | 1.00 |
| High income countries | … | … | … | … | … | … | … | … | … | … | … | … |
| Sub-Saharan Africa | 181 950 | 37 | 62 | 61 | 53 | 59 | 0.68 | 0.74 | 66 | 70 | 0.80 | 0.84 |
| Arab States | 47 603 | -8 | 63 | 66 | 55 | 77 | 0.62 | 0.81 | 74 | 90 | 0.78 | 0.93 |
| Central Asia | 290 | -69 | 77 | 63 | 98 | 100 | 0.98 | 1.00 | 100 | 100 | 1.00 | 1.00 |
| East Asia and the Pacific | 89 478 | -61 | 69 | 71 | 82 | 95 | 0.84 | 0.95 | 95 | 99 | 0.96 | 1.00 |
| South and West Asia | 407 021 | 2 | 60 | 64 | 47 | 63 | 0.57 | 0.70 | 60 | 81 | 0.70 | 0.86 |
| Latin America and the Caribbean | 35 614 | -16 | 55 | 55 | 86 | 92 | 0.97 | 0.99 | 93 | 97 | 1.01 | 1.01 |
| North America and Western Europe | … | … | … | … | … | … | … | … | … | … | … | … |
| Central and Eastern Europe | 4 919 | -59 | 79 | 78 | 96 | 99 | 0.96 | 0.99 | 98 | 99 | 0.98 | 1.00 |

*Notes:* Data are for the most recent year available during the period specified. Gender parity is reached when the gender parity index is between 0.97 and 1.03.
*Sources:* Annex, Statistical Table 2; UIS database.

Universal literacy is fundamental to social and economic progress. Literacy skills are best developed in childhood through good quality education. Few countries have been able to build robust adult education institutions that offer genuine second chances to the majority of illiterate adults. As a result, countries with a legacy of low access to school have so far been unable to eradicate illiteracy among youth and adults.

The number of illiterate adults is estimated to have fallen by 12%, from 880 million in the period 1985–1994 to 774 million in the period 2005–2011. The pace of decline has slowed considerably. Almost all the decline took place in the 1990s. Since 2000 the number of illiterate adults fell by only 1%. In relative terms, the adult illiteracy rate fell from 24% in the earlier period to 16% in the later period. It is projected to fall further to 14% by 2015.

Women make up nearly two-thirds of the total, and since 1990 there has been no progress in reducing this share. Of the 61 countries with data on gender parity in adult literacy for the beginning and end of the decade, 22 were already at parity at the beginning and 6 have achieved it since. It is projected that only 2 more countries will reach parity by 2015, although 10 will be within 2 percentage points from parity.

Since 1990, adult literacy rates have risen fastest in the Arab States, from 55% to 77%. Nevertheless, as a result of population growth, the actual number of illiterate adults has only fallen from 52 million to 48 million. South and West Asia and sub-Saharan Africa account for 76% of the global population of illiterate adults, up from 61% in 1990.

South and West Asia has experienced the second fastest increase in adult literacy rates (from 47% to 63%). Yet it has seen its population of illiterate adults remain stable at just over 400 million. As a result, the region accounts for a higher share of the global population of illiterate adults now (53%) than in 1990 (46%).

In sub-Saharan Africa, where the adult literacy rate has only increased from 53% to 59% since 1990, the actual number of illiterate adults has grown by 37% to 182 million in 2011. By 2015, it is projected that 26% of the world's illiterate adults will live in sub-Saharan Africa, up from 15% in 1990.

In order to move towards universal adult literacy, youth literacy rates need to improve. Globally, the youth literacy rate stands at 89%, 5 percentage points higher than the literacy rate of the entire adult population. In the Arab States, the youth literacy rate of 90% exceeds the adult literacy rate by 13 percentage points. In sub-Saharan Africa, the youth literacy rate exceeds the adult literacy rate by 11 percentage points (70% versus 59%). In Mozambique the difference is 17 percentage points (67% versus 51%). However, in Angola it is only 3 percentage points (73% versus 70%), indicating that universal adult literacy is unlikely to be achieved for at least another generation.

**China has reduced its number of illiterate adults by 130 million over 2 decades**

The countries that appear to have higher youth literacy rates are those that have made greater progress in recent years towards UPE. However, new estimates prepared for this Report, which are based on direct assessments of literacy rather than on self-declarations, suggest that progress towards universal youth literacy may have been overstated (see Chapter 4). West Africa is of particular concern, accounting for 35% of sub-Saharan Africa's young population but 44% of its illiterate youth (Box 1.4.1).

Ten countries account for 557 million, or 72%, of the global population of illiterate adults. These countries have followed very different trajectories. China has made enormous progress, reducing its total by 130 million (or by 71%). India has by far the largest population of illiterate adults, 287 million, amounting to 37% of the global total. Its literacy rate rose from 48% in 1991 to 63% in 2006, the latest year it has available data, but population growth cancelled the gains so there was no change in the number of illiterate adults. And Nigeria had 17 million more illiterate adults in 2008 than in 1991, an increase of 71% (Figure 1.4.2).

Schooling is not the only determinant of literacy skills. The UIS Literacy Assessment and Monitoring Programme (LAMP) draws attention to the key contribution to sustained literacy made by diverse everyday reading practices (Box 1.4.2).

## Box 1.4.1: West Africa accounts for almost half of the region's illiterate adults

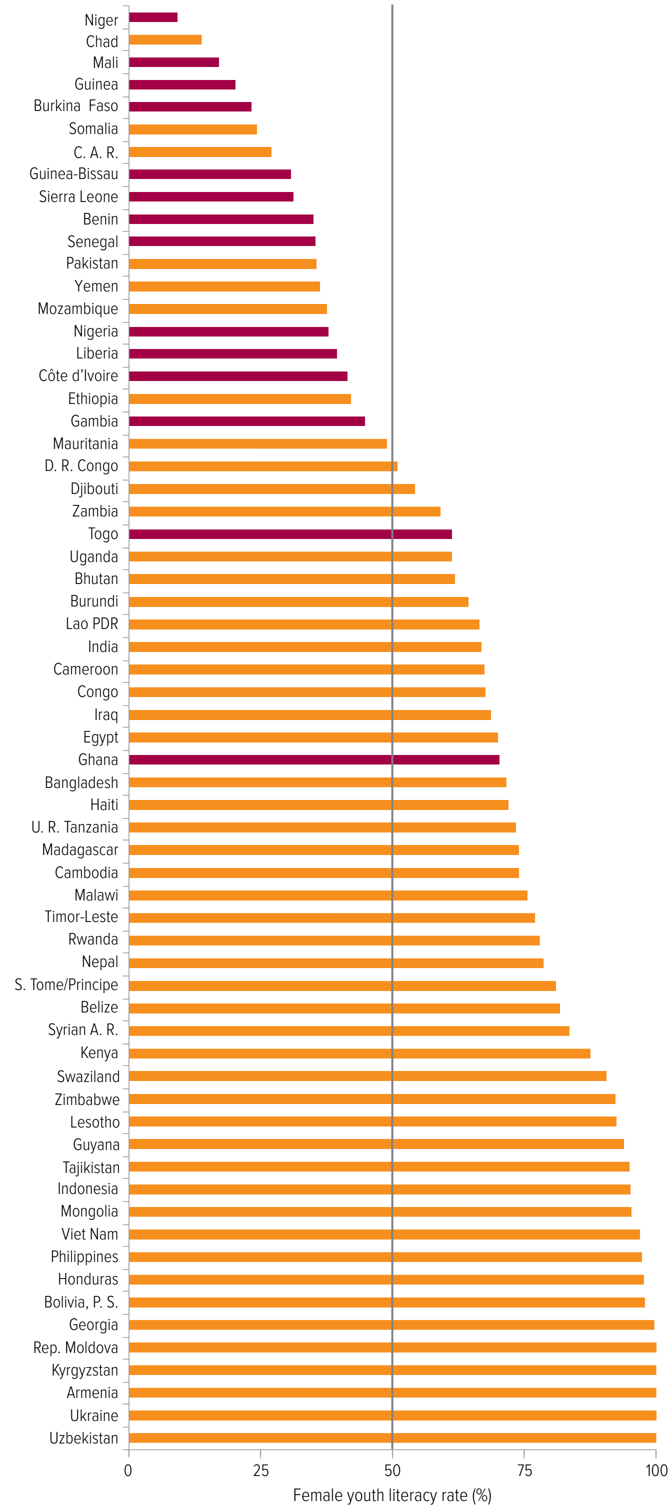
The 15 countries of West Africa are among those with the worst adult literacy rates globally, and include the five countries with the world's lowest literacy rates, below 35%. Those five countries also have the unenviable record of female literacy rates below 25%, compared with an average for sub-Saharan Africa of 50%. Ten of the 15 countries with the lowest female literacy rates are in the subregion.

Some countries have made considerable progress. Cape Verde's literacy rate has reached 85% and Ghana's 71%. From 1988 to 2009, Senegal almost doubled its literacy rate, from 27% to 50%, and more than doubled its female literacy rate, from 18% to 39%. But Benin, Liberia and Nigeria stagnated at low levels. In Benin, the adult literacy rate increased by just two percentage points between 1992 and 2006, to 29%.

Unfortunately, these trends are not likely to improve soon. In 11 out of 14 countries, less than one in four of children complete lower secondary school. Direct assessments of literacy collected through household surveys and analysed for this Report provide further evidence that West Africa has not caught up with the rest of the world. Young women are the least likely to be literate. Twelve West African countries are among the 20 countries with the world's lowest female youth literacy rates, below 50% (Figure 1.4.1). In Mali, for example, only 17% of young women can read a sentence.

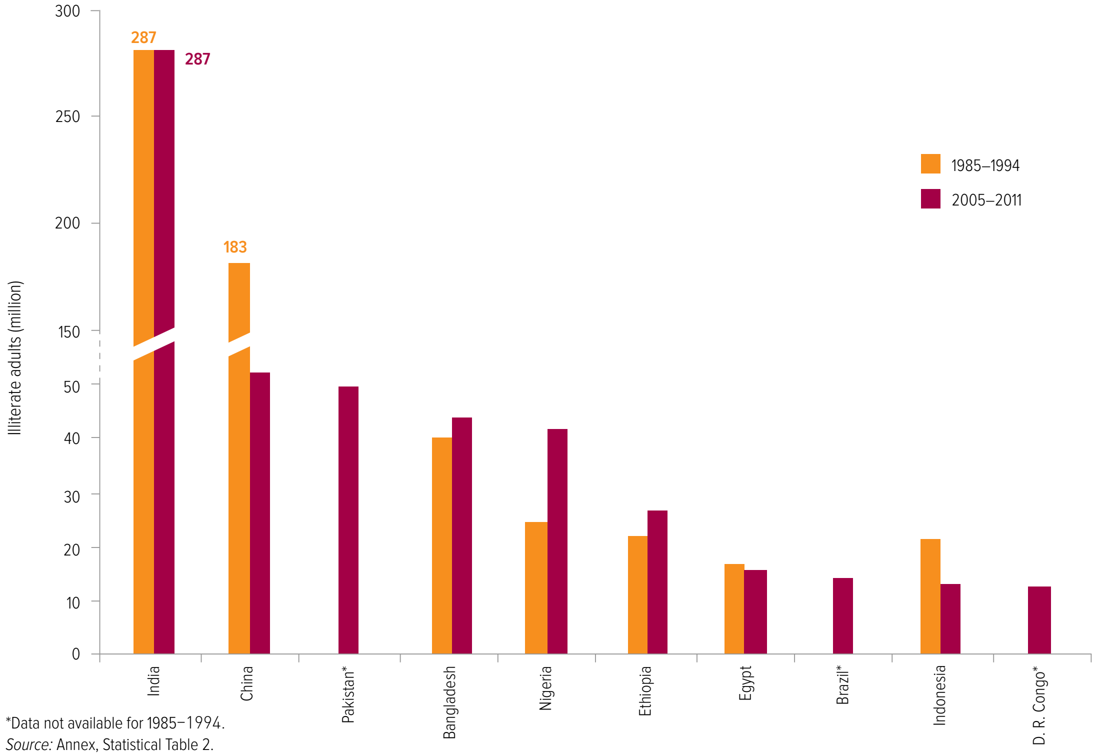
A key reason levels of youth literacy have remained low is that school systems have not expanded quickly enough. Nigeria, for example, has the highest number of out-of-school children in the world and a female youth literacy rate of just 38%. Even those who do attend school fail to acquire basic literacy skills because of the poor quality of education. Almost 80% of young people aged 15 to 24 who left school after five to six years could not read a full sentence. And conflict helps explain why youth literacy rates remain low in countries such as Liberia.

**Figure 1.4.1: In 12 West African countries, less than half of young women are literate**
Female youth literacy rate, selected countries, 2004–2011




*Notes:* A purple bar signifies a country in West Africa. Data are unavailable for Cape Verde.
*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Survey data and Multiple Indicator Cluster Survey data.

**Figure 1.4.2: 10 countries account for 72% of the global population of illiterate adults**
Number of illiterate adults, 10 countries with highest populations of illiterate adults, 1985–1994 and 2005–2011




*Data not available for 1985–1994.
*Source:* Annex, Statistical Table 2.

## Box 1.4.2: Engagement in everyday reading activities helps sustain literacy skills

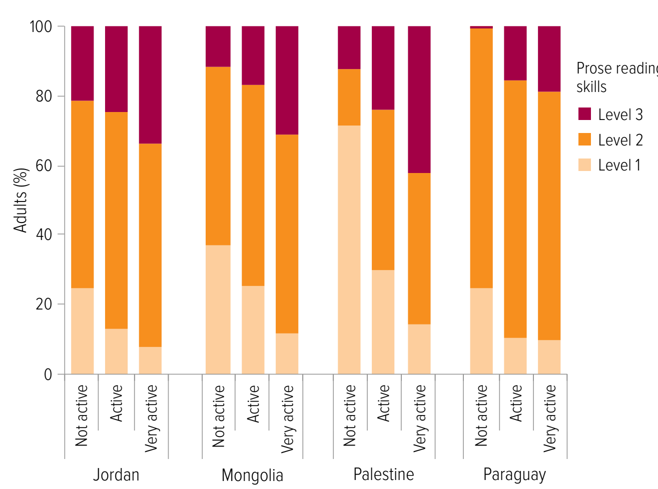
The number of years spent in school is the most important predictor of literacy skills. However, everyday reading activities such as sending text messages or emails and using the Internet can sustain and extend literacy skills beyond what would be expected based on an individual's schooling history. Conversely, individuals who are unable to engage regularly in such activities may lose even the skills they acquired in their schooling.

Recognizing this, the Literacy Assessment and Monitoring Program (LAMP) collected information on everyday reading activities that could be related to literacy in four countries – Jordan, Mongolia, Palestine and Paraguay. Respondents were grouped into three levels of engagement in these activities. At the lowest level, respondents watch television, listen to the radio and use mobile phones only for talking. At the next level, respondents also use their mobile phones to send text messages. At the highest level are computer users, who also send e-mail, search the Internet and use social media. They tend to be highly educated and younger: in all four countries the vast majority of those at the highest level are under 40 years old.

Two people with the same schooling could end up with very different literacy levels depending on how much they use literacy skills in their spare time. In the four countries, at least half of the people with the least engagement in everyday reading activities perform at the lowest level of prose reading (level 1). This means they can, at best, identify a telephone number in a newspaper advertisement, or locate and copy, verbatim, the answer to a simple question in a one-paragraph text.

Those who engage more in everyday reading activities tend to be more educated. Even at a given level of education, however, these activities affect literacy skills. In Mongolia, among those with secondary schooling, only 12% of those who engage in few everyday reading activities score at the highest level in prose reading tests (level 3). The share increased to 31% among those with the highest engagement in everyday reading activities (Figure 1.4.3).

**Figure 1.4.3: Literacy skills differ among people with the same education**
Performance in prose reading, by level of engagement in everyday reading activities, adults with secondary education, 2010–2011




*Source:* UIS calculations (2013), based on Literacy Assessment and Monitoring Programme data.

## How many countries are likely to achieve universal adult literacy by 2015?

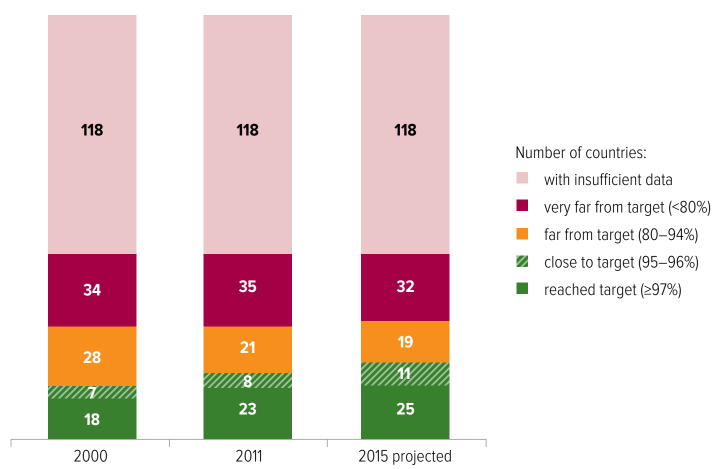
The literacy goal set in 2000 at the World Education Forum in Dakar, Senegal – which restated the commitment made in Jomtien, Thailand, in 1990 – was to halve illiteracy by 2015. The EFA Global Monitoring Report team proposes that after 2015 the goal should be made more ambitious: to achieve universal youth and adult literacy. While UIS projections are available to 2015 for most countries, information is not always available for the starting point in 2000.[1]

New analysis for this Report considers how close countries are likely to get to universal adult literacy by 2015. For the 87 countries with the information, 18 had an adult literacy rate above 97% in 2000, while 7 were close. Between 2000 and 2011, the number of countries that had reached the target increased to 23, and 8 were close. It is projected that by 2015, 25 countries will have achieved universal adult literacy and 11 countries will be close.

By contrast, 34 countries were very far from the target in 2000 and 35 in 2011. Three countries moved out of this group during the period (Burundi, the Islamic Republic of Iran and Saudi Arabia) but four fell back into it (Kenya, Lesotho, Namibia and Sao Tome and Principe). It is projected that by 2015, 32 countries will be very far from the target, while 19 countries will be far from it (Figure 1.4.4).

Of the 32 countries that are expected to be very far from the target in 2015 for which information on trends between 2000 and 2011 is available, three have reduced their illiteracy rate by at least a quarter, and at least five percentage points in absolute terms: Eritrea, Ghana and Timor-Leste. Of the 19 countries that will still be far from the target in 2015, six have reduced the illiteracy rate by at least half: the Plurinational State of Bolivia, Burundi, Malaysia, Mexico, Myanmar and Viet Nam (Table 1.4.2).

**Figure 1.4.4: Goal 4 – At least one in five adults will be illiterate in a third of countries in 2015**
Number of countries by level of adult literacy rate, 2000, 2011 and 2015 (projected)




*Note:* The analysis was conducted on the subset of countries for which a projection was possible; therefore, it covers fewer countries than those for which information is available for either 2000 or 2011.
*Source:* Bruneforth (2013).

1. Note that, unlike with other indicators, almost half the 58 countries for which such projections are not provided are high income countries.

**Table 1.4.2: Likelihood of countries achieving an adult literacy target of at least 95% by 2015**

<table>
<tr><td rowspan="3">Level expected by 2015</td><td>Target reached or close (≥95%) 58</td><td colspan="3">Albania, Argentina, Armenia, Aruba, Azerbaijan, Bahrain, Belarus, Bosnia and Herzegovina, Brunei Darussalam, Bulgaria, Chile, China, Costa Rica, Croatia, Cuba, Cyprus, Democratic People's Republic of Korea, Equatorial Guinea, Estonia, Georgia, Greece, Italy, Kazakhstan, Kuwait, Kyrgyzstan, Latvia, Lithuania, Macao (China), Maldives, Mongolia, Montenegro, Netherlands Antilles, Palestine, Panama, Paraguay, Philippines, Portugal, Qatar, Republic of Moldova, Romania, Russian Federation, Samoa, Serbia, Singapore, Slovenia, Spain, Suriname, Tajikistan, Thailand, The former Yugoslav Republic of Macedonia, Tonga, Trinidad and Tobago, Turkey, Turkmenistan, Ukraine, Uruguay, Uzbekistan, Venezuela (Bolivarian Republic of)</td></tr>
<tr><td>Far from target (80-94%) 37</td><td>Bolivia (Plurinational State of), Burundi, Honduras, Iran (Islamic Republic of), Malaysia, Mexico, Myanmar, Saudi Arabia, Swaziland, Viet Nam</td><td>Algeria, Brazil, Dominican Republic, Ecuador, Mauritius, Namibia, Nicaragua, Sri Lanka, Syrian Arab Republic</td><td>Botswana, Cape Verde, Colombia, El Salvador, Gabon, Guyana, Indonesia, Jamaica, Jordan, Lebanon, Libya, Malta, Oman, Peru, South Africa, Tunisia, United Arab Emirates, Zimbabwe</td></tr>
<tr><td>Very far from target (<80%) 48</td><td>Eritrea, Ghana, Timor-Leste</td><td>Angola, Bangladesh, Benin, Cameroon, Central African Republic, Chad, Comoros, Côte d'Ivoire, Democratic Republic of the Congo, Gambia, Guatemala, Guinea-Bissau, India, Iraq, Kenya, Lao People's Democratic Republic, Lesotho, Madagascar, Mauritania, Nepal, Niger, Papua New Guinea, Rwanda, Sao Tome and Principe, Senegal, Togo, Uganda, United Republic of Tanzania, Zambia</td><td>Bhutan, Burkina Faso, Cambodia, Egypt, Ethiopia, Guinea, Haiti, Liberia, Malawi, Mali, Morocco, Mozambique, Nigeria, Pakistan, Sierra Leone, Yemen</td></tr>
<tr><td></td><td></td><td>Strong relative progress 13</td><td>Slow progress or moving away from target 38</td><td>No evidence on trend 34</td></tr>
<tr><td></td><td></td><td colspan="3">Change between 2000 and 2011</td></tr>
</table>

| | |
|---|---|
| Countries not included in analysis because of insufficient data 62 | Afghanistan, Andorra, Anguilla, Antigua and Barbuda, Australia, Austria, Bahamas, Barbados, Belgium, Belize, Bermuda, British Virgin Islands, Canada, Cayman Islands, Congo, Cook Islands, Czech Republic, Denmark, Djibouti, Dominica, Fiji, Finland, France, Germany, Grenada, Hungary, Iceland, Ireland, Israel, Japan, Kiribati, Luxembourg, Marshall Islands, Micronesia (Federated States of), Monaco, Montserrat, Nauru, Netherlands, New Zealand, Niue, Norway, Palau, Poland, Republic of Korea, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, San Marino, Seychelles, Slovakia, Solomon Islands, Somalia, South Sudan, Sudan, Sweden, Switzerland, Tokelau, Turks and Caicos Islands, Tuvalu, United Kingdom, United States, Vanuatu |

*Note:* This table includes a wider group of countries with projections by the UIS.
*Source:* Bruneforth (2013); UIS database.

# Goal 5 Gender parity and equality

*Eliminating gender disparities in primary and secondary education by 2005, and achieving gender equality in education by 2015, with a focus on ensuring girls' full and equal access to and achievement in basic education of good quality.*

## Highlights

- At the primary level, only 60% of countries with data had achieved gender parity by 2011. Among low income countries, just over a fifth have achieved parity. Of all countries, 17 had fewer than 9 girls enrolled in school for every 10 boys.
- It is projected that by 2015, 112 out of 161 countries will have achieved parity in primary education, but also that 12 countries will still have fewer than 9 girls enrolled in school for every 10 boys.
- At the secondary level, only 38% of countries with data had achieved parity by 2011. There are 30 countries with fewer than 9 girls enrolled in school for every 10 boys, but also 15 countries with fewer than 9 boys enrolled for every 10 girls.
- It is projected that by 2015, 84 out of 150 countries will have achieved parity in lower secondary education, but also that 31 countries will still have severe gender disparities.

**Table 1.5.1: Key indicators for goal 5**

| | Primary education | | | | | Secondary education | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Gender parity achieved in 2011 | | Countries with fewer than 90 girls enrolled for every 100 boys | Gender parity index (GPI) | | Gender parity achieved in 2011 | | Countries with fewer than 90 girls enrolled for every 100 boys | Gender parity index (GPI) | |
| | Total number of countries | Countries with data | | 1999 | 2011 | Total number of countries | Countries with data | | 1999 | 2011 |
| **World** | **104** | **173** | **17** | **0.92** | **0.97** | **59** | **157** | **30** | **0.91** | **0.97** |
| Low income countries | 7 | 32 | 10 | 0.86 | 0.95 | 1 | 27 | 18 | 0.83 | 0.88 |
| Lower middle income countries | 20 | 46 | 6 | 0.86 | 0.96 | 11 | 39 | 9 | 0.80 | 0.92 |
| Upper middle income countries | 33 | 48 | 1 | 0.99 | 1.00 | 17 | 44 | 1 | 0.98 | 1.04 |
| High income countries | 44 | 47 | 0 | 1.00 | 0.99 | 30 | 47 | 2 | 1.01 | 0.99 |
| Sub-Saharan Africa | 12 | 43 | 13 | 0.85 | 0.93 | 1 | 31 | 18 | 0.82 | 0.83 |
| Arab States | 7 | 15 | 2 | 0.87 | 0.92 | 2 | 15 | 6 | 0.88 | 0.93 |
| Central Asia | 6 | 7 | 0 | 0.99 | 0.98 | 5 | 7 | 1 | 1.00 | 0.97 |
| East Asia and the Pacific | 13 | 21 | 0 | 0.99 | 1.02 | 7 | 20 | 2 | 0.94 | 1.03 |
| South and West Asia | 5 | 9 | 2 | 0.83 | 0.98 | 0 | 8 | 2 | 0.75 | 0.92 |
| Latin America and the Caribbean | 17 | 34 | 0 | 0.97 | 0.97 | 11 | 31 | 0 | 1.07 | 1.07 |
| North America and Western Europe | 24 | 24 | 0 | 1.01 | 0.99 | 18 | 25 | 1 | 1.02 | 1.00 |
| Central and Eastern Europe | 20 | 20 | 0 | 0.97 | 1.00 | 15 | 20 | 0 | 0.96 | 0.97 |

*Note:* Gender parity is reached when the gender parity index of the gross enrolment ratio is between 0.97 and 1.03.
*Sources:* Annex, Statistical Tables 5 and 7.

Gender parity – equal enrolment ratios for girls and boys – is just the first step towards the fifth EFA goal of full gender equality in education: a schooling environment that is free of discrimination and provides equal opportunities for boys and girls to realize their potential. Other starting points towards gender equality include making sure the school environment is safe, improving facilities to provide, for example, separate latrines for girls and boys, training teachers in gender sensitivity, achieving gender balance among teachers and rewriting curricula and textbooks to remove gender stereotypes.

Parity in enrolment ratios at both the primary and secondary levels was singled out among all EFA goals to be achieved by 2005. With that early deadline missed, there has been progress towards this goal since, but the achievement of parity remains elusive.

At the primary education level, where disparities remain in 40% of the countries with data, disparity is at the expense of girls in more than 80% of the cases. South and West Asia is home to four of the countries with the highest gender disparities globally. Two of these have very high disparities at the expense of girls: Afghanistan, with 71 girls in school for every 100 boys, and Pakistan, with 82 girls for every 100 boys. Two other countries in the region have high disparities at the expense of boys: Bangladesh, with 94 boys for every 100 girls, and Nepal, with 92 boys for every 100 girls.

Of the 31 countries with fewer than 90 girls for every 100 boys enrolled in 1999, only about half had managed to exit that group by 2011. Others, such as Cameroon and the Central African Republic, made very slow progress towards parity. But even in some countries that made very fast progress towards parity, such as Burkina Faso and Senegal, enrolment ratios have remained among the lowest globally. In addition, gender parity is even more elusive in relation to primary school completion (Box 1.5.1).

In secondary education, gender parity trends vary by region, income group and level. Among countries with data, 38% have achieved parity in secondary education. By level, 42% of countries are at parity in lower secondary education and 22% in upper secondary education. In two-thirds of the countries with gender disparity in lower secondary education, it is at the expense of girls. But this is the case in less than half the countries with gender disparity in upper secondary education.

The most extreme cases of inequality in secondary education continue to afflict girls. Of the 30 countries with fewer than 90 girls for every 100 boys, 18 are in sub-Saharan Africa. Extreme examples from other regions include Afghanistan and Yemen, despite improvements over the decade. In Afghanistan, no girls were in secondary school in 1999. By 2011, the female gross enrolment ratio rose to 34%, increasing the gender parity index to 0.55. In Yemen, the female gross enrolment ratio increased from 21% in 1999 to 35% in 2011, resulting in an improvement in the gender parity index from 0.37 to 0.63.

**Just 20% of low income countries have achieved gender parity at the primary level**

There are also 15 countries with fewer than 90 boys for every 100 girls, about half of which are in Latin America and the Caribbean. In secondary school in Argentina, there were 95 boys for every 100 girls in 1999 and 90 boys for every 100 girls in 2010.

Comparisons by income group show that low income countries differ from middle and high income countries in terms of gender participation in education. Just 20% of low income countries have achieved gender parity at the primary level, 10% at the lower secondary level and 8% at the upper secondary level. In Burundi, while parity in primary education has been achieved, only 77 girls are enrolled for every 100 boys in lower secondary education, and only 62 girls for every 100 boys in upper secondary education.

By contrast, in middle and high income countries, where parity has been achieved in a higher percentage of countries, disparity is often at the expense of boys in lower and upper secondary education (Figure 1.5.3). In Honduras, while parity in primary education has been achieved, only 88 boys are enrolled for every 100 girls in lower secondary education and only 73 boys for every 100 girls in upper secondary education.

## Box 1.5.1: In some sub-Saharan African countries, progress in primary school completion

Sub-Saharan Africa remains the region with the largest number of countries having severe gender disparity in access to primary education. Countries in the region have followed varying trajectories since 1999. However, even where there has been progress in gender parity, this has not necessarily meant getting more children into school, let alone improving equality in completion or learning achievement.

Burkina Faso and the Central African Republic started from the same level of extreme gender disparity, with around 70 girls enrolled for every 100 boys. Gender disparity in the Central African Republic has remained unchanged, with the result that the country now has the second highest level of gender disparity in the world, after Afghanistan. Burkina Faso has made fast progress towards parity, reaching 95 girls for every 100 boys in 2012, although it still has the world's seventh lowest gross enrolment ratio (Figure 1.5.1).

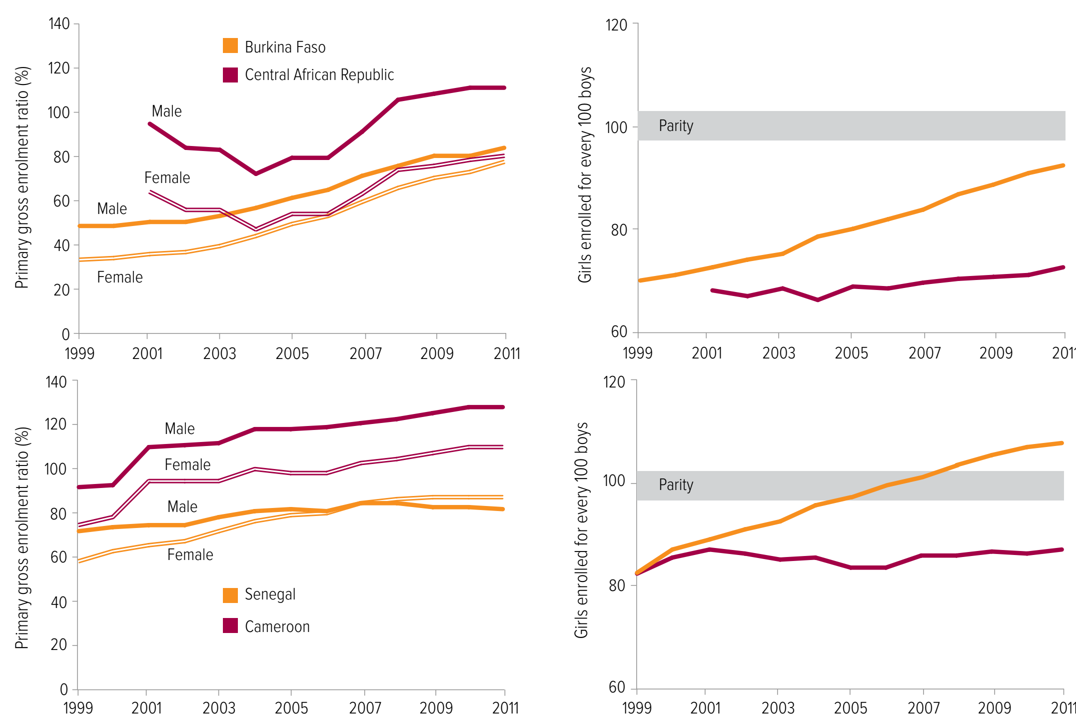
Part of the reason for Burkina Faso's progress was the successful implementation of the Ten-year Plan for the Development of Basic Education 2000–2009, which included a focus on girls' education. Measures, often carried out in collaboration with non-government organizations, included publicity campaigns, targeting girls in disadvantaged areas, and scholarships.

In addition to the government plan, other aid-supported interventions have helped move the primary education system towards gender parity. The Burkinabé Response to Improve Girls' Chances to Succeed programme provided an integrated set of interventions in rural areas, including the construction of schools equipped with boreholes and latrines, an increased number of female teachers and mobilization of community support for girls' education. An evaluation of the programme showed that it increased enrolment by 18 percentage points for boys and 23 percentage points for girls.

Cameroon and Senegal also started from similar levels of severe primary gender disparity, with around 80 girls enrolled for every 100 boys. Cameroon's disparity remained largely unchanged. But its enrolment levels rose continuously throughout the decade. Senegal made fast progress and reached parity in 2006. However, it still has the ninth lowest primary gross enrolment ratio in the world. Its move towards gender parity is a result of extremely slow progress in male enrolment – which is virtually unchanged since 2004 – rather than substantial growth in female enrolment.

**Figure 1.5.1: A move towards gender parity does not always mean access for all**

Primary gross enrolment ratio by gender and gender parity index, 1999–2011




*Source:* UIS database.

## for the poorest girls is too slow

Household survey data allow a more detailed look at trends in primary completion rates between and within these four countries, showing that, in terms of primary completion, much higher levels of gender disparity remain in all four countries.

In Burkina Faso, which has moved towards parity in enrolment, the gender gap in primary school completion remains wide, with 34% of boys and 24% of girls completing primary education in 2010. The gender gap in completion is narrow among the poorest because so few reach this stage: in 2010, just 11% of boys and 7% of girls completed primary school, only a slight increase from 1998. Greater progress in completion is evident in Senegal, but as poor boys have benefited more than poor girls, the gender gap among poor children has widened. In 2005, there was very little difference in poor children's completion rates, which were very low, but in 2010, 20% of boys completed while only 12% of girls did (Figure 1.5.2).

In Cameroon, the improvement in overall completion rates did not filter down to the poorest girls, who were even less likely to complete in 2011 than in 1998, while completion rates for the poorest boys stagnated. As a result, the gap in enrolment widened from 10 percentage points to 20 percentage points over the period.

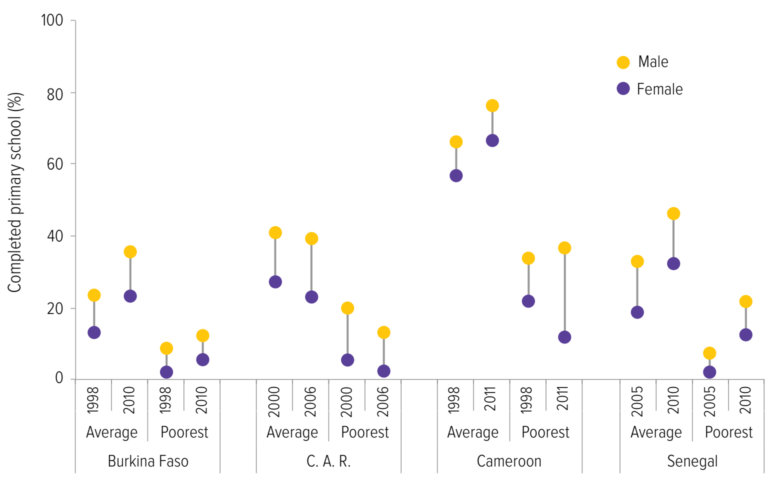
In the Central African Republic, the effects of conflict contributed to a slight reduction in completion rates for both boys and girls, in the population on average as well as for the poorest boys and girls. As a result, by 2006, only 3% of girls were completing primary school.

While the numbers of girls and boys completing primary school give an indication of the degree of gender parity in education, what boys and girls learn in school is a better measure of equality. In the 2006/07 round of the PASEC survey, both Burkina Faso and Senegal had a considerable gender gap in learning outcomes of grade 5 students. For example, in Burkina Faso, 45% of boys and 39% of girls passed the low benchmark in reading, while 53% of boys and 45% of girls passed the low benchmark in mathematics. The gap was almost twice as large in rural areas.

At first sight, it appears that countries such as Burkina Faso and Senegal have made strong progress towards eliminating gender gaps in enrolment. However, even in these countries, policies need to be put in place to ensure that all children, regardless of their gender, both stay in school and learn.

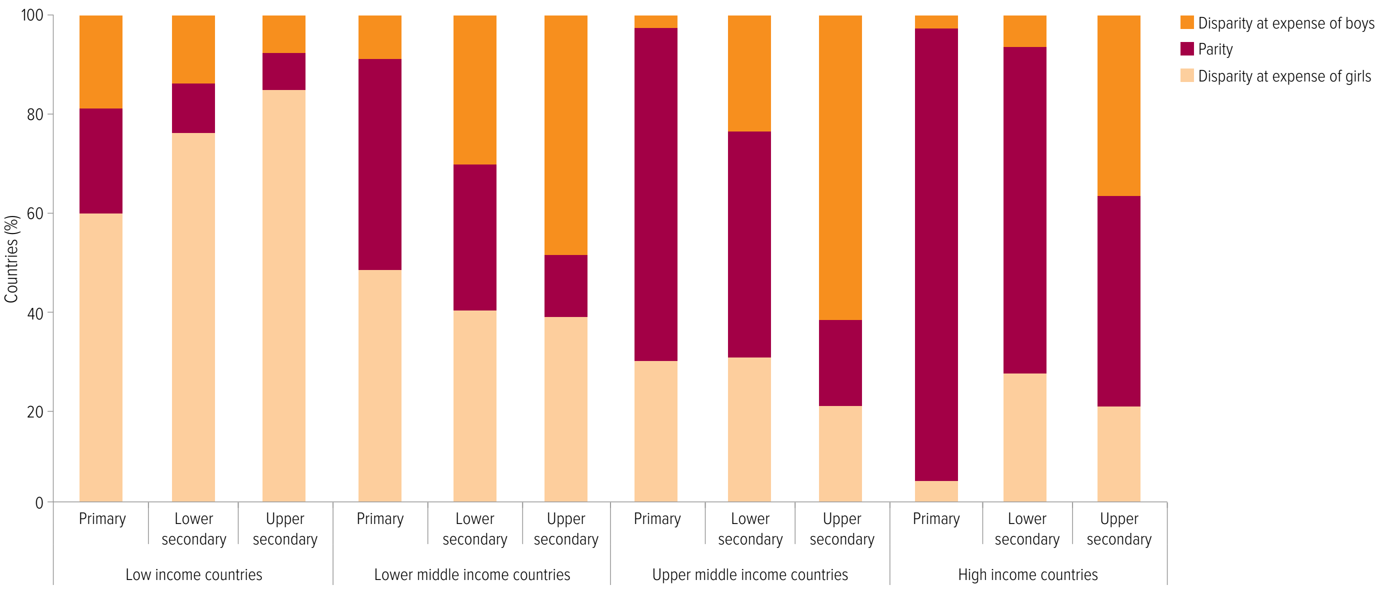
*Sources:* Burkina Faso Ministry of Basic Education and Literacy (1999); Kazianga et al. (2012).

**Figure 1.5.2: The poorest girls have the least chance of completing primary school**
Primary school completion rate by gender, national average and poorest 20% of households




*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Surveys and Multiple Indicator Cluster Surveys.

**Figure 1.5.3: Few low income countries have achieved gender parity at any level of education**
Countries with gender parity in enrolment ratios, by country income group, 2011




*Source:* UIS database.

## How many countries are likely to reach the gender parity goal by 2015?

The Dakar Framework for Action established clear targets on gender parity, with a value of the gender parity index between 0.97 and 1.03 indicating parity. Values below 0.90 and above 1.11 demonstrate severe disparity.

**By 2015, 84 countries are expected to have achieved gender parity in lower secondary enrolment**

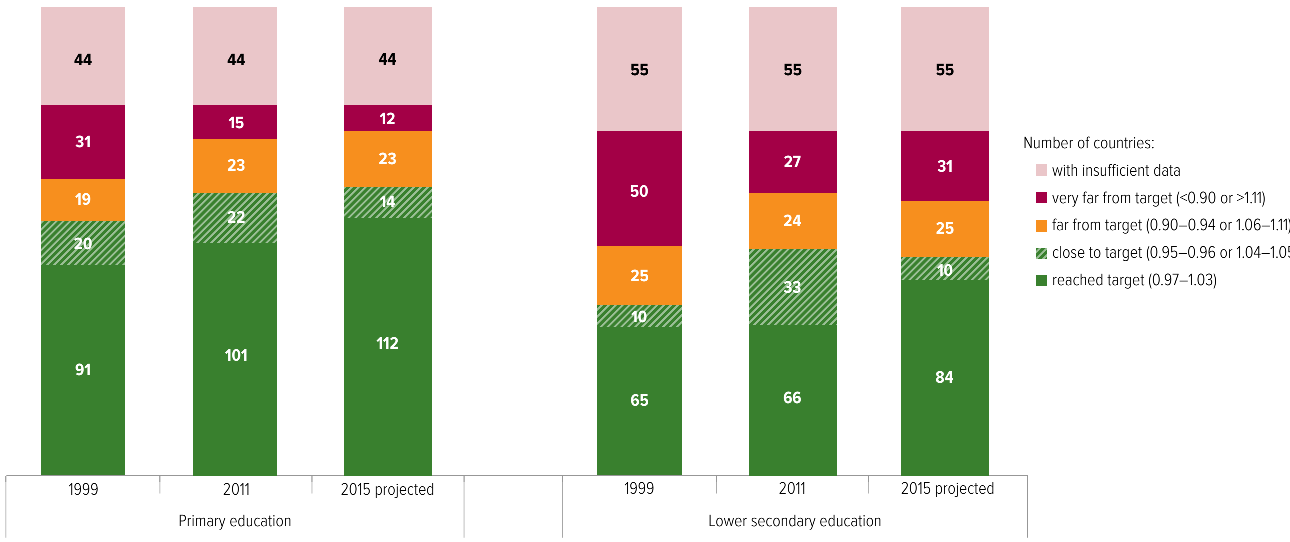
At the primary education level, it is possible to make projections on gender parity to 2015 for 161 countries. In 1999, 91 had achieved gender parity. Between 1999 and 2011, the number of countries that had reached the target increased to 101. It is projected that by 2015, 112 countries will have reached the goal and 14 will be close. However, 23 will be far from the target, and 12 will be very far from it. Of the 35 countries that will still be far or very far from the target, 19 are in sub-Saharan Africa. The number of countries furthest from the target, with severe disparity, fell from 31 in 1999 to 15 in 2011 (Figure 1.5.4).

In assessing the performance of countries that have not achieved parity, it is important to recognize how quickly they have moved towards the target. Of the countries that are not expected to reach or come close to the target, 8 made strong progress nonetheless, increasing their gender parity index by at least 33% between 1999 and 2011 (Table 1.5.2). In Mozambique, the female gross enrolment ratio increased from 59% in 1999 to 105% in 2012, helping to bring about an increase of the gender parity index from 0.74 to 0.91.

At the lower secondary level, it is possible to make projections to 2015 for 150 countries. In 1999, 65 countries had achieved gender parity, increasing by only one, to 66, by 2011. But many came close to the target. It is, therefore, projected that by 2015, 84 of the 150 countries will have achieved the target and 10 will be close (Figure 1.5.4).

Among the 50 countries that were furthest from the target in 1999, in 38 the disparity was at the expense of girls. Of these 38 countries, 17 had moved out of this group by 2011. Although only two achieved parity, several came close despite having started from a gender parity index below 0.75, Turkey being a notable example (Box 1.5.2). It is projected that by 2015, 31 countries will still be very far from the target, and in 22 of these the disparity will be at the expense of girls.

**Figure 1.5.4: Goal 5 – despite progress towards gender parity in education, the goal will not be achieved by 2015**
Number of countries by level of gender parity index in primary and lower secondary education, 1999, 2011 and 2015 (projected)




*Note:* The analysis was conducted on the subset of countries for which a projection was possible; therefore, it covers fewer countries than those for which information is available for either 1999 or 2011.
*Source:* Bruneforth (2013).

**Table 1.5.2: Likelihood of achieving gender parity in primary education by 2015**

| Level expected by 2015 | | Strong relative progress (8) | Slow progress or moving away from target (27) |
|---|---|---|---|
| Level expected by 2015 | Target reached or close (≥95%) 126 | Algeria, Argentina, Australia, Austria, Azerbaijan, Bahamas, Barbados, Belarus, Belgium, Bermuda, Bhutan, Bolivia (Plurinational State of), Brunei Darussalam, Bulgaria, Burundi, Cambodia, Canada, Chile, China, Colombia, Cook Islands, Costa Rica, Croatia, Cuba, Cyprus, Czech Republic, Denmark, Dominica, Ecuador, Egypt, El Salvador, Equatorial Guinea, Estonia, Ethiopia, Fiji, Finland, France, Gabon, Gambia, Georgia, Germany, Ghana, Greece, Grenada, Guatemala, Honduras, Hungary, Iceland, India, Indonesia, Iran (Islamic Republic of), Ireland, Israel, Italy, Japan, Jordan, Kazakhstan, Kenya, Kyrgyzstan, Lao People's Democratic Republic, Latvia, Lebanon, Lesotho, Liberia, Lithuania, Luxembourg, Madagascar, Maldives, Malta, Marshall Islands, Mexico, Mongolia, Morocco, Myanmar, Namibia, Netherlands, New Zealand, Nicaragua, Nigeria, Norway, Oman, Palestine, Panama, Paraguay, Peru, Philippines, Poland, Portugal, Qatar, Republic of Korea, Republic of Moldova, Romania, Russian Federation, Rwanda, Saint Kitts and Nevis, Saint Lucia, Samoa, Sao Tome and Principe, Serbia, Seychelles, Slovakia, Slovenia, Solomon Islands, South Africa, Spain, Sri Lanka, Sweden, Switzerland, Syrian Arab Republic, Tajikistan, Thailand, The former Yugoslav Republic of Macedonia, Togo, Trinidad and Tobago, Tunisia, Turkey, Uganda, Ukraine, United Kingdom, United Republic of Tanzania, United States, Uruguay, Uzbekistan, Venezuela (Bolivarian Republic of), Viet Nam, Zambia | |
| | Far from target (80-94%) 23 | Benin, Burkina Faso, Djibouti, Guinea, Mali, Mozambique, Yemen | Aruba, Belize, Botswana, British Virgin Islands, Cape Verde, Cayman Islands*, Comoros, Congo, Guyana*, Jamaica, Kiribati, Malawi*, Mauritania*, Saint Vincent and the Grenadines, Suriname, Vanuatu |
| | Very far from target (<80%) 12 | Afghanistan | Cameroon, Central African Republic, Chad, Côte d'Ivoire, Democratic Republic of the Congo, Dominican Republic, Eritrea, Niger, Pakistan, Senegal*, Swaziland |
| | | Strong relative progress 8 | Slow progress or moving away from target 27 |
| | | **Change between 1999 and 2011** | |

| | |
|---|---|
| Countries not included in analysis because of insufficient data 44 | Albania, Andorra, Angola, Anguilla, Antigua and Barbuda, Armenia, Bahrain, Bangladesh, Bosnia and Herzegovina, Brazil, Democratic People's Republic of Korea, Guinea-Bissau, Haiti, Iraq, Kuwait, Libya, Macao (China), Malaysia, Mauritius, Micronesia (Federated States of), Monaco, Montenegro, Montserrat, Nauru, Nepal, Netherlands Antilles, Niue, Palau, Papua New Guinea, San Marino, Saudi Arabia, Sierra Leone, Singapore, Somalia, South Sudan, Sudan, Timor-Leste, Tokelau, Tonga, Turkmenistan, Turks and Caicos Islands, Tuvalu, United Arab Emirates, Zimbabwe |

*Note:* An asterisk indicates disparity at the expense of boys.
*Source:* Bruneforth (2013).

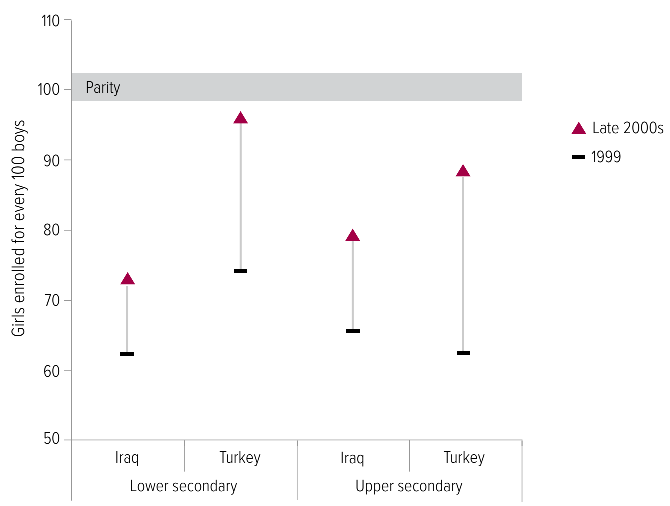
## Box 1.5.2: Challenges in improving access to secondary education for girls in Iraq and Turkey

Iraq and Turkey have moved at different speeds towards gender parity in secondary education. Turkey has made greater progress. In 1999, 87% of boys made it to lower secondary school, compared with 65% of girls. By the end of the decade, this large gap had been almost closed. Gender inequalities remain in upper secondary education but they also narrowed rapidly over the last decade (Figure 1.5.5).

The turning point was the extension of compulsory education from five to eight years in 1997, accompanied by a range of strategies aimed at widening access. A conditional cash transfer programme that provided a larger benefit for girls than for boys also helped close the enrolment gap.

Nevertheless, despite overall progress, problems remain. Girls in rural areas are more disadvantaged and some differences by region have been not only deep but also persistent: in the poor predominantly Kurdish provinces of Siirt, Mus and Bitlis, just 60 girls are enrolled in secondary school for every 100 boys, with little change in recent years.

**Figure 1.5.5: Fast progress towards gender parity in secondary education is possible**
Gender parity index of the secondary education gross enrolment ratio, Iraq, 1999–2007, and Turkey, 1999–2011




*Source:* UIS database.

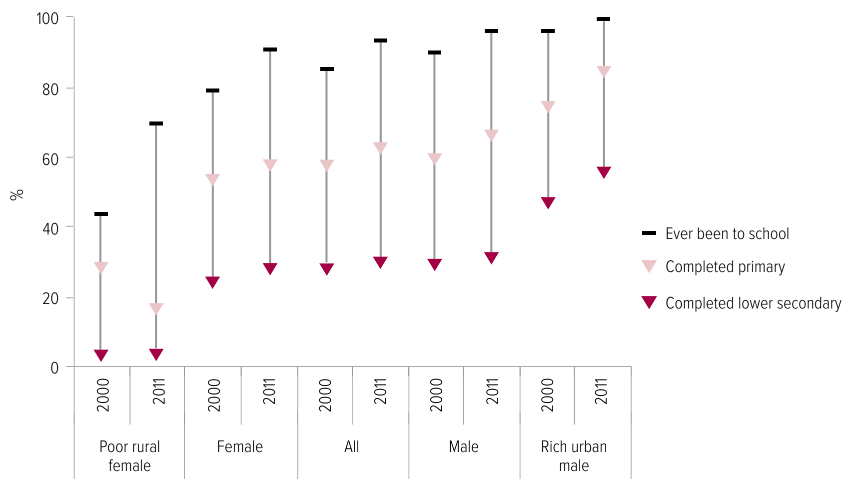
There is continuing commitment to eliminate the remaining disparities: the 2010–2014 strategic plan of the Ministry of National Education aims to lower the gender gap in secondary education enrolment from 8.9% to less than 2%. An amendment to the Education Law introduced in April 2012 that extended compulsory education from 8 to 12 years might help further close the gap at the upper secondary level.

But there is no room for complacency. The continuing very low participation of women in the labour force and their marginalization in the labour market could deter young girls from completing secondary school. More generally, traditional perceptions of gender roles that permeate society filter down to schools. These are issues that neighbouring countries aspiring to gender equality in education need to contend with.

In Iraq, not only has progress towards gender parity been slower, but poor, rural girls have not benefited. The lower secondary completion rate was 58% for rich urban boys and just 3% for poor rural girls in 2011. Safety remains an issue for girls' schooling, particularly in areas of major instability and insecurity (Figure 1.5.6).

*Sources:* Turkey Ministry of National Education (2009, 2013); Uçan (2013); World Bank (2012b).

**Figure 1.5.6: Poor Iraqi girls living in rural areas are far less likely to complete lower secondary school**
Percentage who have ever been to school, completed primary, and completed lower secondary school, by gender, location and wealth, 2000 and 2011




*Source:* EFA Global Monitoring Report team analysis (2013), based on the 2000 and 2011 Iraq Multiple Indicator Cluster Surveys.

# Goal 6 Quality of education

*Improving all aspects of the quality of education and ensuring excellence of all so that recognized and measurable learning outcomes are achieved by all, especially in literacy, numeracy and essential life skills.*

## Highlights

- At the primary education level, the pupil/teacher ratio exceeded 40:1 in 26 of the 162 countries with data in 2011. Less than 75% of primary school teachers are trained according to national standards in around a third of the countries with data.
- At the secondary education level, the pupil/teacher ratio exceeded 30:1 in 14 of the 130 countries with data in 2011. Less than 75% of secondary school teachers are trained according to national standards in half of the countries with data.
- In sub-Saharan Africa there is a lack of female teachers in primary schools that is even more accute in secondary schools. Among the countries with data, female teachers make up less than 40% of the total in 43% of countries at the primary level, in 72% of countries at the lower secondary level and in all countries at the upper secondary level.

**Table 1.6.1: Key indicators for goal 6**

| | Pre-primary education | | | | Primary education | | | | Secondary education | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Teaching staff | | Pupil/teacher ratio | | Teaching staff | | Pupil/teacher ratio | | Teaching staff | | Pupil/teacher ratio | |
| | 2011 (000) | Change since 1999 (%) | 1999 | 2011 | 2011 (000) | Change since 1999 (%) | 1999 | 2011 | 2011 (000) | Change since 1999 (%) | 1999 | 2011 |
| **World** | **8 230** | **53** | **21** | **21** | **28 824** | **16** | **26** | **24** | **31 473** | **28** | **18** | **17** |
| Low income countries | 427 | 100 | 27 | 25 | 2 978 | 70 | 43 | 43 | 1 892 | 95 | 28 | 26 |
| Lower middle income countries | ... | ... | 26 | ... | 9 589 | 24 | 31 | 31 | 9 229 | 61 | 24 | 22 |
| Upper middle income countries | 3 521 | 35 | 19 | 18 | 11 017 | 4 | 24 | 19 | 13 446 | 17 | 16 | 15 |
| High income countries | 1 990 | 43 | 18 | 15 | 5 239 | 10 | 16 | 14 | 6 906 | 9 | 14 | 12 |
| Sub-Saharan Africa | 439 | 123 | 28 | 28 | 3 190 | 62 | 42 | 43 | 1 788 | 115 | 26 | 26 |
| Arab States | 197 | 66 | 20 | 21 | 1 931 | 27 | 23 | 22 | 2 023 | 48 | 16 | 15 |
| Central Asia | 158 | 24 | 10 | 11 | 340 | 3 | 21 | 16 | 873 | 9 | 11 | 12 |
| East Asia and the Pacific | 2 262 | 60 | 26 | 21 | 10 355 | 13 | 24 | 18 | 10 000 | 32 | 17 | 16 |
| South and West Asia | ... | ... | 36 | ... | ... | ... | 36 | ... | 5 428 | 85 | 33 | 27 |
| Latin America and the Caribbean | 1 149 | 53 | 21 | 18 | 3 079 | 13 | 26 | 21 | 3 811 | 26 | 17 | 16 |
| North America and Western Europe | 1 596 | 50 | 18 | 14 | 3 801 | 11 | 15 | 14 | 4 957 | 10 | 14 | 12 |
| Central and Eastern Europe | 1 130 | 1 | 8 | 10 | 1 127 | -17 | 18 | 17 | 2 694 | -23 | 12 | 11 |

*Source:* Annex, Statistical Table 8.

The quality of education was at the heart of goals set at the World Education Forum in Dakar, Senegal, in 2000. Until recently, however, international attention has tended to focus on universal primary education, which is also the second Millennium Development Goal. A shift in emphasis is now discernible towards quality and learning, which are likely to be more central to the post-2015 global framework. Such a shift is vital to improve education opportunities for the 250 million children who have not had the chance to learn the basics, even though 130 million of them have spent at least four years in school.

This Report's thematic part presents analysis on learning disparities (see Chapter 4). It discusses in depth how to improve learning, in particular the vital role that teachers play supported by appropriate curricular and assessment practices (see Chapters 5, 6 and 7). This section examines progress in education quality in selected areas of the learning environment: the pupil/teacher ratio, the share of trained teachers, number of female teachers, availability of learning materials and school infrastructure. In addition, it provides an overview of the role international and regional assessments can play in monitoring progress towards a global goal on education quality and learning beyond 2015.

## Pupil/teacher ratios have changed little

The pupil/teacher ratio has been a key measure for assessing progress towards goal 6 since the EFA goals were set. Globally, between 1999 and 2011, average pupil/teacher ratios have barely changed at the pre-primary, primary and secondary education levels. In pre-primary education, the average pupil/teacher ratio remained at 21:1; in primary education it improved slightly, from 26:1 to 24:1; and in secondary education, from 18:1 to 17:1.

The pupil/teacher ratio has tended to improve only in richer countries. In primary education, it fell by 13% in high income countries and 23% in upper middle income countries, but by only 3% in lower middle income countries. In low income countries it remained at 43 pupils per teacher, three times higher than in high income countries.

Pupil/teacher ratios in sub-Saharan Africa hardly changed at any level of education. In primary education, teacher recruitment grew by 62%, lagging behind enrolment, which grew by 66% over the period. At 43 pupils per teacher, this is now the region with the highest ratio at the primary level.

As of 2011, 26 out of 162 countries with data had a pupil/teacher ratio in primary education exceeding 40:1. Of these, 23 were in sub-Saharan Africa, two in South and West Asia (Afghanistan and Bangladesh) and one in East Asia (Cambodia). Of the countries that had a pupil/teacher ratio above 40:1 in 1999, nine managed to bring the ratio below 40:1 by 2011; in Timor-Leste it halved, from 62:1 in 2001 to 31:1 in 2011. But in eight of these countries, the ratio increased, often because teacher recruitment did not keep pace with increases in enrolment thanks to policies such as fee abolition. In Malawi, the ratio increased by 20% from an already high level, reaching 76:1 in 2011. In Kenya, the ratio was below 40:1 in 1999, but rose by 45% to 47:1 in 2009.

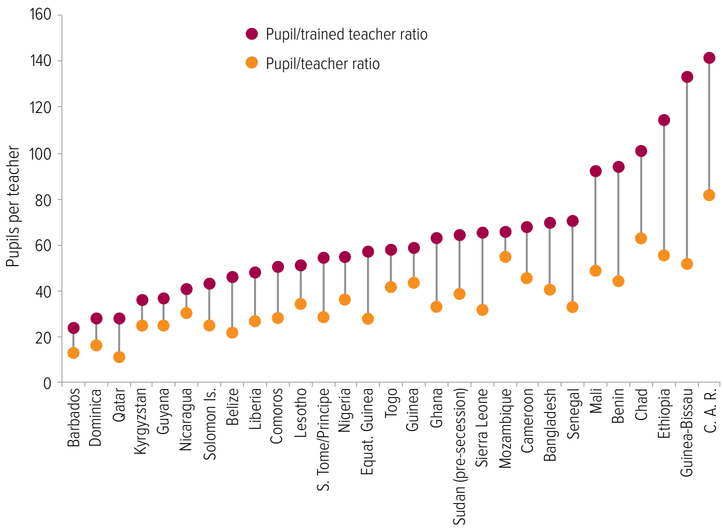
The pupil/teacher ratio in primary education increased by at least 20% in nine countries between 1999 and 2011, including the Democratic Republic of the Congo, Egypt, Pakistan and Yemen. By contrast, it fell by at least 20% in 60 countries, including Georgia, Guatemala, the Republic of Moldova, Nepal, Senegal, Tunisia and Viet Nam. Congo, Ethiopia and Mali more than doubled primary school enrolment while reducing their pupil/teacher ratios by more than 10 pupils per teacher.

**In Rwanda, the share of qualified teachers increased from 49% of the teaching force in 1999 to 98% in 2011**

However, many countries have expanded teacher numbers rapidly by hiring people without the qualifications of a trained teacher. Some countries have even lowered entry requirements to the profession, often out of necessity. For example, in Ghana, while there has been a 54% increase in the number of primary school teachers, keeping the pupil/teacher ratio below 40:1 over the decade, the proportion of trained teachers fell gradually from 72% in 1999 to 52% in 2012. Hiring untrained teachers may well serve to get more children into school, but it can jeopardize education quality. In Rwanda, by contrast, the pupil/teacher ratio remained high, at 58:1 in 2011, but the share of qualified teachers increased from 49% of the teaching force in 1999 to 98% in 2011.

In 34 of the 98 countries with data on trained teachers, less than 75% of teachers are trained

**Figure 1.6.1: In 29 countries, there is a big gap between the number of pupils per teacher and per trained teacher**
Pupil/teacher ratio and pupil/trained teacher ratio, primary education, countries where the pupil/trained teacher ratio exceeds the pupil/teacher ratio by at least 10:1, 2011




*Source:* Annex, Statistical Table 8.

according to national standards, with figures below 50% in Guinea-Bissau, Sao Tome and Principe, Senegal and Sierra Leone. In Guinea-Bissau, only 39% of primary school teachers have minimum qualifications, and the pupil/teacher ratio increased from 44:1 in 2000 to 52:1 in 2010. The ratio of pupils to trained teachers exceeds the pupil/teacher ratio by 10 pupils in 29 of these 98 countries, including 19 in sub-Saharan Africa and 4 in the Caribbean (Figure 1.6.1). In Sierra Leone, for example, the pupil/teacher ratio was 31:1 but the pupil/trained teacher ratio was 65:1 in 2011.

At the secondary level, in 14 of the 130 countries with data, the pupil/teacher ratio exceeds 30:1. In the Central African Republic, Ethiopia and Malawi, the ratio exceeds 40:1. Although the vast majority of the countries facing the largest challenges are in sub-Saharan Africa, the region nevertheless managed to expand the number of secondary teachers by 115% over the period to maintain the pupil/teacher ratio at 26:1.

**In Niger, only 17% of teachers in secondary schools are trained**

Of the 60 countries with data on the share of trained secondary teachers, half had less than 75% of the teaching force trained to the national standard, while 11 had less than 50% trained. Niger had the lowest share of trained teachers in secondary education (17%).

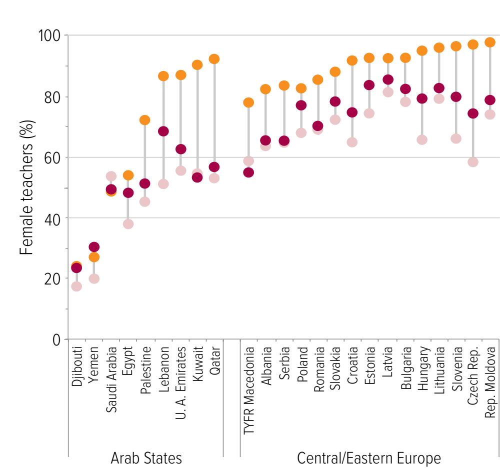
The proportion of teachers trained to national standards is even lower in pre-primary education. Although the number of teachers at this level has increased by 53% since 1999, the share of trained teachers remains very small. In 40 of the 75 countries with data, less than 75% of teachers are trained to the national standard. In Senegal, the share of trained pre-primary school teachers was just 15% in 2011.

## Women teachers are still lacking in some regions

In some contexts, the presence of female teachers is crucial to attract girls to school and improve their learning outcomes. In other contexts, a lack of male teachers can hinder boys' learning. The availability of male and female teachers is heavily unbalanced, however, between levels of education and between regions.

In sub-Saharan Africa, which suffers from gender disparities in schooling at the expense of girls, the lack of female teachers in primary schools is even more acute at secondary school. Among the countries with data, female teachers make up less than 40% of the total in 43% of countries at the primary level, in 72% of countries at the lower secondary level and in all countries at the upper secondary level. In Niger, the share of female teachers falls from

**Figure 1.6.2: The lack of female teachers is marked in**
Percentage of female teachers, primary, lower and upper secondary




*Sources:* Annex, Statistical Table 8 (print) and 10B (web).

46% in primary school to 22% in lower secondary school and to 18% in upper secondary school. The same problem is encountered in South and West Asia: in Nepal, the share of female teachers falls from 42% in primary school to 27% in lower secondary school and to 16% in upper secondary school (Figure 1.6.2).

Women teachers are particularly lacking in countries with wide gender disparity in enrolment. In Djibouti and Eritrea, only about 8 girls were enrolled for every 10 boys in lower secondary school, with very limited progress since 2000. The percentage of female teachers remained at 25% in Djibouti over the period and 14% in Eritrea. By contrast, in Cambodia, where in 1999 the level of gender disparity in lower secondary enrolment was the seventh highest in the world with 53 girls enrolled for every 100 boys, gender parity had almost been achieved by 2011 and the share of female teachers had increased from 30% to 36%.

By contrast, in Latin America and the Caribbean, where more girls tend to be enrolled in school than boys, female teachers make up at least 60% of the total in 70% of countries at the lower secondary level. In Suriname, where there are 91 boys for every 100 girls enrolled, 75% of lower secondary education teachers are female.

## Insufficient textbooks and poor infrastructure hinder learning

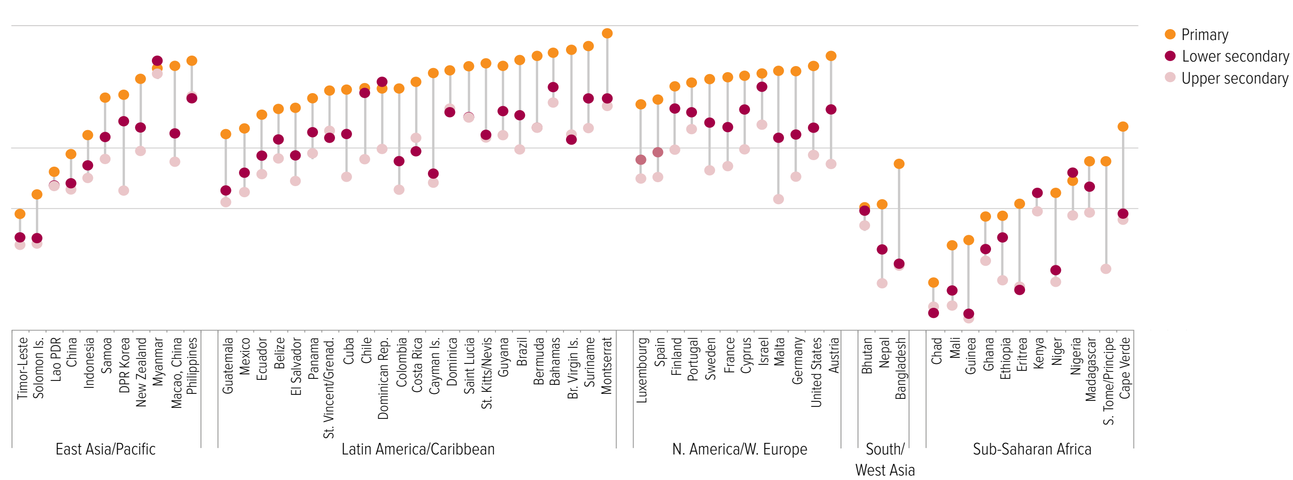
Teachers need good learning materials, such as textbooks, to be effective. Factors affecting the quality of textbooks differ, from content to printing quality and timeliness of distribution. But many students suffer from a very basic problem: they do not have access to textbooks.

In the United Republic of Tanzania, only 3.5% of all grade 6 pupils had sole use of a reading textbook (SACMEQ, 2010). In Cameroon, there were 11 primary school students for every reading textbook and 13 for every mathematics textbook in grade 2. Pupils in early grades were the most disadvantaged. For example, in Zambia there were 3.5 students for every mathematics textbook in grade 2, compared with 2.3 students in grade 5 (UIS, 2012). In Rwanda, where the government target was one textbook for every two pupils, a 2007 study in two-thirds of districts revealed that there were 143 pupils for every Kinyarwanda textbook in grade 1, and 180 pupils for every mathematics textbook (Read and Bontoux, forthcoming).

In some countries, textbooks are becoming even scarcer. Between 2000 and 2007, Kenya, Malawi and Namibia experienced rapid increases in enrolment, but the availability of textbooks did not keep pace. In Malawi, the percentage of students who either had no textbook or had to

**Women teachers are particularly lacking in countries with wide gender disparity in enrolment**

**sub-Saharan Africa**
education, 2011



share with at least two more pupils increased from 28% in 2000 to 63% in 2007 (Figure 1.6.3).

Factors that limit textbook availability include low priority on teaching and learning inputs in countries' education budgets, high textbook costs and wastage due to wear and tear (Box 1.6.1).

Poor physical infrastructure is another problem for students in many parts of sub-Saharan Africa. Children are often squeezed in overcrowded classrooms or learning outside, with those in early grades again particularly disadvantaged. In Malawi, there are 130 children per classroom in grade 1 on average, compared with 64 in grade 6 (Figure 1.6.4). In Chad, only one in seven schools has potable water, and just one in four has a toilet; moreover, only one-third of the toilets that do exist are for girls only (UIS, 2012).

### Box 1.6.1: Reducing the cost of textbooks helps increase their availability

The availability of textbooks could be significantly increased by devoting more resources to teaching and learning materials, and by reducing the cost of each book. Textbooks and other teaching and learning materials accounted for just 6.6% of education budgets in sub-Saharan Africa in 2009. Uganda allocated as little as 1.7% of its primary education recurrent budget on textbooks and learning materials in 2009, which was equivalent to US$1.30 per student. An analysis of six districts in Pakistan found that only 5% of the recurrent budget in 2009/10 was spent on non-salary items, and as little as 0.6% in one district.

After abolishing school fees, some countries introduced grants per pupil to schools to cover non-salary expenditure, including textbooks. The grants are often insufficient and vulnerable to budget cuts, however. In the United Republic of Tanzania, the government shifted the responsibility for procuring textbooks and other materials from the district to the school. It introduced a grant of US$10 per primary school pupil, earmarking 40% for textbooks and teacher guides. But this amount covered only 10% of the cost of a full set of textbooks for a grade 5 pupil. Furthermore, as a result of inflation and budget cuts, by 2011 less than US$2 per primary pupil was reaching schools. In addition, schools received their allocation several months after the school year started, so funds were not available to purchase textbooks in time.

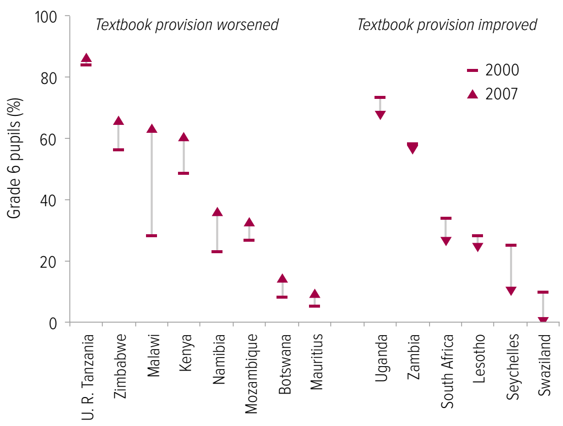
Reducing the cost per textbook would significantly increase their availability, even within existing budgets. In sub-Saharan Africa, a primary school textbook costs around US$4. In Viet Nam, by contrast, the cost is US$0.60 because it is possible to print books in country, with competition between publishers driving prices down. Not all countries have the technical capacity to do so, however. In Timor-Leste, for example, the unit cost would double if books were printed in country rather than in Singapore or Indonesia.

Other options for reducing unit costs include higher printing quality to extend the life of textbooks, printing in black and white instead of colour or increasing print runs. On one estimate, in India if a primary school book's specifications give it a four-year shelf life rather than just one year, the cost per textbook per year falls from US$0.36 to US$0.14.

The logistics of distribution also need to be improved to reduce wastage. Costs increase because of poor security in transport and storage, and because books are stolen and resold to private schools. A tracking survey in Ghana found that 29% of English textbooks could not be accounted for in 2010, for example.

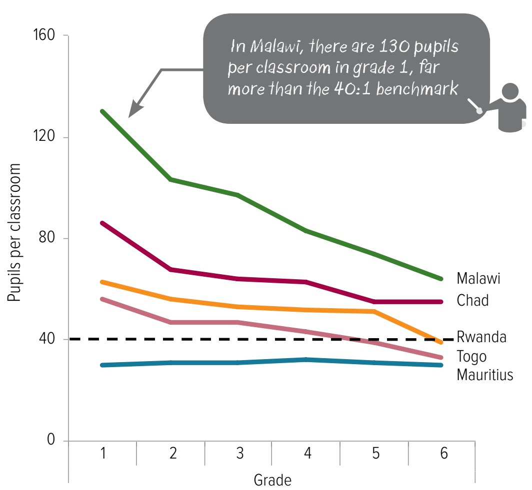
*Sources*: Bontoux (2012); Fredriksen (2012); Read and Bontoux (forthcoming); Twaweza (2012); UIS (2011); UNESCO (2013b); World Bank (2008a).

**Figure 1.6.3: Access to textbooks in some southern and east African countries has worsened**
Percentage of grade 6 pupils without access to a reading textbook or having to share with two or more pupils, selected countries, 2000–2007




*Source:* SACMEQ (2010).

**Figure 1.6.4: Children in early grades often learn in overcrowded classrooms**
Pupil/classroom ratio in public primary schools by grade, selected sub-Saharan African countries, 2011




*Source:* UIS database.

## Strengthening learning assessments to measure progress towards global goals after 2015

With 250 million children not learning the basics and the 2015 deadline for the Education for All goals fast approaching, it is vital for a global post-2015 goal to be set to ensure that, by 2030, all children and youth, regardless of their circumstances, acquire foundation skills in reading, writing and mathematics. Setting a goal is not enough on its own, however; it is also crucial to monitor progress to make sure countries are on track to achieve the goal.

**A global post-2015 goal must be set to ensure that, by 2030, all children and youth, regardless of their circumstances, acquire foundation skills**

International and regional assessments, which have expanded to include more countries over the past two decades, provide a good basis for tracking such progress. Beyond assessing whether children are learning the basics, the Learning Metrics Task Force promotes an expanded vision of learning (Learning Metrics Task Force, 2013). This expanded vision highlights the need for a global measure of whether all children are learning the basics to be complemented by monitoring within countries of progress towards a wider set of learning outcomes.

This section argues that meeting this need is important, but also requires countries to strengthen their national assessment systems and ensure that they are used to inform policy in ways that can help tackle the global learning crisis. In addition, this section reviews the learning assessment instruments available for monitoring progress towards a global learning goal, noting their strengths for monitoring progress in learning and for informing policy, as well as key principles that still need to be addressed to enable global monitoring of learning.

### *National assessments are indispensable for informing policy*

Many governments in low and middle income countries have been paying greater attention in recent years to measuring learning outcomes in an attempt to assess the quality of their education systems and use the results to inform policy. However, policy-makers often consider their public examination system as equivalent to a national assessment system, even though the two serve very different purposes. Public examination systems are used to promote students between levels of education (and so set standards and benchmarks according to places available); national assessments should be a diagnostic tool that can establish whether students achieve the learning standards expected in the curriculum by a particular age or grade, and how this achievement changes over time for subgroups of the population.

**Brazil has used national assessments to significantly improve education quality, especially for disadvantaged groups**

The ultimate test of national assessments as a diagnostic tool is whether the results are used effectively to help education ministries strengthen the policy mix so as to improve education quality and learning outcomes. Many national assessment systems are lacking in this respect. Brazil is an exception, having used national assessments to significantly improve education quality, especially for disadvantaged groups. A national assessment system, Prova Brasil, is used to build an Index of Basic Education Development, combining measurements of students' learning and progress, including repetition rates, grade progression and graduation rates. This has been a key tool in holding schools accountable for the quality of the education they provide. Each school, working with the municipality and monitored by the state, develops a strategic plan for achieving the required improvement in learning. Schools performing poorly attract more support (Bruns et al., 2012).

Government action is not the only route to an effective assessment system for informing national policy. Some civil society organizations have drawn government attention to the need for reforms and supported local communities in their demands for better learning outcomes in schools. In India, for example, the Annual Status of Education Report (ASER) produced by Pratham, an NGO, has been influential in shaping policy and planning to improve education quality. ASER's findings contributed to India's 12th five-year plan (2012–2017), helping to place emphasis on basic learning as an explicit objective of primary education, and on the need for regular learning assessments to make sure quality goals are met. Pratham has also used ASER results to influence education policy and practice at state level. In Rajasthan, for example, ASER results have led the state government to focus on improving instruction in early grades (ASER, 2013).

International aid can be a catalyst for effective national assessment systems. In Liberia, an assessment funded by a USAID project highlighted the low reading ability of children in early grades, prompting the Ministry of Education to institute reforms. These include revising the national curriculum to provide reading as a separate subject and strengthening capacity to train and support teachers in early grade reading approaches (Davidson and Hobbs, 2013). Similarly, Zambia has been benefiting from the Russia Education Aid for Development (READ) Trust Fund, executed by the World Bank, which aims to help low income countries strengthen the capacity of institutions to assess student learning and use the results to improve the quality of education (World Bank, 2013h).

Other countries need to learn from such experiences and use national assessments to monitor learning in ways that can inform policy. To obtain a globally comparable picture of progress in learning, however, international and regional approaches are also needed.

### *Regional and international assessments are vital for global monitoring*

Regional and international assessments have developed considerably since the 1990s, covering an increasing number of countries, subjects and levels (Table 1.6.2). Participating in a regional or international assessment helps mobilize interest in improving learning around the world. Just as improved global monitoring of access has helped maintain pressure on governments to ensure that each child completes primary school, better global monitoring of learning can push governments to make certain that all children not only go to school but are achieving the basics.

Participating in such assessments can present challenges. The need for cross-country comparability may mean that countries are asked to assess students in unfamiliar curriculum areas, so countries may re-orient their systems to fit these international assessments in ways that do not suit their circumstances. In addition, the assessments are sometimes used inappropriately to rank countries, which can discourage participation by poorer countries where fewer children are learning the basics. The cost of participation, moreover, can be substantial for poorer countries, which are likely to need support from international aid agencies.

It is vital nonetheless for all countries to engage in regional and international assessments to track whether all children, regardless of their circumstances, are learning the basics. These assessments promote a culture of transparency, evidence-based public debates on learning outcomes, and better national and international policy-making. They can also help countries develop their capacity for analysing results and assessing a wider range of skills (Bloem, 2013).

Regional and international assessments have been successfully used by some countries to inform national policy. Armenia, which has participated in three rounds of the TIMSS, has regularly and widely disseminated copies of reports on its performance and communicated key findings through press releases, television, radio and newspapers. The Ministry of Education and Science has used the results to track the impact of reforms on student achievement levels, to improve curricula and teacher training, and to inform classroom assessment activities (World Bank, 2013b).

Similarly, Namibia has used information from SACMEQ to improve learning. Namibia recorded one of the highest increases in reading and math scores of any country between 2000 and 2007. The increase of over 40 points in both scores has been attributed to policies that provided additional support for six poor-performing regions (Makuwa, 2010).

For regional and international learning assessments to facilitate monitoring post-2015 global learning goals, three key principles need to be taken into account:

*All children and young people need to be included in evaluation of learning, whether they are in school or not.* Regional and international assessments must increase understanding of the impact of poverty, ethnicity, location and gender on learning outcomes – and how policies can respond. To do so, they need to take into account the most disadvantaged children, who may already be out of the school system. The assessments collect information only from students without including information on children not in school, which can produce misleading results. For example, more than half of children are out of school by grade 5 in the Central African Republic, Mozambique and Niger. While it is likely that the vast majority of these children have not learned the basics, their exclusion means that the scale of the problem is understated, and can also let governments off the hook in addressing the learning needs of these children.

**Namibia has used information from a regional assessment to improve learning**

This issue is also relevant in assessments of older children. Even among the higher income OECD countries participating in PISA, 13% of 15-year-olds are not in school. It is difficult to determine whether adolescents of lower secondary school age who have dropped out of school by age 15 have failed to learn the basics. Only if they are included in assessments can governments form a full picture of learning achievement in the population. One way of including all children, whether they are in school or not, is to test young people's literacy as part of household surveys. This is currently done in the Demographic and Health Surveys and the Multiple Indicator Cluster Surveys (see Chapter 4 for an analysis of these data), but their purpose is only to gather very basic information on literacy.

The OECD is offering an alternative approach, redesigning its PISA surveys to be more relevant to developing countries while producing results that are comparable with the main PISA assessment. PISA for Development intends to explore how best to include out-of-school youth, and its results will help inform future approaches.

**Better information from learning assessments is needed to identify children from disadvantaged backgrounds**

*Better information on background characteristics of students is needed to identify which groups of students are not learning.* All regional and international assessments collect information on children's socio-economic background (see Table 1.6.2). However, some countries opt out of this process, such as Mexico in the SERCE study, reducing the results' comparability and hence their usefulness. Comparability also suffers when surveys fail to distinguish clearly between children from advantaged and disadvantaged households. Analysis for this Report suggests that surveys conducted as part of PASEC show children from the worst-off families in some countries doing as well as children from better-off families, a result which is likely to reflect weaknesses in how socio-economic background information is collected, rather than strong performance by students from poor households.

Collecting information on socio-economic status is not straightforward when the source of the information is a child or youth, who may not know, for example, the family's income or expenditure. Such information is better collected via household surveys. Nevertheless, some robust measures have been developed enabling comparable information to be collected using proxies for socio-economic status, such as parental education, occupation, and ownership of particular items (radios, televisions, cars, mobile phones). It should be possible to develop indicators that are comparable across countries and surveys so as to more accurately identify the most disadvantaged children.

*Information on the quality of education systems should be included as part of the assessments.* To inform policy, it is vital to know not just which students are not learning, but also other factors that can lead to poor learning outcomes. Collecting information on the school environment and teacher quality can help ascertain whether the right strategies are in place. The information on teacher subject knowledge that countries participating in SACMEQ have collected is useful in this respect. The experience of TIMSS could also be drawn upon, as its surveys collect data from students, teachers, principals and curriculum experts on the context of teaching and learning, allowing more in-depth analysis that links individual characteristics to the quality of education systems (Drent et al., 2013).

Existing regional and international assessments differ in many respects, including the target groups and the competencies tested. There is no strong reason to argue that countries should choose to participate in one rather than another. However, if policy-makers around the world are to have a picture of the extent of the global learning crisis and so be able to take measures to tackle it, agencies involved in administering regional and international assessments will need to collaborate to increase the comparability of their instruments. There is some evidence that this is already happening: TIMSS, PASEC and SACMEQ have been coordinating test items to ensure that there is a minimum basis for making their results comparable.

**Table 1.6.2: Major international and regional learning assessment studies**

| General | | | | | | Background data | | |
|---|---|---|---|---|---|---|---|---|
| Assessment | Countries | Target group | Subjects | Frequency | Years | Socio-economic status | Other | Prior achievement |
| International | | | | | | | | |
| **TIMSS** | 76 | Grades 4, 8 | Mathematics Science | 4-year cycle | 1995, 1999, 2003, 2007, **2011**, *2015* | Books, computer, study desk, parental education, parental occupation | Language spoken at home, country of birth of student and parents | Preschool, early literacy and numeracy |
| **PIRLS** | 59 | Grade 4 | Reading | 5-year cycle | 2001, 2006, **2011**, *2016* | | | |
| **PISA** | 73 | 15 year-olds | Reading Mathematics Science | 3-year cycle | 2000, 2003, 2006, 2009, **2012**, *2015* | Index based on parental education, parental occupation and home possessions, including books | Language spoken at home, country of birth of student and parents, reading habits | Grade repetition, self-reported performance, pre-school, extra tuition |
| Regional | | | | | | | | |
| **LLECE** Latin America | 18 | Grades 3, 6 | Reading Mathematics Science | Variable | 1997 **2006** (SERCE) *2013* (TERCE) | Index based on parental education, home facilities, asset ownership, house construction materials, books | Language spoken at home, child labour, reading habits | Grade repetition, age at entry, pre-school |
| **SACMEQ** Anglophone eastern and southern Africa | 14 | Grade 6 | Reading Mathematics | Variable | 1995-1997, 2000-2002, **2007**, *2014* | Index based on parental education, parental occupation, home possessions, books | Language spoken at home, child labour, distance travelled | Grade repetition, absenteeism, pre-school |
| **PASEC** Francophone sub-Saharan Africa and South East Asia | 24 | Grades 2, 5; (as of 2014) Grade 6 | Reading Mathematics | Variable | 1993-1995, 1997-2001, **2004-2010**, *2014* | Home facilities (water, sanitation, electricity), consumption of meat, parental literacy, books | Language spoken at home | Same students tested over time, grade repetition, absenteeism, pre-school |

Note: Dates in bold are the most recent rounds; dates in italics are the forthcoming rounds.

# Leaving no one behind — how long will it take?

After 2015, unfinished business will remain across the six EFA goals, while new priorities are likely to emerge. This section looks across three areas: universal primary school completion, universal lower secondary school completion and universal youth literacy. It assesses how long it will take to achieve these three goals, in particular from the perspective of reaching the most marginalized.[1]

This analysis paints a worrying picture. In many countries, the last mile to universal primary education will not be covered in this generation unless concerted efforts are taken to support the children who are the most disadvantaged. On recent trends, it may be only in the last quarter of this century that all of the poorest boys and girls in more than 20 countries will graduate from primary education – and only next century that they will all complete lower secondary education.

## Achieving universal primary completion

**By 2060, universal primary completion will not have been achieved for poor girls in 24 low income countries**

Many of the world's poorest countries have made considerable progress in expanding access to primary school since 2000. This progress tends to be measured using the adjusted primary net enrolment ratio. As the analysis of goal 2 has shown, this indicator can provide a snapshot of whether children of primary school age are at school, but fails to show whether children actually complete primary education. Household survey data on the percentage of young people who have completed primary education give a clearer picture of actual progress towards the target – and whether this progress is shared across population groups.

Seventeen mostly upper middle income countries, out of the 74 analysed, have reached universal primary completion or are expected to reach it by 2015 if recent trends continue. Even by 2030, however, only 26 of these countries will have met the goal. The majority of lower middle income countries are not expected to reach universal primary completion until the 2030s or 2040s. For low income countries, the picture is even worse: on recent trends, they are only expected to start achieving the target from the 2040s onwards. Four West African countries, Burkina Faso, Mali, Niger and Senegal, will not reach it before 2070.

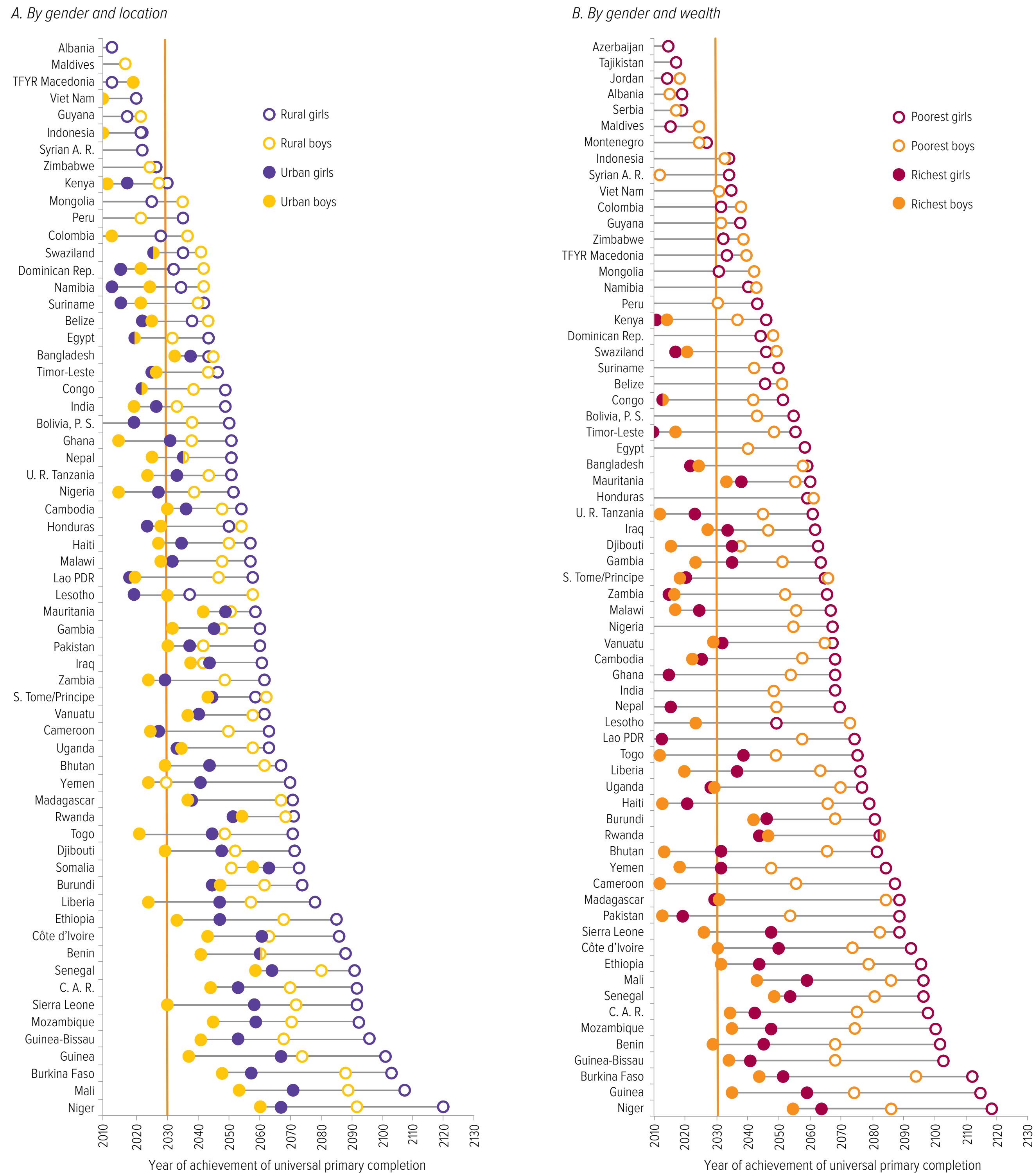
Disadvantaged groups face an even harsher reality. Among the low income sub-Saharan African countries in the sample, Zimbabwe and Kenya are the first expected to reach universal primary completion, by 2019 and 2026 respectively. For children in urban areas, the target has already been achieved in Zimbabwe and is expected to be reached in Kenya by 2018. But for children in rural areas of both countries, the goal will not be achieved before the late 2020s (Figure 1.7.1A). The period that will elapse between the achievement of the goal among boys in urban areas and among girls in rural areas is projected to be considerably wider in other countries: 39 years in the Lao People's Democratic Republic, 46 years in Yemen, 52 years in Ethiopia and 64 years in Guinea.

The differences are even starker when looking at the projected achievement patterns by gender and family income. While rich boys are expected to reach the goal by 2030 in 56 of the 74 countries, this is the case for poor girls in 7 countries. Even by 2060, universal primary completion will not have been achieved for poor girls in 24 of the 28 low income countries in the sample.

Lower middle income countries with large populations also face considerable challenges. In Nigeria, for example, rich boys already complete primary school, but it may be another three generations before poor girls do. In Pakistan, rich boys and girls are expected to complete primary school by 2020, but on recent trends poor boys will reach this fundamental target only in the late 2050s and poor girls just before the end of the century. In the Central African Republic, rich boys are expected to complete primary school by 2037 but poor girls only after 2100 (Figure 1.7.1B).

1. Household surveys covering 74 low and middle income countries were included in this analysis (Lange, 2013).

**Figure 1.7.1: On recent trends, universal primary completion will not be achieved for the poor in some countries for at least another two generations**
Projected year of achieving a primary completion rate in excess of 97%, selected countries




*Source:* EFA Global Monitoring Report team analysis (2013), based on Lange (2013).

## Achieving universal lower secondary completion

Post-2015 global education goals are widely expected to include achieving universal lower secondary completion by 2030. To inform forthcoming decisions on this target, it is vital to consider the results of projections on how long it would take countries to reach it.

Of the 74 countries analysed, in only six will all children be completing lower secondary school by 2015 on recent trends. In 25 countries, less than half of adolescents will be completing lower secondary school by this date, which shows that there will be a considerable distance to travel after 2015.

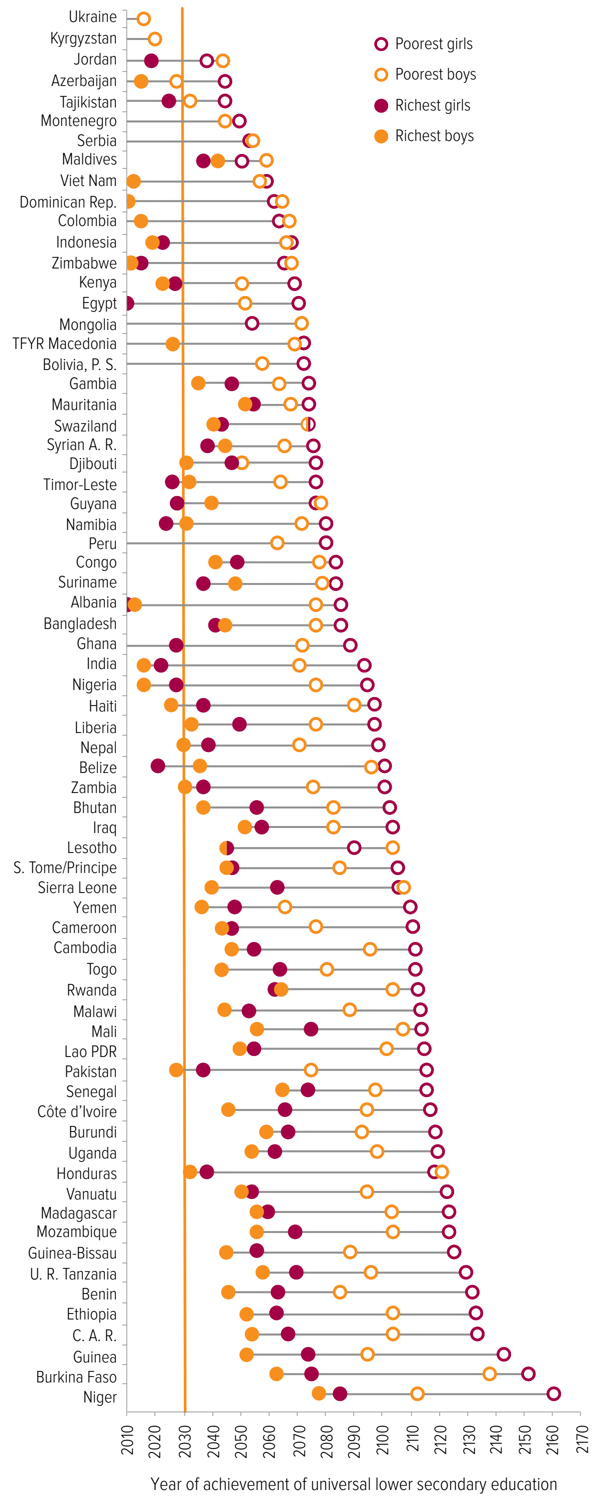
Taking the projections to 2030, 11 countries in the sample, most of them upper middle income countries, are expected to have reached universal lower secondary completion by that year. Of the 31 lower middle income countries in the sample, 14 are not expected to reach universal lower secondary until 2060. And only six low income countries are expected to achieve the target before 2060. On recent trends, the average gap between achieving universal completion of primary education and lower secondary education is projected to be 19 years in low income countries and 17 years in lower middle income countries.

**In 44 countries, there is at least a 50-year gap between all the richest boys completing lower secondary school and all the poorest girls doing so**

The gap between better-performing and worse-performing groups is likely to be even wider for achieving universal lower secondary completion than for achieving universal primary completion. In 44 of the 74 countries, there is at least a 50-year gap between all the richest boys completing lower secondary school and all the poorest girls doing so. In low income countries, the average gap is 63 years (Figure 1.7.2).

In Honduras, it is projected that the target will be achieved in the 2030s for the richest boys and girls but almost 100 years later among the poorest boys and girls. In 2011/12, 84% of the richest but only 10% of the poorest boys and girls completed lower secondary school (Honduras Ministry of Health et al., 2013). In Niger, the gap in achievement between the most and least advantaged groups will also span a century, with a massive gender gap: all the poorest girls are projected to be completing lower secondary school almost half a century after all the poorest boys if current trends continue.

**Figure 1.7.2: Achieving universal lower secondary completion will require more concerted efforts**
Projected year of achieving a lower secondary completion rate in excess of 97%, by gender and wealth, selected countries




*Source:* EFA Global Monitoring Report team analysis (2013), based on Lange (2013).

## How long will it take to achieve universal completion in sub-Saharan Africa?

Available data for primary and lower secondary completion cover more than four-fifths of the population in sub-Saharan Africa, so it is possible to project average achievement rates for different population groups for the entire region (Figure 1.7.3). On average, if recent trends continue, the region will not achieve universal primary completion until 2052, more than 35 years after the Dakar target and two decades after the likely target date for the post-2015 goals. While boys are expected to reach the target by 2046, on average, the richest boys are expected to achieve it in 2021 – before the post-2015 target – but the poorest boys only by 2069. Girls are further behind: on average they will reach it by 2057, with rich girls getting to zero by 2029. The poorest girls will, however, only reach the target by 2086.

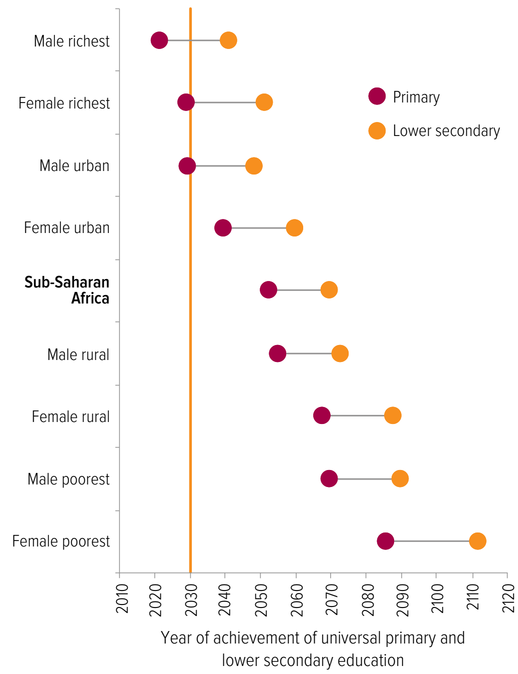
On average, if recent trends continue, lower secondary school completion will be achieved in 2069 in sub-Saharan Africa, several decades after the target dates currently under discussion. Girls will achieve universal lower secondary completion by 2075, on average. Girls from the richest fifth of the population will reach the target by 2051, but girls from the poorest fifth of families are currently projected to achieve it only by 2111.

Overall, it is projected that, on average, all children in low income countries will be completing primary school by 2053 and lower secondary school by 2072. Lower middle income countries will achieve these targets about two decades earlier, in 2036 for primary school and in 2053 for secondary school.

## Achieving universal youth literacy

Even children who complete primary school are not necessarily able to read and write (see Chapter 4). To assess how long it will take to achieve universal youth literacy, projections were made based on household survey data available in 37 countries for the whole population and in 68 countries for the female population only.

**Figure 1.7.3: The achievement of universal primary and lower secondary education lies well into the future for sub-Saharan Africa**
Projected year of achieving primary and lower secondary completion in excess of 97%, by population group, sub-Saharan Africa




*Source:* EFA Global Monitoring Report team analysis (2013), based on Lange (2013).

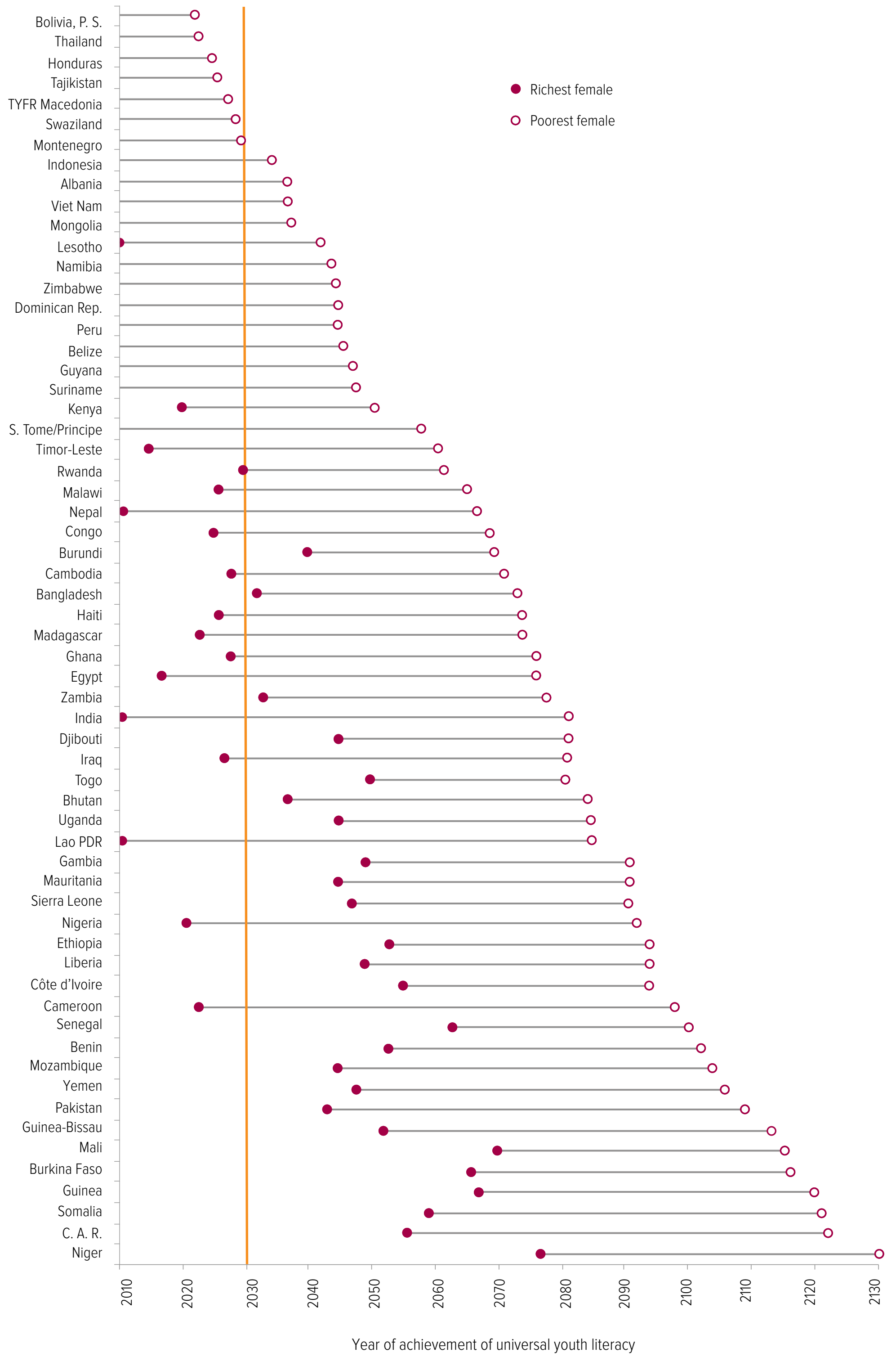
Among the 37 countries for which information is available for the whole population, universal youth literacy may not be achieved before 2060 in 11 countries, most of them low and lower middle income countries, if recent trends continue. And among the larger group of 68 countries with only a sample of the female population, there are 22 low and lower middle income countries where universal youth literacy may not be achieved before 2060.

**The poorest young women in low and middle income countries will only achieve universal literacy by 2072**

The situation is worse for the poorest women: across the 68 countries, the poorest young women will only achieve universal literacy by 2072, on average. In India and the Lao People's Democratic Republic, the richest young women have already achieved universal literacy but the poorest will only do so around 2080. In Nigeria, while universal youth literacy has almost been achieved for the richest women aged 15 to 24, their poorest counterparts will need to wait 70 years to realize this fundamental right if no active steps are taken to fight illiteracy among the most disadvantaged groups (Figure 1.7.4).

**Figure 1.7.4: Literacy is a distant dream for the most vulnerable young women**

Projected year of achieving a female youth literacy rate in excess of 97%, by wealth, selected countries




*Source:* EFA Global Monitoring Report team analysis (2013), based on Lange (2013).

## Conclusion

These projections of the time it could take to achieve universal primary and secondary education and youth literacy are extremely disconcerting. However, they are based on recent trends that can be changed if governments, aid donors and the international education community take concerted action to make education available to all, including the marginalized. The key message that emerges from the analysis in this section is that it is vital to track progress towards education goals for the most disadvantaged groups after 2015, and to put policies in place that maintain and accelerate progress by redressing imbalances.

## Monitoring global education targets after 2015

Since the six Education for All goals were adopted in Dakar, Senegal, in 2000, a lack of precise targets and indicators has prevented some education priorities from receiving the attention they deserve. To ensure that nobody is left behind after 2015, it is vital for the global framework to include targets and indicators, tracking the progress of the most disadvantaged.

As the new global development framework takes shape, a consensus is emerging over post-2015 education aims (UNESCO and UNICEF, 2013; United Nations, 2013b, 2013c). Common priorities include:

- addressing unfinished business on primary school completion;
- extending goals to include universal access to early childhood care and education programmes, and lower secondary school completion;
- ensuring that children are not only in school but are also learning;
- making sure all young people and adults have the skills needed for the world of work.

**Better data are needed to help governments track progress of disadvantaged groups**

These goals are a foundation on which to build more ambitious goals, including those set nationally.

The High Level Panel on the Post-2015 Development Agenda further noted the need for a data revolution to improve the quality of statistics and information available to citizens. It emphasized that data need to be disaggregated – broken down according to factors such as gender, geography, income or disability – to make sure no group is left behind due to circumstances of birth. It underlined the need for better data not only to help governments track progress and make evidence-based decisions,
but also to strengthen accountability.

These issues have been at the heart of the *EFA Global Monitoring Report* over the past decade. Building on this experience, the Report team proposes five principles for setting goals after 2015 (Box 1.8.1).

Each of the EFA goals specified a commitment to equity – the basic principle of making sure every person has an equal chance of obtaining an education. Over the years, however, the *EFA Global Monitoring Report* has shown that progress in narrowing inequality in education has been limited. One possible reason is that the MDGs, which have dominated development planning, did not incorporate equity as a core principle.

### Box 1.8.1: Guiding principles for setting education goals after 2015

The following five principles should guide post-2015 education goals:

- The right to an education, as guaranteed under international and national laws and conventions, must be at the core of the goals. As a right, education should be free and compulsory. It should help people fulfil their potential and should foster the well-being and prosperity of individuals and society.
- Ensuring that all people have an equal chance of education, regardless of their circumstances, must be at the heart of every goal. No person should be denied access to good quality education because of factors such as poverty, gender, location, ethnicity or disability.
- The goals should recognize the learning needs at each stage of a person's life and the fact that learning takes place in non-formal as well as formal settings.
- One core set of goals should encapsulate the overall ambition for education as part of the broader post-2015 global development framework replacing the MDGs. The goals should be universally applicable and accompanied by a more detailed set of targets and indicators that make up a post-2015 education framework.
- The goals should enable governments and the international community to be held to account for their education commitments. Each goal must have a specific deadline, be worded clearly and simply, be measurable, and have the ambition of leaving no one behind. The framework should include a commitment to independent regular and rigorous monitoring of the goals at the global and national levels.

Another is that disaggregated data were scarce when the goals were established, so targets and indicators were not developed that might have enabled national and global tracking of progress for different population groups. The growing availability of household survey data in recent years has eased the constraints on comparing groups within countries and over time. It is increasingly possible to measure equity targets, and incorporating them in post-2015 goals is now essential.

Equity is vital not just in access to education, however; it also needs to extend to education quality, so that each child has an equal chance to learn. While the sixth EFA goal focuses on education quality, an absence of concrete targets and of internationally comparable data has undermined efforts to monitor progress towards the goal. The neglect of learning outcomes in the MDGs also diminished attention to education quality. The international development agenda is now giving increasing recognition to learning outcomes, as the expansion of regional and international surveys has revealed low levels of attainment and extreme inequality in learning. While more coordination is needed before commonly agreed measures can be reached (see Goal 6), it is time to include learning outcomes in an improved global education monitoring framework.

Within this framework, goals must be simple, clear and measurable. One challenge the *EFA Global Monitoring Report* has faced over the past decade is that indicators were not identified for many of the goals and targets. As a result, progress towards the EFA goals and MDGs was measured using indicators that were not always compatible. In addition, some goals failed to receive the attention they deserved because they could not easily be measured, either with existing data or with data that could be collected within the timeframe. One example is goal 3 on skills, the focus of the 2012 *EFA Global Monitoring Report*. Such gaps need to be avoided in a post-2015 framework.

## How far are we from achieving equality in education goals?

It is not acceptable for any child or adolescent to remain out of school or to be in school but not learning, or for any young person or adult to lack the skills needed to get decent jobs and lead fulfilling lives. The EFA Global Monitoring Report team proposes, therefore, that each global education goal should aim for no one to be left behind by 2030.

Targets should be set to achieve equality, taking into account that characteristics of disadvantage often interact: girls from poor households in rural areas, for example, are usually among the most marginalized. Each goal should be tracked not only overall but also according to the progress of the lowest performing groups in each country to ensure that these groups reach the target by 2030.

**Collecting data on children with disabilities needs to be improved urgently**

Among the lowest performing groups, children with disabilities are likely to face the most severe discrimination and exclusion, which in many contexts keep them out of school. Collecting data on children with different types and severity of impairments needs to be improved urgently so that policy-makers can be held accountable for making sure these children's right to education is fulfilled.

To inform post-2015 monitoring, this section analyses the progress made by those who were most behind in 2000 on selected goals that are likely to be set for 2030.[1] Through this analysis, several data limitations are identified that need to be overcome (Box 1.8.2).

1. For the analysis in this section, data from Demographic and Health Surveys and Multiple Indicator Cluster Surveys have been grouped into two waves: surveys undertaken between 1998 and 2003, and surveys undertaken since 2005. The reported results are based on 62 countries for which a survey is available in each period. The surveys are at least five years apart for each country.

### Box 1.8.2: Data needs for monitoring education goals after 2015

Several difficulties in measuring progress in priority areas after 2015 deserve urgent attention:

- The number of children of primary school age who are not learning the basics, whether they are in school or not, is a key indicator that needs to be updated regularly. Its importance is underlined by the fact that the number is estimated at 250 million in this Report. But to measure progress, it is vital to have information from more countries on this indicator, as current surveys of learning achievement do not provide the necessary data.
- To enable better tracking of progress among the children who are the most disadvantaged – whether by poverty, gender, location or disability – comparable household surveys need to be conducted more frequently and should cover more countries, with sufficient observations to allow analysis of population subgroups.
- Despite progress since 2000, there are still major gaps for key indicators reported by the UNESCO Institute for Statistics. Extending goals, for example to include lower secondary completion, will require much better information from administrative data sources.
- For some indicators, there are still no data available systematically:
  - *Early childhood care and education programmes.* A survey should be developed that collects information on access to and quality of these programmes.
  - *Skills domains for young people and adults.* Indicators should be agreed for measuring foundation skills in literacy and numeracy, transferable skills, and technical and vocational skills, based on the analysis in the 2012 *EFA Global Monitoring Report*. Currently data are fairly widely available on foundation skills, but only from selected countries for some transferable skills, chiefly problem-solving (from the OECD PIAAC and World Bank STEP programmes). It is also vital to collect data on technical and vocational training programmes for youth and adults outside of the formal education system. The International Labour Organization collects some information on these programmes, but does not collate the data systematically.

**By 2010, poor rural young women had spent fewer than 3 years in school in low and lower middle income countries**

### *Which young people are spending at least nine years in school?*

The number of years young people have spent in school is one measure of overall progress in access to education across countries.[2] For universal lower secondary education completion to be achieved, young people would need to have stayed in school for around nine years.

Between 2000 and 2010, the number of years spent in school increased from 4.8 to 6 in low income countries and from 7.1 to 8 in lower middle income countries. While this is a big achievement, the gap between advantaged and disadvantaged groups has remained wide.

In the early part of the decade, the richest urban young men had already spent 9 years in school in low income countries and 11.5 years in lower middle income countries. But the poorest rural young women had spent only 2.3 years in school in low income countries and 2.6 years in lower middle income countries.

By the latter part of the decade, the richest urban young men had spent more than 9.5 years in school in low income countries and more than 12 years in lower middle income countries, well above the 9-year target suggested for 2030. But the poorest rural young women had still spent fewer than 3 years in school in both low and lower middle income countries, leaving them well below even a 6-year target that is associated with universal primary completion, which is supposed to be achieved by 2015. This leaves a wide gap between the time they spent in school and that spent by the richest urban young men. In sub-Saharan Africa, the difference between the time the poorest rural females and the richest urban males spent in school actually widened between 2000 and 2010, from 6.9 years to 8.3 years (Figure 1.8.1).

Looking across the 62 low and middle income countries with data at the beginning and end of the decade, the lack of progress for the disadvantaged becomes even clearer. The poorest rural young women spent more than 6 years in school in 18 of the 62 countries in 2000, increasing to 21 countries in 2010. They spent more than 9 years in just 9 countries in 2000, mostly in Central and Eastern Europe and

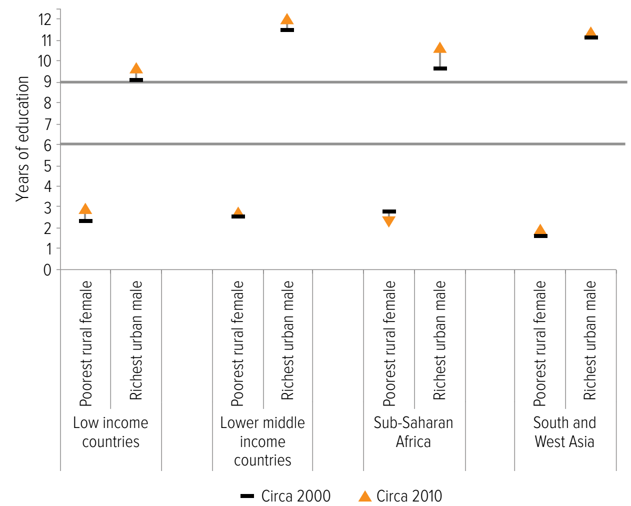
2. The indicator used in this section measures the number of years of schooling completed among 20- to 24-year-olds and is net of years spent repeating grades. While many young people continue their education until their late 20s, the age group is sufficiently indicative of progress in the average population.

Central Asia, with no improvement by 2010. By contrast, the richest urban young men spent more than 6 years in school in all but one country, Niger, in 2000. And they reached more than 9 years in 42 countries in 2000, increasing to 51 countries in 2010. Progress has often been faster for the more advantaged group: between 2000 and 2010, the gap in years spent in school has widened by at least half a year in 29 countries of the 62 countries, and by at least one year in 19 of these countries.

Data for 11 sub-Saharan African countries confirm this general pattern: while the average number of years spent in school increased across the board, in all but two countries the progress was faster for the richest urban young men. For example, in Mozambique attainment among the richest urban young men increased by 3.5 years, to 9.6 years, while among the poorest rural young women it increased by less than a year, to 1.9 years (Figure 1.8.2).

**Figure 1.8.1: Inequality in educational attainment remained unchanged over the last decade**

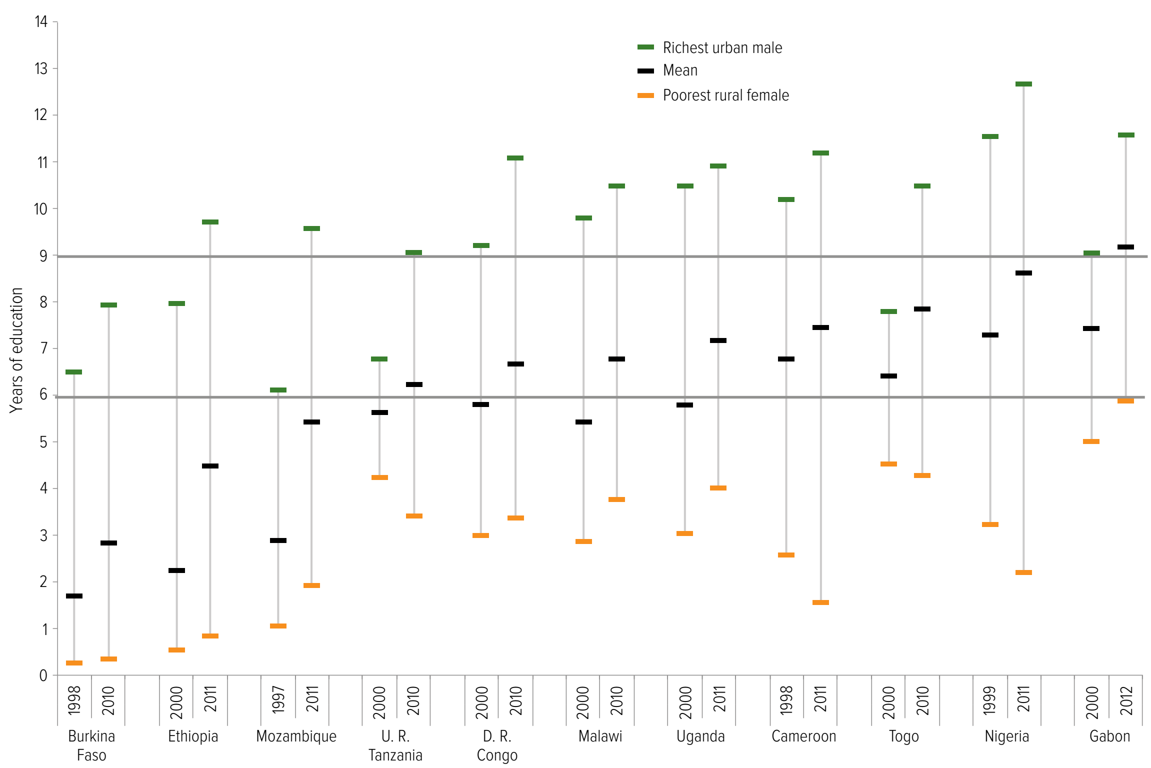
Years of education, 20- to 24-year-olds, circa 2000 and 2010




*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Surveys and Multiple Indicator Cluster Surveys.

**Figure 1.8.2: In sub-Saharan Africa, the richest urban young men increased the number of years in school faster than the poorest rural young women**

Years of education, 20- to 24-year-olds, selected countries, circa 2000 and 2010




*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Surveys and Multiple Indicator Cluster Surveys.

### *Which children complete primary and lower secondary school?*

Over the past decade there has been more progress in getting children into primary school than in making sure children complete primary or lower secondary education. In addition, the progress is unequally distributed; extreme inequality persists and in some cases has widened.

**By 2010, only 23% of the poorest rural girls completed primary school in sub-Saharan Africa**

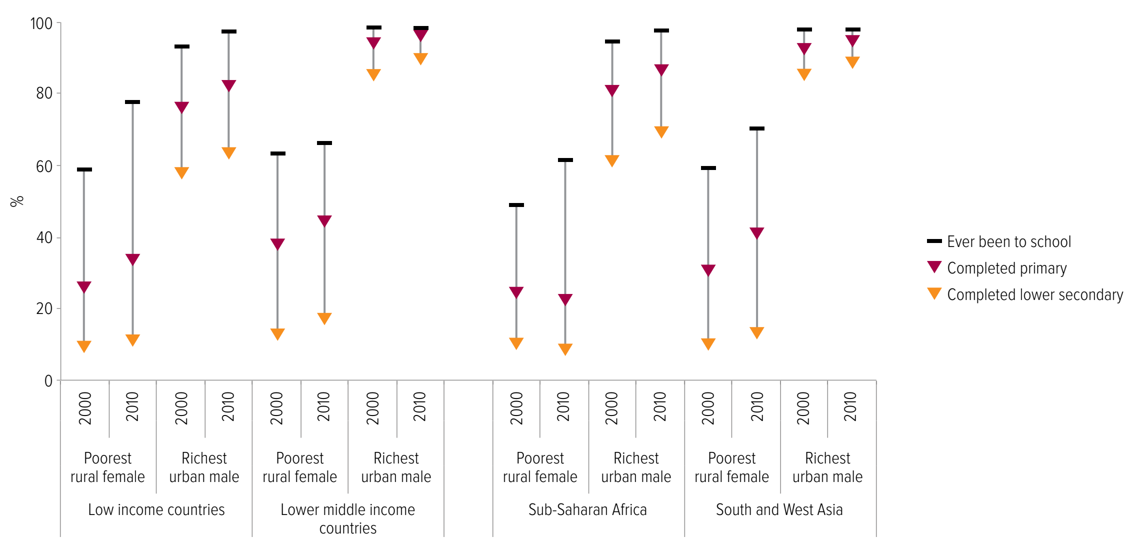
One accomplishment since 2000 is the improvement in the proportion of children who enter school, which rose from 73% to 87% in low income countries. There is still, however, a long way to go in primary school completion, which increased from 45% to 60% in low income countries and from 72% to 79% in lower middle income countries. The proportion of adolescents completing lower secondary education increased more slowly over the decade, from 27% to 37% in low income countries and from 50% to 58% in lower middle income countries.

Insufficient attention to inequality in entry to and completion of school over the decade means that gaps remain wide. In sub-Saharan Africa, for example, almost all the richest boys in urban areas entered school in 2000. Moreover, 82% completed primary education and 62% completed lower secondary education, and these rates had improved by the end of the decade to 87% for primary school completion and 70% for lower secondary school completion.

By contrast, among the poorest girls in rural areas, 49% entered school at the beginning of the decade, increasing to 61% by the end of the decade. Only 25% completed primary education and 11% completed lower secondary education in 2000, and these rates had fallen by the end of the decade to 23% and 9%, respectively. Inequality in South and West Asia is also wide and largely unchanged: by the end of the decade, while 89% of the richest urban adolescent boys completed lower secondary school, only 13% of the poorest rural girls did so (Figure 1.8.3).

Across the 62 countries, a lack of progress in primary and lower secondary completion among the disadvantaged is clearly visible. At least half the poorest rural girls completed primary school in 2000 in only 23 countries, increasing to 29 in 2010. And at least half of them had completed lower secondary school in 2000 in just 10 countries, with no change over the decade. By contrast, at least half the richest urban boys completed primary school in 2000 in all countries except Burundi and Mozambique, and were doing so there as well by 2010. The number of countries where at least half the richest urban adolescent boys completed lower

**Figure 1.8.3: Over the decade, disadvantaged groups mainly improved their access to school rather than their completion rates**
Percentage of poorest rural girls and richest urban boys who have ever been to school, completed primary education, and completed lower secondary education, circa 2000 and 2010




*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Surveys and Multiple Indicator Cluster Surveys.

secondary school increased from 39 in 2000 to 50 in 2010; the remaining 12 countries were all in sub-Saharan Africa.

Comparing these two population groups in 12 countries from five regions further shows that progress has often been slowest for the most disadvantaged. Progress in the Democratic Republic of the Congo has largely benefited the more advantaged: the proportion of the poorest rural girls completing primary school improved by only three percentage points over the decade, from 20% to 23%, while the corresponding share among the richest urban boys increased by eight percentage points, from 84% to 92%.

Similarly, among the poorest rural adolescent girls the proportion completing lower secondary school improved by seven percentage points over the decade, from 7% to 14%, while among the richest urban adolescent boys it increased by twelve percentage points, from 66% to 78%. In Burkina Faso, Ethiopia, Iraq, Mozambique and the United Republic of Tanzania, hardly any of the poorest rural females completed lower secondary school in 2010 (Figure 1.8.4).

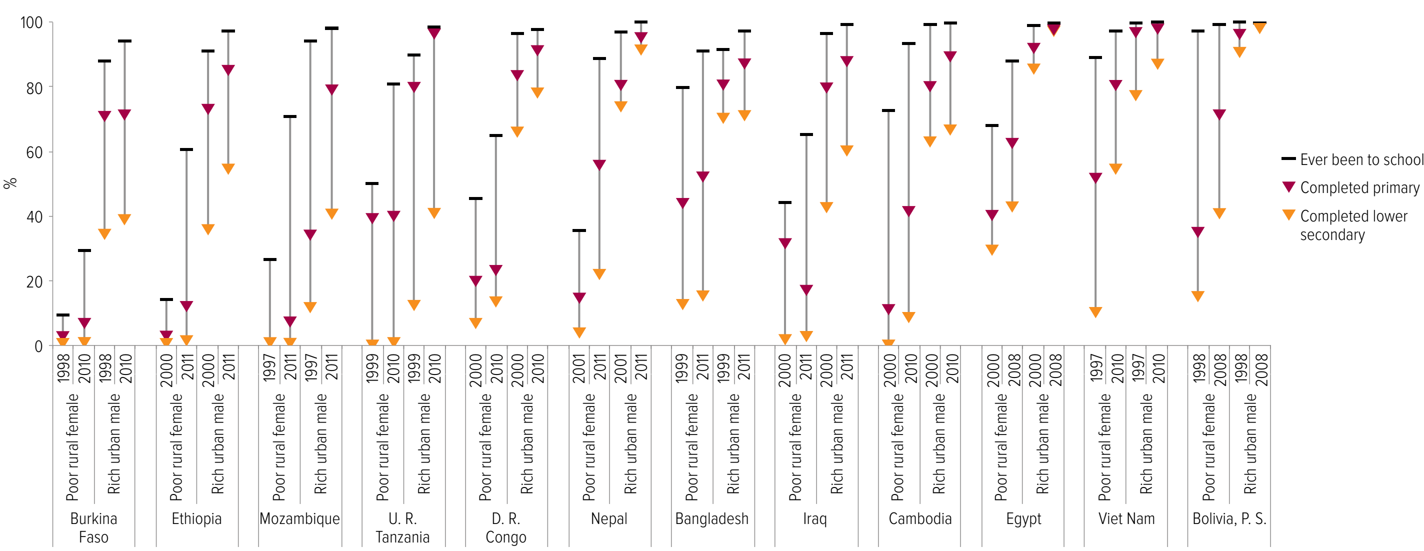
In some countries, disadvantaged groups have made progress, albeit from a very low starting point. Countries including the Plurinational State of Bolivia, Nepal and Viet Nam achieved rapid progress for poor rural girls, showing that such change is possible. In Nepal, between 2001 and 2011, there were marked increases not only in the proportion of the poorest rural girls entering school, from 36% to 89%, but also in the share of those completing primary school, from 15% to 56%. Primary completion among the richest urban boys also increased, from 80% to 95%. There is still unfinished business, however: while nearly all the richest urban males complete primary school in the Plurinational State of Bolivia and in Viet Nam, only 72% of the poorest rural girls complete school in the Plurinational State of Bolivia and only 80% in Viet Nam.

**In Iraq and Mozambique, hardly any of the poorest rural females completed lower secondary school in 2010**

### *Which children are achieving minimum learning standards?*

A goal on learning is an indispensable part of a future global education monitoring framework, but focusing only on children who take part in learning assessments can be misleading if large numbers of children never make it to the grade where skills are tested. For this reason, the EFA Global Monitoring Report team has argued that all children should be taken into account, not just those who take part in learning assessments.

**Figure 1.8.4: Ensuring that the poorest girls complete lower secondary school remains a major challenge**
Percentage of poorest rural girls and richest urban boys who have ever been to school, and completed primary and lower secondary education, selected countries, circa 2000 and 2010




*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Surveys and Multiple Indicator Cluster Surveys.

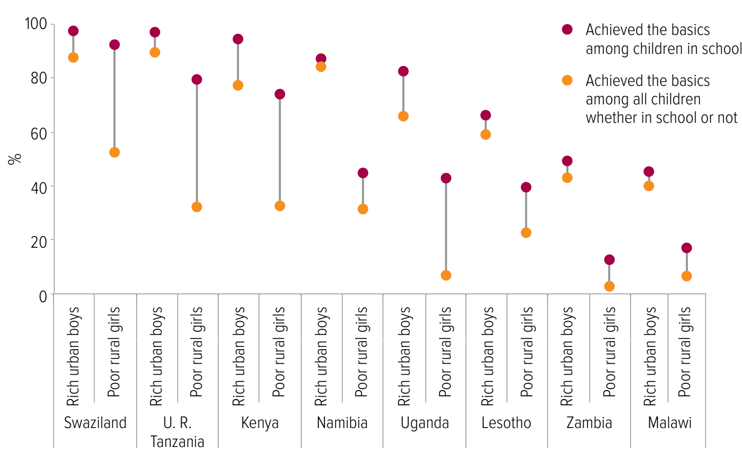
Data from SACMEQ show how much difference it can make if out-of-school children are included when assessing the numbers who are learning the basics (Figure 1.8.5). In the United Republic of Tanzania, the proportion of children in grade 6 who achieved a minimum standard in reading was 90% in 2007, ranging from 97% among the richest urban boys to 80% among the poorest rural girls. However, while 92% of the richest urban boys of grade 6 age had reached that grade, only 40% of the poorest rural girls had done so. If it is assumed that children who did not reach grade 6 could not have achieved the minimum standard, then the proportions among children of that cohort who learned the basics were 90% for the richest urban boys and 32% for the poorest rural girls.

**In Malawi, Uganda and Zambia, less than 10% of the poorest rural girls had learned the basics**

In Malawi, Uganda and Zambia, less than 10% of the poorest rural girls learned the basics. These striking statistics emphasize the need for monitoring learning of all children, a point reiterated in Part 3 of this Report.

**Figure 1.8.5: Assessing the number of children learning should not exclude those not tested**

Percentage of children who achieved minimum learning standard in reading, selected countries, southern and eastern Africa, 2007




*Note:* The percentage of children in school was calculated using Demographic and Health Survey data from the nearest year to 2007 and projected to 2007 using a linear trend based on earlier surveys. It is assumed that children not in school have not achieved the minimum learning standards.
*Source:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Surveys and SACMEQ data.

## Few countries monitor education inequality

To make sure inequality in education access and learning is overcome by 2030, country plans need to include specific targets to monitor progress in education participation and learning for individual population groups. A review of the education plans of 53 countries for this Report shows that very few currently do so, although there are some encouraging exceptions (Box 1.8.3).

In terms of education participation, gender is the dimension of potential disadvantage covered best in education plans. Even so, only 24 of the 53 plans included gender equality targets on participation in primary and lower secondary education. While most of these targets were restricted to enrolment ratios, some countries included gender targets in other aspects. For example, Ghana targeted gender parity in primary and lower secondary education enrolment, attendance and completion (Ghana Ministry of Education, 2010).

By contrast, only three countries specifically aimed to reduce disparity in access between rural and urban areas in primary and lower secondary education enrolment. Indonesia, for example, sought to reduce enrolment disparity between cities and rural areas by 36% over five years (Indonesia Ministry of National Education, 2005). Otherwise, countries aimed to increase enrolment in rural areas but without setting a target or timeline. Twelve countries included indicators on primary and lower secondary education enrolment by region, but only five of these specified a target for inequality.

Of the 53 plans, only 4 included indicators on participation by particular ethnic groups. But most of these plans focused only on increases in enrolment, and no countries set specific targets on how much to reduce disparity. Only Bangladesh, Kenya and the Lao People's Democratic Republic had an enrolment indicator that differentiated between poorer and richer children.

Measures of progress in learning are not common. Of the 53 countries surveyed, 31 included indicators on learning outcomes at either the primary or lower secondary level. Of these countries, only 19 specified a target.

Even fewer countries measured inequality in learning outcomes. Only eight countries do so at the primary level and eight at the lower secondary level, and in most cases monitoring is restricted to gender inequality. At the secondary level, six countries aim to reduce gender inequality in learning to some extent, although only Rwanda had a target and timeline: 90% of boys and girls should pass the national lower secondary school grade 3 examination by 2015, compared with a baseline of 56% for boys and 44% for girls in 2009.

**Box 1.8.3: Some countries take a close interest in education inequality**

Some countries stand out for including indicators that pay attention to disadvantaged groups:

- In Bangladesh, the monitoring framework of the Third Primary Education Development Programme proposed monitoring enrolment ratios across wealth quintiles, using household survey data. The monitoring framework also included an index that would allow planners to assess the relative performance of the approximately 500 subdistricts. A target is set of narrowing the gap in the value of this index between the 10 top performing and 10 lowest performing subdistricts by one-third over five years. The index includes subdistrict performance in the end of primary school examination.
- In the Lao People's Democratic Republic, the 2009–2015 Education Sector Development Framework includes indicators on enrolment at the national, provincial and district levels, by gender and location. In addition, minimum learning standards are to be defined for grades 3, 5 and 9 and then assessed through sample surveys of schools.
- In Namibia, the Education and Training Sector Improvement Programme set targets on how much children should have learned by the end of primary education. Linked to SACMEQ, the targets included reducing the regional dispersion of reading scores by 4%. In terms of access, the document expressed a commitment to monitor participation for orphans and other vulnerable children as well as children from marginalized groups. For example, a target was set for 80% of orphans and other vulnerable children from each region to be enrolled in primary and secondary education by the final year of the plan.

*Sources:* Bangladesh Ministry of Primary and Mass Education (2011b1); Lao PDR Ministry of Education (2009); Namibia Ministry of Education (2007).

Only four countries aim to monitor inequality in learning beyond gender at the primary or lower secondary level: Bangladesh, Belize, Namibia and Sri Lanka. In Belize, scores on primary and secondary examinations are to be disaggregated by district and urban/rural location (Belize Ministry of Education, 2011). In Sri Lanka, regional targets are set for achievement scores in mathematics and native language. On average, scores are targeted to increase by four percentage points over the duration of the plan, but with higher increases envisaged for lower performing regions (Sri Lanka Ministry of Education, 2006).

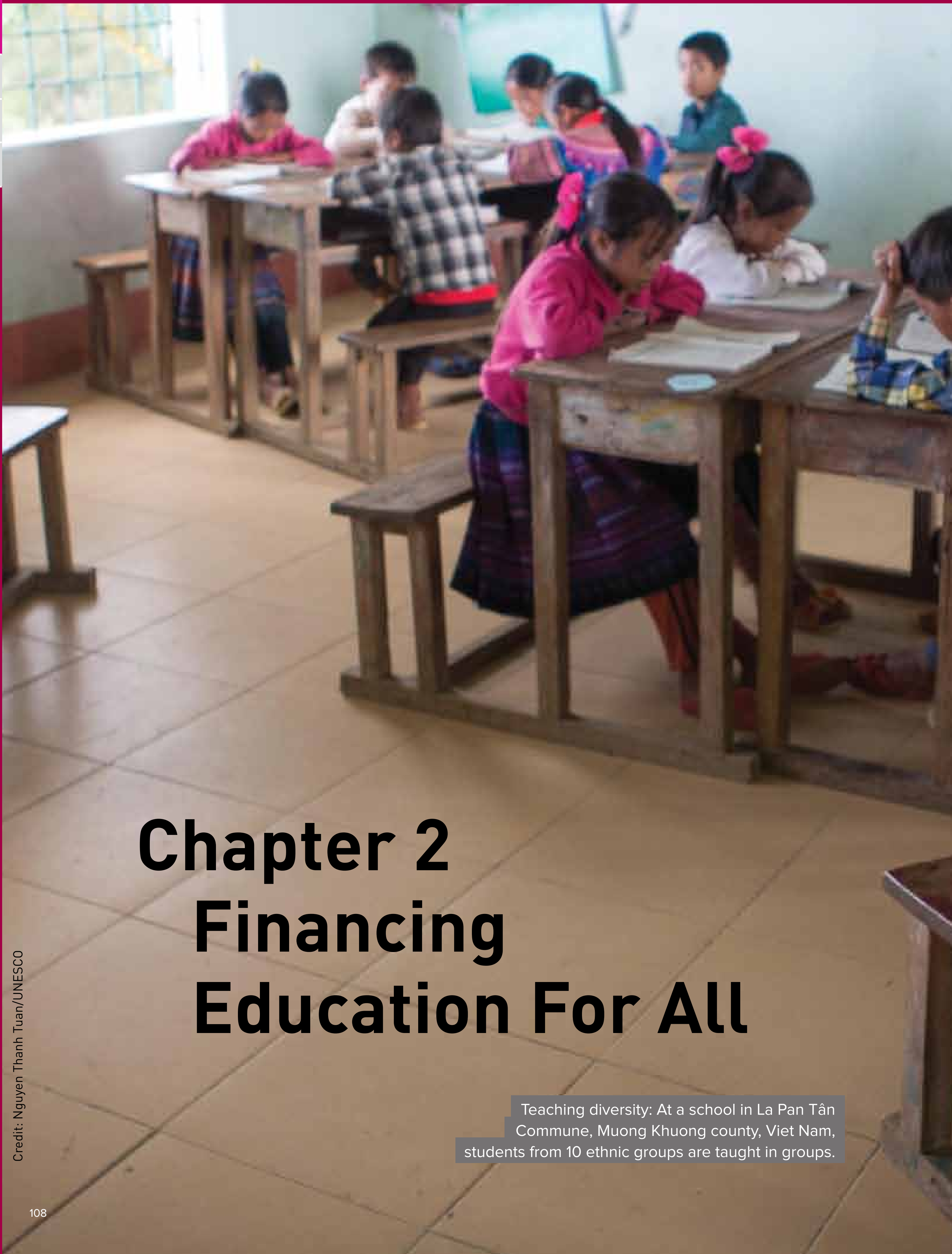
## Conclusion

**If post-2015 goals are to be achieved, regular independent monitoring is essential**

The failure over the past decade to assess progress in education goals by various population subgroups has concealed wide inequality. The invisibility of this inequality is further reflected in the lack of national targets in country plans for assessing progress in narrowing inequality gaps in access or learning.

Post-2015 goals need to include a commitment to make sure the most disadvantaged groups achieve benchmarks set for goals. Failure to do so could mean that measurement of progress continues to mask the fact that the advantaged benefit the most.

It is not enough just to set goals. If they are to be achieved, regular independent monitoring is essential so that progress can be tracked, policies that have facilitated progress can be identified and governments and the international community can be held to account for their promises. The *EFA Global Monitoring Report* has put education ahead of many other sectors in this regard. It is crucial for such independent monitoring to continue after 2015.



# Chapter 2
# Financing Education For All

Teaching diversity: At a school in La Pan Tân Commune, Muong Khuong county, Viet Nam, students from 10 ethnic groups are taught in groups.

Credit: Nguyen Thanh Tuan/UNESCO



Unless urgent action is taken to increase aid to education, the goal of ensuring that every child is in school and learning by 2015 will be seriously jeopardized. With little time left, closing the financing gap may seem impossible. But this chapter shows that the gap could be filled by raising more domestic revenue, devoting a fair share of existing and projected government resources to education, and sharpening the focus of aid spending.

# Trends in financing Education for All

## Highlights

- Many governments, particularly in poorer countries, have increased their commitment to education: 30 low and middle income countries increased their spending on education by more than 1 percentage point of GNP between 1999 and 2011.
- After 2015, a common financing target should be set for countries to allocate at least 6% of GNP on education. Of the 150 countries with data, only 41 had reached this level by 2011.
- Many countries have the opportunity to expand their tax base. For a group of 67 low and middle income countries, a modest increase in tax-raising efforts, and allocating 20% of the funds raised to education, could increase education spending by US$153 billion, or 72%, in 2015. This would raise the average share of GDP spent on education by these countries to 6%.
- Aid to basic education fell by 6% between 2010 and 2011, its first decrease since 2002.
- Low income countries were particularly hard hit by the reduction in aid to basic education: 19 of them experienced a cut, and 13 of those were in sub-Saharan Africa. The reduction was a result of 24 donors, including 9 of the 15 largest donors, reducing their spending.

**Table 2.1: Public spending on education, by region and income level, 1999 and 2011**

| | Public education spending | | | | | |
|---|---|---|---|---|---|---|
| | % of GNP | | % of government expenditure on education | | Per capita (primary education) (PPP constant 2010 US$) | |
| | 1999 | 2011 | 1999 | 2011 | 1999 | 2011 |
| **World** | **4.6** | **5.1** | **15.0** | **15.5** | **2 149** | **3 089** |
| Low income | 3.1 | 4.1 | 16.4 | 18.3 | 102 | 115 |
| Lower middle income | 4.6 | 5.1 | 15.9 | 16.9 | 356 | 545 |
| Upper middle income | 4.8 | 5.1 | 15.8 | 15.5 | 1 117 | 1 745 |
| High income | 5.3 | 5.6 | 13.3 | 13.2 | 4 752 | 6 721 |
| Sub-Saharan Africa | 4.0 | 5.0 | 17.1 | 18.7 | 345 | 468 |
| Arab States | 5.3 | 4.8 | 21.0 | 18.1 | 822 | 1 338 |
| Central Asia | 3.4 | 4.1 | 15.4 | 12.3 | … | … |
| East Asia and the Pacific | 3.9 | 4.4 | 15.0 | 16.6 | 2 216 | 3 245 |
| South and West Asia | 3.9 | 3.7 | 14.6 | 15.0 | 297 | 573 |
| Latin America and the Caribbean | 5.0 | 5.5 | 14.4 | 16.2 | 1 142 | 1 753 |
| Central and Eastern Europe | 4.8 | 5.2 | 12.4 | 12.2 | 1 813 | 3 846 |
| North America and Western Europe | 5.6 | 6.2 | 13.3 | 13.1 | 5 990 | 8 039 |

*Note:* World, regional and income values are means for countries with data in both 1999 and 2011, and may therefore not match what is reported in Statistical Table 9.
*Source:* EFA Global Monitoring Report team calculations (2013), based on UIS database.

**Donors have not met their commitment to ensure that no country would be prevented from achieving Education for All due to lack of resources**

Insufficient financing, particularly by aid donors, has been one of the main obstacles to achieving the Education for All goals. In 2010, the EFA Global Monitoring Report team calculated that it would take another US$16 billion per year in external financing to achieve good quality basic education for all in 46 low and lower middle income countries by 2015. Rather than increasing, however, aid has stagnated in recent years. Donors have not met the commitment they made at the World Education Forum in Dakar in 2000 to ensure that no country would be prevented from achieving Education for All due to lack of resources.

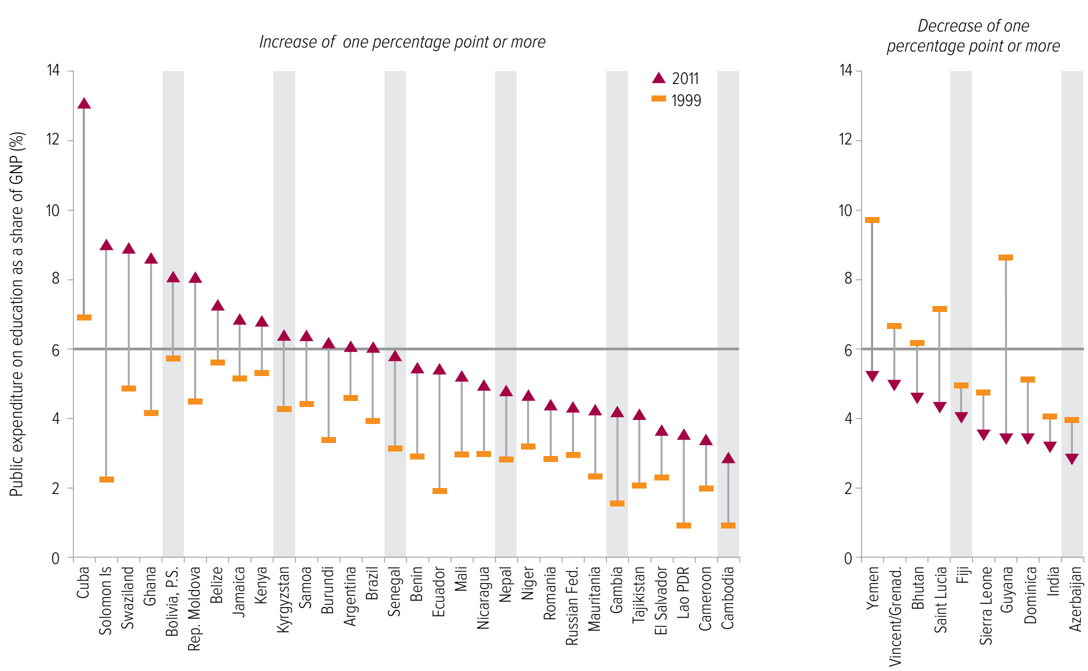
The EFA Global Monitoring Report team has calculated that, as a result of this stagnation, it would now take US$29 billion per year between 2012 and 2015, in addition to the amount that governments are spending, to achieve basic education for all. Taking into account the US$3 billion currently provided by donors to the 46 countries, this leaves an annual financing gap of US$26 billion (UNESCO, 2013c). Unfortunately, it seems that donors are more likely to reduce their aid than increase it in coming years. Unless urgent action is taken to change aid patterns, the goal of ensuring that every child is in school and learning by 2015 will be seriously jeopardized.

With little time left before 2015, closing the financing gap might seem impossible. But analysis in this chapter shows that the gap could be filled by raising more domestic revenue, devoting an adequate share of existing and projected government resources to education, and sharpening the focus of external assistance.

Post-2015 global education goals are expected to be more ambitious than the EFA goals, extending to lower secondary education. The shortfall in the financing necessary to achieve universal basic and lower secondary education by 2015 is estimated at US$38 billion annually, a gap that improvement in domestic spending alone cannot bridge.

Those who control global resources need to recommit themselves to international education goals, agree to be accountable for their financial commitments and deliver on them in a transparent way, so that they contribute to common education objectives. The post-2015 framework should therefore include explicit financing targets so that all funders can be held to account for their promises.

**Figure 2.1: Most low and middle income countries have increased education spending since 1999**
Public expenditure on education as percentage of GNP, low and middle income countries, 1999 and 2011




*Source:* Annex, Statistical Table 9.

# Many countries far from EFA need to spend more on education

Domestic spending on education has increased in recent years, particularly in low and lower middle income countries, partly because their economic growth has been improving. In richer countries, by contrast, the economic downturn has hit government budgets for education.

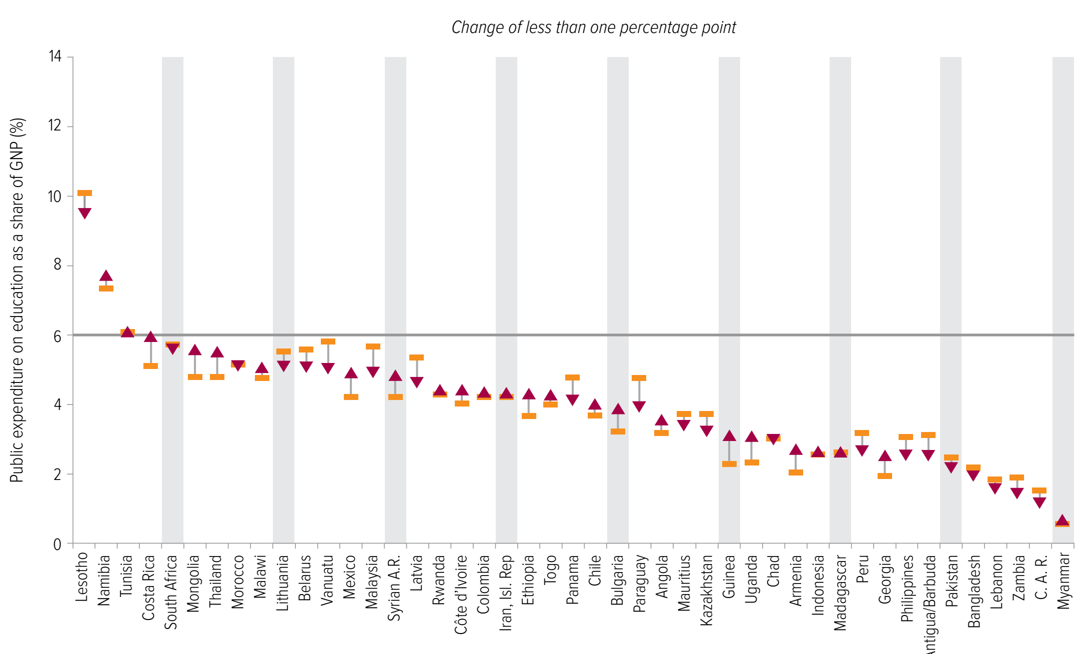
Many governments, particularly in poorer countries, have also increased their commitment to education. Globally, the amount devoted to education rose from 4.6% of gross national product (GNP) in 1999 to 5.1% in 2011 (Table 2.1). In low and middle income countries it rose faster: 30 of these countries increased their spending on education by one percentage point of GNP or more between 1999 and 2011 (Figure 2.1).

The Dakar Framework for Action did not establish financing targets for education. This has resulted in wide differences in government spending on education, so children's chances of being in school and learning continue to depend on where they happen to be born.

The failure to set a common financing target for the EFA goals should be addressed after 2015, with a specific goal that countries should allocate at least 6% of GNP to education. Some countries, such as the United Republic of Tanzania, already spend more than 6% of GNP on education, showing that such a target is feasible. Of the 150 countries with data, however, only 41 spent 6% or more of GNP on education in 2011. It is of particular concern that 10 low and middle income countries reduced their education spending as a percentage of GNP by one percentage point or more over the decade. India, for example, decreased its spending on education from 4.4% of GNP in 1999 to 3.3% in 2010, jeopardizing the huge progress it has made in getting more children into school, and its prospects for improving its poor quality of education.

**The United Republic of Tanzania spends more than 6% of GNP on education**

It is unacceptable that 25 countries, including Bangladesh, the Central African Republic, the Democratic Republic of the Congo and Pakistan – most of which are still a long way from achieving EFA – dedicate less than 3% of GNP to education. It is particularly worrying that some countries that were already spending a small proportion of GNP on education, such as Bangladesh, have



reduced their spending further. Pakistan, home to 10% of the world's out-of-school children, cut spending on education from 2.6% of GNP in 1999 to 2.3% in 2010.

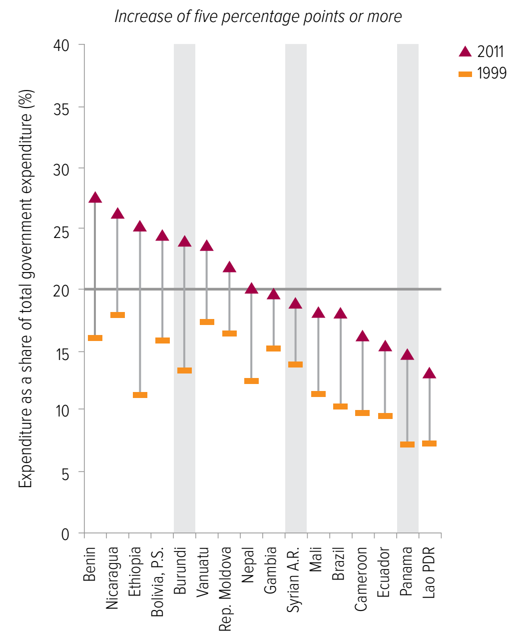
If governments are to meet a target of spending 6% of GNP on education, they must not only raise sufficient revenue from tax as a share of national income, but also allocate a sufficient proportion of domestic spending to education. It is widely accepted that countries should allocate at least 20% of their budget to education, and the EFA Global Monitoring Report team recommends making this an explicit target for the goals set after 2015. Yet only 15% of government expenditure globally was directed to education in 2011, a proportion little changed since 1999.

Low income countries increased the share of government spending on education the most over the decade, having raised their spending from around 16% in 1999 to 18% in 2011. It is lower middle and upper middle income countries that are furthest from the target of 20%: while they were at a similar level to low income countries in 1999, their spending has hardly changed since. Lower middle income countries spent 17% on education in 2011 and upper middle income countries 15%. The latter, in particular, should do far more to improve children's education chances through their own spending, leaving more external funds for the countries most in need.

**Only 25 countries spent more than 20% on education in 2011**

Of the 138 countries with data, only 25 spent more than 20% on education in 2011, including 17 low and lower middle income countries and 8 upper middle and high income countries. Nineteen of the 25 countries spending more than 20% on education in 2011 had comparable data for 1999; of these, 12 countries, including Nepal, started below the 20% benchmark in 1999 but increased their allocation. Indeed, some of the poorest countries started from a low base in 1999 but were able to allocate more than 20% of their budget to education in 2011 (Figure 2.2). These countries have seen fast progress in education in recent years. Participation in primary schooling in Ethiopia, for instance, has achieved impressive improvement: the net enrolment ratio, which was just 37% in 1999, had increased to 87% by 2011, far above the sub-Saharan African regional average. Burundi

**Figure 2.2: Only a few countries spend at least one-fifth of**
Public expenditure on education as percentage of government



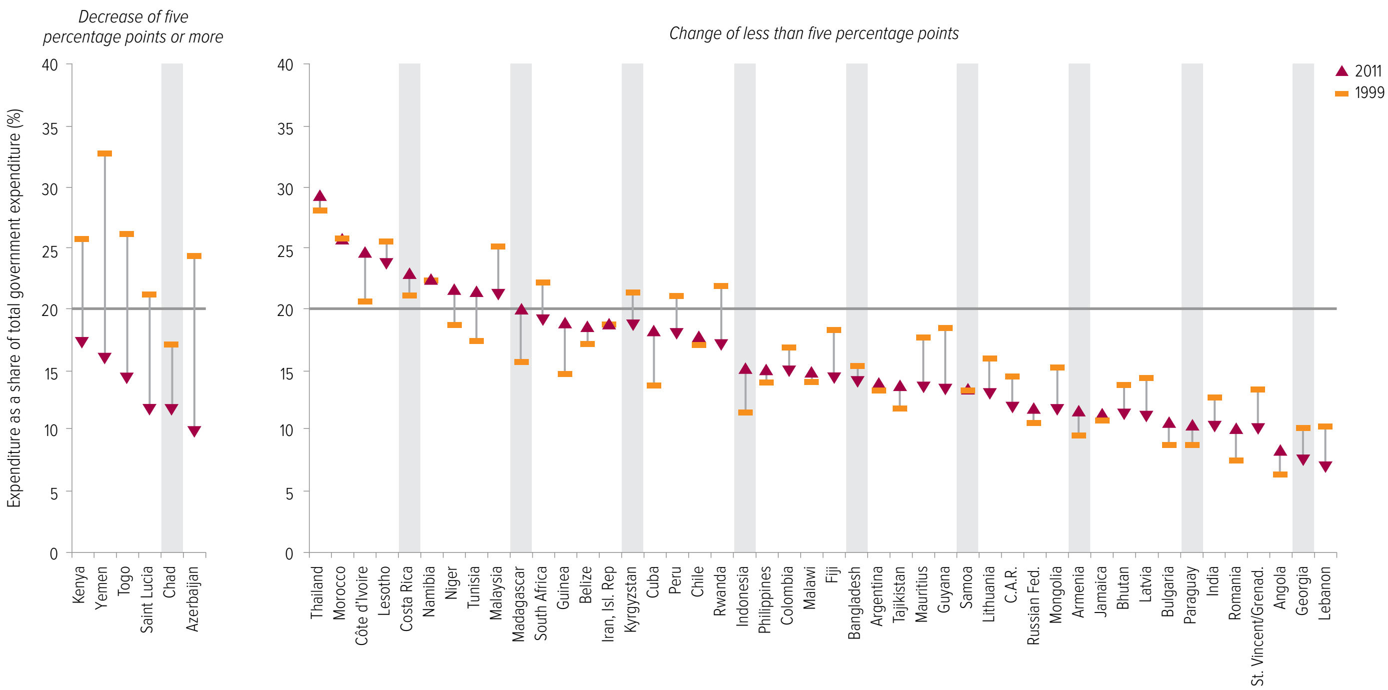

*Source:* Annex, Statistical Table 9.

had attained gender parity in primary school participation by 2011, in contrast to 1999 when it had only 8 girls in school for every 10 boys.

At the other extreme, it is worrying that the Democratic Republic of the Congo spent less than 9% of its budget on education in 2010, even though it is estimated to have well over 2.4 million children out of school (UNESCO, 2012). It is also of concern that at least six low and middle income countries decreased spending on education as a share of total government expenditure between 1999 and 2011 by five percentage points or more. While some of these countries had been allocating more than 20% to education, their spending has now fallen below this threshold. Chad's education spending, already below 20% in 1999, had fallen to 12% by 2011, contributing to it having some of the lowest education indicators in the world. In 2010, just 21% of children completed a cycle of primary schooling. India, which faces huge challenges in improving the quality of its education, spent 10% of its government budget on education in 2011, a reduction from 13% in 1999.

**their budget on education**

expenditure, low and middle income countries, 1999 and 2011



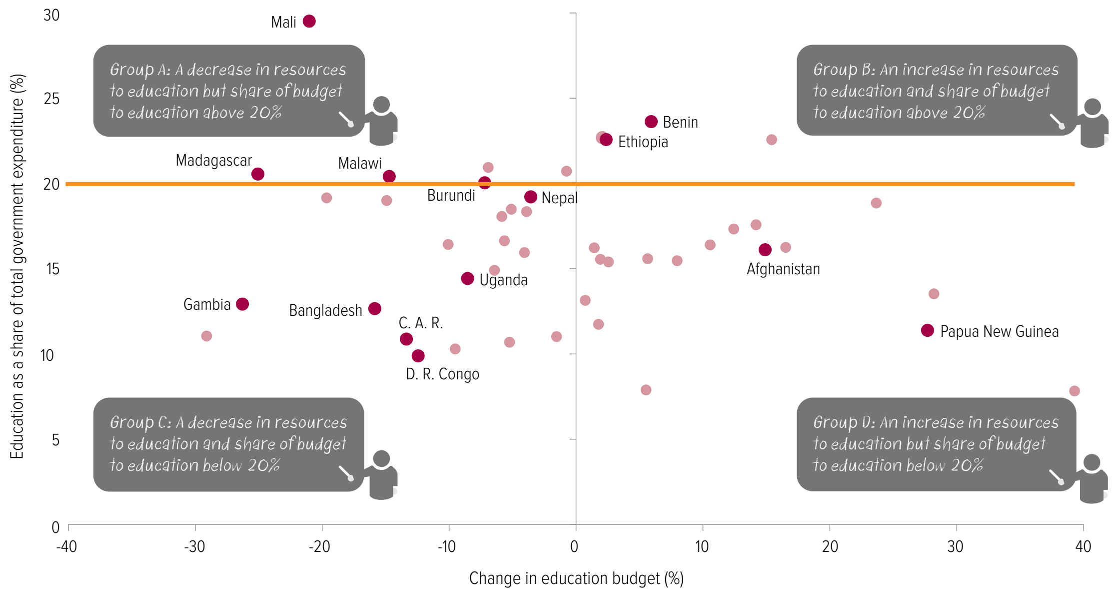

Countries' financial commitment to education can be gauged not only by looking at their education spending as a percentage of GNP and of the total government budget, but also by tracking the evolution of their education budgets in real terms – and here again there is cause for concern. Of the 49 countries with data, 25 planned to shrink their education budgets in real terms between 2011 and 2012 (Figure 2.3, Groups A and C). Of these 25 countries, 16 were in sub-Saharan Africa; they included Burundi, the Democratic Republic of the Congo, Madagascar, Malawi and Uganda – countries that are still a long way from the EFA goals. The Central African Republic's education spending was expected to decline by 13% between 2011 and 2012, resulting in it reaching just 11% as a proportion of the government budget in 2012.

Beyond sub-Saharan Africa, Bangladesh and Nepal were also expected to decrease their education budgets in real terms. Bangladesh planned to reduce its budget by 16% in 2012, resulting in its spending on education reaching only 13% of the government budget.

Some countries are resisting this negative trend, however. Afghanistan was expected to increase its spending on education by 15% in real terms between 2011 and 2012, although the share of the education budget as a proportion of government spending was likely to remain below 20% (Group D). Benin and Ethiopia are among just a handful of countries with data available where governments planned to increase resources to the education sector in real terms and where the sector is already well served by government spending (Group B).

**25 countries planned to shrink their education budgets between 2011 and 2012**

**Figure 2.3: Many countries that are far from reaching EFA goals cut their education budgets in 2012**
Percentage change in education budget, 2011 to 2012, and education as a share of government expenditure




*Source:* Development Finance International and Oxfam (2013).

**The benefits of Africa's growth have not yet been fairly distributed**

## Domestic resources can help bridge the education financing gap

Sustained economic growth in many of the world's poorest countries has increased the resources that governments can raise domestically to finance their education strategies. Many of the countries furthest from the EFA goals, however, do not sufficiently tap their tax base or devote an adequate share of their revenue to education.

Over the past decade, economic growth has enabled countries such as Ghana and India to cross the threshold from low income to lower middle income status, while countries such as Angola have moved to upper middle income status. Sub-Saharan Africa has achieved annual economic growth of around 5.4% in recent years, with the economies of poor countries such as Ethiopia growing by as much as 9.9% per year (World Bank, 2013c).

The benefits of this growth have not yet been fairly distributed, however. Nigeria's economy, for example, grew by at least 5% per year since 2003 but its net enrolment ratio has fallen from 61% in 1999 to 58% in 2010 (World Bank, 2013f). To guarantee their citizens' right to education and tap into education's power to transform lives, it is vital that countries put in place strong fiscal policies, backed by budget policy reforms to allocate an adequate share of public spending to education and promote equity in its distribution.

Strengthening tax systems is not only crucial for wider development but also an essential condition for achieving Education for All. This section shows that if governments in 67 low and middle income countries modestly increased their tax-raising efforts and devoted a fifth of their budget to education, they could raise an additional US$153 billion for education spending in 2015, amounting to a total of US$365 billion in 2015. This would increase the average share of GDP spent on education from 3% to 6% by 2015.

## Raising taxes and allocating an adequate share to education

It is estimated that countries need to raise 20% of their GDP in taxes to achieve the Millennium Development Goals (MDGs) (IMF et al., 2011). Few low and middle income countries manage to mobilize domestic resources on this scale, however. Among those that do, many do not allocate a sufficient proportion to education. Of the 67 countries for which data are available on tax revenue as a proportion of GDP as well as on the allocation of government revenue to education, only 7 reach the 20% threshold on both indicators (Figure 2.4). Namibia, which raises 24% of its GDP in taxes and allocates 22% of its government budget to education, shows that such goals are attainable.

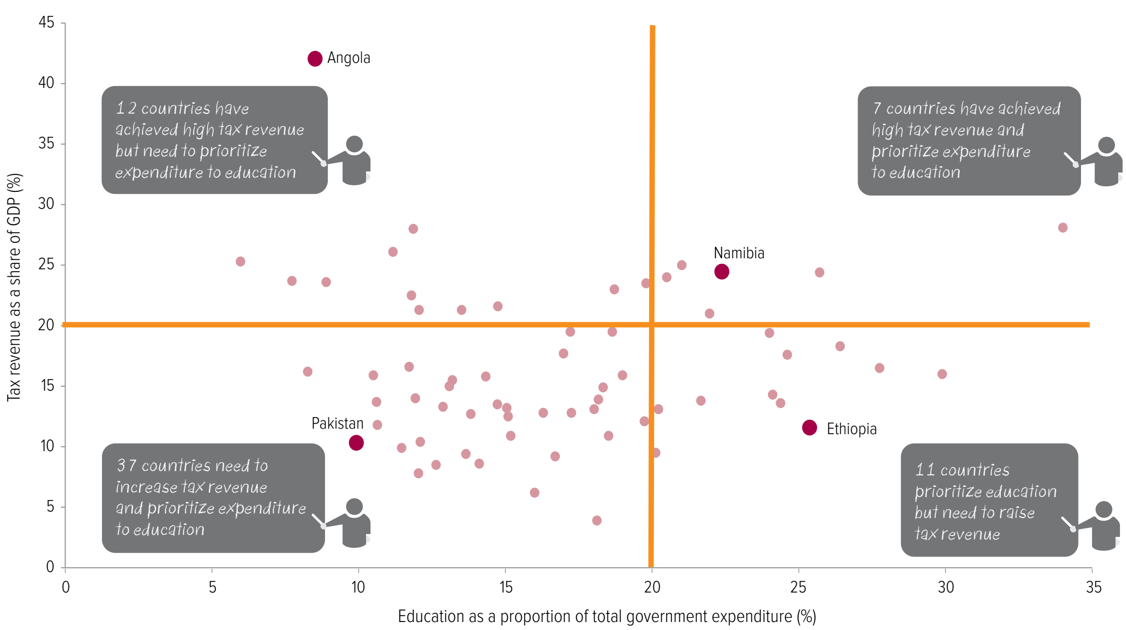
More commonly, countries' revenue from taxes is inadequate and education receives insufficient resources. Of the 67 countries for which data are available, 37 are below the 20% threshold on both indicators. In Pakistan, for example, tax revenue is just 10% of GDP and education receives only around 10% of government expenditure. If the government increased its tax revenue to 14% of GDP by 2015 and allocated one-fifth of this to education, it could raise sufficient funds to get all of Pakistan's children and adolescents into school.

Ethiopia is one of 11 among the 67 countries that has been successful in prioritizing education in its government budget but could do far more to maximize revenue from taxation, which would further increase the resources available for education. In 2011, the government received 12% of GDP from taxes, on average. If this proportion were to increase to 16% by 2015, and 25% continued to be allocated to education, the sector would receive 18% more resources, equivalent to US$435 million – enabling US$19 more to be spent per primary school age child.

By contrast, Angola has succeeded in converting much of its vast natural resource wealth into government revenue, with tax revenue representing 42% of GDP, but it only spends 9% of these funds on education, one of the lowest proportions in the world. Raising the share to 20% would increase resources to education almost two and a half times, or by US$7 billion. Assuming half of this is allocated to primary education, it could more than double

**Namibia raises 24% of its GDP in taxes and allocates 22% of its government budget to education**

**Figure 2.4: Countries need to both mobilize resources and prioritize education**
Tax revenue as percentage of GDP and education expenditure as percentage of total government expenditure, selected countries, 2011




*Sources:* IMF (2012); Annex, Statistical Table 9.

resources spent per primary school age child. In total, 12 of the 67 countries raise 20% or more of GDP through taxes but devote less than 20% of government spending to education.

Tax revenue as a share of GDP grew by 0.44% per year in low and lower middle income countries between 2002 and 2009 (World Bank, 2013f), but many countries need to make much faster progress. At this rate of growth, only 4 of the 48 countries currently raising less than 20% of GDP through taxes would reach the 20% threshold by 2015, and Pakistan – to give one example of a country far from the target – would not reach 20% until 2034.

**In some middle income countries there is far greater potential to mobilize domestic resources for education**

## Taxation as a pillar of development and education progress

A well-functioning taxation system enables governments to support the education system with domestic revenue instead of borrowing or relying on external finance. In high income countries in North America and Western Europe, tax revenue amounted to 27% of GDP in 2011. Most of these funds came from tax on income (13% of GDP), the majority of which is taxation of individuals.

By contrast, in sub-Saharan Africa, tax revenue accounted for 18% of GDP, with taxes on both individual and corporate income at 7% of GDP. In South and West Asia, the tax share was even lower: 12% of GDP, with revenue from individual and corporate income tax making up 4% of GDP. Unlike in North America and Western Europe, a quarter of the tax revenue in sub-Saharan Africa and South and West Asia comes from international trade and transactions (IMF, 2012).

While low and lower middle income countries are more dependent than richer countries on international and national corporate taxes for government revenue, they receive only 22% of the total annual corporation tax revenue collected globally. One study estimates that a percentage point increase could raise revenue by US$10 billion per year for these countries (Hearson, 2013).[1] If 20% of this additional income were allocated to education, an additional US$2 billion could be mobilized annually to help fill the financing gap.

1. This estimate assumes that two-thirds of corporation tax revenue comes from multinational corporations.

Many of the world's poorest countries cannot expect domestic taxes alone to provide the financing needed to meet the EFA goals in the near future. In some middle income countries, however, such as Egypt, India and the Philippines, there is far greater potential to mobilize domestic resources for education. India became the world's 10th largest economy in 2011, but tax revenue was equivalent to only 16% of GDP, and government expenditure per person was just US$409. By contrast, in Brazil – the world's sixth largest economy – tax revenue was equivalent to 24% of GDP and expenditure per person was US$4,952 (IMF, 2013).

This huge difference is a key reason Brazil has managed to go further in improving education quality and narrowing learning inequality. The levels of current spending on education as a share of total government expenditure in the two countries also reflect the greater priority that Brazil affords to the education sector. In 2011, government spending on education in Brazil was 18% of total government expenditure, with US$2,218 being spent on each primary school child. India devoted 10% of the government budget to education, with US$212 spent per primary school child. If India reduced tax exemptions, tackled tax evasion and diversified its tax base, it could greatly change this picture.

### *Limit tax exemptions*

While low and middle income countries as a group rely heavily on tax revenue from corporations, many of them forgo considerable revenue from businesses by granting too many tax exemptions. In much of sub-Saharan Africa, these exemptions can amount to the equivalent of 5% of GDP (UK House of Commons International Development Committee, 2012). In the United Republic of Tanzania, for instance, tax exemptions were equivalent to around 4% of GDP between 2005/06 and 2007/08; it is estimated that if these taxes had been collected, they could have provided 40% more resources for education (Uwazi, 2010).

Despite committing a large share of government expenditure to education, Ethiopia has one of the lowest tax/GDP ratios of all developing countries, reaching just 12% of GDP. This is largely due to generous tax exemptions, which amounted to about 4.2% of GDP in 2008/09 (Abay, 2010). If Ethiopia eliminated these exemptions and devoted 10% of the resulting revenue to basic education, then a country with 1.7 million out-of-school children would have an additional US$133 million available, enough to get approximately 1.4 million more children into school.

Countries in South Asia have some of the world's lowest tax/GDP ratios, mainly because large tax exemptions are granted to strong domestic lobby groups, such as landowners. In Pakistan, the tax/GDP ratio of 10% can be partly explained by the political influence of the agricultural lobby in tax rate negotiations. While the agricultural sector makes up 22.5% of Pakistan's GDP, its share in tax revenue is just 1.2% (Asad, 2012). Total tax exemptions amounted to the equivalent of 3% of GDP (Pasha, 2010).

In India, the majority of tax revenue forgone is due to exemptions from custom and excise duties, and, to a lesser extent, from corporate income tax. The revenue lost to exemptions came to the equivalent of 5.7% of GDP in 2012/13 (Bandyopadhyay, 2013); if 20% of this had been earmarked for education, the sector would have received an additional US$22.5 billion in 2013, increasing funding by almost 40% compared with the current education budget.

Losses occur not only when governments grant exemptions, but also when they sell natural resource concessions for less than their true value. One analysis concluded that the Democratic Republic of the Congo incurred losses of US$1.36 billion from its dealings with five mining companies over three years between 2010 and 2012 (African Progress Panel, 2013). This is the same amount as allocated to the education sector over two years between 2010 and 2011 (Development Finance International and Oxfam, 2013).

Some governments have started reviewing the terms and conditions of concession agreements. When Liberia reviewed 105 agreements signed between 2003 and 2006, it determined that 36 should be cancelled outright and 14 needed to be renegotiated (African Progress Panel, 2013).

### *Fight tax evasion*

For many of the world's poorest countries, tax evasion results in resources being used to build personal fortunes for the minority elite, rather than strong education systems for the benefit of the majority.

Some individuals and companies avoid taxes legally by moving money to tax havens. The Tax Justice Network estimates that between US$21 trillion and US$32 trillion is hidden by rich individuals in more than 80 tax havens. Taxing capital gains on this wealth at 30%, would generate revenue of between US$190 billion and US$280 billion a year (Tax Justice Network, 2012). If 20% of this revenue were allocated to education, it would add between US$38 billion and US$56 billion to funding for the sector. Tax avoidance by individuals is another important reason for the low amount spent on education in Pakistan. The Pakistan Federal Board of Revenue estimates that only 0.57% of Pakistanis – just 768,000 individuals – paid income tax in 2012 (Economist, 2012).

Tax avoidance practices of multinational corporations also raise grave concerns. Some companies shift profits to subsidiaries in countries with low or zero tax rates. While such tax havens are a legal way for companies to avoid paying taxes, illicit capital flight also occurs through corruption and illegal mispricing practices in some multinational companies. In 2010, developing countries lost an estimated US$859 billion through illegal practices (Kar and Freitas, 2012). This was 64 times the amount that countries received in aid for the education sector in 2011. It is estimated that African governments alone lost US$38 billion annually from such practices between 2008 and 2010. A further US$25 billion is thought to be lost annually through other tax practices related to corruption and criminal activities (African Progress Panel, 2013). If these illegal practices were halted and 20% of the resulting government income was spent on education, the sector would receive US$13 billion in additional resources each year.

**In Africa, if illegal tax practices were halted education could receive US$13 billion in additional resources each year**

Another stark example of the scale of the losses involved comes from the tax practices of SABMiller, a multinational drinks company, which is estimated to have deprived governments in Ghana, India, Mozambique, South Africa, the United Republic of Tanzania and Zambia of up to US$30 million in tax revenue (ActionAid, 2012).

Some governments are beginning to challenge the status quo. In Africa, 21 countries have agreed to a legal framework for the pursuit of tax avoiders and evaders across borders (Crotty, 2013). The government of India, which claims Vodafone India owes US$2.5 billion in tax (equivalent to around 4.5% of the country's education budget in 2011), recently issued a high profile tax demand against Vodafone and other multinational companies, including Shell and Nokia (Development Finance International and Oxfam, 2013; Heikkila, 2013).

**Governments could raise additional revenue by increasing the amount they receive in taxes from corporations**

### *Diversify the tax base*

Governments could raise considerable additional revenue by increasing the amount they receive in taxes from corporations, particularly in the natural resource extraction industry (UNESCO, 2012). Governments need to avoid dependence on a single source of income, however, and should plan for uncertainty. It has been estimated that a 1% increase in the share of natural resource rents in government revenue lowers the fiscal capacity of a country by 1.4% because there are fewer incentives to collect taxes from other sources (Besley and Persson, 2013).

In recent years sub-Saharan Africa has relied heavily on natural resources, which represented 46% of the region's tax revenue in 2008 (Bhushan and Samy, 2012). There has been little progress made in broadening the tax base beyond these resources (AfDB et al., 2010; DiJohn, 2010). After Chad began extracting oil in 2003, tax revenue from oil surged, but other tax income fell from 6.6% to 5.2% of GDP between 2003 and 2010 (AfDB et al., 2012). Meanwhile, education expenditure, at 3% of GDP in 2011, was unchanged since 1999.

In some countries, taxes tend to penalize the poorest. In India, direct taxes, such as personal income tax, make up 5.5% of GDP while indirect taxes, such as value-added tax – which are regressive, imposing a greater burden on the poor than on the rich – account for 9.3% (Hui, 2012). Moreover, just 3.3% of GDP was converted into education spending in 2010.

By contrast, Ecuador has initiated measures to expand its tax base and reduce dependence on rents: non-oil revenue as a share of government revenue rose from 70% in 2001–2005 to 74% in 2006–2010 (Ghosh, 2012). Viet Nam, which has one of the highest tax/GDP ratios in East Asia and the Pacific, has also broadened its tax base. Direct taxes, for instance, account for 8.2% of GDP, a share that has increased primarily because of government commitment to taxing corporate income from the oil sector and foreign-owned companies (McKinley and Kyrili, 2009).

The informal sector is another potential source of tax revenue. On some estimates, the informal sector accounts for 55% of GDP in sub-Saharan Africa, and revenue forgone by not taxing it can be equivalent to at least 35% of total tax revenue (Ncube, 2013; OECD, 2012b). As the majority of the poorest work in the informal sector, governments need to ensure that taxation of the sector is not regressive. However, the sector also includes prosperous small to medium-sized businesses that often pay little or no tax. Some governments are introducing measures to register them.

The Mozambique Revenue Authority introduced a highly simplified tax regime for micro- and small enterprises in 2009 that registered 40,000 taxpayers in a year (OECD, 2012b). In the United Republic of Tanzania, the government introduced a system to register eligible small and medium-sized businesses, and 41% of new companies were registered through this system by 2009 (Joshi et al., 2012). In 2004, the Malawi Revenue Authority encouraged tax compliance by giving compliant businesses an annual certificate documenting their tax status, which banks started to use in loan transactions (OECD, 2008). Such measures can generate substantial revenue, some of which could be used for education.

### *External assistance is needed to strengthen tax systems*

While domestic political will needs to be the main force behind tax reform and increased allocations to education, donors can play an important complementary role. Between 2002 and 2011, just 1% of total aid was directed to public financial management and less than 0.1% of total aid supported tax programmes. Yet by one estimate, every US$1 of aid to strengthen tax regimes could generate up to US$350 in tax revenue (OECD-DAC, 2013; African Tax Administration Forum and OECD, 2013).

The foundations of a long-term tax development strategy need to be put in place. It took European economies a century to increase their tax revenue from 12% to 46% of GDP by developing new taxes (Besley and Persson, 2013). There are signs of a similar shift in poorer countries. The Rwanda Revenue Authority, which has received long-term support from the UK Department for International Development (DFID), increased the share of tax revenue as a proportion of GDP from 10% in 1998 to 13% in 2011 (IMF, 2012). The additional resources mobilized through better tax collection is equivalent to recovering the full value of DFID's 10-year support programme every three weeks (Rwanda Revenue Authority, 2012). This helped raise expenditure per primary school child from US$72 in 1999 to US$81 in 2011. Similarly, the Norwegian Tax Administration assists the tax authority in Mozambique in auditing international oil companies, and the German Agency for International Cooperation has helped tax authorities in Ghana, for example, build capacity and introduce legislation on transfer pricing (Fontana and Hansen-Shino, 2012).

Part of the problem governments face in raising tax revenue is lack of transparency by companies. The support of international partners can be valuable in changing this. In June 2013, France, the United Kingdom and the United States said they intended to implement the Extractive Industries Transparency Initiative, the global standard for transparency in natural resource revenue, with which 23 countries are classed as compliant and 16 countries have been accepted as candidates. At the 2013 G8 summit, partnerships were announced with nine countries to help them support extractive industry governance and increase tax collection capacity (G8, 2013).

Similarly, while poorer countries need to strengthen their institutions to prevent tax evasion, the problem cannot be addressed without the support of the international community.

The Africa Progress Panel has called for a stronger multilateral tax transparency regime to tackle unethical aspects of tax avoidance (Africa Progress Panel, 2013). In addition, governments in high income countries can put pressure on corporations registered in their countries. For example, they can require them to publish a full list of their subsidiaries and the revenue, profit and taxes paid in all jurisdictions.

The United States made a start by introducing taxation transparency requirements for 1,100 oil, gas and mineral companies in 2010. The Cardin-Lugar amendment to the Dodd-Frank law, which took effect in September 2013, requires companies listed on the stock market and their subsidiaries to make information public on profit accrued and taxes paid (Jackson, 2013). If the European Union, as expected, also starts requiring oil, gas, mining and logging companies to declare payments to governments, such legislation will cover up to 90% of the world's international extractive industries (Publish What You Pay, 2013).

**67 low and middle income countries could increase education resources by US$153 billion in 2015 through tax reforms**

## Estimating the potential increase in domestic resources for education

Increasing tax revenue and allocating an adequate share to education could raise considerable extra resources for the sector in a short time. The EFA Global Monitoring Report team estimates that 67 low and middle income countries could increase education resources by US$153 billion, or 72%, in 2015 through reforms to raise tax/GDP ratios and public expenditure on education (Table 2.2).

These additions to domestic resources could meet 56% of the US$26 billion average annual financing gap in basic education for 46 low and lower middle income countries, or 54% of the US$38 billion gap in basic and lower secondary education.[2]

Overall, the necessary reforms would more than double resources available for education in 13 countries. The increases would be particularly important for countries that now spend very little on education per school age child, allowing

2. This assumes that 50% of the funds are allocated to basic education and 20% to lower secondary education.

**Table 2.2: Countries can dig deeper to fund education from domestic resources**

| | | Current situation (2011) | | | Potential situation (2015) | | | Unit cost (2015) | |
|---|---|---|---|---|---|---|---|---|---|
| | | Education as a share of GDP | Education as a share of total government spending | Tax/GDP ratio | Education as a share of GDP | Tax/GDP ratio in 2015 | Total potential additional funding in 2015 | Expenditure per primary school child (stagnant) | Expenditure per primary school child (tax mobilization and education prioritized) |
| | | % | % | % | % | % | US$ millions | US$ | US$ |
| Current tax/GDP ratio of less than 10% | Afghanistan | 3.5 | 16.7 | 9.2 | 5.6 | 14.2 | 500 | 67 | 161 |
| | Bangladesh | 2.2 | 14.1 | 8.6 | 4.2 | 13.6 | 3 198 | 101 | 216 |
| | Bhutan | 4.7 | 11.5 | 9.9 | 7.9 | 14.9 | 94 | 255 | 924 |
| | Central African Republic | 1.2 | 12.0 | 7.8 | 3.6 | 12.8 | 66 | 44 | 95 |
| | Eritrea | 2.1 | 7.5 | 8.4 | 6.7 | 13.4 | 201 | 57 | 191 |
| | Guinea-Bissau | 2.6 | 12.6 | 8.5 | 5.1 | 13.5 | 28 | 48 | 101 |
| | Madagascar | 2.8 | 20.1 | 9.5 | 3.8 | 14.5 | 119 | 62 | 85 |
| | Myanmar | 0.8 | 18.1 | 3.9 | 2.3 | 8.9 | 1 000 | 389 | 513 |
| | Sierra Leone | 3.6 | 13.7 | 9.4 | 5.6 | 14.4 | 117 | 60 | 118 |
| | Yemen | 5.2 | 16.0 | 6.2 | 7.4 | 11.2 | 998 | 251 | 362 |
| Current tax/GDP ratio of ≥10% - <12.5% | Cambodia | 2.6 | 12.1 | 10.4 | 4.9 | 14.4 | 438 | 82 | 232 |
| | Ethiopia | 4.7 | 25.4 | 11.5 | 5.5 | 15.5 | 435 | 106 | 125 |
| | Gambia | 3.9 | 19.7 | 12.1 | 4.8 | 16.1 | 10 | 106 | 123 |
| | Guatemala | 2.8 | 18.5 | 10.9 | 3.8 | 14.9 | 605 | 365 | 500 |
| | Haiti | 3.6 | 10.6 | 11.8 | 6.9 | 15.8 | 336 | 65 | 213 |
| | Indonesia | 3.0 | 15.2 | 10.9 | 4.7 | 14.9 | 19 506 | 526 | 1 049 |
| | Pakistan | 2.4 | 9.9 | 10.2 | 5.2 | 14.2 | 7 241 | 62 | 262 |
| Current tax/GDP ratio of ≥12.5% - <15% | Burkina Faso | 4.0 | 18.0 | 13.1 | 5.1 | 16.1 | 157 | 125 | 154 |
| | Burundi | 6.1 | 24.1 | 14.3 | 6.7 | 17.3 | 19 | 86 | 94 |
| | Cameroon | 3.2 | 16.3 | 12.8 | 4.6 | 15.8 | 461 | 115 | 241 |
| | Egypt | 3.8 | 11.9 | 14.0 | 7.1 | 17.0 | 9 592 | 520 | 948 |
| | Ghana | 8.2 | 24.4 | 13.6 | 8.8 | 16.6 | 317 | 365 | 567 |
| | Mali | 4.8 | 18.2 | 13.9 | 5.9 | 16.9 | 139 | 81 | 124 |
| | Mauritania | 3.9 | 14.7 | 13.5 | 5.8 | 16.5 | 114 | 187 | 292 |
| | Nepal | 4.7 | 20.2 | 13.1 | 5.3 | 16.1 | 144 | 160 | 185 |
| | Niger | 4.5 | 21.7 | 13.8 | 5.1 | 16.8 | 50 | 92 | 100 |
| | Paraguay | 4.1 | 10.6 | 13.7 | 6.8 | 16.7 | 968 | 384 | 1 037 |
| | Philippines | 2.7 | 15.0 | 13.2 | 4.2 | 16.2 | 5 361 | 442 | 679 |
| | Rwanda | 4.8 | 17.2 | 12.8 | 6.0 | 15.8 | 116 | 78 | 137 |
| | Sri Lanka | 2.0 | 12.9 | 13.3 | 3.9 | 16.3 | 1 509 | 292 | 970 |
| | Tajikistan | 3.9 | 13.8 | 12.7 | 6.1 | 15.7 | 219 | 262 | 424 |
| | Uganda | 3.3 | 15.1 | 12.5 | 4.9 | 15.5 | 387 | 51 | 77 |
| | United Republic of Tanzania | 6.2 | 18.3 | 14.9 | 7.2 | 17.9 | 383 | 84 | 109 |
| Current tax/GDP ratio of ≥15% - <20% | Armenia | 3.2 | 11.7 | 16.6 | 5.8 | 18.6 | 269 | 1 232 | 2 266 |
| | Benin | 5.3 | 27.8 | 16.5 | 5.7 | 18.5 | 37 | 169 | 182 |
| | Côte d'Ivoire | 4.6 | 24.6 | 17.6 | 5.0 | 19.6 | 137 | 285 | 322 |
| | Guinea | 3.1 | 19.0 | 15.9 | 3.7 | 17.9 | 44 | 66 | 107 |
| | Honduras | 6.5 | 29.9 | 16.0 | 6.9 | 18.0 | 82 | 484 | 776 |
| | India | 3.3 | 10.5 | 15.9 | 6.4 | 17.9 | 70 529 | 157 | 558 |
| | Kenya | 6.7 | 17.2 | 19.5 | 7.9 | 21.5 | 710 | 181 | 238 |
| | Kyrgyzstan | 5.8 | 18.6 | 19.5 | 6.7 | 21.5 | 76 | 662 | 751 |
| | Lao People's Democratic Republic | 3.3 | 13.2 | 15.5 | 5.2 | 17.5 | 229 | 161 | 377 |
| | Mozambique | 6.4 | 17.0 | 17.7 | 7.9 | 19.7 | 274 | 103 | 137 |
| | Nicaragua | 4.7 | 26.4 | 18.3 | 5.1 | 20.3 | 50 | 526 | 639 |
| | Senegal | 5.6 | 24.0 | 19.4 | 6.0 | 21.4 | 66 | 188 | 249 |
| | Togo | 4.6 | 14.3 | 15.8 | 6.6 | 17.8 | 94 | 80 | 128 |
| | Zambia | 1.3 | 13.1 | 15.0 | 3.5 | 17.0 | 622 | 155 | 271 |
| | Zimbabwe | 2.5 | 8.3 | 16.2 | 6.8 | 18.2 | 588 | 86 | 221 |
| Current tax/GDP ratio of more than 20% | Angola | 3.5 | 8.5 | 41.9 | 8.2 | 43.7 | 6 819 | 588 | 1 534 |
| | Belize | 6.6 | 18.7 | 23.0 | 7.3 | 24.8 | 13 | 701 | 1 215 |
| | Chad | 2.9 | 11.8 | 22.5 | 5.0 | 24.3 | 317 | 94 | 171 |
| | Democratic Republic of the Congo | 2.5 | 8.9 | 23.6 | 6.5 | 25.4 | 885 | 18 | 59 |
| | Georgia | 2.7 | 7.7 | 23.7 | 6.6 | 25.5 | 734 | 580 | 2 082 |
| | Guyana | 3.6 | 13.5 | 21.3 | 5.9 | 23.1 | 83 | 544 | 1 415 |
| | Lesotho | 13.0 | 23.7 | 60.1 | 13.3 | 61.9 | 11 | 340 | 492 |
| | Liberia | 3.3 | 12.1 | 21.3 | 6.1 | 23.1 | 69 | 60 | 111 |
| | Malawi | 5.4 | 14.7 | 21.6 | 7.7 | 23.4 | 106 | 30 | 58 |
| | Mongolia | 5.5 | 11.9 | 28.0 | 8.9 | 29.8 | 539 | 953 | 2 391 |
| | Morocco | 5.4 | 25.7 | 24.4 | 5.7 | 26.2 | 431 | 1 428 | 1 490 |
| | Namibia | 8.3 | 22.4 | 24.4 | 8.7 | 26.2 | 50 | 1 093 | 1 406 |
| | Nigeria | 1.5 | 6.0 | 25.3 | 5.6 | 27.1 | 13 090 | 87 | 330 |
| | Papua New Guinea | 3.4 | 11.2 | 26.1 | 5.9 | 27.9 | 633 | 217 | 590 |
| | Republic of Moldova | 8.6 | 22.0 | 21.0 | 8.9 | 22.8 | 32 | 991 | 2 691 |
| | Solomon Islands | 7.3 | 34.0 | 28.1 | 7.6 | 29.9 | 4 | 838 | 1 208 |
| | Swaziland | 8.2 | 21.0 | 25.0 | 8.6 | 26.8 | 14 | 809 | 863 |
| | Uzbekistan | 6.4 | 20.5 | 24.0 | 6.8 | 25.8 | 234 | 1 170 | 1 228 |
| | Viet Nam | 6.6 | 19.8 | 23.5 | 7.0 | 25.3 | 750 | 446 | 732 |
| | **Total for all 67 countries** | **3.4** | **13.1** | **15.7** | **5.8** | **18.4** | **153 451** | **209** | **466** |

*Notes:* Countries were ranked in five groups according to their initial tax/GDP ratio. At one end, countries where the ratio was already at least 20% were projected to increase their effort by 0.44 percentage point per year and to allocate 20% of the budget to education if they were not already doing so. At the opposite end, countries with a tax/GDP ratio of less than 10% were projected to increase the ratio by 1.25 percentage points annually and to allocate 20% of the budget to education if they were not already doing so.

The calculations assume five different rates at which a country's tax/GDP ratio can grow, depending on its starting point. Countries starting at a tax/GDP ratio of (a) <10% should aim to increase the tax/GDP ratio by 1.25 percentage points per year, (b) ≥10% to <12.5% should aim to increase the ratio by 1 percentage point per year, (c) ≥12.5% to <15% should aim to increase it by 0.75 percentage point per year, (d) ≥15% to <20% should aim for an increase by 0.5 percentage point per year and (e) ≥20% should aim to increase the ratio by 0.44 percentage point per year.

*Sources:* EFA Global Monitoring Report team calculations (2013), based on UIS database; Development Finance International and Oxfam (2013); IMF (2012, 2013).

education quality to improve. Across the 67 countries, spending per primary school age child would increase from US$209 to US$466 in 2015. Of these countries which are low income, the amount spent per primary school age child would increase from US$102 to US$158. Bangladesh, for example, could allocate an extra US$3.2 billion to education in 2015, increasing the amount available to spend on each school age child from US$101 to US$216.[3]

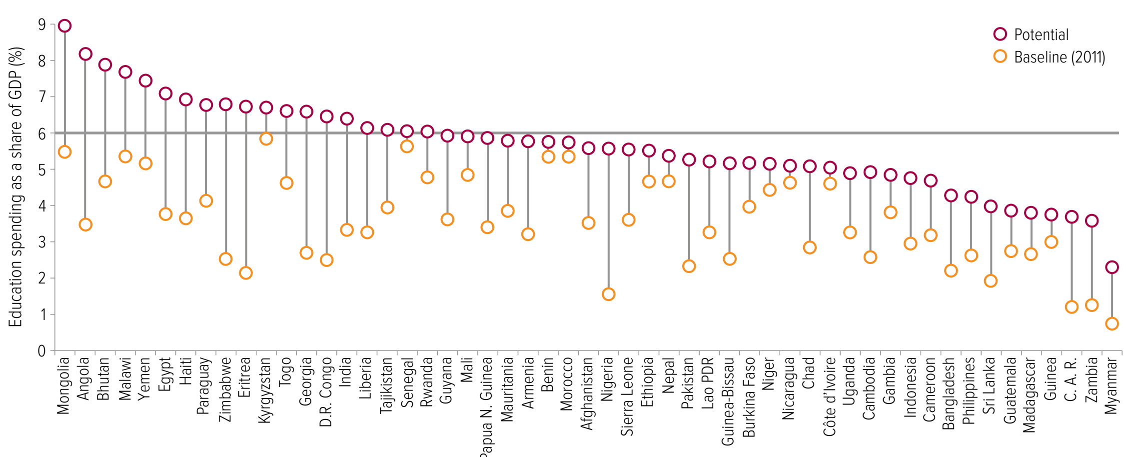
Pakistan, where 10% of the world's out-of-school children are concentrated, spends 3% of GDP on the military (World Bank, 2013f). This is more than the amount it spends on education. If it maximized tax revenue – especially by reversing the huge exemptions it grants – and spent 20% of its budget on education, the government could add US$7.2 billion to its education budget in 2015, increasing spending per school age child from US$62 to US$262.[4]

The Central African Republic has the potential to raise an additional US$66 million in 2015 if it increased the tax/GDP ratio from its current level of 8% to 13% and at the same time prioritized the education sector within government expenditure from the current level of 12% to 20%. This could potentially more than double expenditure per primary school child, from US$44 to US$95. Such increases may seem large but are not without precedent. Through effective tax mobilization policies, Ecuador, for instance, tripled its education expenditure from US$225 million in 2003–2006 to US$941 million in 2007–2010 (Ghosh, 2012).

Calculations by the EFA Global Monitoring Report team show that it is possible for countries to spend 6% of GDP on education. Among the 67 countries analysed, 14 have already attained this target. Of the 53 countries that have not yet reached the target, 19 would be able to achieve it by expanding and diversifying the tax base and prioritizing education spending by 2015 (Figure 2.5). Such efforts would go a long way towards ensuring that children are in school and learning by 2015, and would provide a solid base for funding more ambitious goals after 2015.

**Tax reforms could increase spending per primary school age child from US$209 to US$466 in 2015**

**Figure 2.5: Modest increases in tax effort and prioritizing education spending could significantly increase resources**
Expenditure on education as percentage of GDP in 2015 if the tax/GDP ratio grew and the budget share of education increased




*Sources:* EFA Global Monitoring Report team calculations (2013), based on UIS database; Development Finance International and Oxfam (2013); IMF (2012, 2013).

3. At current levels of prioritization of education and tax effort, and with future levels of GDP and government expenditure as projected by the IMF, Bangladesh would spend US$3.9 billion on education in 2015. If it increased the share of education in its budget from 14% to 20% in 2015, the sector would have an extra US$1.6 billion at its disposal. If in addition it increased the share of tax to GDP by 1.25% per year, that ratio would rise from 8.6% in 2011 to 13.6% in 2015, making an extra US$3.2 billion, in all, available for education.

4. Expenditure on education as a share of the government budget was 10% in 2011. If the share rose to 20%, and tax as a share of GDP rose from its current rate of around 10% to 14%, the government could raise an additional US$7.2 billion for education.

## Targeting the marginalized through education expenditure

**In low income sub-Saharan African countries, 43% of public spending on education is received by the most educated 10%**

To achieve Education for All, it is necessary not only to increase domestic resources for education but also to ensure that they are spent on improving education opportunities for the most marginalized. The children and young people who are hardest to reach – such as the poor, those who live in remote locations, members of ethnic and linguistic minorities, and those with disabilities – are often the last to benefit from education spending, so special efforts are needed to make sure these funds reach them. In addition, the cost of teaching these students is likely to be far higher than the average cost per student because of the expense involved in mitigating the disadvantages they face. This means redistributing domestic resources to those most in need.

More often, however, resources are skewed towards the most privileged. In low income sub-Saharan African countries, 43% of public spending on education is received by the most educated 10%; in middle income countries, the top 10% receive 25% of public spending on education (Majgaard and Mingat, 2012). In Malawi, where the level of public spending per primary school child is among the world's lowest, 73% of public resources allocated to the education sector benefit the most educated 10% (World Bank, 2010).

Similarly, public expenditure on education is often skewed towards urban areas, even where the majority of the school age population resides in rural areas – which are also often home to the poorest households. In the United Republic of Tanzania, for instance, nearly three-quarters of the population resides in rural areas, yet only 47% of public education resources were allocated to these areas in 2009 (U. R. Tanzania Ministry of Education and Vocational Training, 2011).

Recognizing the need to target the most disadvantaged, many countries in Latin America have established social protection programmes that transfer cash to poor households on the condition that their children attended school (UNESCO, 2010). Their success has led to similar programmes in other countries, including Malawi and the Philippines. Malawi's Social Cash Transfer Scheme provides US$14 per month to more than 26,000 households to fight poverty and hunger and help families send their children to school (UN, 2013). In the Philippines, Pantawid Pamilya, a conditional cash transfer programme, reaches 7.5 million children nationwide in an effort to keep them in school and in good health (World Bank, 2012a).

Although many social protection programmes address constraints that poor households face in getting children into school, they generally do not include strategies to improve the quality of education once children are enrolled. In addition, such programmes often represent a tiny fraction of the education budget. More effective targeting of education spending is vital for improving the education chances of the disadvantaged. For example, a cash transfer programme targeting orphans and vulnerable children in Kenya costs the equivalent of 0.12% of GDP (Bryant, 2009). By contrast, the government spent 6.7% of GDP on education in 2010.

To shift education spending in favour of the marginalized, many governments have introduced funding formulas that allocate more resources to parts of the country or groups of schools that need greater support to overcome educational deprivation and inequality. While some of these efforts have produced positive results, they have not always improved learning as much as desired, in some cases because redistribution has been too limited and in others because programmes have not focused sufficiently on improving education quality. Some programmes, moreover, have been hampered by administrative challenges and by weak capacity on the part of the lower tiers of government that are expected to implement the reforms.

### *How do countries redistribute education expenditure to address inequality?*

Countries adopt different methods of redistributing resources to disadvantaged areas and schools, depending on their ability to identify and target those most in need. India has aimed to redistribute resources to the poorer states with the worst education outcomes. Under the Sarva Shiksha Abhiyan (Education for All) programme, districts were identified in 2006 to receive additional funding on the basis of out-of-school population, gender disparity, infrastructure conditions and minority populations (Jhingran and Sankar, 2009).

Sri Lanka, lacking sufficiently disaggregated information on poverty, took a different approach to target disadvantaged schools: in its Education Quality Input system, it skews non-salary expenditure in favour of smaller schools. These schools have higher fixed operating costs and are located in rural areas, where poverty tends to be higher, making it hard for them to mobilize their own resources. School with fewer than 100 students receive about 53% of their funding from this programme, compared with 9% for schools with more than 2,000 students, which tend to be in richer, more urban areas (Arunatilake and Jayawardena, 2013).

South Africa's redistribution reforms have aimed to reverse the legacy of the apartheid schooling system. In 2007, the National Norms and Standards for School Funding were implemented and a 'no fee schools' policy was introduced. Catchment areas were ranked according to income, unemployment and education level, and grouped into quintiles. Schools in the bottom quintile were designated as no fee schools. The policy had been extended to the bottom three quintiles by 2011. Schools in the three lowest quintiles are eligible for an allocation to cover non-salary expenditure to offset the loss of fee income. In 2009, schools in the poorest quintile received a per student allocation that was six times higher than the allocation to schools in the richest quintile (Sayed and Motala, 2012).

Indonesia's centrally funded capitation grant to schools, the School Operational Assistance programme, was found to discriminate against schools whose costs were high because of their location or the characteristics of the population they served. To address this, about half of district governments provide supplementary local grants. The majority of school funds, excluding salaries, still come from central grants, with 15% coming from local grants. In a few pilot cases, districts have attempted to favour schools that are potentially in more need of support, and there are indications that these schools have used the additional district funds to provide extra student support, along with teaching and learning materials (World Bank, 2013e).

Brazil's redistribution of funds to poorer and more marginalized parts of the country has contributed to improvements in school attendance, pupil/teacher ratios and learning outcomes (Box 2.1).

**Sri Lanka skews non-salary expenditure in favour of smaller schools in rural areas**

**Box 2.1: Brazil's reforms reduce regional education inequality**

Greater equity in national spending has been at the heart of Brazil's reforms to tackle widespread education inequality between states. In the poorer northern states, income is less than half the level in the richer southern states, so tax revenue and spending per pupil are lower.

In the mid-1990s, the government introduced the Fund for Primary Education Administration and Development for the Enhancement of Teacher Status (FUNDEF), which guaranteed a certain minimum spending level per pupil by complementing state spending with federal allocations. Schools in rural areas were generally favoured over urban schools, with greater weight given to highly marginalized indigenous groups. Of the funding distributed, 60% was earmarked for teacher salaries and 40% for school operations. The salary component allowed teachers in poor northern states to upgrade qualifications, so that by 2002 almost all teachers had acquired minimum required training, and it ensured an influx of fully qualified teachers in those areas, allowing for an increase of around one-fifth in the teaching workforce between 1997 and 2002.

In 2006, FUNDEF was replaced by the Fund for the Development of Basic Education and Appreciation of the Teaching Profession (FUNDEB), also with the aim of establishing a minimum allocation per student. Average school attendance among children from the poorest 20% of families, which had been four years in the mid-1990s, had risen to eight years.

FUNDEF led to rapid and substantial improvement in northern Brazil. Between 1997 and 2002, average enrolment increased by 61% in the North-East region and 32% in the North region. Mathematics scores for grade 4 students have increased in northern states since 2001, though they continue to lag behind those in other regions – suggesting a need for the reforms to continue and be further strengthened.

*Sources:* Bruns et al. (2012); OECD (2011); UNESCO (2010).

*What are the lessons of redistribution policies?*

Redistribution policies are vital for achieving more equal education outcomes. However, they have often not gone far enough. Experience from these initiatives offers some important lessons on how to strengthen their design to improve learning for marginalized children.

**Kenya's capitation grant disadvantages arid and semi-arid areas that are home to 46% of the out-of-school population**

One of the biggest criticisms of redistribution initiatives is that even these higher allocations per child often do not adequately equalize spending. In Brazil, for example, it is estimated that US$971 per pupil is required to attain a minimum level of quality for grades 1 to 4, but in 2009 the government allocated US$611 per pupil in the North-East region, about half as much as in the wealthier South-East region (PREAL and Lemann Foundation, 2009). This is one area where the reform needs to be strengthened to further narrow the gap in learning outcomes between regions.

In India, despite increased resources for Sarva Shiksha Abhiyan, allocations are still not sufficiently reaching the states most in need. In 2012/13, total expenditure per elementary pupil from both central and state funds was still much lower in states where education indicators were worse than in the states with some of the best education indicators. In one of India's wealthier states, Kerala, education spending per pupil was about US$685. Similarly, in Himachal Pradesh it was US$542. By contrast, in West Bengal it was US$127 and in Bihar US$100. Increased financial allocations are still insufficient to translate into improved learning outcomes, suggesting that far more needs to be done. In Bihar, for example, where spending rose by 61% between 2010/11 and 2012/13 but remained low, only 48% of Standard 3 to 5 students could read a Standard 1 text in 2012 (Accountability Initiative, 2013).

Another limitation of redistribution measures is that they largely focus on non-salary expenditure – often a small part of total expenditure – and thus do not allow for using the funds to implement teacher reforms that are vital to improving quality. In Sri Lanka, Education Quality Input funds amounted to just 2% of the total recurrent budget for education (Arunatilake, 2007). There are exceptions; Brazil, for instance, earmarked 60% of FUNDEF funds for teacher salaries. Salaries overall rose by 13% at the national level, but in the North and North-East they rose by 60% because all teachers were upgraded with higher education qualifications (OECD, 2011).

Rigid earmarking can prevent schools from spending funds in areas that could have greater impact on learning. In India, for example, the Sarva Shiksha Abhiyan grants are delivered to schools in the form of three separate components earmarked for maintenance, development and teaching-learning, which may not necessarily reflect the needs of a given school (Accountability Initiative, 2013).

Poorer countries can find it difficult to identify and target the groups most in need. Many base allocations on enrolment figures, to the detriment of areas where large numbers of children are out of school. In Kenya, for example, the capitation grant is distributed on the basis of number of students enrolled, a disadvantage for the 12 counties in the arid and semi-arid areas that are home to 46% of the out-of-school population. Children in these areas who do enter school tend to be first-generation learners from non-literate home environments, and so need additional support in the form of higher spending per pupil (Watkins and Alemayehu, 2012).

Similarly, in Bangladesh almost two-thirds of primary schools received a grant of about US$300 per school in 2010 to help fund improvement plans. However, the amount was the same for every school, regardless of size or location, and the grant was not directed at activities aimed at improving the quality of teaching and learning (Bangladesh Ministry of Primary and Mass Education, 2011a; Bernard, 2010).

To address the need to target out-of-school populations, Brazil complemented FUNDEB with the Bolsa Familia programme, which provides a cash transfer to compensate for the loss of children's labour, conditional on children attending school (Bruns et al., 2012).

Implementation problems also plague redistribution efforts. In many countries, schools do not receive as much funding as they expect, or do not receive it on time. In Sri Lanka in 2011, less than one-third of schools had received their Education Quality Input funds halfway through the school year.

Late arrival of funds, together with limited implementation capacity, is particularly detrimental for smaller schools, given their greater reliance on such funding. In response, the government has introduced a policy measure allowing schools to carry unspent funds into the next financial year (Arunatilake and Jayawardena, 2013).

Schools may lack the capacity to spend the resources they receive. In India, Bihar managed to spend only 38% of its Sarva Shiksha Abhiyan funds in 2011/12, while the national average was 62% (Accountability Initiative, 2013). This low spending is likely to be a reason for the continued poor learning outcomes in the state. Disbursement systems in decentralized countries can further stretch school capacity. In Indonesia, schools receive funds from eight different sources and four different budgets, making it difficult to plan (World Bank, 2013e).

In summary, redistribution measures are vital to ensure that domestic resources are used to equalize education opportunities, but they need to cover the full cost of delivering quality education to the most vulnerable. They also need to be combined with reforms that strengthen education systems' capacity to implement such measures, and they should complement other interventions aimed at making sure all children are in school and learning.

## Trends in aid to education

Government spending provides the largest contribution to education, but aid is vital for many poorer countries. However, aid to basic education is declining, jeopardizing the chances of schooling for millions of children. With improvement in the numbers of children out of school stagnating, a final push is needed to ensure that all children are in school by 2015, but the fall in aid will increase the difficulty of this task.

Even before the economic downturn, donors were off track to fulfil the promise they made in 2000 at the World Education Forum in Dakar, Senegal: that no country would be left behind in education due to a lack of resources. Economic austerity should not be an excuse for donors to abandon their pledges to the world's poorest. Recipient countries need predictable financing for their national education plans. Donors urgently need to reconsider their aid cuts to ensure that they meet their commitment to the world's children.

**In 2011, aid to basic education fell for the first time since 2002**

### Aid to education is falling

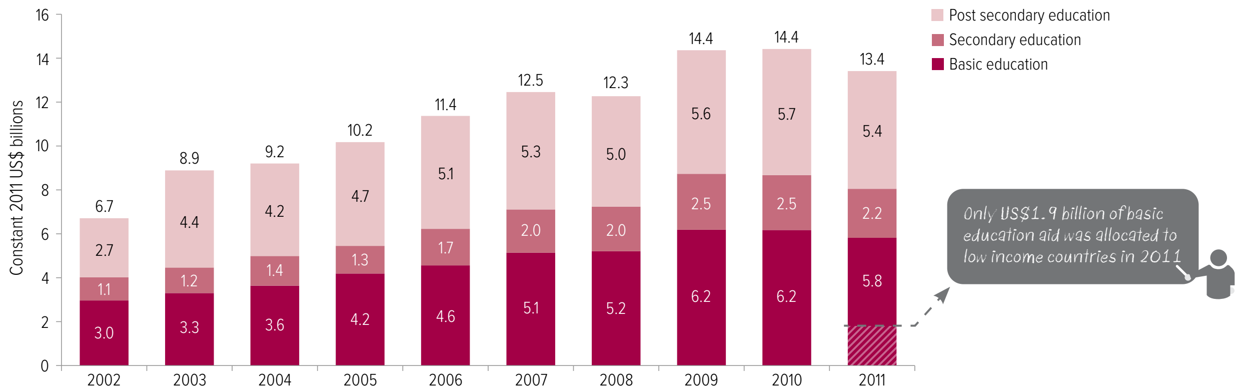
While aid to education increased steadily after 2002, it peaked in 2010 and is now falling: total aid to all levels of education declined by 7% (US$1 billion) between 2010 and 2011 (Table 2.3, Figure 2.6). Aid to basic education fell for the first time since 2002, by 6%: from US$6.2 billion in 2010 to US$5.8 billion in 2011,

**Table 2.3: Total aid disbursements to education and basic education, by region and income level, 2002–2011**

| | Total aid to education | | | Total aid to basic education | | | |
|---|---|---|---|---|---|---|---|
| | Constant 2011 US$ millions | | | Constant 2011 US$ millions | | | Per capita |
| | 2002 | 2010 | 2011 | 2002 | 2010 | 2011 | 2011 |
| **World** | **7 799** | **14 419** | **13 413** | **3 133** | **6 174** | **5 819** | **9** |
| Low income | 2 145 | 3 796 | 3 461 | 1 240 | 2 047 | 1 858 | 16 |
| Lower middle income | 3 012 | 5 407 | 5 371 | 1 290 | 2 451 | 2 607 | 9 |
| Upper middle income | 1 652 | 2 800 | 2 641 | 302 | 595 | 579 | 3 |
| High income | 25 | 36 | 13 | 6 | 9 | 6 | 1 |
| Unallocated by income | 964 | 2 379 | 1 926 | 296 | 1 072 | 769 | … |
| Arab States | 1 053 | 1 939 | 1 922 | 221 | 825 | 845 | 20 |
| Central/Eastern Europe | 305 | 574 | 517 | 90 | 80 | 64 | 6 |
| Central Asia | 130 | 331 | 346 | 43 | 99 | 101 | 18 |
| East Asia/Pacific | 1 155 | 2 309 | 2 060 | 253 | 687 | 552 | 4 |
| Latin America and the Caribbean | 560 | 1 110 | 948 | 226 | 438 | 381 | 6 |
| South and West Asia | 967 | 2 267 | 2 417 | 597 | 1 309 | 1 445 | 8 |
| Sub-Saharan Africa | 2 816 | 3 959 | 3 647 | 1 490 | 1 891 | 1 757 | 13 |
| Overseas territories | 254 | 523 | 74 | 127 | 243 | 26 | … |
| Unallocated by region or country | 559 | 1 406 | 1 481 | 86 | 602 | 648 | … |

*Notes:* The 2002 figure is an average for 2002–2003. Aid per capita refers to the amount a primary school age child received in aid to basic education in 2011.
*Source:* Annex, Aid Table 3.

**Figure 2.6: Aid to education fell by US$1 billion between 2010 and 2011**
Total aid to education disbursements, 2002–2011




*Source:* OECD-DAC (2013).

**The reduction in basic education aid to sub-Saharan Africa could have funded good quality school places for over 1 million children**

putting at risk the chances of meeting the 2015 goals. Aid to total secondary education also declined between 2010 and 2011, by 11%, from an already low level. This means that hopes of extending global goals to include universal lower secondary education after 2015 could be difficult to meet unless this changes.

### *Low income countries are bearing the brunt of aid cuts*

Although many low income countries are making commendable efforts to scale up domestic resources for education, a funding gap remains that needs to be addressed urgently. Aid is a crucial source of funding, equivalent to a quarter of education budgets in nine countries (UNESCO, 2012). Higher levels of education aid in the last decade have boosted enrolment (Birchler and Michaelowa, 2013).

Despite such benefits, low income countries, which only receive around one-third of aid to basic education, witnessed a larger decrease in aid to basic education than middle income countries. Aid fell by 9% in low income countries between 2010 and 2011 – from US$2.05 billion to US$1.86 billion. In sub-Saharan Africa, home to over half the world's out-of-school population, aid to basic education declined by 7% between 2010 and 2011, from US$1.89 billion to US$1.76 billion. The US$134 million reduction in basic education aid to the region would have been enough to fund good quality school places for over 1 million children.

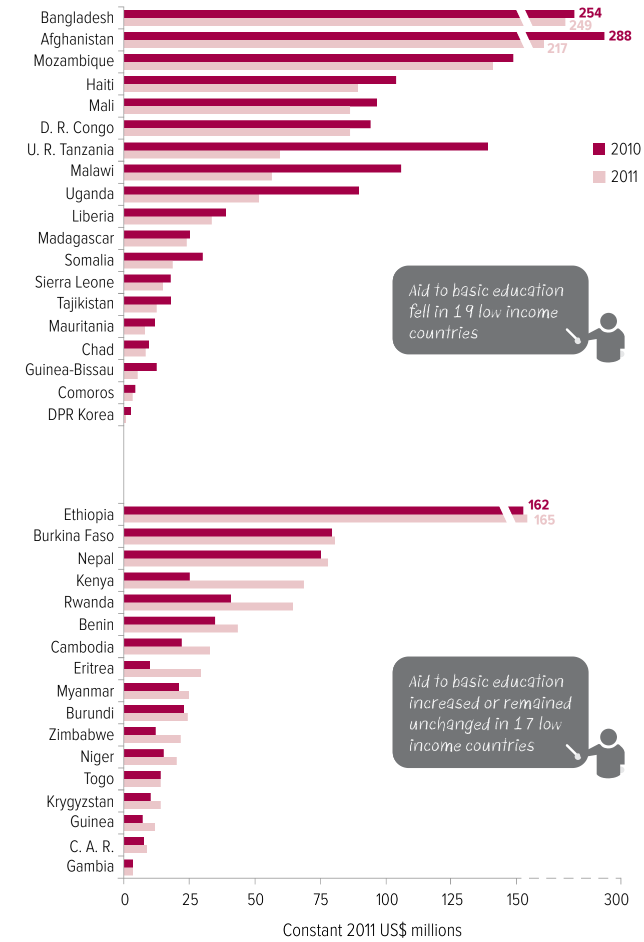
Aid to basic education fell between 2010 and 2011 in 19 low income countries, 13 of which are in sub-Saharan Africa (Figure 2.7). In Malawi, for instance, aid to basic education almost halved over the space of one year, largely due to a political impasse between the donor community and the government at that time. While annual fluctuations in aid may not be uncommon, such changes make it difficult for countries to plan. Given that a large proportion of education spending is on teacher salaries, sudden reductions in aid can mean that teachers are not paid on time, or that teachers leaving the profession are not replaced, seriously harming the quality of education.

In some countries, aid has been falling for more than a year. In the Democratic Republic of the Congo, Mali and the United Republic of Tanzania, aid to basic education fell considerably in both 2010 and 2011. Aid has played a key part in supporting efforts to get more children into school in the United Republic of Tanzania, but it fell by 12% between 2009 and 2010 and by a further 57% in 2011, in the latter case largely because of reductions by Canada and the World Bank. These cuts endanger the progress that has been made and could thwart efforts to strengthen the quality of education.

The fall in basic education aid to low income countries has resulted in the resources available per child falling from US$18 in 2010 to US$16 in 2011. The United Republic of Tanzania received US$7 per child in 2011 – US$13 less than in 2009.

**Figure 2.7: Aid to basic education fell in 19 low income countries between 2010 and 2011**

Total aid to basic education in low income countries, 2010 and 2011




*Source:* OECD-DAC (2013).

## *Out-of-school children need the support of aid, regardless of where they live*

Of the 10 countries with the most children out of school, 6 are in lower middle income countries. Of these, only two were among the top 10 recipients of aid to basic education in 2011: India and Pakistan. Nigeria, home to the world's largest number of children out of school, does not figure among the top 10 recipients of aid to basic education, and the aid it does receive decreased by nearly 28% between 2010 and 2011. While these countries need to do far more to increase their own domestic spending on education, a lack of resources should not prevent the most disadvantaged children from going to school because of where they live (Box 2.2).

### Box 2.2: Poor children out of school in some lower middle income countries also need aid

The world's poorest children, who are the most likely to be out of school, live not only in low income countries but also in lower middle income countries. Since 2000, 25 countries have graduated to this group, which now comprises 54 countries, while 36 are classified as low income. In 1999, 84% of the world's out-of-school children lived in low income countries and 12% lived in lower middle income countries, but by 2011, 37% lived in low income countries and 49% in lower middle income countries, due in particular to the graduation to lower middle income status of some large-population countries, such as India, Nigeria and Pakistan.

The current income thresholds used by the World Bank to classify countries, which determine their eligibility for concessional loans and grants from the bank's International Development Association (IDA), influence heavily the allocation decisions of other major donors. The lower middle income group now comprises countries with annual income per capita between US$1,026 and US$4,035. These countries differ tremendously in the obstacles they face in reaching the Education for All goals and other development targets. They include some sub-Saharan African countries and countries affected by conflict. Their levels of income differ considerably, from Egypt, Indonesia and Morocco with higher income per capita to ones closer to the level of low income countries, such as Cameroon, Côte d'Ivoire, Senegal and Yemen. The more populous countries, India, Nigeria and Pakistan, fall at the lower end of the lower middle income group and are home to 54% of the developing world's population living below US$1.25 per day, based on 2010 population estimates.

Some lower middle income countries could do far more to raise their own resources for education and ensure that these resources reach those most in need. The necessary domestic tax reforms are likely to take time, however, so these countries will need aid in the coming years if another generation is not to be denied its right to education. In India, for example, which became a lower middle income country in 2007, there is an elite large enough to provide sufficient domestic taxes to give all those from poor households the chance to learn, but redistribution to the poorest parts of the country takes time and will not be straightforward, given its size. If each Indian state were a country, Uttar Pradesh would have the world's second largest concentration of poor people (after China), and Bihar the sixth largest. To make sure aid targets the poor, donors should direct aid to the areas in lower middle income countries where poverty is concentrated.

*Sources:* World Bank (2013f); Oxford Poverty and Human Development Initiative (2013).

### *The donor landscape is changing*

The fall in aid to education reflects changing spending patterns among many donors. Direct aid to education fell slightly more than overall aid to other sectors between 2010 and 2011, and thus the share of aid to education declined from 12% to 11%. Canada, France, the Netherlands and the United States, in particular, cut spending on education more than they reduced overall aid. Between 2010 and 2011, 21 bilateral and multilateral donors reduced their aid disbursements to basic education. The largest decreases in volume terms were from Canada, the European Union, France, Japan, the Netherlands, Spain and the United States, which together accounted for 90% of the reduction in aid to basic education.[5]

**Between 2010 and 2011, 24 donors reduced their aid to low income countries**

The United States, formerly the largest bilateral donor to basic education in absolute terms, cut its aid in this area so much that it fell to second place. As a consequence the United Kingdom overtook the United States as the largest bilateral donor, thanks to its commitment to increase overall aid to the target agreed by European donors of 0.7% of gross national income (GNI) by 2015, as well as its prioritization of the education sector. In 2012, the United Kingdom allocated 0.56% of GNI to aid. By contrast, the United States, which has not set a similar target, devoted 0.19% of GNI to aid in 2012. Of the decline in the United States's total aid to basic education between 2010 and 2011, 94% is accounted for by large falls in its spending in Afghanistan, Iraq and Pakistan.

The Netherlands decided in 2011 that it would phase out education programmes that did not contribute directly to its foreign policy priorities. Consequently its aid to basic education fell by over a third between 2010 and 2011; it had been the largest donor to basic education in 2007, but by 2011 was in eleventh place. The Netherlands was a key funder and policy leader for education, so its shift away from the sector is cause for concern. Its reductions in aid have particularly affected Mali, Mozambique and Uganda: in all three, aid to basic education declined between 2010 and 2011, suggesting that the Netherlands did not succeed in its aim of withdrawing without harming education in affected countries, as other donors have not stepped in to fill the gap.

Australia increased its basic education aid disbursements by 49% between 2010 and 2011, though the increase was largely concentrated in lower middle income countries it considers strategically important, including Indonesia, Papua New Guinea and the Philippines. In 2011, 68% of Australia's aid disbursements to basic education went to the East Asia and the Pacific region, which is likely to continue to be a top priority. While sub-Saharan Africa received just 0.3% of Australia's total bilateral aid to basic education, the previous Australian government had committed to join the African Development Bank with initial contribution and payments for the 13th and 14th replenishment set at US$161 million for the six-year replenishment period; on average the yearly contribution to the bank would be equivalent to around 12% of its basic education aid in 2011 (Parmanand, 2013).

The United Kingdom's increase in aid to basic education between 2010 and 2011 benefited low income countries. However, 24 donors reduced their aid to these countries over this period, including nine of the 15 largest donors to low income countries. The biggest cuts were made by EU institutions, the World Bank and the Netherlands (Figure 2.8). These donors, along with Canada and Spain, were also responsible for the overall reduction in education aid to sub-Saharan Africa.

For the main donors, the cut in aid to low income and sub-Saharan African countries between 2010 and 2011 was part of an overall reduction in basic education aid. EU institutions, for example, reduced their overall aid to basic education by 31%, which resulted in a reduction of 36% for low income countries, with Bangladesh, the Democratic Republic of the Congo, Malawi and Nepal among the most affected. Australia, the World Bank and the IMF increased their overall aid to basic education between 2010 and 2011 but reduced their spending in low income countries.

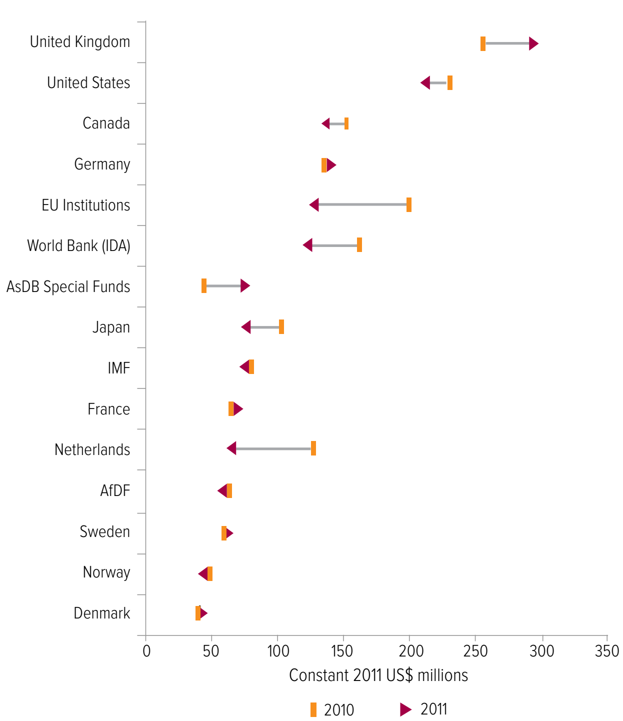
World Bank aid to basic education increased by 13% overall, but fell by 23% in low income countries; the largest cuts affected Haiti and

5. The decrease in France's aid to basic education is largely due to Mayotte no longer being classed by the OECD as ODA-eligible as from 2011.

the United Republic of Tanzania. World Bank aid to basic education for lower middle income countries rose by 23%, largely due to increases in disbursements to India and Pakistan. The World Bank's aid disbursements to basic education fell or remained steady in most sub-Saharan African countries, while South and West Asia experienced an increase, reflecting a trend that has been apparent over the past decade.

Spending by the Global Partnership for Education (GPE) is unlikely to have filled the gap left by the World Bank's reduction in aid to low income countries. Uganda, for instance, was the World Bank's second largest recipient of aid to basic education in 2002, after India, receiving US$113 million for education; by 2011 disbursements were zero, even though Uganda was still classed as a low income country. Yet Uganda has not received any funding from the GPE. Similarly, the United Republic of Tanzania, another large recipient of World Bank aid disbursements to basic education in the early 2000s, saw disbursements fall from US$88 million in 2002 to less than US$0.3 million in 2011. The United Republic of Tanzania became a GPE partner in 2013 with a US$5.2 million grant for its education plan. All this, however, is allocated to Zanzibar, and the amount is small compared with what the country received from the World Bank early in the 2000s.[6] Nevertheless, the GPE is an important source of financing in some low income countries, although information is not currently publicly available in a format to enable tracking of spending patterns (Box 2.3).

**Figure 2.8: Nine of the 15 largest donors reduced their aid to basic education for low income countries between 2010 and 2011**
Total disbursements of aid to basic education for low income countries, 2010 and 2011




*Source:* OECD-DAC (2013).

## Aid forecasts remain bleak

There is no sign that overall aid will stop declining before the 2015 deadline for education goals is reached. From 2011 to 2012, total aid fell by 4% in real terms. Italy and Spain accounted for most of the US$5.2 billion decrease in aid by DAC donors. The Netherlands, the United Kingdom and the United States also reported large decreases, although the UK reduction did not represent a fall in the proportion of GNI allocated to aid, as GNI also fell. It is particularly worrying that less developed countries are expected to bear the brunt of these cuts: bilateral aid to these countries fell by 12.8% from 2011 to 2012 (OECD, 2013a).

**If EU countries kept their aid promises this could provide US$9 billion to fill the education financing gap in 2015**

Many donors are expected to reduce their aid further in coming years (Table 2.4). In all, 16 bilateral DAC donors decreased their aid between 2011 and 2012, not only because of the economic downturn but also because they made aid a lower priority: 13 donors decreased aid as a proportion of GNI. Only five donors, including Australia and Luxembourg, disbursed more aid as a proportion of GNI in 2012 than in 2011.

The 15 countries that were members of the European Union in early 2004 agreed in 2005 to increase their aid to 0.7% of GNI by 2015, but projections by the European Commission indicate just five of the countries are expected to meet this commitment. The overall shortfall in aid due to EU donors' unmet promises is expected to be US$43 billion, the bulk of it attributed to France, Germany, Italy and Spain (European Commission, 2013). If EU countries raised their aid to the levels needed to meet their 2015 pledge, this would provide US$9 billion to fill the education financing gap in 2015 if 20% was allocated for the education sector.

6. According to the GPE, the United Republic of Tanzania is eligible to apply for a US$100 million program implementation grant (Global Partnership for Education, 2013b).

## Box 2.3: How important is the Global Partnership for Education for funding basic education in poor countries?

The GPE is an important source of external financing for education for some low and lower middle income countries, although currently it only accounts for a small proportion of education aid. Between 2004 and 2011, donors paid in US$2 billion to the GPE. By comparison, donors spent US$32 billion in aid to basic education to low and lower middle income countries over the same period. However, the GPE's influence appears to be increasing over time.

In 2011, the GPE disbursed US$385 million to basic education – an all-time high – making it the fourth largest donor to low and lower middle income countries that year. For the 31 countries that had a programme implementation grant in 2011, 24% of basic education aid was disbursed by the GPE.

For three of these 31 countries, in 2011 the GPE provided at least half of external finance for basic education, either because aid from other donors was negligible or because the GPE had channelled significant volumes to some countries. GPE funding to the Central African Republic, whose grant agreement was signed in 2009, was US$13.2 million in 2011, or 60% of the country's aid to basic education that year. Eleven other donors gave small amounts. For 10 of the 28 countries receiving GPE funds in 2011, these funds constituted no more than one-fifth of aid to basic education. In Niger, whose grant agreement was also signed in 2009, the GPE disbursed US$1.5 million, less than 10% of external funding for basic education, while aid from DAC donors came to US$20.2 million (Figure 2.9).

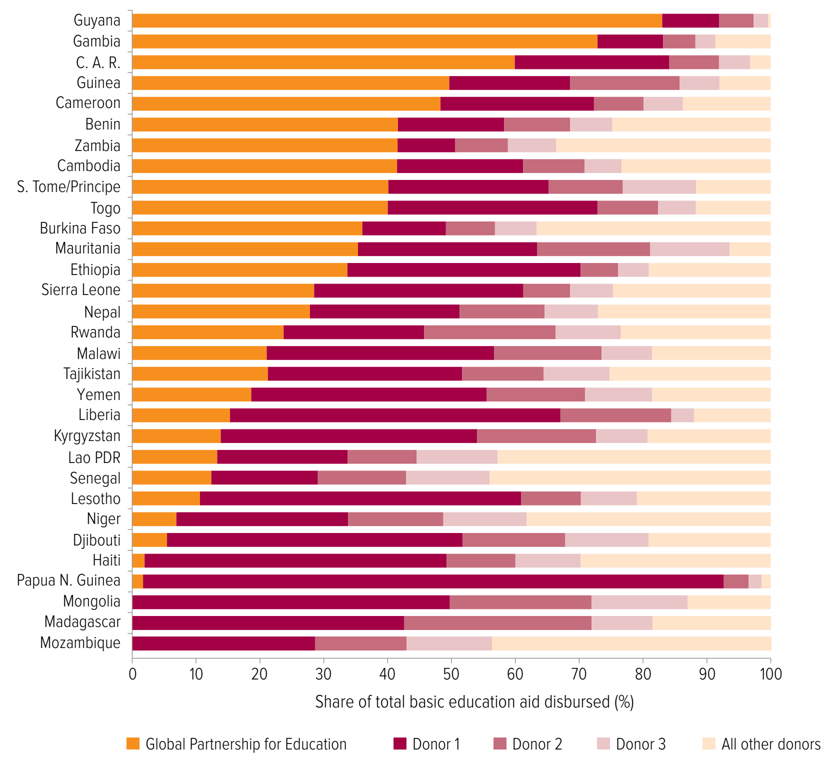
Because of the importance of the GPE in financing education in some low income countries, it is vital for its disbursements to be timely and predictable. While much criticism was levelled at the EFA Fast Track Initiative (which the GPE replaced) for poor disbursement rates from its Catalytic Fund, the GPE committed itself to more timely and predictable disbursements.

To keep track of these commitments, the GPE needs to report its aid flows to the OECD-DAC, just as global health funds such as the GAVI Alliance and the Global Fund to Fight AIDS, Tuberculosis and Malaria do. The Global Fund also provides details such as whether funds go directly to governments or through multilateral organizations or civil society organizations, a level of disaggregation that is equally necessary for GPE disbursements.

*Sources:* Global Partnership for Education (2013a); OECD-DAC (2013).

**Figure 2.9: In some countries, the Global Partnership for Education is a large donor for aid to basic education**

Shares of the GPE and other donors in aid disbursements to basic education for countries with a programme implementation grant in 2011




*Note:* Information on GPE disbursements is from the GPE; aid disbursement information for all other donors is from the OECD's Creditor Reporting System (CRS). When reporting their GPE contributions to the CRS, most donors put them in the category: 'bilateral country unspecified'. Where donors reported GPE contributions to the CRS as direct allocations to recipient countries, these were deducted to avoid double counting.

Spending projections for DAC donors' country programmable aid is also likely to decline. It is of particular concern that the countries most in need of external resources to help them meet the EFA funding gap will be hit hard: country programmable aid is expected to fall in 31 of the 36 low income countries, the majority of which are in sub-Saharan Africa, where education resources fall far short of requirements to achieve EFA (OECD, 2012a).

Projections raise further concern as to whether the remaining resources will go to the countries most in need. Any increases in country programmable aid are expected to be directed largely to middle income countries, including China and Uzbekistan, while disbursements to the countries furthest off track to reach the MDGs will fall by the equivalent of half a billion dollars. These countries include Burundi, Chad, Malawi and Niger, which depend on aid to fund education (OECD, 2013d).

**Table 2.4: Many donors expect to reduce aid further in coming years**

| Donor | Donor's share of total aid to basic education, 2002–2011 | Aid forecasts |
|---|---|---|
| World Bank | 16% | New commitments for education totalled US$3 billion in the 2012 financial year. This is a sharp increase from US$1.8 billion in 2011, largely due to increased investment in primary and lower secondary education. South and West Asia is expected to be the main beneficiary of the additional resources. |
| United States | 10% | The international affairs budget is expected to stay the same in 2014 as 2013. However, the bilateral aid programme is set to decrease in 58 out of 100 countries, including 17 in sub-Saharan Africa, with a shift in focus to East Asia and the Pacific. In addition, the request for basic education is 37% lower in 2014 than in 2013; total resources for basic education have been cut by half since 2010. |
| United Kingdom | 9% | Between 2011 and 2012, bilateral aid was expected to increase from US$8.8 billion to US$8.9 billion, but aid to education was expected to fall from US$1.0 billion to US$0.8 billion. |
| European Union | 8% | The European Development Fund budget will fall by US$4.3 billion between 2014 and 2020 as part of general cuts to the EU budget. |
| Netherlands | 7% | Aid will fall by US$1.3 billion between 2014 and 2017, from 0.63% to 0.55% of GNI. Aid to basic education is expected to decrease from US$243 million to US$76 million between 2011 and 2014. |
| Japan | 6% | Japan has pledged to increase funding for basic education between 2011 and 2014. |
| Canada | 4% | After Canada decided in 2012 to cut aid by 7.5% by 2015, aid is expected to fall by US$650 million by 2013/14 and US$781 million by 2015/16. The Canadian International Aid Agency will close and aid will be managed by the Department of Foreign Affairs and International Trade. |
| Germany | 4% | Aid in 2013 is US$159 million less than was planned and US$112 million less than in 2012. Further cuts are planned for 2014. Germany has committed to increasing aid for basic education, however, and has committed to supporting education in additional countries by 2013. |
| Australia | 3% | The recently elected coalition government has pushed back on the country's commitment to reach the 2015 0.5% aid target pledge for an indefinite period. It has announced that, over the next four years, US$4.2 billion will be cut from the foreign aid budget. This means that aid as a share of GNI is expected to fall from a projected 0.37% in 2013 to 0.32% in 2016/17. |
| Denmark | 1% | Bilateral aid to education is being phased out, with 11 programmes closing between 2011 and 2015. To keep to its promise of maintaining aid levels to basic education, Denmark would need to increase the levels it channels through multilateral mechanisms. |

*Sources:* Beckett (2013); Bernard Van Leer Foundation (2013); CIDA (2013); Deutsche Welle (2012); DFID (2013); Gavas (2013); Global Partnership for Education (2012); Piccio (2013); Robinson and Barder (2013); Taylor (2013); World Bank (2013a).

It is also worrying that education's decline in importance for many donors means that cuts for education could become even more significant. At the GPE Replenishment Conference in 2011, only five donors pledged to increase aid to basic education in low income countries between 2011 and 2014 (Global Partnership for Education, 2011).

## Humanitarian aid appeals neglect education needs

With half the world's out-of-school children residing in conflict-affected countries, education in these countries should be a priority for donors. Recognizing this, the UN Secretary-General's Global Education First Initiative (GEFI) set a target of 4% for education's share of short-term humanitarian aid. While this sounds modest, it is unfortunately far beyond the actual share in 2012: 1.4%, down from 2.2% in 2009. Education is the sector receiving the smallest proportion of requests for humanitarian aid, and just 26% of the amounts requested are actually covered (Figure 2.10).

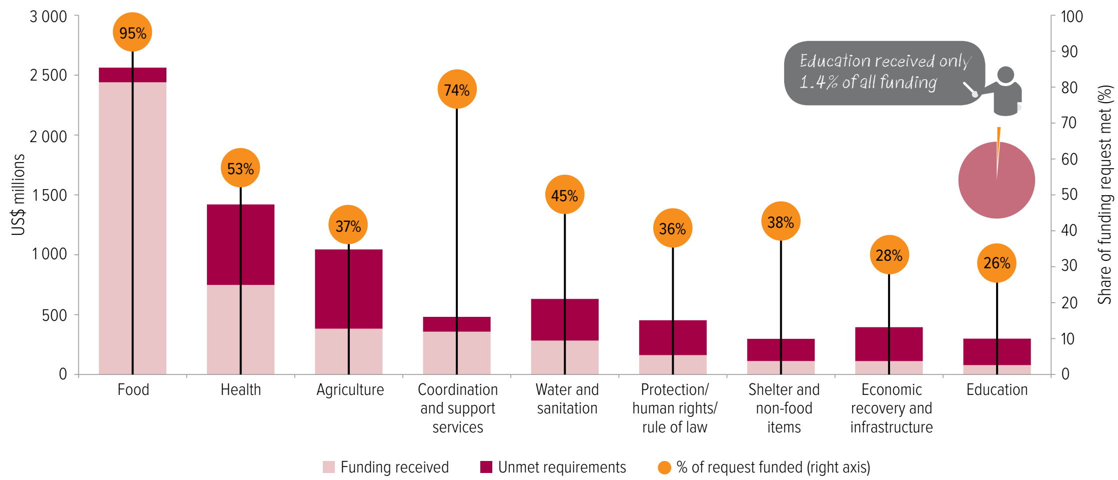
The funds requested for the education sector in Yemen's consolidated appeal, for instance, was the lowest for any sector (3.2% of the total requested), and despite this just a quarter of the request was met (Figure 2.11). As a share of total funding, education received just 1.4% of the Yemen humanitarian appeal. Even in countries where requirements defined as education received a large portion of total humanitarian funding, it does not necessarily improve access to quality education. In the Central African Republic humanitarian appeal, for instance, education received 7% of total funding, a higher share than education receives in most appeals. However, 61% was for school feeding and deworming programmes administered by the World Food Programme.

In humanitarian work plans for 2013, 4.3% of total requirements were earmarked for education – matching the GEFI target – but if education receives the lowest funding relative to requests, as in previous years, its share of overall humanitarian aid is likely to reach no more than 2% once again.

**Education only received 1.4% of humanitarian aid in 2012, down from 2.2% in 2009**

**Figure 2.10: Education's double disadvantage in humanitarian aid: a small share of requests and the smallest share of requests that get funded**

Consolidated and flash appeal requests and funding by sector, 2012




*Source:* OCHA (2013).

Of the 17 consolidated humanitarian appeals to the United Nations Office for the Coordination of Humanitarian Affairs for 2013, seven proposed that 4% or more of total humanitarian funds requested should be earmarked for the education sector. In Mali, where most schools in the north have been closed due to conflict, education makes up 5% of requests for the 2013 appeal, or US$21.6 million; by September 2013, however, the sector had received just 15% of the funds requested, despite the growing education crisis.

**A quarter of direct aid to education is allocated to scholarships and student imputed costs**

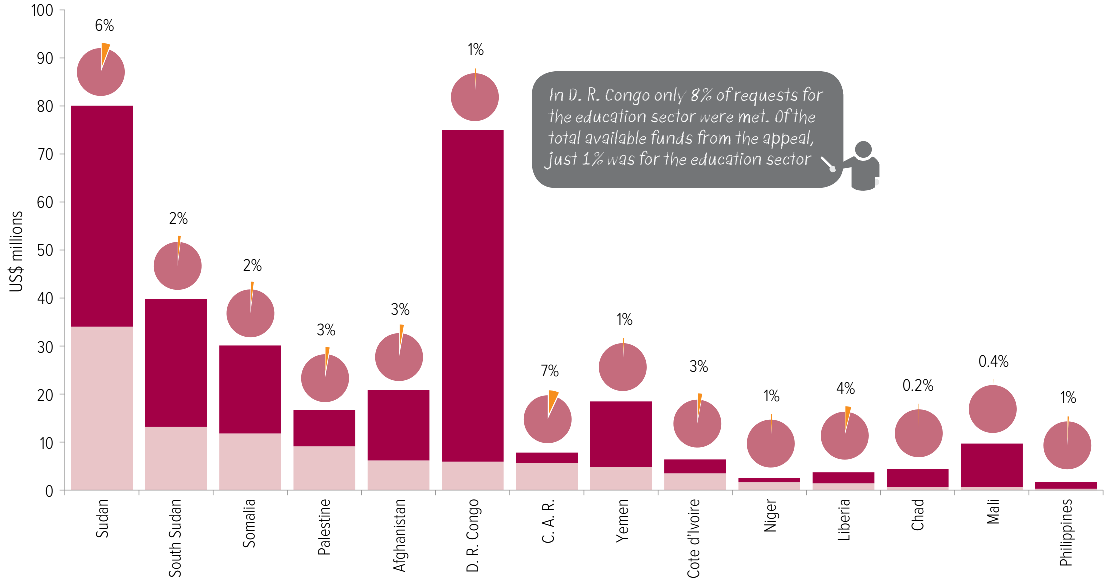
Similarly, 3% of the Syrian Arab Republic's Humanitarian Assistance Response Plan for 2013 was earmarked for education, with 36% of the resources requested for education having been pledged by September 2013 (Office for the Coordination of Humanitarian Affairs, 2013), even though one in five schools have been destroyed in some areas. In and around Aleppo, where fighting has been intense, only 6% of children were attending schools. Children comprise almost 50% of those in need of urgent humanitarian assistance in this conflict, which is in its third year (UNICEF, 2013b, 2013c). While these countries might receive more of the requested funds in the latter part of 2013, they would come too late for the millions of children who have had to drop out of school due to conflict.

## Making sure education aid is spent effectively

It is not only the amount of aid that counts but also whether it is used most effectively to ensure that disadvantaged children are in school and learning. These children do not receive all the aid available, however: a quarter of direct aid to education is spent on students studying in universities in rich countries. In addition, 15% of aid is in the form of loans that countries have to pay back at concessional interest rates, depriving them of resources that could be spent domestically on education in the future (Figure 2.12).

In 2010–2011, an average of US$3.2 billion of aid was allocated annually to scholarships and student imputed costs, equivalent to a quarter of total aid to education. Of this, 81% is spent by four donors: Canada, France, Germany and Japan. Such training is no doubt beneficial, but the spending is not 'real' aid according to the OECD definition. For this reason, some countries do not register the spending as aid. The United States, for example, includes it in the Bureau of Educational and Cultural Affairs budget administered by the Department of State (US Department of State, 2013).

**Figure 2.11: Conflict-affected countries receive only a tiny share of their requests for humanitarian education funding**
Consolidated and flash appeal requests and funding for education, selected conflict-affected countries, 2012

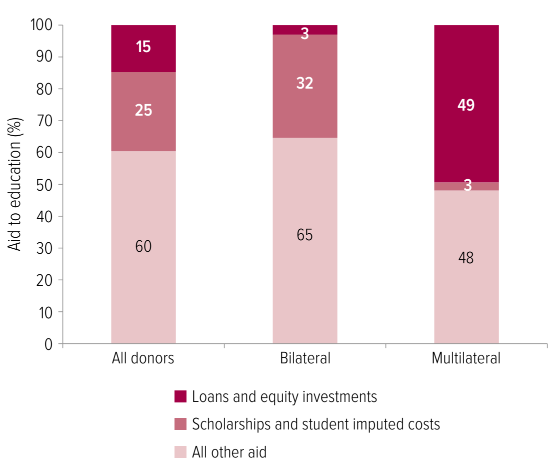



*Source:* Office for the Coordination of Humanitarian Affairs (2013).

Even though scholarships and imputed student costs may be vital to strengthen human resource capacity in low income countries, most of this funding actually goes to upper middle income countries, with China the largest recipient, receiving 21% of the total (Figure 2.13). Such 'aid' to China exceeds the aid received by some of the poorest countries for basic education. For instance, on average over 2010–2011, donors – primarily Germany and Japan – disbursed US$656 million per year to China for scholarships and student imputed costs, which was 77 times the amount of aid disbursed to Chad for basic education over the same period, and 37 times the amount given to Niger. The total funding in the form of imputed student costs and scholarships received annually by Algeria, China, Morocco, Tunisia and Turkey was equivalent to the total amount of direct aid to basic education for all 36 low income countries in 2010–2011, on average.

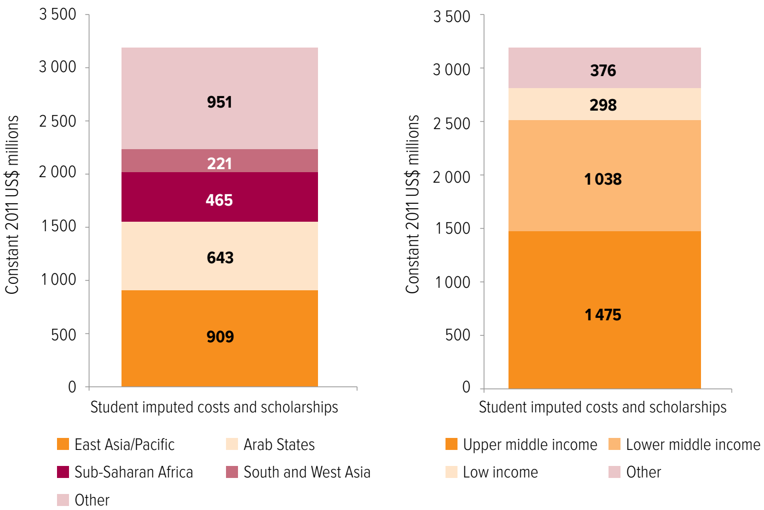
While most direct aid to education is in the form of grants, in 2011, US$2.0 billion was in the form of concessional loans, of which US$0.6 billion was spent on basic education. The overwhelming majority of such loans are disbursed by the World Bank, but some bilateral donors, notably France and Japan, also provide some education aid in this way. Japan's loans to the education sector go largely to middle income countries such as China and Indonesia, while the majority of France's have been directed to North African countries such as Morocco.

**Figure 2.12: 40% of education aid does not leave donor countries, or is returned to them**
Distribution of direct aid to education by type, 2010–2011




*Source:* OECD-DAC (2013).

**Figure 2.13: Middle income countries receive almost 80% of aid to scholarships and imputed student costs**
Education aid spent on scholarships and student imputed costs, 2010–2011




*Note:* For region groups, aid to 'other' includes disbursements to Central and Eastern Europe, Central Asia, Latin America and the Caribbean and aid that is geographically unallocated. For income groups, 'other' refers to high income and unallocated by income.
*Source:* OECD-DAC (2013).

Excluding amounts spent on students studying in donor countries and loans, Germany would drop four places from being the largest donor in direct aid to education in 2010–2011 to being the fifth largest, and France would drop two places to fourth largest. The World Bank would fall from third to fourteenth place, since a large share of its funding is in the form of loans. The United Kingdom and United States, which give negligible amounts of aid to education in the form of loans or student imputed costs, would jump from sixth and seventh place to first and second if these aid items were excluded.

The European Union, World Bank and African Development Fund are the largest multilateral donors to education, the latter two providing the majority of education aid in the form of concessional loans. Where bilateral donors are absent in providing education financing, there is a danger of low income countries becoming overdependent on concessional loans from international finance institutions, leading to indebtedness that could restrict countries' ability to fund education from their own resources.

## Accounting for all education expenditure

Governments are the largest source of education financing, with donors making up a significant proportion of resources in some of the poorest countries. One important source of education finance is often overlooked, however: the amount that households themselves contribute. More broadly, information on the whole spectrum of education financing is often insufficient and fragmented, resulting in partial analysis and diagnosis of who bears the cost for education.

Since 2000, the OECD's tracking of aid from DAC donors has improved considerably, with more attention not only to commitments but also to disbursements, and to identifying 'real' aid. However, the collection of national finance data remains weak. Data on private spending from household surveys often do not provide sufficient detail on spending by level of education. To the extent that such information is available, it is not sufficiently used by policy-makers to get a comprehensive picture of education spending.

Such limitations could have consequences for the achievement of Education for All and the extension of education goals after 2015. Analysis for this Report of data from seven countries shows that the share of education expenditure borne by households ranges from 14% to 37% in primary education and from 30% to 58% in secondary education.[7] These findings highlight the importance of building a comprehensive national education accounts system, which could be modelled on the experience in health.

### Learning from national health accounts

The idea of a comprehensive framework to account for all expenditure flowing into a sector is not new. It has been applied successfully in the health sector, starting with high income countries in the 1970s and spreading to middle and low income countries since the 1990s.

National health accounts systems have developed in recognition of the fact that informed decision-making requires reliable information on the quantity of financial

7. See technical note on the EFA Global Monitoring Report website for further information (www.efareport.unesco.org).

resources used, their sources and the way they are used. The systems aim to provide evidence to allow monitoring of trends in health spending, whether by government, donors or households. Drawing on this information, Eurostat, the OECD and WHO have established global standards to systematize how such information is collected, leading to the International Classification for Health Accounts. This has enabled countries to compare their performance and improve effectiveness, efficiency and equity (OECD, 2000; OECD et al., 2011; WHO et al., 2003).

Evidence from national health accounts has had far-reaching consequences for policy decisions. In Rwanda, the revelation that HIV/AIDS-related health services were starved of public funding, with households making 93% of the necessary expenditure, prompted the donor community to triple funding between 1998 and 2000 (Schneider et al., 2001). In Burkina Faso, the unequal distribution of expenditure shown by publication of national health accounts triggered the introduction of free subsidized public services, with the result that the household share of health spending fell from 50% in 2003 to 38% in 2008 (Zida et al., 2010).

The OECD has introduced a similar classification of education expenditure for OECD member states that permits useful comparisons across countries and over time. It shows, for example, that households bear the brunt of pre-primary education costs in some of these countries, such as Australia, where they account for 44% of spending, and Japan, at 38%. Even in primary and secondary education, households cover more than 20% of total expenditure in Chile and the Republic of Korea. The data also reveal that a share of education costs was shifted from government to households in the 2000s (OECD, 2013b).

Using a similar methodology, the joint OECD/UIS World Education Indicators programme provided information for 19 middle income countries, showing that a higher share of primary and secondary education costs was borne by households in these countries than in the richer OECD countries. For example, 28% of costs were met by households in India and 50% in Jamaica (UIS, 2006).

Several efforts in the last 10 years have bolstered the spread of national education accounts. Pilot accounting exercises have been carried out in El Salvador, Guatemala, Morocco and Turkey, with support in some cases from the World Bank and in others from USAID. They have lacked the common framework that has lent credibility and relevance to the national health accounts, however, and information has not yet emerged in a systematic way for poorer countries.

## Piecing together the financing puzzle

Adopting a national accounts approach to education for this Report shows what is possible from existing data across a diverse set of countries. Expenditure patterns were analysed for seven countries – Albania, Bangladesh, Indonesia, Malawi, Nicaragua, Rwanda and Tajikistan – drawing on data from 2007–2011.

Three types of data sources were used. First, to estimate public expenditure by education level and type, data from ministries of education and finance or international organization reports were used. Second, the OECD Creditor Reporting System database was drawn upon to identify aid data. Third, income and expenditure data from household surveys were used to estimate private education expenditure. The analysis shows that private spending is often much higher than has been recognized, which penalizes the poorest households; and that aid provides an important contribution to education financing in some of the poorest countries.

**Households in poorer countries bear a larger share of the responsibility for education spending**

Across the seven countries, governments are responsible, on average, for 65% of primary education spending, 51% at secondary level and 57% for tertiary institutions, with governments responsible for a larger share of overall spending in the richer countries in the sample – Albania and Indonesia – than in the poorer countries. As a result, households in poorer countries bear a larger share of the responsibility for education spending: the share of private expenditure in primary education ranges from 14% in Indonesia to 37% in Bangladesh. Households play an even more important role at secondary level, where the share of private expenditure ranges from 30% in Indonesia to 58% in Bangladesh (Figure 2.14).[8]

8. A comparison of household education spending in 15 African countries similarly showed that, despite the abolition of fees, household expenditure amounted to 33% of government spending on primary education (Foko et al., 2012).

By comparison, the share of private expenditure in primary and secondary education in high income OECD countries is 8.5% (OECD, 2013b). Education is therefore far from free – the goal envisaged in the Dakar Framework for Action – with households in poorer countries having to devote a particularly large share of their resources to education.

**In Malawi, households are responsible for 32% of secondary education expenditure**

The breakdown at tertiary level is further cause for concern: because higher education is subsidized in some poorer countries, those reaching this level, predominantly the better-off, benefit from government spending more than those who do not make it this far. In Malawi, for example, households are responsible for 32% of secondary education expenditure, but 21% at tertiary level.

Closer inspection shows the degree to which poorer and richer households benefit from public expenditure and the amount of extra expenditure they incur. In the case of primary education, government expenditure is progressive. As most children in the seven countries attend primary school, and as more of the richer families send their children to private school, poorer families benefit slightly more in relative terms from public expenditure.

In secondary education, however, the picture changes considerably. As fewer children of poorer families stay in school after primary education, they benefit less from government spending at secondary level. For example, in Malawi the richest 20% of families receive a public subsidy for secondary education that is almost six times as much as that received by the poorest 20% of families. In Indonesia, the ratio is almost three times. When public spending and private spending are added, a child from the richest 20% gets four times as much as a

**Figure 2.14: In low and middle income countries, families pay a heavy price for education**
Share of total education expenditure borne by government, donors and households, selected countries, 2007–2011


*Sources:* EFA Global Monitoring Report team analysis, based on data from OECD-DAC (aid), ministries of education and finance (public expenditure) and the following household surveys (private expenditure): 2008 Albania Living Standards Measurement Survey; 2010 Bangladesh Household Income and Expenditure Survey; 2009 Indonesia National Socio-economic Survey; 2010–11 Malawi Integrated Household Survey; 2009 Nicaragua National Living Standards Survey; 2011 Rwanda Integrated Household Living Conditions Survey; and 2007 Tajikistan Living Standards Measurement Survey.

child from the poorest 20% in Indonesia and Nicaragua, six times as much in Bangladesh and Rwanda and 10 times as much in Malawi.

One reason for the divergence between household contributions for primary and secondary schooling is that fees are usually still charged for secondary schools, while they have been abolished in many countries at primary level. Tuition fees in public secondary schools account for 64% of spending by the poorest households in Malawi compared with 45% spent by the richest households.

Spending by poor households on fees reduces the amount they can spend on items to improve the quality of their children's education, such as books and school supplies. In Bangladesh and Indonesia, the richest households spend four times as much on books as the poorest households in public lower secondary schools. In Bangladesh, the richest households spend 15 times as much on private tuition as poorer households for secondary education.

Analysing education spending across different sources reveals that, in the case of the two sub-Saharan African countries included, external assistance is a vital component of education financing. In Malawi, aid accounted for 23% of total spending on education in 2010/11. The share provided by donors was higher for primary education (30%) than secondary (16%) or tertiary (23%). In Rwanda, aid was equivalent to 22% of total government expenditure in 2010/11. By contrast, the share of external assistance in total education spending was 13% in Nicaragua, 9% in Albania (concentrated in tertiary education), 6% in Bangladesh, 4% in Tajikistan and less than 2% in Indonesia.

This preliminary analysis shows that a national education accounts system can be established using existing data – although improvements need to be made in terms of coverage and comparability between countries. The results can reveal whether resources are being used equitably. It is now time to build a comprehensive system to make sure that deficits in data do not thwart efforts to ensure that no child is denied the right to education after 2015. As a lead multilateral agency responsible for ensuring sufficient financing of education, the GPE could play a coordinating role in taking this forward.

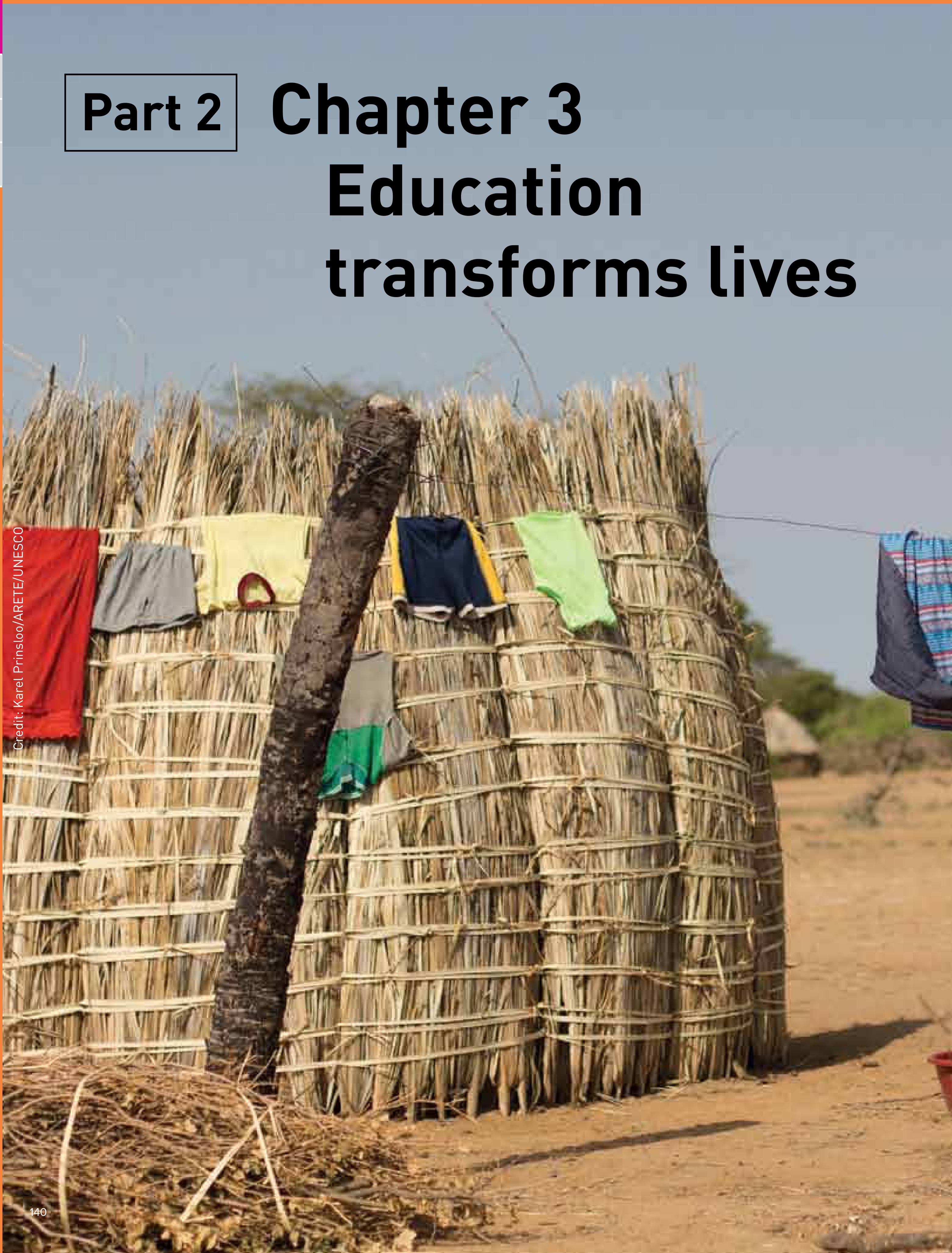
## Conclusion

One of the biggest failures of the EFA period has been fulfilling the pledge that no country would be thwarted in achieving its goals due to lack of resources. To avoid this happening after 2015, national governments, aid donors and other education funders need to be held to account for their commitments to provide the resources necessary to reach education goals. Post-2015 education goals must include a specific target for financing by governments and aid donors. Otherwise, children will continue to pay the price.

**Post-2015 education goals must include a specific target for financing by governments and aid donors**

Drawing on analysis included in *EFA Global Monitoring Reports* over the years, the Report team proposes that a target should be set for national governments to allocate at least 6% of their GNP to education. Targets for governments and aid donors should also include commitments for them to spend at least 20% of their budgets on education. Setting these targets, and making sure governments and aid donors keep to them, will be an important contribution to the education opportunities of a future generation of children and young people.

# Part 2 Chapter 3 Education transforms lives



Credit: Karel Prinsloo/ARETE/UNESCO

Educating mothers: In Kenyan villages like Nagis, Turkana, educating mothers can help reduce high levels of infant mortality.



Education lights every stage of the journey to a better life. To unlock the wider benefits of education, all children need access to both primary and lower secondary education of good quality. Special efforts are needed to ensure that all children and young people – regardless of their family income, where they live, their gender, their ethnicity, whether they are disabled – can benefit equally from its transformative power. This chapter provides comprehensive evidence that progress in education is vital to achieve all new development goals after 2015.

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## Introduction

Treaties and laws worldwide recognize that education is a fundamental human right. The Education for All goals and the Millennium Development Goals acknowledge its indispensable role in imparting the knowledge and skills that enable people to realize their full potential. But the international community and national governments have so far failed to sufficiently recognize and exploit education's considerable power as a catalyst for other development goals.

As a result, education has been slipping down the global agenda and some donors have moved funds elsewhere, at the very time when education's wider benefits are sorely needed to help countries get back on track to reach other development goals. Education's power to accelerate the achievement of wider goals needs to be much better recognized in the post-2015 development framework. This chapter shows why, drawing on new analysis carried out for this Report and a thorough survey of other research to build a comprehensive overview of the numerous ways in which education advances wider development aims.

This analysis sheds new light on education's better-known benefits: the first part of the chapter examines its power to reduce poverty, boost job opportunities and drive economic growth, and the second part shows how it increases people's chances of leading a healthy life. In the third part the chapter goes beyond these questions to look in depth at education's contribution to social goals that are increasingly being recognized as vital elements of the post-2015 framework: deepening democratic institutions, protecting the environment and adapting to climate change, and empowering women.

Several common themes emerge from the evidence laid out in this chapter, each of which must be recognized and acted upon by governments and their development partners in order to realize the wider benefits of education. One persistent thread is the transformative power of girls' education. As well as boosting their own chances of getting jobs, staying healthy and participating fully in society, educating girls and young women has a marked impact on the health of their children and on accelerating countries' transition to stable population growth with low birth and death rates.

Another common thread is that universal primary education is vital – but is often not enough. To unlock the wider benefits of education, especially its power to save lives, universal access needs to be extended at least to lower secondary school. And access alone is not enough either: the education that children receive needs to be of good quality so that they actually learn the basic literacy and numeracy skills that are necessary to acquire further skills.

Above all, the evidence assembled here underlines the imperative that good quality education must be made accessible to all, regardless of their income, where they live, their gender, their ethnicity, whether they are disabled, and other factors that can contribute to disadvantage. Education's unique potential to boost wider development goals can only be fully realized if education is equitable, which means making special efforts to ensure that the marginalized can benefit equally from its transformative power.

**Education's power to accelerate the achievement of wider goals needs to be much better recognized in the post-2015 framework**

## Education reduces poverty and boosts jobs and growth

Education is a key way of tackling poverty, and makes it more likely for men and women not just to be employed, but to hold jobs that are more secure and provide good working conditions and decent pay. It also lays the foundations for more robust and longer-term economic growth.

### Education offers the poor a route to a better life

**If all students in low income countries left school with basic reading skills, 171 million people could be lifted out of poverty**

For the poor, education is one of the most powerful routes to a better future. As the evidence in this section shows, education enables people to escape from the trap of chronic poverty and prevents the transmission of poverty between generations. Good quality education that improves learning outcomes increases economic growth (Hanushek and Woessmann, 2012b). In turn, economic growth reduces poverty because it tends to increase wages and the amount people can earn from work in agriculture and the urban informal sector (Ravallion, 2001). Based on these two core assumptions, EFA Global Monitoring Report team calculations show that if all students in low income countries left school with basic reading skills, 171 million people could be lifted out of poverty, which would be equivalent to a 12% cut in world poverty.

The persistence of poverty makes it vital for policy-makers to take into account education's power to help reduce it. The proportion of the world's people living on less than US$1.25 a day fell from 47% in 1990 to 22% in 2010 (United Nations, 2013). However, almost 1 billion people are still likely to be living below this poverty line in 2015. In sub-Saharan Africa, almost half the population was living in poverty in 2010, and the number of poor is growing: by 2015, around 40% more people are expected to be living in poverty than in 1990. Sub-Saharan Africa's experience is in stark contrast to the experience of East Asia and the Pacific, where about 56% of the population lived below the poverty line in 1990. By 2010, the share had fallen to just 12.5%, due in part to the availability of good quality education (World Bank, 2013a).

The Millennium Development Goal (MDG) of halving the proportion of people living in poverty between 1990 and 2015 has been met, but many are proposing the eradication of poverty by 2030 as a new post-2015 goal. For the reasons outlined here, it is crucial for education to be an integral part of all plans to make poverty a thing of the past.

#### *Education accelerates an escape from chronic poverty*

For some people, poverty is transitory: they may experience brief periods of being poor or frequently move into or out of poverty. But for the more vulnerable, poverty is a chronic state: they remain poor for long periods, even all their lives, passing on their poverty to their children. For example, in Kenya three-quarters of rural households that were poor in 2000 were still poor in 2009 (Radeny et al., 2012).

Education is a key way of reducing chronic poverty, because increased levels of education not only help lift households out of poverty permanently but also guard against them falling into poverty. Evidence from diverse settings confirms that even after taking account of other factors that can have an influence, such as household land holdings and other assets, education at all levels reduces the chance of people living in chronic poverty:

- Although Ethiopia has reduced poverty by half since 1995, 31% of the population still lives in poverty (World Bank, 2013b). Raising levels of education, which are particularly low in rural areas, can make a clear difference. Between 1994 and 2009, for example, rural households where the household head had completed primary education were 16% less likely to be chronically poor (Dercon et al., 2012).

- The probability that poor households in Uganda would remain poor over a seven-year period fell by 16% for each year the household head had spent in secondary school and by a further 5% for each year the spouse had spent in primary school (Lawson et al., 2006).

- In KwaZulu-Natal province of South Africa, where education helped people escape poverty after the apartheid era, an additional year of schooling increased consumption expenditure by 11% (May et al., 2011).

- Extra years in school are more likely to help lift people out of poverty when schooling is of good enough quality to give people the basic skills they need, such as literacy. In Nepal, where more than half those in poverty remained poor over a 10-year period, 42% of adults were literate in households that remained above the poverty line throughout the period, 23% were literate in households that made a transition into or out of poverty, but only 15% were literate in chronically poor households (Bhatta and Sharma, 2011).

- Education's power to help people escape poverty even in the face of adversity is evident in a rural district in Sindh province of Pakistan, where poverty increased over a 17-year period due to drought and water shortages. The heads of households who remained poor throughout had both the lowest initial level of education (by 1 year, on average) and the lowest increase over the period (less than 1 year). By contrast, the heads of households that escaped poverty had a higher initial level of education (1.8 years) as well as the highest increase over the period (by 2 years) (Lohano, 2011).

- Getting at least as far as lower secondary school has a particularly strong effect, in a wide range of settings. Among households in rural Viet Nam, those whose heads had lower secondary education were 24% more likely not to be poor between 2002 and 2006 than households with no schooling, and likelihood for those with upper secondary education was 31% higher (Baulch and Dat, 2011). In rural Nicaragua, the chronic poverty rate was 22% for households with no educated adult, 7% for households where adults averaged three years of schooling and 1% for households where adults had six years of education or more (Stampini and Davis, 2006). In urban Brazil between 1993 and 2003, chronic poverty was 29% for those with no education, 22% for those with incomplete primary education, 12% for those with primary education and 7.5% for those with at least lower secondary education (Ribas and Machado, 2007).

- Education's power to keep people out of poverty is evident in rural Indonesia. Of all the factors making it more likely that a household would exit poverty over a seven-year period, education was the most important. Each additional year of schooling increased income growth by 6% over the seven years (McCulloch et al., 2007). Completing lower secondary education not only more than doubled the probability of escaping poverty, but also reduced by a quarter the probability of falling into poverty.

**In rural Indonesia, completing lower secondary education more than doubled the probability of escaping poverty**

### *Education prevents the transmission of poverty between generations*

Children whose parents have little or no schooling are more likely to be poorly educated themselves. This is one of the ways poverty is perpetuated, so raising education levels is key to breaking the cycle of chronic poverty. The right policies can ensure that the benefits are passed on equitably. New analysis for this Report, based on 142 Demographic and Health Surveys from 56 countries between 1990 and 2009, looks at how parental education affects the number of years of education attained by household members aged 15 to 18. For each additional year of mother's education, the average child attained an extra 0.32 years, and for girls the benefit was slightly larger (Bhalotra et al., 2013b).

In Guatemala, higher levels of education and cognitive skills among women increased the number of years their children spent in school. In turn, each grade completed raised the wages of these children once they became adults by 10%, while an increase in the reading comprehension test score from 14 points to the mean of 36 points raised their wages by 35% (Behrman et al., 2009, 2010).

In poor countries parental education can even be more powerful than other possible factors, such as inheriting assets. In Senegal, for example, inheriting land or a house did not increase wealth, but children whose parents had some formal education were more likely to find off-farm employment and so escape poverty. In particular, the sons of educated mothers in rural areas were 27% more likely to find off-farm employment (Lambert et al., 2011).

### *How education reduces poverty*

The main way education reduces poverty is by increasing people's income. It enables those in paid formal employment to earn higher wages, and offers better livelihoods for those who work in the urban informal sector or in rural areas.[1]

*Education helps provide a decent wage.* Globally an estimated 400 million people – or 15% of all workers – are paid less than US$1.25 per day, too little to enable them to lift themselves and their families out of poverty (ILO, 2013a). Young people are particularly vulnerable to working in poverty, with 28% in this situation (Understanding Children's Work, 2012).

**In Pakistan, the wages of a literate person are 23% higher than those of an illiterate person**

Education can help people escape from working poverty. In the United Republic of Tanzania, 82% of workers who had less than primary education were earning below the poverty line. By contrast, working men and women with primary education were 20% less likely to be poor, while secondary education reduced the chances of being poor by almost 60%. In Brazil, while working in poverty is less common – and almost non-existent for those with at least secondary education – 13% of those lacking primary education are consigned to this situation (Understanding Children's Work, 2013).

Better-educated individuals in wage employment are paid more to reward them for their higher productivity. On average, one year of education is estimated to increase wage earnings by 10%; in sub-Saharan Africa, as much as 13% (Montenegro and Patrinos, 2012). And, on average, returns to investment in education increase by level of education, from 8% at primary level to 13% at secondary level and 16% at tertiary level (Colclough et al., 2010).

In some countries, education has a stronger effect on the wages of the highest earners. In Colombia, for example, the return to investment in education for those whose wages are in the bottom one-tenth, or decile, is 9%, compared with 14% for those in the top decile. This may reflect the better quality of education that wealthier people receive. By contrast, in Viet Nam, where the education system is more equal, returns to education are higher for the bottom decile, with returns estimated at 10%, compared with 6% for the top decile (Fasih et al., 2012).

It is necessary to assess the returns to skills acquired in school, such as literacy and numeracy, rather than focusing only on time spent in school. Analysis in 13 richer countries showed that for every 60-point increase on a 500-point literacy skills scale, wage earnings rose from about 5% in Italy to 15% in Chile (Hanushek and Zhang, 2006).

Similarly, in Pakistan the wages of a literate person are 23% higher than those of an illiterate person. Improved literacy can have a particularly strong effect on women's earnings, suggesting that investing in women's education can pay dividends. Working women with a high level of literacy skills earned 95% more than women with weak or no literacy skills, whereas the differential was only 33% among men (Aslam et al., 2012).

*Education increases earnings from informal work.* In urban areas, many of the poorest are involved in informal sector work. Education can help lift these people out of poverty by increasing their earnings and enabling them to benefit more from entrepreneurial activity. More educated people are more likely to start a business, and their businesses are likely to be more profitable.

In eight sub-Saharan African countries, for example, people who had completed primary education were more likely to own household enterprises or microenterprises than those with less education. In Uganda, owners of household enterprises who had completed primary education earned 36% more than those with no education, and those who had completed lower secondary education earned 56% more (Fox and Sohnesen, 2012). In Angola, an additional year of schooling increased profits of small businesses by 7% to 9.5% (Wiig and Kolstad, 2013).

In Viet Nam, a survey of 1,400 new businesses, of which 91% were micro or small and 61% were household enterprises, showed that having at

1. Education is also linked to other factors that lead to poverty reduction. For example, it reduces fertility, which leads to fewer dependents per family.

least secondary education raised profits by 34% (Santarelli and Tran, 2013). In Thailand, a year of education increased returns to household assets by 7%, primarily because educated households tended to invest the profits (Pawasutipaisit and Townsend, 2011).

*Education increases earnings from agriculture.* In low and middle income countries, many of the poorest depend on farming for their livelihoods (World Bank, 2013a). Education can offer these people a crucial route to a better life.

Literacy and numeracy skills increase farmers' ability to interpret and respond to new information. Educated farmers make better use of fertilizers, seed varieties and farming technologies to increase the productivity of traditional crops. They are also more likely to diversify into higher value crops.

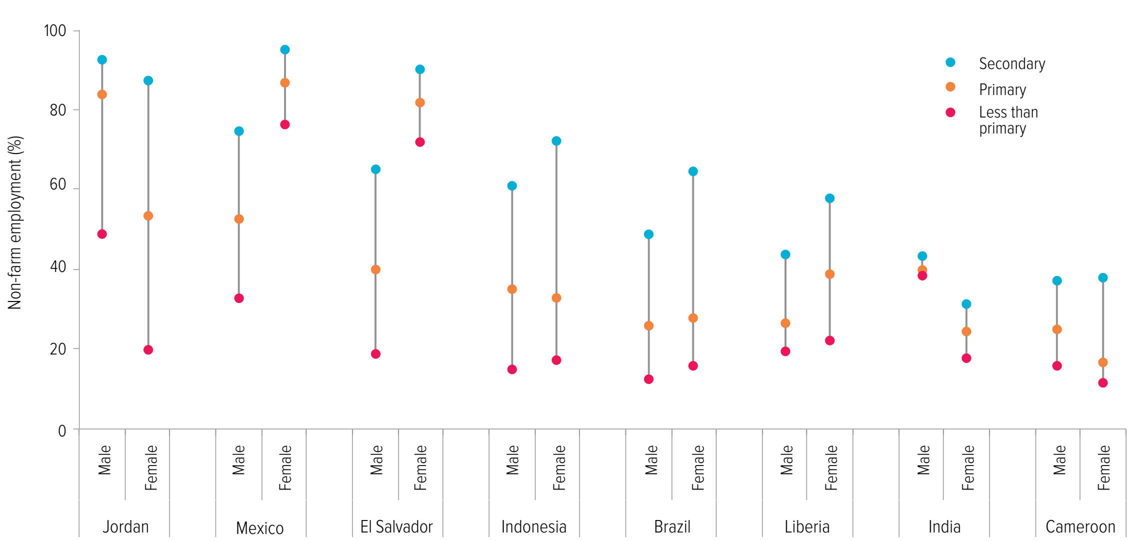
Examples from around the world have shown the power of education to boost farmers' productivity in ways that can help lift them out of poverty. In semi-arid areas of China, educated farmers were more likely to use rainwater harvesting and supplementary irrigation technology to alleviate water shortages (He et al., 2007). In Ethiopia, education of household members other than the household head led to increased fertilizer use (Asfaw and Admassie, 2004). In Mozambique, literate farmers were 26 percentage points more likely than non-literate ones to cultivate cash crops (Bandiera and Rasul, 2006). In Nepal, a household where a farmer had completed primary education was about 20 percentage points more likely to adopt soil conservation and erosion-control measures (Tiwari et al., 2008).

The returns to education are larger where farmers are able to make use of new technologies. In northern Nigeria, when the household head had four years of education, production of cowpeas increased by a quarter with modern technology but was unchanged where traditional methods were used (Alene and Manyong, 2007).

*Education enables rural households to diversify their income sources.* Many of the world's poor living in rural areas have no choice other than to work on small farms. Education allows households to respond flexibly and diversify their income-earning opportunities. It improves their chances of obtaining non-farm work – which tends to be more lucrative – as analysis of labour force survey data for this Report shows. For example, in Indonesia, where half the population lives in rural areas, the share of rural dwellers employed in non-farm work is 15% of men and 17% of women with no education, but among those with secondary education increases to 61% of men and 72% of women (Figure 3.1).

**In Nepal, a household where a farmer had completed primary education was more likely to adopt soil conservation measures**

**Figure 3.1: Education opens doors to non-farm employment**
Non-farm employment rate among people aged 15 to 64 in rural areas, by education level and gender, selected countries, 2007–2011




*Source:* Understanding Children's Work (2013).

## Education transforms employment prospects

With more young people than ever entering the world of work, the poorest and most vulnerable risk being left unemployed or working below the poverty line in insecure work, because they are the least likely to have completed school. Even in richer countries, the economic downturn is leaving those with less education facing the prospect of long-term unemployment.

Education makes it more likely for men and women not just to be employed, but to hold jobs that are more secure and provide good working conditions and decent pay. Education is particularly vital for women so that they can benefit from decent jobs, becoming able to make decisions about spending resources and hence gaining more control over their own lives.

**Education plays a strong role in determining women's engagement in the labour force in Latin American countries**

The importance of jobs in reducing poverty was recognized eight years after the MDG framework was established, when a new target was added as part of the first MDG: to achieve full and productive employment and decent work for all, including women and young people. An additional indicator for women's engagement in wage employment was also included under the third MDG, on gender equality, in recognition of the fact that better education improves women's livelihoods.

As a way of improving people's well-being and eradicating poverty, jobs are likely to feature in the post-2015 development framework. The International Labour Organization argues that employment should be given an even stronger focus than in the MDGs, with a goal in its own right this time (ILO, 2013b). This argument is reinforced by the views of poor people, who see jobs and education as two of the most important ways to improve their lives (Bergh and Melamed, 2012).

### *Education protects against unemployment in rich countries*

Globally, 193 million people were estimated to be unemployed in 2011, including 73 million young people – equivalent to 1 in 8 of the youth population (ILO, 2013a). Unemployment has soared in some richer countries that have borne the brunt of the economic downturn. Young people in particular have been severely hit, with their unemployment rates reaching over 50% in Greece and Spain.

Those with higher levels of education suffer less from the effects of the economic downturn: in Spain, while unemployment rates for those with less than secondary education rose from around 20% in 2007 to 60% in 2012 , they increased from 14% to 40% for those with higher levels of education (Eurostat, 2013).

### *Education increases women's chances of participating in the labour force*

As countries develop, education becomes a passport for women to enter the labour force. When society becomes more accepting of women's formal employment, women with more education are in a stronger position to get paid work (Gaddis and Klasen, 2012). By enabling women's participation in the labour market, education contributes to their empowerment and to their country's prosperity (Kabeer, 2012).

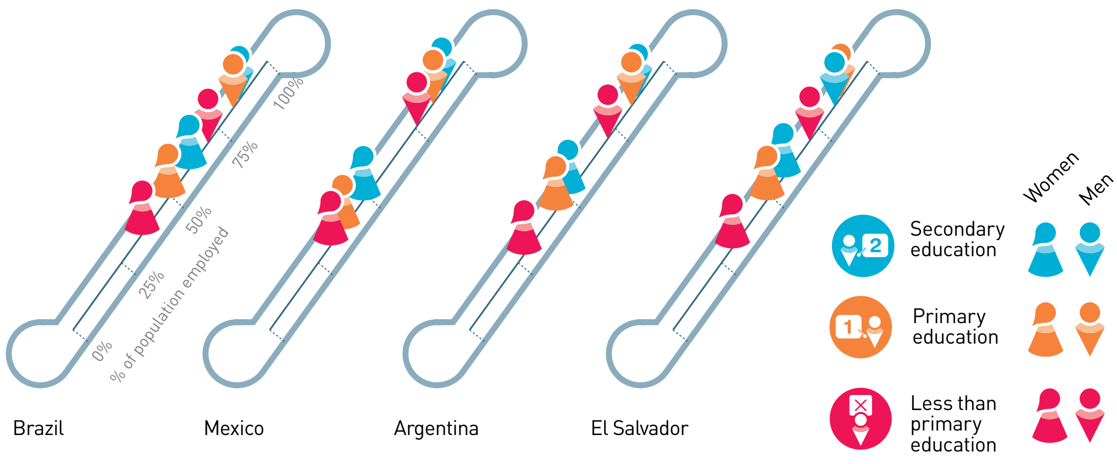
In middle income countries in Latin America, such as Argentina, Brazil, El Salvador and Mexico, the proportion of women in paid employment increases sharply as women's education level rises, according to analysis of labour force survey data for this Report (Infographic: Job search). In Mexico, while 39% of women with primary education are employed, the proportion rises to 48% of those with secondary education.

Education plays a much stronger role in determining women's engagement in the labour force than it does for men in these Latin American countries. Men who have not even completed primary school are as likely to be engaged in the labour market as women with tertiary education in all four countries. And there is very little difference in participation between men with primary and secondary education: it is over 80% regardless of level of education achieved.

In poorer countries, cultural factors and a lack of affordable child care facilities and transport continue to prevent women from taking paid jobs. In India and Pakistan, for example, women are less likely to be counted as participating in the labour market whether they have been to

# JOB SEARCH

Educated men and women are more likely to find work




*Source:* Understanding Children's Work (2013).

school or not. The only exception is the extremely small numbers of women who make it to upper secondary school in Pakistan. Only around one in five make it to this level, and they are likely to be from more privileged backgrounds (Aslam, 2013). By contrast, the vast majority of men are in the labour force in the two countries, regardless of their education.

Women are kept out of the labour force not only by cultural stigma associated with taking paid employment but also by social expectations related to family size and household chores, which mean they often put in long hours in work that is less visible to policy-makers, particularly in poorer countries (Bloom et al., 2009). For example, in Jordan, 25% of rural women with only primary education work for no pay, compared with 7% of those with secondary education.

### *Education increases the chances of better work conditions*

Those with more education tend to enjoy better work conditions, including opportunities to work full time and secure contracts. Although some choose part-time jobs, for the most vulnerable part-time work is low paid and insecure.

Education increases the chance of getting full-time work, particularly for women. Among working women in Pakistan, while one-third of those with primary education work full time, one-half of women with secondary education have full-time jobs. By contrast, almost all men work full time, regardless of their level of education.

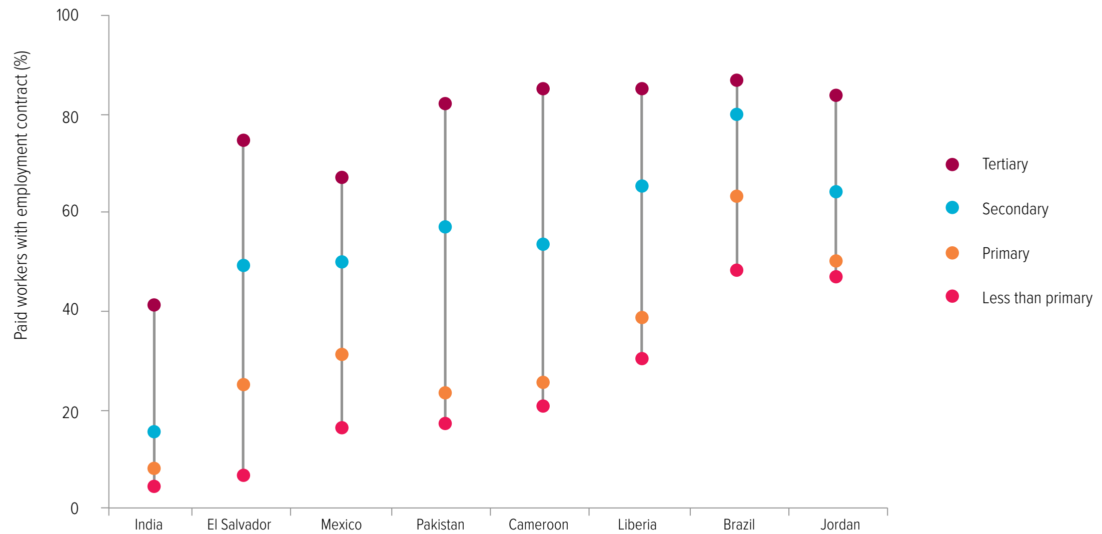
Education helps protect working men and women from exploitation by increasing their opportunities to obtain secure contracts. In urban El Salvador, only 7% of working women and men with less than primary education have an employment contract, leaving them very vulnerable. By contrast, 49% of those with secondary education have signed a contract (Figure 3.2).

### *Education closes gender wage gaps*

Globally, women are paid less than men for comparable work. Even though this gap has been narrowing in some parts of the world, it remains a cause for concern (OECD, 2012). The higher the level of education, the lower the gap, even in countries where discrimination in the labour force means gender differences remain entrenched, as analysis of gender wage gaps in 64 countries shows. Education makes a

**In Jordan, 25% of rural women with only primary education work for no pay, compared with 7% of those with secondary education**

**Figure 3.2: Education leads to more secure jobs**
Percentage of paid workers aged 15 to 64 in urban labour markets with an employment contract, by education level, selected countries, 2007–2011




*Note:* The figures for Pakistan refer to both urban and rural areas.
*Source:* Understanding Children's Work (2013).

**In the Arab States, women with secondary education earn 87% of the wages of men, compared with 60% for those with primary education**

particular difference in the Arab States, where women with secondary education earn 87% of the wages of men, compared with 60% for those with primary education (Ñopo et al., 2011). In sub-Saharan Africa, men earn twice as much as women on average, but education has a strong effect on closing the earnings gap. In Ghana, among those with no education, men earn 57% more than women, but the gap shrinks to 24% among those with primary education and 16% with secondary education (Kolev and Sirven, 2010).

These regional patterns are confirmed by analysis for this Report of data from nine countries in diverse settings. In Argentina and Jordan, among people with primary education, women earn around half the average wage of men, while among those with secondary education, women earn around two-thirds of men's wages (Infographic: Wage gaps).

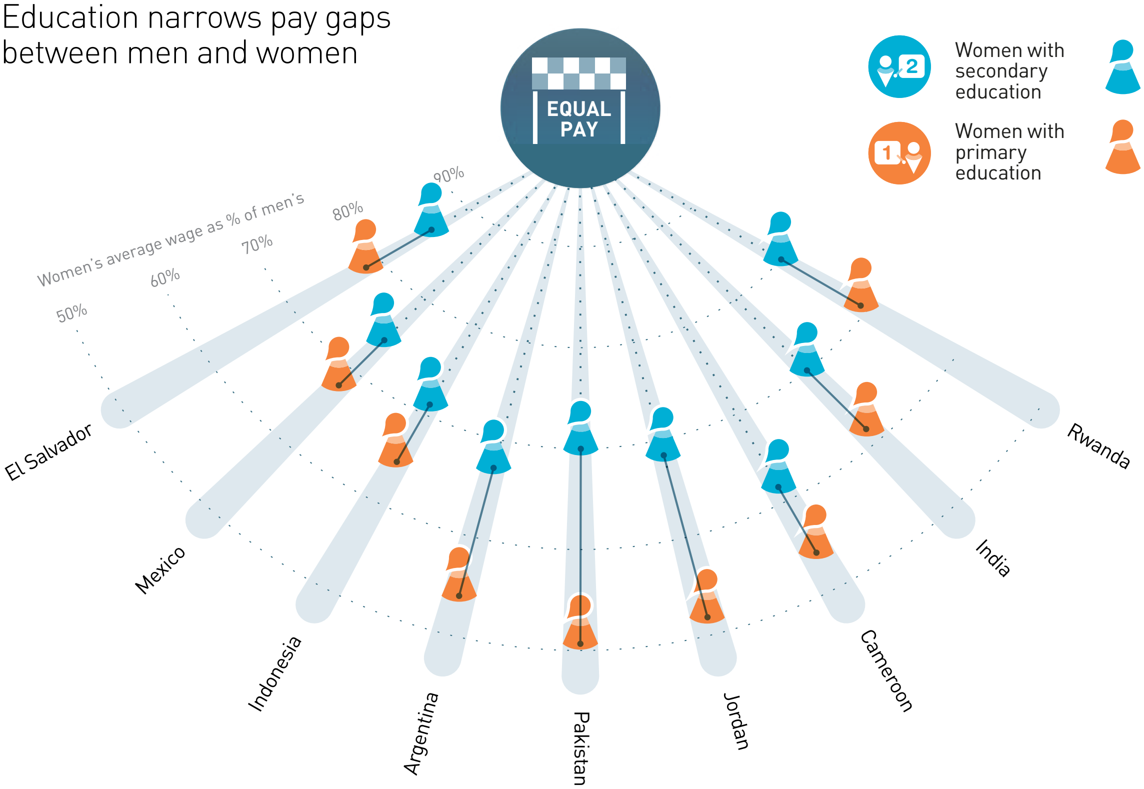
## Education boosts prosperity

Education not only helps individuals escape poverty by developing the skills they need to improve their livelihoods, but also generates productivity gains that fuel economic growth. For growth to reduce poverty, education needs to overcome inequality. Expanding access to education alone is not enough, however. Equitable learning for all is key to shared national prosperity for all.

Some analysts have expressed reservations about education's effect on economic growth (Pritchett, 2006). But when improved data and methods are used, and the impact of education inequality and education quality is taken into account, it is clear that education is a key force for national prosperity (Castelló-Climent, 2010; Hanushek and Woessmann, 2008; Krueger and Lindahl, 2001).

# WAGE GAPS

Education narrows pay gaps between men and women




*Source:* Understanding Children's Work (2013).

*Education can fuel economic growth ...*

Education's power to boost growth substantially is underlined by new analysis for this Report. An increase in the average educational attainment of a country's population by one year increases annual per capita GDP growth from 2% to 2.5%.[2] These estimates take into account factors such as the level of income at the beginning of the period, the level of inflation, the share of the public sector in the economy and the degree of openness to trade (Castelló-Climent, 2013).

The key role of education is evident in the way it helps explain differences between regions in the pace of economic growth. In 1965, the average level of schooling was 2.7 years higher in East Asia and the Pacific than in sub-Saharan Africa. Over the following 45-year period, average annual growth in income per capita was 3.4% in East Asia and the Pacific but only 0.8% in sub-Saharan Africa. The difference in initial education levels could help explain about half the difference in growth rates.

Comparing experiences within regions further illustrates the importance of education. In Latin America and the Caribbean, the average number of years that adults had spent at school rose from 3.6 in 1965 to 7.5 in 2005. This is estimated to have contributed two-thirds of the average annual growth rate in GDP per capita of 2.8% between 2005 and 2010. But not all countries in the region kept pace. In Guatemala, adults had just 3.6 years of schooling on average in 2005, equivalent to Côte d'Ivoire in sub-Saharan Africa, and on average schooling increased by only 2.3 years in the country from 1965 to 2005, the second lowest rate in the region, after Belize. If Guatemala had matched the regional average, it could have more than doubled its average

**An increase in the average educational attainment of a country's population by one year increases annual per capita GDP growth from 2% to 2.5%**

2. The effect is broadly compatible with the earlier, much quoted result that each additional year of average education attainment increased per capita income by 12% among 73 countries over 1960–1990 (Cohen and Soto, 2007).

annual growth rate between 2005 and 2010, from 1.7% to 3.6%, equivalent to an additional US$500 per person (Castelló-Climent, 2013). A major reason for Guatemala's poor performance is that members of indigenous groups have historically received half as many years of schooling as non-indigenous people (Shapiro, 2006).

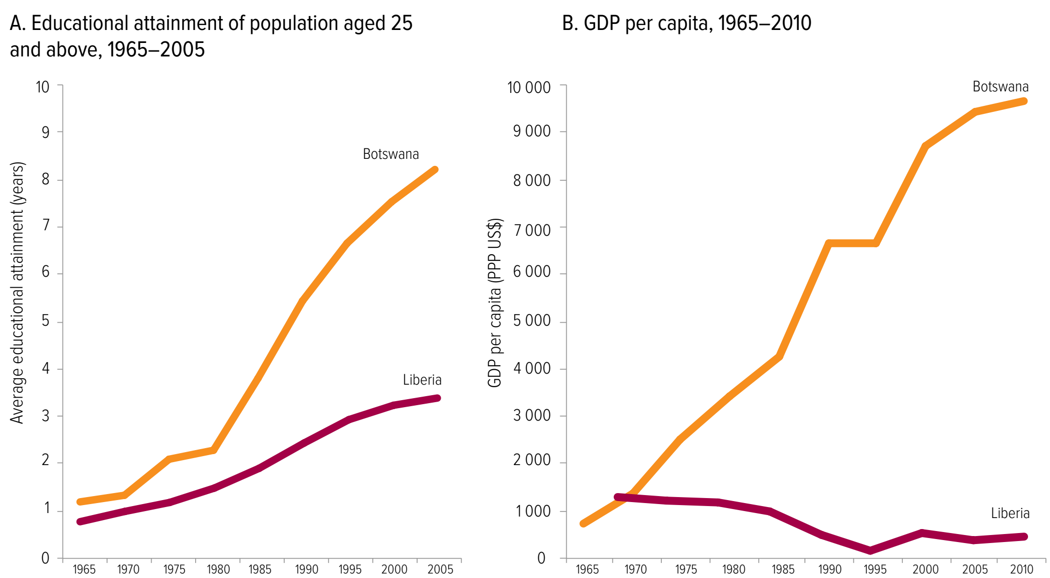
**If Thailand had matched the regional average of years in school, its annual growth rate between 2005 and 2010 could have reached 3.9% instead of 2.9%**

To take another example, average schooling attainment in Haiti lagged behind the regional average by three years throughout the analysis period. If it had been equal to the average for Latin America and the Caribbean, per capita income could have grown annually by 1.5% to almost double between 1965 and 2010, when in fact it stagnated (Castelló-Climent, 2013). This stagnation has meant that Haiti has remained the poorest country in the region, while it could have achieved the same level of income per capita as Nicaragua. Though its more recent problems are due to the effects of natural disasters, including the devastating earthquake in 2010 which destroyed much of the country's education infrastructure, its inability to invest in education adequately has meant it continues to face significant challenges in rebuilding its economy. By 2012, 51% of women and 46% of men aged 15 to 49 had no more than primary education (Institut Haïtien de l'Enfance and ICF International, 2012).

Even in East Asia and the Pacific, where growth has been high thanks to investment in education as well as economic and institutional reform, countries whose education spending has been falling – such as Thailand – risk prolonged periods of low growth (Eichengreen et al., 2013). In 2005, the average number of years that adults in Thailand had spent at school was 5.9, two years below the regional average, having increased by 2.8 years since 1965. If Thailand had matched the regional average, its average annual growth rate between 2005 and 2010 could have reached 3.9% instead of 2.9% (Castelló-Climent, 2013).

In sub-Saharan Africa, the diverging paths of Botswana and Liberia, which started at similar levels of economic and educational development and are both rich in diamonds, provide striking evidence of the benefits of education. In 1970, income per capita was about the same in the two countries. On average, adults in both countries had spent about one year in school, just below the regional average. While Botswana invested its natural resource wealth in education and other social sectors, Liberia's was until recently largely squandered, partly on financing two civil wars. The difference is reflected in spending on education: in 2009, Botswana spent 8.2% of GNP on education, twice the level of Liberia. As a result, over a 40-year period, the level of schooling increased by eight years in Botswana but only three years in Liberia, well below the average for the region. By 2010, GDP per capita was more than 20 times higher in Botswana (Figure 3.3).

**Figure 3.3: Investment in education helps deliver growth**




*Source:* Barro and Lee (2013).

*Source:* Heston et al. (2012).

These experiences highlight why it is a mistake for countries to reduce their investment in education, even during economic downturns. Doing so not only denies people their right to education, but also jeopardizes future prosperity. Yet countries including Greece, Italy and Portugal reduced education spending by more than 5% between 2010 and 2012, at the time when their economies needed this investment the most (European Commission et al., 2013).

### *... but education must be equitable to reap economic rewards ...*

Poor countries cannot escape the low income trap just by increasing the average years of education. New analysis for this Report shows that only by investing in equitable education – making sure that the poorest, as well as the richest, complete more years in school – can countries achieve the kind of growth that banishes poverty. The countries that have experienced the highest levels of economic growth, including China, the Republic of Korea and Taiwan Province of China, have done so while reducing education inequality.

The new analysis shows that education inequality substantially hinders a country's growth prospects. Education equality can be measured using the Gini coefficient, which ranges from zero, indicating perfect equality, to one, indicating maximum inequality. An improvement in the Gini coefficient by 0.1 accelerates growth by half a percentage point, which increases income per capita by 23% over a 40-year period.

Education in sub-Saharan Africa is particularly unequal. If the region's education Gini coefficient of 0.49 had improved to the level in Latin America and the Caribbean (0.27), the annual growth rate in GDP per capita over 2005–2010 could have been 47% higher (from 2.4% to 3.5%) and per capita income could have grown by US$82 per capita over the period.

Differences in education inequality are one reason for the wide variations in growth rates among countries in the East Asia and the Pacific region over the past 40 years. The Republic of Korea reduced inequality in education 50% faster than countries such as the Philippines. This has resulted in very different paces of economic growth. Over the period, average annual growth in GDP per capita was 5.9% in the Republic of Korea but 1.5% in the Philippines.

Comparing Pakistan and Viet Nam illustrates starkly the importance of equitable education. In 2005, the average number of years adults had spent at school was similar: 4.5 in Pakistan and 4.9 in Viet Nam. Education levels, however, were very unequally distributed in Pakistan, where 51% of the population had no education, compared with only 8% in Viet Nam. By contrast, 33% had post-primary education in Pakistan compared with 21% in Viet Nam. The Gini coefficient for education inequality was 0.60 in Pakistan, more than double the level in Viet Nam (0.25).

The difference in education inequality between the two countries accounts for 60% of the difference in their per capita growth between 2005 and 2010. Viet Nam's per capita income, which was around 40% below Pakistan's in the 1990s, not only caught up with Pakistan's but was 20% higher by 2010, and a continued widening of the gap is seen as likely (Infographic: Educated growth).

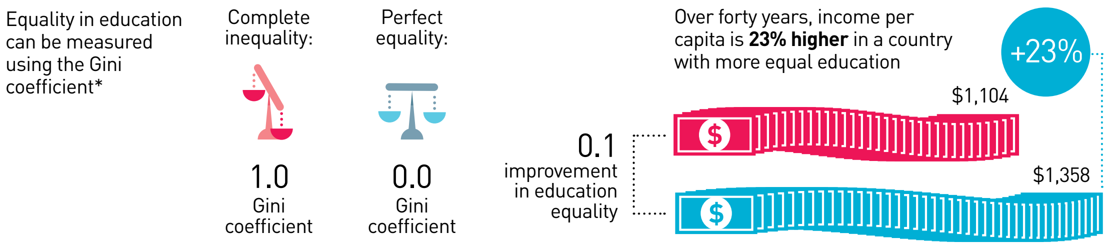
While there are still pockets of inequality in access to education at the secondary level in Viet Nam, with ethnic minority children lagging behind the national average, almost all children complete primary school regardless of background (Viet Nam General Statistical Office, 2011). Among the poorest females aged 15 to 24, about 17% had not completed primary school and 37% had not completed lower secondary school in 2010. By contrast, in Pakistan in 2006, the corresponding figures were 89% and 96% (UNESCO, 2013).

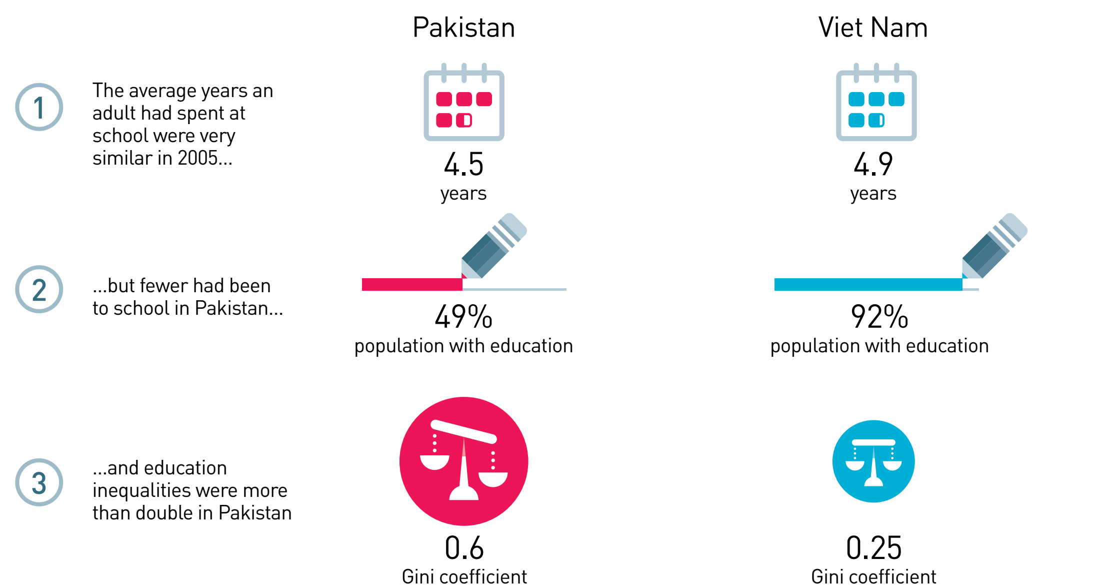
If Pakistan were to halve inequality in access to education to the level of Viet Nam, it would increase its economic growth by 1.7 percentage points; its actual average growth in GDP per capita between 2005 and 2010 was 2.5%. This comparison highlights the urgent need for increased government investment in education in Pakistan, which devoted less than 10% of its budget to education in 2011. Spending needs to be targeted at the poorest girls in rural areas in particular, not only to benefit the girls and their families, but also for the sake of the country's prosperity.

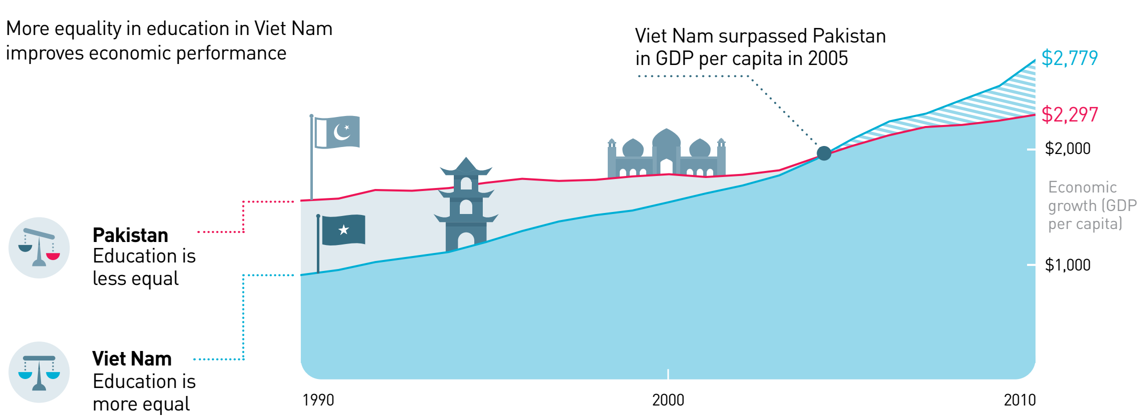
**The Republic of Korea reduced inequality in education 50% faster than the Philippines resulting in very different paces of economic growth**

# EDUCATED GROWTH

## Education equality accelerates prosperity




*A statistical measure of inequality. Perfect equality (where everyone goes to the school for the same amount of time) would equal 0 and perfect inequality (where only one person goes to school) would equal 1

*Sources:* Barro and Lee (2013); Castelló-Climent (2013); Heston et al. (2012).

*... and quality of education is vital for economic growth*

Spending more time in school is not enough – children need to be learning. Where quality of education is low, the skills base of the economy is low and cannot become an engine of growth.

In recent years, increases in education quality as measured by test scores have been linked to increases in annual per capita growth rates. If Mexico had raised its mathematics score in the OECD Programme for International Student Assessment (PISA) by 70 points, to reach the OECD average, this would have boosted its annual per capita growth rate by about 1.4 percentage points (Hanushek and Woessmann, 2012a). As its growth rate was 1.5% between 1990 and 2010, this means the rate would have almost doubled.

The returns to increasing learning outcomes make investment in quality very attractive. Estimates made for the Report, linking improvement in educational outcomes with economic growth, have attempted to quantify the impact of quality-enhancing investment (Hanushek and Woessmann, 2012b). The starting point is a reform in the education system costing the equivalent of 1.8% of GDP that would increase PISA scores by an estimated 50 points (or the equivalent on other assessment tests). The improved learning outcomes, in turn, would raise the per capita income growth rate by 1 percentage point per year: income per capita would rise by more than 60% over the 50 years after completion of the reform. If, as a result of the reform, 75% of students who took the PISA test passed the 400-point benchmark, rather than about 10% to 15% (which would be similar to Kyrgyzstan or Tamil Nadu, scoring at the lowest end, reaching the level of Italy or Russia, scoring just below the average), it would mean every US$1 invested in education yielded US$10 to US$15 in higher national income.

## Education improves people's chances of a healthier life

Education is one of the most powerful ways of improving people's health – and of making sure the benefits are passed on to future generations. It saves the lives of millions of mothers and children, helps prevent and contain disease, and is an essential element of efforts to reduce malnutrition. But this key role is seldom appreciated. Policy-makers focusing on health often neglect the fact that education should be regarded as a vital health intervention in itself – and that without it, other health interventions may not be as effective.

The complementarity between education and health also works in the other direction: people who are healthier are more likely to be better educated. Even taking these links into account, however, there is strong evidence, presented in this section, that education consistently increases people's chances of leading a healthy life.

Globally there has been significant progress towards the health targets laid out in the MDGs – reducing child mortality (goal 4), reducing maternal mortality (goal 5) and combating AIDS, malaria and other diseases (goal 6), as well as reducing hunger, which is part of the core goal on poverty reduction (goal 1). As this section shows, education has contributed to these advances. But more could have been achieved if education's power had been better tapped.

Education will continue to be crucial in achieving global post-2015 health targets, which are likely to go beyond the MDGs to include elimination of hunger and of preventable child and maternal deaths. 'Healthy life expectancy', the overarching post-2015 health goal preferred by many experts, implies reducing not only mortality but also the incidence of disease and disability (WHO, 2012a). These are threats that education can help hold at bay.

**Investing the equivalent of 1.8% GDP on improving learning would raise the per capita income growth rate by 1 percentage point per year**

**If all women completed secondary education, the under-5 mortality rate would fall by 49% in low and lower middle income countries**

*How education contributes to better health*

Schooling increases people's chances of leading a healthy life in many different ways. Educated people are better informed about specific diseases, so they can take measures to prevent them. They can recognize signs of illness early, seek advice and act on it. And they also tend to use health care services more effectively. Education also strengthens people's belief in their ability to achieve goals, so they are likely to be more confident that they can make necessary lifestyle changes and cope with treatment of disease.

Educated people tend to earn more, increasing the amount they can spend on health care and on measures that help prevent illness. They also tend to be less exposed to work and living environments that can jeopardize health – not only physical factors but also psychological stress such as that caused by discrimination and exclusion (Feinstein et al., 2006; Grossman, 2006).

Most of all, educated people – and women in particular – tend to have healthier children. The education of girls and young women is a vital goal in itself, as well as a fundamental human right – but it also makes a particularly strong and effective contribution to health. Making sure that girls enter and complete lower secondary school is the key to unlocking education's health benefits.

This section shows how mothers' education improves their own health and that of their children, then explores ways in which education helps limit disease and eliminate hunger.

## Mothers' education has saved millions of children's lives

There are few more dramatic illustrations of the power of education than the estimate that 2.1 million lives of children under 5 were saved between 1990 and 2009 because of improvements in the education of women of reproductive age. That is more than half the total of 4 million lives saved by reducing child mortality during the period. By contrast, economic growth accounted for less than 10% of the total (Gakidou et al., 2010).

Such progress pales beside the remaining challenge, however. The fourth MDG, which aimed to reduce the number of child deaths by two-thirds between 1990 and 2015, will probably not be achieved, even though the annual rate of reduction in under-5 deaths accelerated from 1.7% in 1990–2000 to 3.8% in 2000–2012. In 2012, 6.6 million children under 5 died, of whom 5.7 million were in low and lower middle income countries (Inter-agency Group for Child Mortality Estimation, 2013).

Many of these deaths could have been avoided through prevention and treatment measures such as making sure that there is a skilled attendant at birth, that children receive basic immunizations – which also help protect against pneumonia – and that oral rehydration treatment is given for diarrhoea (Infographic: Educated mothers, healthy children). Not only are most of these measures cheap and effective, but all are more likely to be taken when mothers are educated.

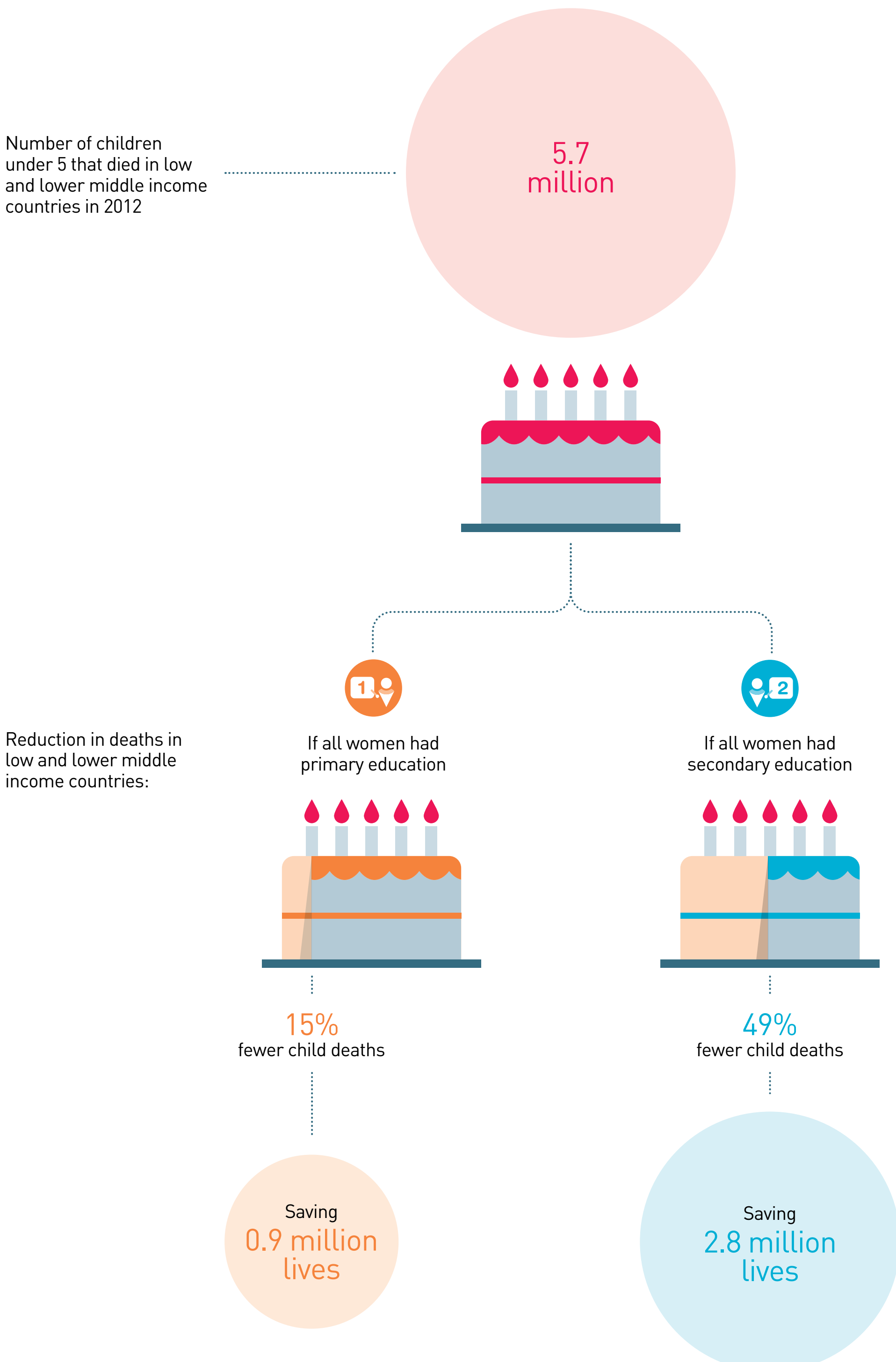
*Extending girls' education could save many more lives*

The scale of the impact that education – particularly of mothers – can have on child mortality is demonstrated by an analysis for this Report of 139 Demographic and Health Surveys from 58 countries. If all women completed primary education, the under-5 mortality rate would fall by 15% in low and lower middle income countries, saving almost a million children's lives every year. Secondary education has an even greater impact: if all women in these countries completed secondary education, the under-5 mortality rate would fall by 49% – an annual savings of 3 million lives (Infographic: Saving children's lives). Fathers' education has a smaller impact: if both women and men had secondary education, the under-5 mortality rate would fall by 54% in these countries.

To eliminate preventable child deaths by 2030 – which is likely to become a new global health target – urgent action is needed, and boosting secondary enrolment must be part of it. Over 9,000 children die every day in sub-Saharan Africa. The region has the lowest secondary enrolment rates in the world, and just 37% of girls are enrolled at this level. Ensuring

# SAVING CHILDREN'S LIVES

A higher level of education reduces preventable child deaths




*Sources:* Gakidou (2013); Inter-agency Group for Child Mortality Estimation (2013).

# EDUCATED MOTHERS, HEALTHY CHILDREN

Higher levels of education for mothers lead to improved child survival rates

## Pneumonia

One additional year of maternal education would decrease child deaths from pneumonia by:

14%

Equivalent to:

160,000
lives saved per year

**Maternal education reduces factors putting children at risk of pneumonia such as:**

1. malnutrition and low birth weight
2. failing to carry out measles vaccination in the first 12 months
3. burning fuel that gives off harmful smoke

## Birth complications

A literate mother is on average:

23%
more likely to seek support from a skilled birth attendant

## Diarrhoea

Reduction in diarrhoea in low and lower middle income countries if all mothers had primary education:

8%

Reduction in diarrhoea if all mothers had secondary education:

30%

**Educated mothers are more likely to:**

1. properly purify water
2. seek care from a health provider when a child has diarrhoea
3. administer rehydration solutions, increase fluids, and continue feeding

## Malaria

In areas of high transmission, the odds of children carrying malaria parasites is **22% lower** if their mothers have primary education than if their mothers have no education

down 22%

In areas of high transmission, the odds of children carrying malaria parasites is **36% lower** if their mothers have secondary education than if their mothers have no education

down 36%

## Immunization

Increase in vaccination for diphtheria, tetanus, and whooping cough (DTP3) in low and lower middle income countries if all mothers had primary education:

10%

Increase in DTP3 vaccination if all mothers had secondary education:

43%

*Sources:* EFA Global Monitoring Report team calculations (2013), based on Demographic and Health Survey data from 2005-2011; Fullman et al. (2013); Gakidou (2013).

that all girls achieve secondary education would mean 1.5 million more children surviving to their fifth birthday. In South and West Asia, the under-5 mortality rate would fall by 62% if all girls reached secondary school, saving 1.3 million lives.

Some of the countries with the highest child mortality rates and the lowest levels of education stand to reap the greatest benefits. In Burkina Faso, the under-5 mortality rate stood at 102 deaths per 1,000 live births in 2012, while the average for all low income countries was 82 deaths per 1,000 live births. The country's secondary gross enrolment ratio for girls is one of the world's lowest, just 25% in 2012. If all women completed primary school, child mortality would fall by 46%. If they all completed secondary school, it would fall by 76% (Gakidou, 2013).

Improving access to quality education could save an enormous number of lives in India and Nigeria, which together account for more than a third of child deaths. In 2012, 1.41 million children under 5 died in India and 0.83 million in Nigeria. If all women had completed primary education, the under-5 mortality rate would have been 13% lower in India and 11% lower in Nigeria. If all women had completed secondary education, it would have been 61% lower in India and 43% lower in Nigeria, saving 1.23 million children's lives.

Education helps reduce child deaths across all groups within countries. In Ethiopia, women of reproductive age tend to have very little schooling. Even richer women have only spent two years in school, on average, while those from poorer households have not been to school at all. Projections for this Report, taking into account various child, household and community factors, show that if all women in Ethiopia were to complete eight years of schooling, equivalent to completing primary school, the under-5 mortality rate would fall by over 40 deaths per 1,000 live births for both rich and poor households (Gakidou, 2013). Since child mortality is higher among poorer households, educating the poorest mothers saves the most lives.

Education's role in reducing child mortality can be assessed by looking at individual countries in more detail. In northern India, analysis based on the Annual Health Survey and the census in 2011 showed that female literacy was strongly linked to child mortality, even after taking into account access to reproductive and child health services. An increase in the female literacy rate from 58%, the current average in the districts surveyed, to 100% would lead to a reduction in the under-5 mortality rate from 81 to 55 deaths per 1,000 live births (Kumar et al., 2012).

Education is consistently found to have a strong effect on reducing mortality before children reach their first birthday – when the majority of child deaths occur – even taking household wealth into account. An analysis using Demographic and Health Surveys showed this to be the case in 18 out of 27 countries. The odds that a child born to a mother with education above the median level would die after the first year was lower by 46% in the Plurinational State of Bolivia, 43% in Cambodia and 39% in the Democratic Republic of the Congo (Fuchs et al., 2010).

**In South and West Asia, the under-5 mortality rate would fall by 62% if all girls reached secondary school**

In Indonesia, the under-5 mortality rate was more than twice as high among mothers with no education than among those who had at least some secondary education. Even after controlling for wealth, region and location, maternal education helped explain child survival. By contrast, household wealth was not a determinant of child survival once the other factors were taken into account (Houweling et al., 2006).

### *Literate mothers are more likely to seek support from a skilled birth attendant*

Analysis of the channels through which education saves children's lives reveals that mothers who are more educated are more likely to give birth with the help of a skilled birth attendant, which means their children are more likely to survive. Around 40% of all under-5 deaths occur within the first 28 days of life, the majority being due to complications during delivery (Liu et al., 2012). Yet there were no skilled birth attendants present in over half the 70 million births per year in sub-Saharan Africa and South Asia in 2006–2010 (UNICEF, 2012a).

Across 57 countries, analysis of Demographic and Health Surveys for this Report shows that a literate mother is, on average, 23% more likely to have a skilled attendant at birth (Infographic:

Educated mothers, healthy children). In Mali, where an estimated 53,000 infants died in 2012, a literate mother was more than three times as likely to have a skilled attendant on hand. In Nepal, 49% of literate mothers have a skilled attendant at birth, compared with 18% of mothers who are not literate. The effect of literacy on the presence of a skilled attendant exceeds 30 percentage points in Niger and Nigeria, countries where the infant mortality rate is around 70 children per 1,000 live births.

The benefits of being literate on having a skilled attendant at birth can be far greater for mothers from poor households. In Cameroon, 54% of literate mothers from poor households have a skilled attendant, compared with 19% of those who are illiterate or semi-literate.

**One extra year of maternal education can lead to a 14% decrease in the pneumonia death rate**

### *Educated mothers ensure their children are vaccinated*

Since 2000, the GAVI Alliance has supported vaccination against preventable diseases for 370 million children in the world's poorest countries, saving an estimated 5.5 million lives (GAVI, 2013). This is a tremendous impact. There is a missing ingredient, however, in GAVI's strategy that would allow for its success to be transmitted across generations: investment in girls' education.

Analysis of data from Demographic and Health Surveys for this Report shows that if all women in low and lower middle income countries completed primary education, the probability of a child receiving immunization against diphtheria, tetanus and whooping cough – a triple vaccination known as DTP3 – would increase by 10%. If they completed secondary education, it would increase by 43% (Infographic: Educated mothers, healthy children). This is a very large potential increase, given that it takes account of other factors, including household size, household wealth, paternal education and the average community education level. The increase would be as high as 80% for sub-Saharan Africa. In Haiti, where immunization rates are low, if all women completed primary education, DTP3 immunization rates would increase from 59% to 78% (Gakidou, 2013; WHO and UNICEF, 2013).

### *Mothers' education helps avert pneumonia*

Pneumonia is the largest cause of child deaths, accounting for 1.1 million or 17% of the total worldwide. Many pneumonia deaths could be prevented through breastfeeding, adequate nutrition, vaccination, safe drinking water and basic sanitation – and several of these factors are influenced by maternal education. UNICEF identifies pneumonia as a 'disease of poverty' (UNICEF, 2012b, p. 17). But lowering poverty only reduces pneumonia if mothers' education is improved at the same time.

As little as one extra year of maternal education can lead to a 14% decrease in the pneumonia death rate – equivalent to 160,000 child lives saved every year – according to new analysis for this Report based on estimates of under-5 pneumonia death rates from the Global Burden of Disease study in 137 countries between 1980 and 2010 (Infographic: Educated mothers, healthy children). The decrease would be 12% in South and West Asia, East Asia and the Pacific, and Latin America and the Caribbean, and 23% in the Arab States (Gakidou, 2013). In sub-Saharan Africa, however, an extra year of maternal education does not seem to reduce pneumonia death rates significantly, possibly because educated mothers do not have access to sufficient health care for their families.

Maternal education reduces all the factors that put children most at risk of dying from pneumonia, such as failure to carry out measles vaccination in the first 12 months, malnutrition and low birth weight (Rudan et al., 2008). Children's risk of pneumonia is also higher when they live in poorly ventilated homes and their families use traditional cooking stoves that burn solid fuel, giving off harmful smoke and fine particles. A review of 32 studies in poorer countries showed that maternal education contributed to the choice of improved fuels and stoves (Lewis and Pattanayak, 2012). In Bangladesh, women with some education were 37% more likely to select an improved cooking stove. Women who had more education than their husbands were 42% more likely to make this choice (Miller and Mobarak, 2013).

*Educating mothers helps prevent and treat childhood diarrhoea*

Diarrhoea is the fourth biggest child killer, accounting for 9% of child deaths, many of which should be easily preventable (Inter-agency Group for Child Mortality Estimation, 2013). If all women completed secondary education, the reported incidence of diarrhoea would fall by 30% in low and lower middle income countries because better-educated mothers are more likely to take prevention and treatment measures (Infographic: Educated mothers, healthy children).

In terms of prevention, education affects household decisions to purify water through filtering, boiling or other methods. In urban India, the probability of purification increased by 9% when the most educated adult had completed primary education and by 22% when the most educated adult had completed secondary education, even once household wealth is accounted for (Jalan et al., 2009).

In terms of treatment, an educated mother whose child has symptoms of diarrhoea is more likely to seek appropriate health care. In low income countries, mothers who had completed primary school were 12% more likely than mothers with no education to take such action, according to analysis for this Report based on Demographic and Health Surveys. In Niger, 14% of under-5 deaths are due to diarrhoeal diseases (Child Health Epidemiology Reference Group, 2012). About 46% of mothers with primary education sought help when their children had diarrhoea, compared with 33% of mothers with no education.

Educated mothers are also more likely to treat children's diarrhoea by administering oral rehydration solutions, increasing fluids and continuing feeding. In 28 countries in sub-Saharan Africa, the percentage of children under 3 with diarrhoea who received oral rehydration increased by 18 percentage points in rural areas when mothers had at least some secondary education, compared with those without education. In West and Central African countries, the increase was from 26% to 46% (Stallings, 2004).

## Education is a key way of saving mothers' lives

A mother's education is just as crucial for her own health as it is for her offspring's. Greater investment in female education, particularly at lower secondary level, would have helped accelerate progress towards the fifth MDG, improving maternal mortality, one of the goals that is most off track.

Maternal mortality is defined as the death of a woman while pregnant, or within 42 days of termination of pregnancy, from any cause related to or aggravated by the pregnancy or its management, though not from accidental or incidental causes. Between 1990 and 2010, the number of such deaths worldwide almost halved. While this is impressive, the maternal mortality ratio – the number of maternal deaths per 100,000 live births – fell by only 3.1% per year on average, well below the annual decline of 5.5% required to achieve the fifth MDG.

Every day, almost 800 women die from preventable causes related to pregnancy and childbirth. Overall, 99% of maternal deaths occur in developing countries. The maternal mortality ratio in these countries is 240 deaths per 100,000 live births, compared with 16 in developed countries. Over half the deaths are in sub-Saharan Africa and over a quarter in South Asia (WHO, 2012b).

Mothers die because of complications during pregnancy, such as pre-eclampsia, bleeding and infections, and because of unsafe abortion. Educated women are more likely to avoid these dangers by adopting simple and low cost practices to maintain hygiene, by reacting to symptoms such as bleeding or high blood pressure, by assessing how and where to have an abortion, by accepting treatment and by making sure a skilled attendant is present at birth.

Policy-makers seldom see education as a way of reducing maternal mortality, however. The World Health Organization's efforts to reduce maternal mortality focus on increasing skilled birth attendance, but education has an equally important role, according to analysis for this Report using data for 108 countries and new estimates of maternal mortality ratios for 1990–2010 (Bhalotra and Clarke, 2013).

**In sub-Saharan Africa, young children in rural areas with diarrhoea are more likely to receive oral rehydration when mothers have some secondary education**

**If all women completed primary education, maternal mortality would fall by 66%**

If all women completed primary education, maternal mortality would fall from 210 to 71 deaths per 100,000 births, or by 66%. This would save the lives of 189,000 women every year. If all women in sub-Saharan Africa completed primary education, the maternal mortality ratio would fall from 500 to 150 deaths per 100,000 births, or by 70% (Infographic: A matter of life and death).

At least two-fifths of the effect of education is indirect: educated women are more likely to use public health care services, to have fewer children and not to give birth as teenagers – all factors that reduce maternal mortality. Across the 108 countries studied, 6 out of 100 births were among women aged 15 to 19. If the rate of teenage births were halved, the maternal mortality ratio would fall by more than a third (Bhalotra and Clarke, 2013).

Education's impressive power to reduce maternal mortality can be gauged by looking at generations of young women who have benefited from education progress. In the 1970s, education reforms led to young women receiving 2.2 more years of schooling on average in Nigeria. With these women now approaching the end of their reproductive cycle, analysis for this Report was able to determine that the expansion of education accounted for a 29% decline in the maternal mortality ratio in Nigeria. However, there is a danger that the slowing of education progress in the 2000s – affecting women born in the late 1980s who are now entering their reproductive phase – will lead to a slowdown in the reduction of maternal mortality (Bhalotra and Clarke, 2013).

## Education plays a major role in containing disease

Infectious diseases, such as HIV/AIDS, and parasitic diseases, such as malaria, as well as non-communicable conditions, such as heart disease and cancer, pose some of the gravest threats to health – but improving education is a powerful way to help reduce their incidence. It is vital for policy-makers to take this into account. Although there has been progress towards the sixth MDG, which aims to reverse the spread of HIV/AIDS, malaria and other major diseases, large numbers of people are still dying from preventable diseases, and unfinished business will remain to be addressed after 2015.

One broad guide to the prevalence of disease is whether people consider themselves to be in good health. People with more education were consistently more likely to do so, according to the World Health Survey. An additional year of education increased the odds of not reporting poor health by 7.6%, and the globally representative survey showed that education had a significant effect on self-rated health in 48 out of 69 countries (Subramanian et al., 2010). Taking into account the characteristics of each country, completing lower secondary school increased the odds of not reporting poor health by 18%, while completing upper secondary school increased the odds by 59%, compared with having no education or less than primary education (Witvliet et al., 2012).

By looking at how specific health risks affect different populations in different ways, it is possible to see how increasing people's access to quality education can protect them from disease. This section examines in particular the way literacy can help people avoid HIV/AIDS, how education combats malaria by helping people correctly identify its cause, symptoms and treatment, and how education lowers people's risk of dying from non-communicable diseases.

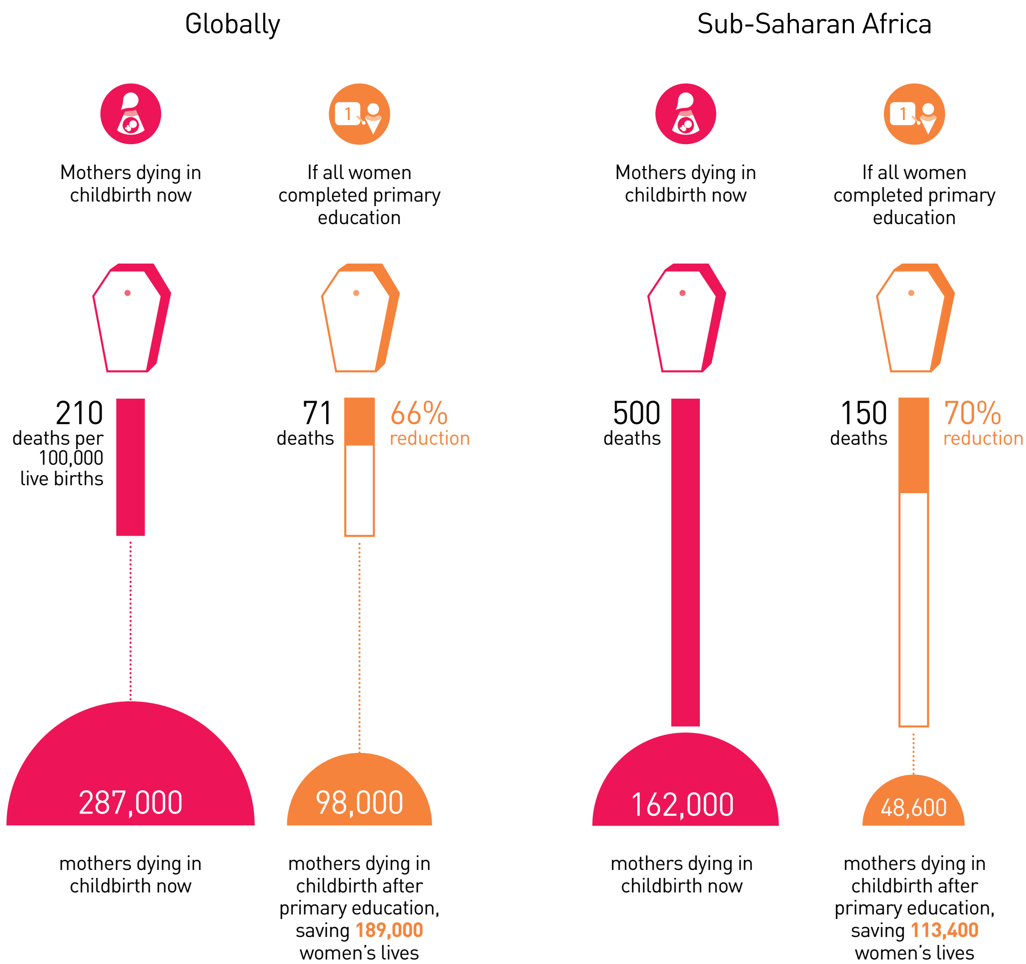
### *Literacy improves knowledge about HIV/AIDS*

Education provides a window of opportunity to increase awareness about HIV prevention among young people and so avert new infections among future generations. Reaching the young is particularly important: while the rate of new infections fell by 27% among those aged 15 to 24 between 2001 and 2011, young people still account for about 40% of new infections. Globally, the incidence of HIV/AIDS is declining, but there were still an estimated 2.5 million new HIV infections and 1.7 million AIDS-related deaths in 2011 (UNAIDS, 2012).

In the early phases of the AIDS epidemic, when knowledge about HIV was scarce, the better educated were more vulnerable to the virus. Since then, however, those with more education have tended to avoid risky behaviour because they understand its consequences better, and

# A MATTER OF LIFE AND DEATH

Educated mothers are less likely to die in childbirth




## Why does education reduce maternal deaths?

**Educated women are more likely to avoid complications during pregnancy, such as pre-eclampsia, bleeding and infections by:**

1. adopting simple and low-cost practices to maintain hygiene
2. reacting to symptoms such as bleeding or high blood pressure
3. making sure a skilled attendant is present at birth

**Educated women are more likely to:**

1. use public health care services
2. not give birth as teenagers
3. have fewer children

Note: Maternal mortality is defined as the death of a woman while pregnant, or within 42 days of termination of pregnancy, from any cause related to or aggravated by the pregnancy or its management, though not from accidental or incidental causes.

*Sources:* Bhalotra and Clarke (2013); WHO (2012b).

**In South and West Asia, 81% of literate women know that HIV is not spread by sharing food, compared with 57% of those who are not literate**

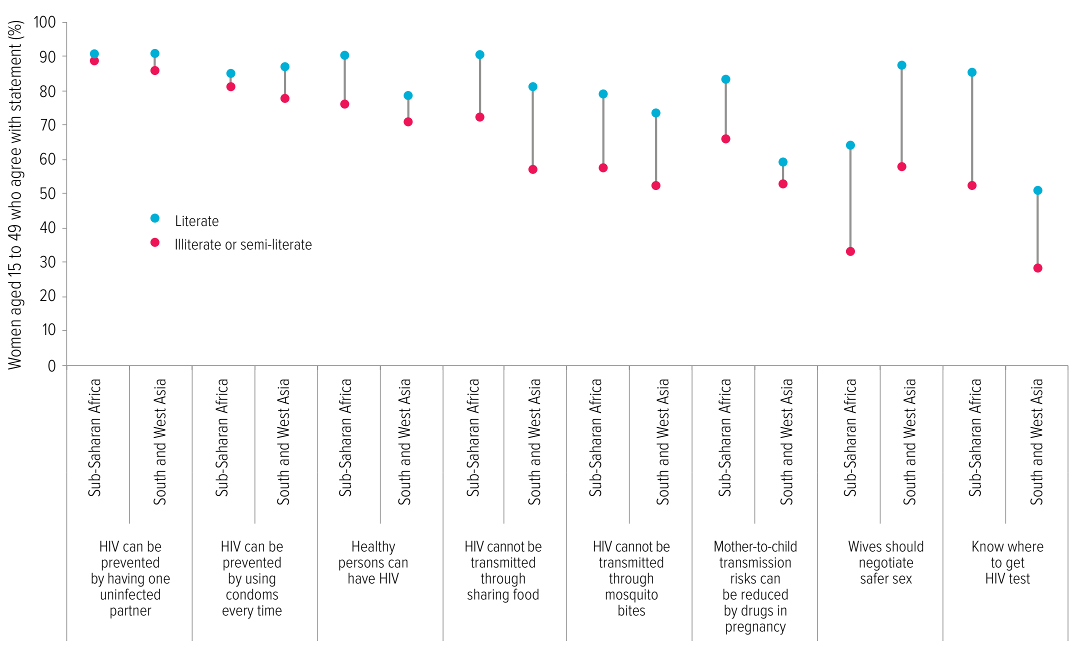
women have been able to exercise more control over their sexual relationships (Hargreaves et al., 2008; Jukes et al., 2008). In the later phases of the epidemic, the better educated have had a lower chance of being infected in 17 sub-Saharan African countries (Iorio and Santaeulàlia-Llopis, 2011).

Education's role in HIV prevention is illustrated by an analysis of 26 countries in sub-Saharan Africa and 5 in South and West Asia for this Report. It demonstrates the importance of literacy skills in improving people's knowledge of how HIV is transmitted (Figure 3.4). These countries account for around half of all new infections among adults. In sub-Saharan Africa, 91% of literate women know that HIV is not transmitted through sharing food, compared with 72% of those who are not literate. In South and West Asia, where the infection rate is still increasing in countries such as Bangladesh and Sri Lanka, the gap in knowledge between those who are literate and those who are not is even wider: 81% of literate women know that HIV is not spread by sharing food, compared with 57% of those who are not literate. Misconceptions that HIV can be contracted through mosquito bites are also more widespread among those who are not literate in both South and West Asia and sub-Saharan Africa. Similar patterns hold for young men.

An educated woman is more likely to be aware that she has a right to negotiate safer sex – to refuse sex or request condom use – if she knows that her partner has a sexually transmitted disease. In both South and West Asia and sub-Saharan Africa, literate women are as much as 30 percentage points more likely to be aware of this right, compared with those who are not literate. In sub-Saharan Africa, two-thirds of literate women are aware of it, compared with only a third of illiterate or semi-literate women. In Burundi, Nepal and Senegal, poor women who are literate are more aware of their right to safer sex than richer women who are not literate.

People who are more literate are more likely to be better informed on a wide range of other beliefs and facts about HIV/AIDS. In Niger, 47% of illiterate and semi-literate women believed a healthy-looking person cannot be infected with HIV, compared with 18% of literate women. In Mali, 52% of illiterate and semi-literate women

**Figure 3.4: Literacy enhances understanding of how to prevent and respond to HIV/AIDS**
Percentage of women aged 15 to 49 agreeing with selected statements, sub-Saharan Africa and South and West Asia, 2005–2011




*Note:* Estimates are based on 26 countries in sub-Saharan Africa and 5 countries in South and West Asia.
*Source:* EFA Global Monitoring Report team calculations (2013), based on Demographic and Health Surveys.

were not aware that the risk of transmission could be reduced if they took appropriate medication during pregnancy, compared with 20% of literate women.

Knowing where to get tested for HIV is a first step to receiving treatment. Yet only 52% of illiterate or semi-literate women in sub-Saharan Africa, and 28% in South Asia, know where to get tested, compared with 85% and 51% of those who are literate. In Nigeria, 73% of literate young women knew where to get an HIV test, compared with 36% of those who were not literate.

Education's strong contribution to HIV prevention has been put forward as an explanation for the remarkably fast decline in infection rates in Zimbabwe, one of the countries hardest hit by the HIV/AIDS epidemic. By 2010, 75% of women aged 15 to 24 in Zimbabwe had completed lower secondary school, and the HIV prevalence rate had fallen from its peak of 29% in 1997 to under 14%, declining four times faster than in neighbouring Malawi and Zambia where education levels are lower (Halperin et al., 2011). The percentage of women aged 15 to 24 who had completed lower secondary school was 49% in Zambia in 2007 and 42% in Malawi in 2010. In rural parts of South Africa – another country where infection rates are very high – every additional year of education was linked with a 7% lower probability of becoming infected (Bärnighausen et al., 2007).

### *Education boosts treatment and prevention of malaria*

Malaria is one of the world's deadliest but most preventable diseases. Education helps people identify its cause and symptoms and take steps to prevent and treat it. About half the world population is at risk of malaria. Those most at risk are children in Africa, where a child dies of malaria every minute. Concerted efforts have helped save over 1 million lives in the past decade, but there are signs that global funding for malaria prevention and control is stagnating, highlighting the importance of keeping the spotlight on the disease and ways to combat it (WHO, 2012c).

Improved access to quality education cannot replace the need for investment in drugs and in bed nets treated with insecticide – one of the most cost-effective ways to prevent malaria – but it has a crucial role to play in complementing and promoting these measures. In India, literate people with schooling up to lower secondary level were more than twice as likely as illiterate people to know that mosquitoes are the transmitters of malaria. They were also about 45% more likely to know that malaria can be prevented by draining stagnant water (Sharma et al., 2007).

**In India, people with schooling up to lower secondary level were more than twice as likely to know that mosquitoes are the transmitters of malaria**

The more schooling people have, the more likely they are to use bed nets, as studies have shown in the Democratic Republic of the Congo, where a fifth of the world's malaria-related deaths occur. In a rural study, in a group of which only 44% had spent the previous night under a bed net, if the household head had completed primary education this increased the odds of bed net use by about 75%, even with other possible factors taken into account (Ndjinga and Minakawa, 2010). In an urban study of a group of pregnant women making an antenatal care visit, only a quarter reported having slept under a bed net the previous night. The odds of having used a net among women with at least secondary education were almost three times as high as those with less than secondary education (Pettifor et al., 2008). In rural Kenya, the odds that children whose mothers had at least secondary education had slept under a net were up to three times as high as those whose mothers were not educated (Noor et al., 2006).

Because those with more education are more likely to have taken preventive measures, they are less likely to contract malaria, even after household wealth is taken into account. In Cameroon, where the female secondary gross enrolment ratio was 47% in 2011, if all women had had secondary education, the incidence of malaria would have dropped from 28% to 19% (Gakidou, 2013).

Children of educated mothers are much less likely to contract malaria, as is shown by an analysis of Malaria Indicator Surveys in Angola, Liberia, Madagascar, Nigeria, Rwanda, Senegal, Uganda and the United Republic of Tanzania. For example, the odds of children carrying malaria parasites was 44% lower if the mother had secondary education than if she had no education. Taking into account factors such as whether the child had slept under a bed net the previous night, children of mothers with

**In 11 sub-Saharan African countries, the odds of malaria parasites in children were 36% lower when mothers had secondary education**

secondary education in rural areas of these eight countries were 16% less likely than those with uneducated mothers to be infected with the malaria parasite (Siri, 2012).

Maternal education contributes even more to preventing malaria in areas where the risk of transmission is high. An analysis of 11 sub-Saharan African countries showed that in such areas the odds of malaria parasites in children were 22% lower when mothers had primary education and 36% lower when mothers had secondary education (Infographic: Educated mothers, healthy children). This is after taking into account whether children had slept under a net and whether the household had sprayed indoors with insecticide over the previous year – measures which are both related to education in the first place (Fullman et al., 2013).

### *Education is needed to fight non-communicable diseases*

Education also has a key role to play in preventing early death from non-communicable diseases. While infectious and parasitic diseases take their greatest toll in poorer countries, non-communicable diseases pose a major challenge for all countries. According to the Global Burden of Disease 2010 study, ischaemic heart disease was the first or second cause of death in all regions except sub-Saharan Africa. Lung cancer was the fifth-highest cause for men and tenth for women. Diabetes was the ninth-highest cause for men and sixth for women (Salomon et al., 2012). The global nature of this concern is reflected in the proposal for a universal health goal after 2015 that would be measured using healthy life expectancy, a criterion that covers individuals in all countries regardless of the level of development.

While non-communicable diseases are the leading causes of death in high income countries, the death rate for most of them is actually lower in high income than in low and middle income countries, where early detection and treatment are lacking. Contrary to popular perception, their incidence generally does not increase as countries get richer. If adults in low and middle income countries had the same mortality from cancer, cardiovascular disease, chronic respiratory disease and diabetes as those in high income countries, global mortality from these four diseases would be lower by more than a quarter (Di Cesare et al., 2013).

Education plays an important role in preventing non-communicable diseases by increasing awareness of the long-term consequences of smoking on health. Tobacco use is a key risk factor for cancer, as well as respiratory and cardiovascular disease – and the leading cause of preventable deaths worldwide. Up to half of current users will die of a tobacco-related disease. Nevertheless, consumption of tobacco products is increasing globally, especially in low and middle income countries (WHO, 2013).

In the United States, one of the first countries where smoking was explicitly linked to non-communicable diseases, the gradual arrival of information about the harm caused by smoking led to major behavioural changes – with the more educated responding faster and more emphatically. Before 1957, the more educated were more likely to smoke. By 2000, however, they were less likely to smoke than the less educated by at least 10 percentage points. Completing four years of tertiary education increased the probability of quitting by 18% (de Walque, 2007, 2010).

Similar effects are observed in low and middle income countries. World Health Survey data from 48 low and middle income countries showed that, taking age, income and employment into account, the odds that men with only primary education would smoke were almost 90% higher than of those with higher education (Hosseinpoor et al., 2011). The Global Adult Tobacco Survey showed that in Bangladesh, Egypt and the Philippines, the odds that those with less than secondary education would smoke were over twice as high as of those with tertiary education (Palipudi et al., 2012).

The power of education to improve health by lowering tobacco use is going untapped in many countries. This should be of urgent concern for policy-makers. After taking into account national GDP levels and the diffusion of cigarettes in the population, the gap in smoking between less and more educated young men increases from 16 percentage points in low income countries to 23 percentage points in lower middle income countries and 29 percentage points in upper middle income countries (Pampel et al., 2011).

## Hunger will not be eliminated without education

Malnutrition, the underlying cause in 45% of child deaths globally (Inter-agency Group for Child Mortality Estimation, 2013), is not just about the availability of food. To eliminate malnutrition in the long term, education – especially education that empowers women – is vital. Mothers who have been to school are more likely to ensure that their children receive the best nutrients to help them prevent or fight off ill health, even in families that are constrained financially. Educated mothers know more about appropriate health and hygiene practices at home, thereby ensuring their children are healthy enough to benefit fully from their food intake. And they have more power to allocate household resources so that children's nutrition needs are met.

There has been significant global progress in improving nutrition. Eliminating hunger within a generation is feasible – there is enough food in the world to feed everyone (Hoddinott et al., 2012). But in some parts of the world people are finding it harder to get the food they need, partly because of climate change. The first MDG aimed to halve the proportion of people who suffer from hunger by 2015, from a starting point of 23% in 1990–1992. But in 2010–2012 some 15% were still malnourished (FAO et al., 2012). Eliminating hunger deserves to be an even stronger focus of global development efforts after 2015.

The global extent of chronic malnutrition is revealed by the fact that one in four children under the age of 5 suffers from moderate or severe stunting – that is, they are short for their age. Three-quarters of these children live in sub-Saharan Africa and South Asia. The odds that a severely stunted child will die are four times higher than for a well-nourished child, while the odds that children who are severely wasted (underweight for their height) would die are nine times higher (Black et al., 2008). Chronic malnutrition affects children's brain development and their ability to learn. The link between malnutrition and cognitive development is also why early childhood care and education is the first goal in the Education for All framework.

Cross-country comparisons show that increasing the percentage of women who attend secondary school from 50% to 60% would result in a decline in the stunting rate by 1.3% after controlling for wealth, fertility and access to health services (Headey, 2013). The stunting rate in Bangladesh fell from 70% to 48% between 1994 and 2005. Over about the same period, the share of women with at least secondary education doubled. Education could explain more than a fifth of the reduction in stunting. If education's effect on fertility reduction were also taken into account, its influence would be even larger.

Education's power to help reduce malnutrition is apparent in 37 countries where the chance of a child being stunted is lower among those whose mothers have higher levels of education. In seven countries, including Ethiopia, Haiti and Honduras, a child whose mother has reached secondary school is at least half as likely to be stunted as one with a mother who has only primary education. In Honduras, for example, the rate of stunting is 54% for children born to mothers with less than primary education, 33% for those whose mothers have primary education and 10% for children of mothers with at least secondary education.

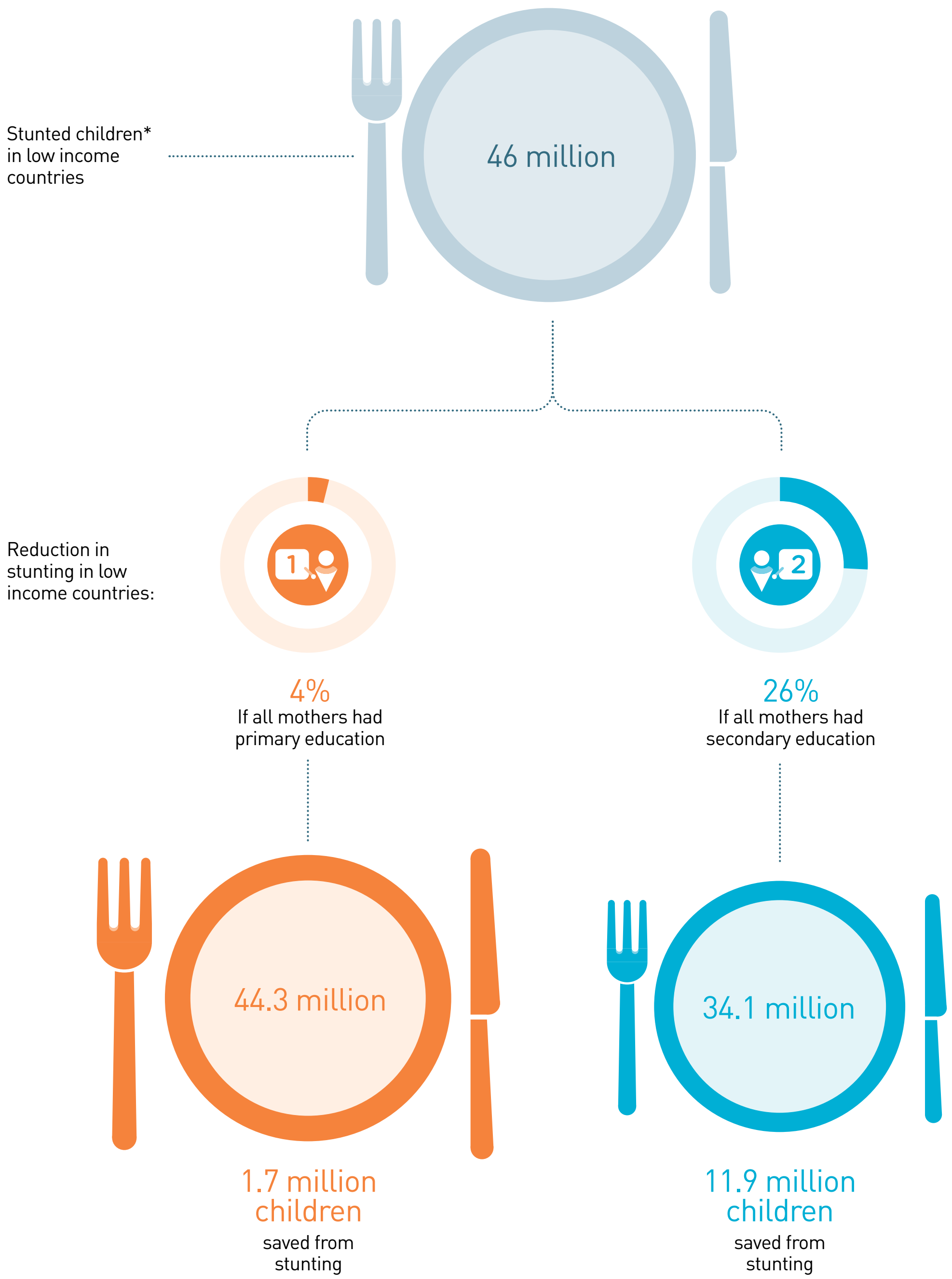
In low income countries, 46 million children suffer from stunting. If all women completed primary education, 1.7 million fewer children would be in this situation. The number rises to 11.9 million if all women complete secondary education, equivalent to 26% less children affected with stunting (Infographic: Education keeps hunger away). In South Asia, 20 million fewer children would be stunted if all mothers reached secondary education.

Detailed analysis within countries that has tracked children over time provides even stronger evidence that mothers' education improves child nutrition, even after taking into account other factors linked to better nutrition, such as mother's height, breastfeeding practices, water and sanitation, and household wealth. A study commissioned for this Report showed that by age 1 – when adverse effects of malnutrition on life prospects are likely to be irreversible – infants whose mothers had reached lower secondary education were less likely to be stunted by 33% in Ethiopia, 48% in

**In Viet Nam, infants whose mothers had reached lower secondary education were 67% less likely to be stunted**

# EDUCATION KEEPS HUNGER AWAY

Mothers' education improves children's nutrition




*Stunting is a manifestation of malnutrition in early childhood.

*Sources:* EFA Global Monitoring Report team analysis (2013), based on Demographic and Health Survey data from 2005-2011; UNICEF et al. (2013).

the state of Andhra Pradesh in India, 60% in Peru and 67% in Viet Nam, compared with those whose mothers had no education.

This new research also highlights the importance of complementing education expansion with reforms in health care to increase the extent to which children benefit from investment in their mothers' education. In Ethiopia, children whose mothers had primary school together with access to antenatal care were 39% less likely to be stunted at age 1 than children whose mothers had primary education but little or no access to antenatal care. For children whose mothers had completed lower secondary school, the difference in likelihood of being stunted at age 1 was 26% between those whose mothers had access to antenatal care and those who had no or limited access to antenatal care (Sabates, 2013).

These impressive findings are supported by evidence from other countries that monitored the nutritional habits of hundreds of thousands of households. In Bangladesh, where 41% of children were stunted in 2011, the odds of a child being stunted were lower by 22% if the mother had primary education, 24% if both parents had primary education and 54% if both parents had secondary education or above. In Indonesia, the odds of stunting fell by 26%, 35% and 59%, respectively, after accounting for maternal height, location and household expenditure (Semba et al., 2008).

A key reason children of educated women are less likely to be stunted is that their mothers have more power to act for the benefit of their children. In rural India, mothers' education has been shown to improve their mobility and their ability to make decisions on seeking care when a child is sick – and infant children of women with such increased autonomy are taller for their age (Shroff et al., 2011).

Malnutrition is caused not only by having too little food to eat, but also by a lack of micronutrients in the diet. Young children lacking vitamin A and iron are more likely to be malnourished and more prone to infections (such as measles and diarrhoeal diseases) and anaemia, which affects their cognitive development. Education helps ensure a varied diet that includes vital micronutrients.

Maternal education is associated with a higher probability of children aged 6 to 23 months consuming food rich in micronutrients in 12 countries analysed for this Report using Demographic and Health Surveys for 2009–2011. In the United Republic of Tanzania, for example, children of this age whose mothers had at least secondary education were almost twice as likely to consume food rich in micronutrients as children whose mothers had less than primary education.

**Maternal education is associated with a higher probability of infants consuming food rich in micronutrients**

Studies within countries provide further evidence of how education influences diet in ways that can prevent malnutrition, even once other factors are taken into account. In Bangladesh, when both parents had some secondary education, food group diversity in the family diet was 10% greater than when neither parent had any education (Rashid et al., 2011). In Indonesia, only 51% of households where mothers had no education used iodized salt, compared with 95% of households where mothers had completed lower secondary education. Similarly, only 41% of households where mothers had no education provided vitamin A supplements to their children within the past half year, compared with 61% of households where mothers had completed lower secondary education (Semba et al., 2008).

## Education promotes healthy societies

**Education plays a key role in promoting democratic values**

As well as improving individual lives and overall economic prosperity, education has an indispensable role in strengthening the bonds that hold communities and societies together. The need to protect and promote these connections is increasingly being recognized as a vital element of the post-2015 development framework. This section examines three key ways in which education contributes to healthy societies: expanding democratic engagement, protecting the environment and empowering women.

Education helps people understand democracy, promotes the tolerance and trust that underpin it, and motivates people to participate in politics. Education has a vital role in preventing environmental degradation and limiting the causes and effects of climate change. And it can empower vulnerable people to overcome discrimination that prevents them from getting a fair share of the fruits of overall progress when it comes to reducing poverty and improving health.

### Education builds the foundations of democracy and good governance

Education's vital role in promoting human rights and the rule of law is enshrined in the Universal Declaration of Human Rights, which states that 'every individual and every organ of society … shall strive by teaching and education to promote respect for these rights and freedoms'. Good quality education helps people make informed judgements about issues that concern them and engage more actively in national and local political debate.

In many parts of the world, however, human rights and citizens' confidence in government are jeopardized by unfair elections, corrupt officials, weak justice systems and other failures of democracy. When disenfranchized groups feel they have no means to voice their concerns, such failures can lead to conflict.

The critical function of democratic and accountable systems of governance in fulfilling human rights and advancing globally agreed development goals has come to the fore in recent discussions on post-2015 development ambitions. Some observers have argued that a democratic governance goal should have been included in the MDGs and deserves a place in the post-2015 framework (UNDP, 2012).

Education's key role in promoting democratic values is confirmed by analysis over the period from 1960 to 2000 showing that a 10% increase in secondary enrolment is associated with a 1.8-point increase in the Polity IV index, which measures the extent of democracy on a scale ranging from -10 (least democratic) to 10 (most democratic) (Glaeser et al., 2006). Democracy emerges slowly, so a longer period of observation is required to establish education's contribution. Studies of political transitions over more than a century show that universal primary education and literacy have been vital for countries to move from authoritarianism to democracy (Murtin and Wacziarg, 2011).

There is even stronger evidence that education plays a key part in building the social and cultural foundations of democracy, even after taking into account other factors that can have an influence, such as wealth, location and occupation. As this section shows, education helps people understand how democracy functions and what its benefits are. It also promotes values and norms that are crucial ingredients of democracy, notably tolerance and trust. Finally, it spurs individuals to participate actively in politics.

#### *Education improves political knowledge*

Education improves knowledge about politics, as evidence from European countries shows: people who have four more years of schooling spend 50% more time, on average, acquiring information by reading newspaper articles on politics and current affairs (Borgonovi et al., 2010).

Education is similarly linked with higher levels of political knowledge in low and middle income countries that are at an earlier stage of the democratic transition. Across 12 sub-Saharan African countries, while 63% of individuals without formal schooling had an understanding of democracy, the share was 71% for those with primary education and 86% for those with secondary education (Bratton et al., 2005). To take one example, five years after Malawi held its first democratic elections, the share of people of voting age who could not provide a definition of democracy was 23% among those who had not received any education but 3% among those with primary schooling (Evans and Rose, 2007).

The degree of political knowledge depends not only on the amount of time a child spends in school, but also on whether education is of sufficient quality to encourage key traits such as critical thinking. Among lower secondary school students in 34 countries, for example, levels of political knowledge were higher where political and social issues were more frequently discussed and where students felt freer to express their opinions, according to the 2009 International Civic and Citizenship Education Study (Schulz et al., 2010).

New analysis for this Report based on the World Values Survey shows that, in all regions, one reason for better knowledge among those with higher levels of education is that they are more interested in politics and so more likely to seek information. In Turkey in 2007, for example, the share of those who said they were interested in politics rose from 41% among those with primary education to 52% among those with secondary education (Chzhen, 2013).

Beyond the formal education system, civic education programmes can help students increase their knowledge. In Kenya, the National Civic Education Programme reached up to 15% of voting age citizens in the run-up to the 2002 elections. An evaluation showed that the programme improved political knowledge, particularly for those with less education (Finkel and Smith, 2011).

### *Education strengthens support for democracy*

Education increases people's support for democracy. Across 18 sub-Saharan African countries, those of voting age with primary education were 1.5 times more likely to express support for democracy than those with no education, and the level doubled among those who had completed secondary education, even once wealth, location and occupation were taken into account (Evans and Rose, 2012).

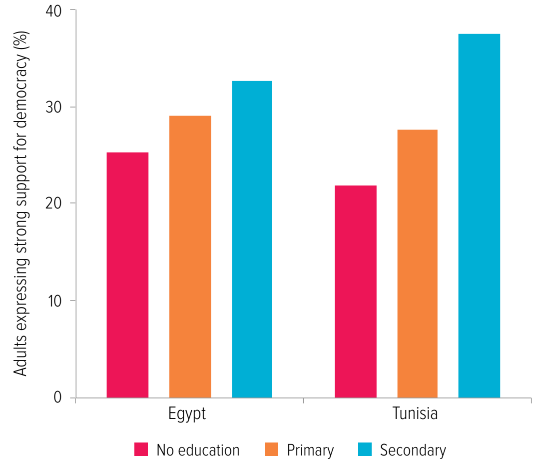
In the Arab States, notably Egypt and Tunisia, where the Arab Spring originated, a major educational expansion over the last 30 years has given rise to strong democratic aspirations. In Tunisia, while only 22% of those with less than primary education agree that democracy, despite its drawbacks, is the best system of governance, 38% of those with secondary education do so (Figure 3.5).

Education is also found to encourage support for democracy in other predominantly Muslim countries, including Jordan, Lebanon and Pakistan. In urban Pakistan, while only 35% of respondents overall supported democracy, those with secondary education were 15 percentage points more likely to do so than those with less than primary education (Shafiq, 2010). This finding reinforces the urgency of increased investment in education in Pakistan, where current levels remain among the lowest in the world and fully fledged democracy has yet to take hold.

In 17 Latin American countries, many of which have recently gone through democratic transitions, education boosts support for democracy or rejection of authoritarian alternatives, according to evidence from Latinobarómetro surveys. An increase in education level from primary to secondary raised support for democracy by five percentage points among men and eight percentage points among women (Walker and Kehoe, 2013).

**In sub-Saharan African countries, those of voting age with primary education were 1.5 times more likely to express support for democracy**

**Figure 3.5: More education is linked with stronger support for democracy**
Percentage of adults who strongly agree that 'A democratic system may have problems, yet it is better than other systems', by education level, Egypt and Tunisia, 2010–2011




*Source:* Prepared for *EFA Global Monitoring Report 2013/4* by the Arab Barometer team.

Around the world, educated people tend to be the most critical of their government regimes, as well as the strongest supporters of democracy. For example, in Nicaragua in 2009, support for democracy rose from 54% of young people with less than secondary education to 70% of those who had completed secondary school. At the same time, satisfaction with how democracy functioned fell from 43% of those with less than secondary education to 31% of those who had completed secondary education (Chzhen, 2012).

### *Education raises political participation*

Education not only increases people's political knowledge, but also makes them more likely to participate in political processes such as voting or standing for office, as well as signing petitions or joining demonstrations.

**In Africa, Asia and Latin America, education increases the likelihood of turning out to vote**

*Educated people are more likely to vote.* Education leads to a higher rate of voting in democratic elections (Nevitte et al., 2009). This is the case in low and middle income countries in particular. Public opinion surveys in 36 countries in Africa, Asia and Latin America showed that education increased the likelihood of turning out to vote (Bratton et al., 2010). In Yemen, a rise of 20 percentage points in the provincial literacy rate boosted voter registration by 12 points (Pintor and Gratschew, 2002).

In 14 Latin American countries in 2010, the probability of voting was five percentage points higher for those with primary education and nine points higher for individuals with secondary education, compared with those with no education. Education tends to boost voting more in countries where average levels of education are lower, such as El Salvador, Guatemala and Paraguay, than in countries with higher levels of education, such as Argentina and Chile (Carreras and Castañeda-Angarita, 2013).

In high income countries with considerable inequality in election participation, such as the United States, education also increases the likelihood that people will vote (Milligan et al., 2004; Sondheimer and Green, 2010). It also helps young people recognize the importance of voting (Campbell, 2006). According to one study, an additional year of schooling raised voter participation by seven percentage points (Dee, 2004).

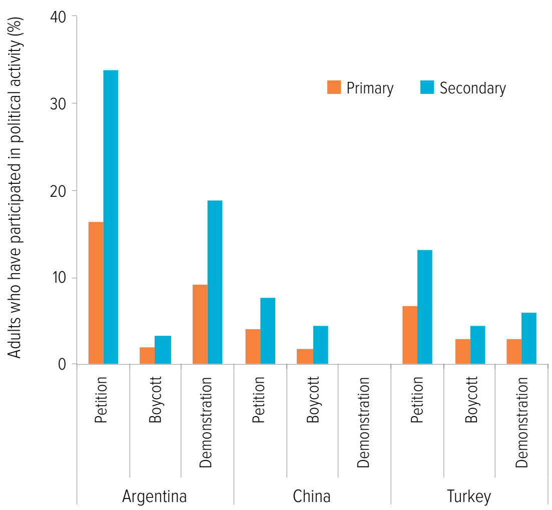
*Education boosts wider democratic activity.* Education contributes to other forms of political participation. In rural areas of the states of Madhya Pradesh and Rajasthan in India, education was positively associated with campaigning, discussing electoral issues, attending rallies and establishing contacts with local government officials (Krishna, 2006). In the state of West Bengal in India, a survey of 85 villages showed that the higher the level of household education, the more likely people were to attend the biannual gram sabha, or village forum, and, especially, to ask questions at the meetings (Bardhan et al., 2009).

Participation also includes joining, and campaigning for, political parties. In Benin, people who had had the chance to attend the first schools that opened in some rural parts of the country were 32% more likely to become party members and 34% more likely to become party campaigners than their peers who did not go to school (Wantchekon et al., 2012).

Education's role in promoting wider forms of participation is also apparent in high income countries. Evidence from the first three rounds of the European Social Survey (2002–2006) shows that each step up the education ladder, from primary to lower secondary, upper secondary and tertiary education, increases the chance of people participating in groups and associations by 10 percentage points (Borgonovi and Miyamoto, 2010).

*Education supports alternative forms of political participation.* Education increases the likelihood that citizens will make their voices heard in other ways, such as signing petitions, boycotting products or taking part in peaceful demonstrations, according to a new analysis for this Report of 26 low and middle income countries that participated in the 2005–2008 World Values Survey. In Argentina, China and Turkey, for example, citizens were around twice as likely to sign a petition if they had secondary education, compared with those who had only primary schooling. Similarly, in Argentina and Turkey those with secondary education were twice as likely to have taken part in a peaceful demonstration as those with only primary schooling (Figure 3.6).

**Figure 3.6: Education leads to more engagement in alternative forms of political participation**
Percentage of adults who signed a petition, boycotted a product or took part in a peaceful demonstration, by education level, selected countries, 2005–2008




* Respondents in China were not asked the question about involvement in peaceful demonstrations.
*Note:* The results control for age, sex, labour market status and country GDP per capita.
*Source:* Chzhen (2013), based on the 2005–2008 World Values Survey.

Education also changes people's attitudes to authority and traditional forms of allegiance. Educated individuals increasingly base their sense of citizenship less on duty and more on their ability and desire to participate directly in decisions that affect their lives (Dalton, 2008). In Kenya, a follow-up to the National Civic Education Programme reached at least 20% of voting-age citizens before the 2007 elections through various forms of public meetings. Interviews with 3,600 people showed that the programme increased their participation in local politics by 10% (Finkel et al., 2012).

### *Education promotes tolerance and social cohesion*

Education is a key mechanism promoting tolerance, as new analysis of the World Values Survey for this Report shows. In Latin America, where levels of tolerance are much higher overall than in the Arab States, people with secondary education were less likely than those with primary education to express intolerance by 47% for people of different race, 39% for people of a different religion, 32% in the case of homosexuals and 45% towards those with HIV/AIDS (Infographic: Love thy neighbour).

Education can play a vital role where intolerance is a particular problem. In the Arab States, for example, people with secondary education were 14% less likely than those with only primary education to express intolerance towards people of a different religion.

In sub-Saharan Africa, those affected by HIV/AIDS often face stigma, which itself can contribute to the spread of the disease by discouraging people from taking preventive measures or seeking treatment. Education can help reduce stigma: compared with those who had not completed primary school, those who had completed primary education were 10% less likely to express intolerance towards people with HIV infection, and those with secondary education were 23% less likely to do so. In Central and Eastern Europe, where intolerance towards immigrants is a cause for concern, those with secondary education were 16% less likely than those who had not completed secondary education to express such intolerance.

In many parts of the world, people remain intransigent in their attitudes towards homosexuality. In the Arab States, as many as 9 out of 10 people express intolerance to homosexuality, regardless of their education. This suggests that education can take time to have an impact on ingrained attitudes, and that specific policy measures are needed to ensure that children learn in school the importance of tolerance. Argentina shows what a difference education can make: those with secondary education were 21% less likely to express intolerant attitudes towards homosexuals than those with only primary education.

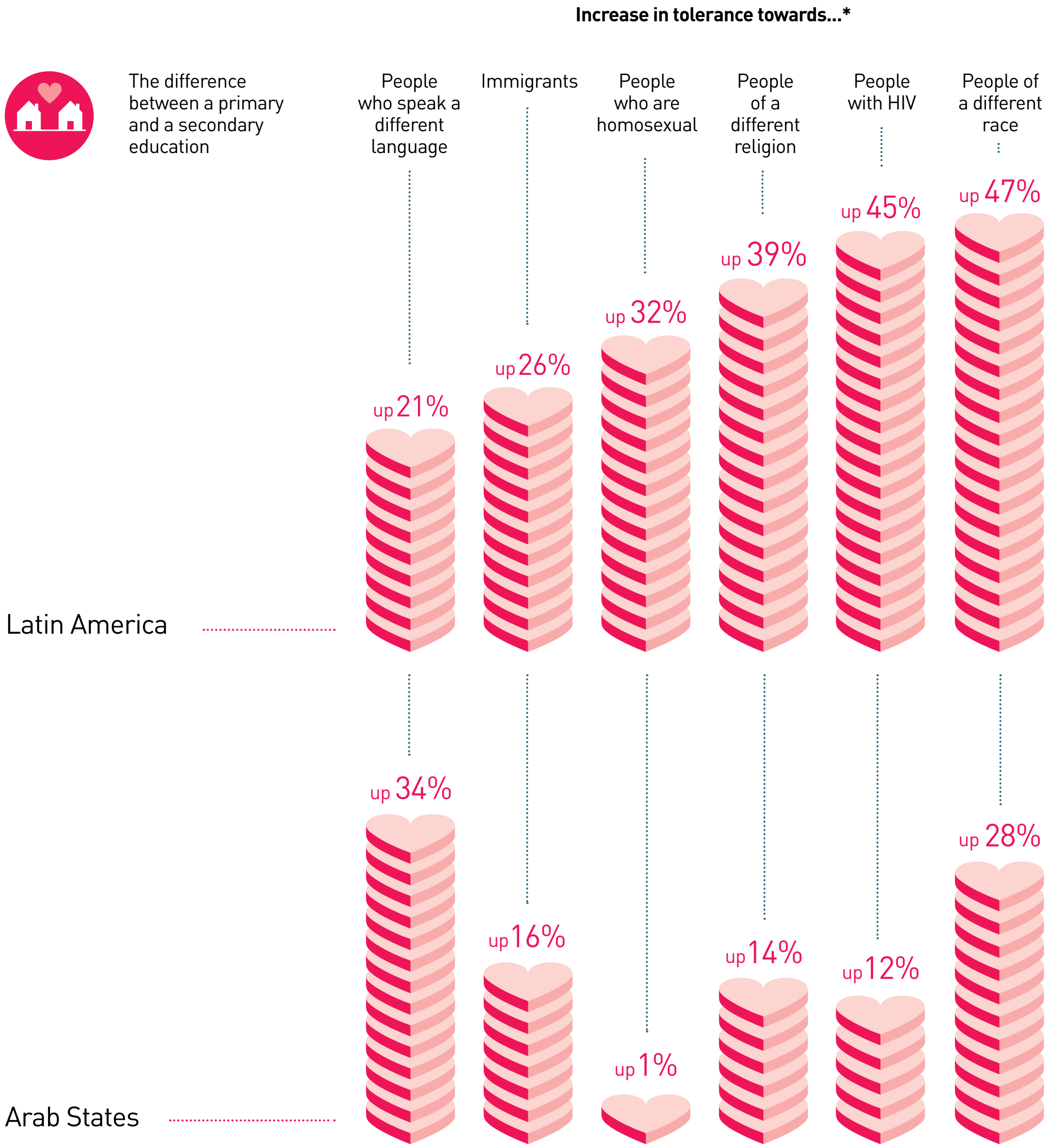
In parts of India, animosity among ethnic and linguistic groups can spark violence, so there is an urgent need to increase tolerance through education. Those with secondary education were 19% less likely to express intolerance towards people speaking a different language than those with less than primary education.

By altering attitudes, education also leads eventually to political changes, such as more democratic representation. New research for this Report shows the importance of equitable education on democratic development (Box 3.1).

**In the Arab States, people with secondary education were 39% less likely to express intolerance for people of a different religion**

# LOVE THY NEIGHBOUR?

Education increases tolerance




*Answers are in response to the question:

"Who would you prefer not to live next door to?"

*Source:* Chzhen (2013), based on the 2005–2008 World Values Survey.

**Box 3.1: In India, education boosts women's role in politics**

India, the world's largest democracy, appears to suffer from voter bias. Over the last three decades, only about 4% of all candidates for state assembly elections were female, and any female candidate received only about 5% of overall votes. New analysis for this Report, which combined information on state assembly elections between 1980 and 2007 with information on literacy rates across 287 districts, shows that narrowing the gender literacy gap raises women's participation and competitiveness in politics. Average literacy over the period was 34% for women and 55% for men. It is estimated that raising the female literacy rate to 42% would increase the share of female candidates by 16%, the share of votes obtained by women candidates by 13% and female voter turnout by 4%.

Improving male literacy also has a positive impact on women's political participation, perhaps because literate men are more likely to vote for women candidates and, as party leaders, to field women candidates.

Raising women's participation in politics is vital not only to achieve gender equality but also because evidence shows that women politicians tend to be both less corrupt and more proactive in representing the interests of children's well-being. Therefore, by stimulating increases in women's political engagement, investing in education improves democratic governance.

*Sources:* Afridi et al. (2013); Beaman et al. (2009, 2012); Bhalotra et al. (2013a); Brollo and Troiano (2013).

By promoting tolerance, education also builds values, attitudes, norms and beliefs that improve interpersonal trust and increase civic engagement, which are pillars of democracy. A review of estimates from various studies suggested that one year of schooling increases the probability of trusting people by 2.4 percentage points and the probability of civic participation by 2.8 percentage points (Huang et al., 2009).

To be successful in boosting trust, education systems need to ensure equal access to all children and young adults regardless of background, make sure there is no discrimination in the classroom and support those who are failing to learn. Within a group of 15 high income countries, social cohesion – in terms of interpersonal trust, trust in political institutions, attitudes to tax evasion, attitudes to cheating on public transport, and violent crime – was lower in those countries with more unequal distribution of educational outcomes (Green et al., 2003).

Inclusive education policies are just as important in instilling trust in poorer countries. In Kerala state, India, religious fragmentation and caste-based social organization could have been a source of conflict, as they have been in some other parts of the world. The early establishment of universal education helped overcome these challenges (Oommen, 2009). By contrast, in Argentina, which is not only a monolingual society but also richer and less characterized by factors that lead to social rifts, levels of social cohesion have declined, according to World Values Survey results over three decades. A gradual increase in income inequality since the 1970s has gone hand in hand with increased segregation in education and the movement of children from all but the poorest households to private schools between 1992 and 2010. This segregation is confirmed by the 2009 PISA results, which showed that social inclusion rates in schools in Argentina were among the lowest measured (OECD, 2010).

**By promoting tolerance, education also builds values and attitudes that improve interpersonal trust and increase civic engagement**

### *Education helps prevent conflict and heal its consequences*

If global development goals are to be achieved, it is vital to reduce conflict, which has held back progress towards the MDGs – and education is a key way of doing so. By accelerating growth and promoting employment, education dampens incentives for disaffected young men to engage in armed violence. And tackling educational inequality will help lessen the chances of conflict, because such inequality fosters a sense of injustice that has fuelled many conflicts.

While a low level of education does not automatically lead to conflict, it is an important risk factor. An influential study found that if the male secondary school enrolment ratio were 10 percentage points higher than average, the risk of war would decline by a quarter (Collier and Hoeffler, 2004). The expected risk of conflict is highest in countries that have both a large youth population and low education. For example, in a country with a high ratio of youth to adult population at 38%, doubling the percentage of youth with secondary education, from 30% to 60%, would halve the risk of conflict (Barakat and Urdal, 2009).

**An increase in educational expenditure from 2.2% to 6.3% of GDP is estimated to lead to a 54% decrease in the likelihood of civil war**

Government commitment to expanding education helps reduce the risk of conflict. Globally, an increase in primary enrolment from 67% to 100% would have resulted in a 35% decrease in the probability of civil war over 1980–1999. Higher spending on education can be a strong signal that the government cares for all citizens. An increase in educational expenditure from 2.2% to 6.3% of GDP is estimated to lead to a 54% decrease in the likelihood of civil war (Thyne, 2006).

Education access and spending need to be increased equally for all population groups, because perceived unfairness can reinforce disillusionment with central authority. A study of 55 low and middle income countries over 1986–2003 showed that if the level of educational inequality doubled, the probability of conflict more than doubled, from 3.8% to 9.5% (Østby, 2008). Similarly, an analysis of the use of force between organized armed groups not involving the government between 1990 and 2008 in sub-Saharan Africa showed that the risk of conflict rose by 83% between a region at the bottom quarter and a region at the top quarter of education inequality (Fjelde and Østby, 2012).

Education not only reduces armed conflict, but also lowers civil strife more broadly. An analysis of violence in 55 major cities in sub-Saharan Africa and Asia between 1960 and 2006 found that an increase in the percentage of the male youth population with secondary education could be expected to lead to a reduction in the number of lethal events (Urdal and Hoelscher, 2009).

Although the frequency of civil conflict has decreased in recent years, new conflicts have emerged or heightened, as in Mali and the Syrian Arab Republic. Urgent action is needed to help heal divisions and resolve the fundamental causes of tension. Expanding educational opportunities and reducing inequality can play a key role in reconstruction efforts, along with changing what is taught and how it is taught (UNESCO, 2011). Such reforms on their own may be insufficient if inequality continues to be a feature of the education system. In addition, separating children according to group identity can perpetuate negative attitudes, as the example of Lebanon shows (Box 3.2).

### Box 3.2: Educating to avert conflict in Lebanon

Lebanon is riven by deep sectarian divisions and sharp inequality between its communities, which are further exacerbated by wider tensions across the Middle East. The devastating war between 1976 and 1990 came to an end with the signing of the Taif Agreement, which recognized education as a means of moving towards reconciliation. This prompted large-scale education reform, including changes in curriculum, textbooks and teacher training.

Some materials, such as the teachers' guide for peace and democratic behaviour, were considered models of their kind. But other reforms were thwarted by political considerations; for instance, there is no common history textbook. A key feature of the education system is that most secondary schools are private and segregated along religious lines. In these schools, communities maintain control over the interpretation of events taught in classrooms, which often reflects this segregation.

Even in public schools, the teaching of civic education faces challenges. One study found that most public secondary schools were characterized by a subject-based rather than cross-cutting approach to civic education, and that the classroom and school environment was authoritarian and hierarchical. Some schools even applied admission policies that were not inclusive or restricted what issues teachers could discuss in class. Their students were accordingly found to be less open and trustful of members of other groups. For example, while about 36% of grade 11 students in schools with a passive approach to civic education said they trusted sectarian parties, only 18% did so in schools with an active approach to civic education.

A citizenship education reform is now trying to build on these lessons by emphasizing collaboration, dialogue, student participation, community service and parent councils.

*Sources:* Frayha (2004); Lebanon Centre for Research and Educational Development (2013); Shuayb (2012); UNDP (2008).

### *Education helps reduce corruption*

A broad-based free schooling system strengthens the foundations of democracy by fostering support for the institutional checks and balances that are necessary to detect and punish abuses of office, and by lowering tolerance towards corruption. An analysis of 78 countries shows that the level of educational attainment is the most powerful factor predicting lower levels of corruption (Rothstein and Uslaner, 2012).

In Brazil, for example, while 53% of voters with no education said they would support a corrupt but competent politician, only 25% of respondents with at least some college education agreed (Pereira et al., 2011).

Better-educated citizens are more likely to stand up to corruption by complaining to government agencies, primarily because they have information about how to complain and defend themselves. In 31 countries that took part in the World Justice Project survey of 2009–2011, those with secondary education were one-sixth more likely than average to complain about deficient government services, and those with tertiary education one-third more likely to do so (Botero et al., 2012).

### *Education is essential for the justice system to function*

Less educated people lack the ability to claim their rights and are often excluded from the legal system (Abregú, 2001). For example, in Sierra Leone, many people cannot use the formal court system because it operates in English, which only people with a higher level of education speak. Translators sometimes interpret into Krio, the lingua franca, but some people only speak local languages, for which interpreters are not available. Accused persons who are less educated can easily be isolated by a system that should support them (Castillejo, 2009).

Even non-formal courts intended to improve less educated people's access to the justice system are burdened by illiteracy. In Eritrea, village courts were set up to help settle cases amicably, as the lowest tier of the court system, but several of the elders appointed as judges were illiterate and lacked basic legal training. The result is that many decisions fell between the two systems, being based neither on customary law nor on national laws (Andemariam, 2011).

Problems can be particularly acute for women when their education levels are lower than men's. In Kenya, lack of knowledge about laws and dependence on male relatives for assistance and resources can prevent women from turning to the formal justice system, for example to resolve disputes such as property conflicts (International Development Law Organization, 2013).

## Education needs to be part of the solution to global environmental problems

Education's vital role in preventing environmental degradation and limiting the causes and effects of climate change has not been sufficiently acknowledged or exploited. By improving knowledge, instilling values, fostering beliefs and shifting attitudes, education has considerable power to change lifestyles and behaviours that are harmful for the environment.

There is an urgent need to identify the best ways to tap this potential as it becomes increasingly clear how much human action has led to environmental degradation and climate change, especially through the release of greenhouse gases. As well as altering the balance of nature, consequences such as extreme weather patterns and loss of biodiversity could reverse progress in improving living standards – especially for poorer and more vulnerable populations, even though they are not the ones responsible for environmental degradation.

**Education's role in mitigating the effects of climate change needs to be fully acknowledged and exploited**

Increased levels of education do not automatically translate into more responsible behaviour towards the environment. But as the influential Stern Review on climate change noted: 'Governments can be a catalyst for dialogue through evidence, education, persuasion and discussion. Educating those currently at school about climate change will help to shape and sustain future policy-making, and a broad public and international debate will support today's policy-makers in taking strong action now' (Stern, 2006, p. xxi).

Education's role in mitigating the effects of climate change has often been played down in discussions on sustainable development. It needs to be fully acknowledged and exploited so that education's benefits are realized. At the same time, education needs to adapt to the challenge of spreading the message of environmental responsibility, as Chapter 6 shows.

Identifying how education can best play this role is urgent. Even though the seventh MDG was intended to ensure environmental sustainability, it is widely acknowledged that it has received insufficient attention. Debates on how to frame global development goals after 2015 have centred on whether the overarching concern should be poverty eradication or sustainable development, while recognizing that the two aims complement each other. One thing is certain: environmental sustainability is a universal issue, affecting people around the world with unprecedented speed. While threats from hunger and poverty need urgent national action, people need to be educated about environmental degradation on a global scale, particularly in rich countries, to make sure those most responsible do not endanger lives in other parts of the world.

### *Galvanizing the potential of education for tackling climate change*

People who are more educated often tend to maintain lifestyles that burden the environment. One reason is that the consequences of climate change are not yet perceptible to the vast majority of people, and many still see it as a distant threat (Weber and Stern, 2011). But experience shows that, when populations are confronted by major challenges, overcoming the inertia of past attitudes is possible – and people with more education respond first. In the case of health threats such as HIV/AIDS, for example, educated people were originally more likely to engage in harmful behaviour such as unprotected sex, but as the dangers became known they were the first to change their behaviour.

**Students who scored higher in environmental science also reported being more aware of complex environmental issues**

Historically, rich countries that were the first to industrialize are the most responsible for environmental degradation and also have higher education levels, so the relationship would seem to work in the opposite direction. In addition, among the six nations that emit the most carbon dioxide, as education levels increase, emissions also tend to increase. A closer look at the data, however, reveals notable differences and exceptions showing that more education does not necessarily lead to increased emissions. In China in 2008, when the average level of education was seven years, the level of emissions per capita was one-third of what the level in the United States was at a similar average level of education, in 1950 (Box 3.3). And while the average level of education in Germany has increased by five years in the last two decades, emissions per capita have declined by almost 20% (Barro and Lee, 2013; World Resources Institute, 2012).

**Box 3.3: Education reform is needed to reduce the United States' impact on the environment**

The United States is the world's biggest polluter, accounting for 20% of total and 30% of transport emissions of greenhouse gases in the world. Reducing emissions will depend on changing public attitudes, including through education, as well as on technological progress. Education has been shown to change car use in a country that is highly dependent on private cars: in parts of the country, those with higher levels of education are more likely to drive smaller and more fuel-efficient vehicles. But far more is needed to reduce emissions by up to 65% compared with 2010 levels, the scale of reduction that high income countries need to achieve to limit the average temperature increase to 2°C in this century.

Schools need to strengthen their efforts to influence attitudes to the environment. Experience from US schools that have exemplary environmental education programmes shows that such change is possible: in comparison with a representative sample of US schools, they did significantly better in environmental literacy, in particular due to greater sensitivity towards the environment and improved self-reported environmental behaviour.

Schools with well-developed environmental programmes need to become the norm rather than the exception. And young people need to be taught the science behind climate change and other environmental problems. This can only happen if recent reforms to the science curriculum, aimed at preparing students to make better decisions about scientific and technical issues and to apply science to their daily lives, are not only fully implemented but also become mandatory.

*Sources:* Choo and Mokhtarian (2004); Flamm (2009); Greene and Plotkin (2011); McBeth et al. (2011); Meinshausen (2007); Pampel and Hunter (2012); World Resources Institute (2012).

### *Education improves knowledge and understanding of the environment*

A key way in which education increases environmental awareness and concern is by improving understanding of the science behind climate change and other environmental issues. Students who scored higher in environmental science across the 57 countries participating in the 2006 PISA also reported being more aware of complex environmental issues. For example, in the 30 OECD countries that took part in the survey, an increase of one unit of the awareness index was associated with an increase of 35 points in the environmental science performance index (OECD, 2009).

The higher the level of education, the more likely it is that people express concern for the environment. In 47 countries covered by the 2005–2008 World Values Survey, 60% of respondents considered global warming a very serious problem. But the positive effect of education was considerable: a person with secondary education was about 10 percentage points more likely to express such concern than a person with primary education (Kvaløy et al., 2012).

Data from the International Social Survey Programme on 29 mostly high income countries similarly showed that the share of those disagreeing that people worry too much about the environment rose from 25% of those with less than secondary education to 37% of people with secondary education and 46% of people with tertiary education, according to a new analysis for this Report (National Centre for Social Research, 2013). In Germany, people with secondary education were twice as likely as those with less than secondary education to express concern, and those with tertiary education three times as likely (Infographic: Schooling can save the planet).

### *Education promotes political activism that influences policy change*

People with more education tend not only to be more concerned about the environment, but also to follow up that concern with activism that promotes and supports political decisions that protect the environment. Such pressure is a vital way of pushing governments towards the type of binding agreement that is needed to control emission levels.

Analysis for this Report of the 2010 round of the International Social Survey Programme showed that in almost all participating countries, respondents with more education were more likely to have signed a petition, given money or taken part in a protest or demonstration, in relation to the environment, over the past five years. The influence of education is apparent in Germany, one of the countries with the highest emissions levels: while 12% of respondents with less than secondary education had taken such political action, the share rose to 26% of those with secondary education and 46% of those with tertiary education (National Centre for Social Research, 2013). These results are corroborated by a household survey in 10 OECD countries, where people with more education – especially those with a university degree – tended strongly to express pro-environment values and to belong to environmental organizations (OECD, 2011). An analysis of the Global Warming Citizen Survey in the United States also showed that the higher the education level of respondents, the greater their activism in terms of policy support, environmental political participation and environment-friendly behaviour (Lubell et al., 2007).

### *Education can promote more environment-friendly behaviour*

By increasing awareness and concern, education can encourage people to reduce their impact on the environment by taking action such as using energy and water more efficiently or recycling household waste. Such behaviour becomes increasingly important as people in high income countries are called upon to modify their consumption and take other measures that limit environmental harm.

In the Netherlands, people with a higher level of education tend to use less energy in the home, even taking account of income (Poortinga et al., 2004). A study of households in 10 OECD countries found that those with more education tended to save water (OECD, 2011), and there have been similar findings in Spain (Aisa and Larramona, 2012). In a group of countries including France, Mexico and the Republic of Korea, people without a high school diploma were found to recycle less glass, plastic, aluminium and paper than people with a post-graduate degree (Ferrara and Missios, 2011).

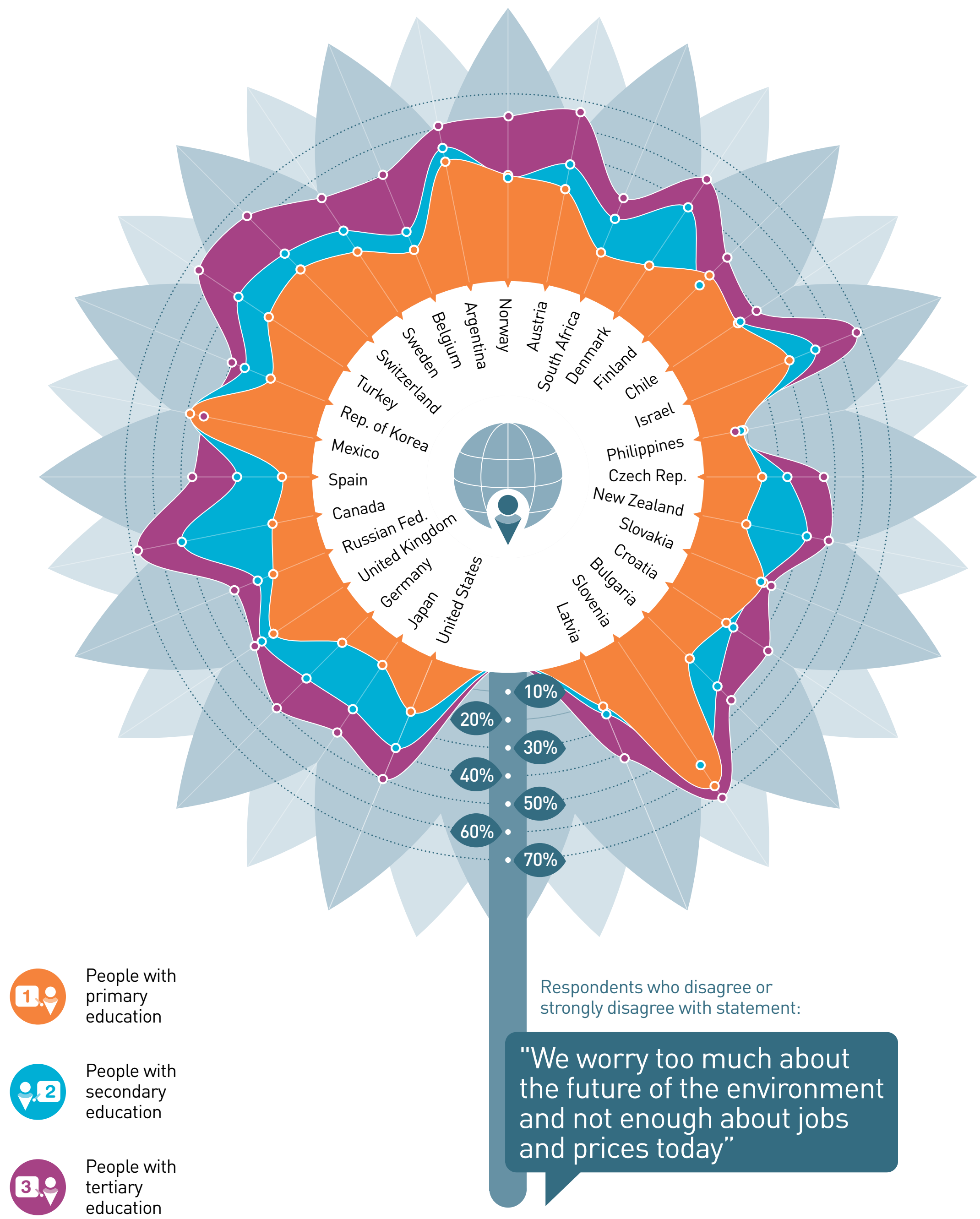
### *Education will help the most vulnerable adapt to climate change*

As well as encouraging people to take action that reduces greenhouse gas emissions, education can help people adapt to the consequences of climate change. The need for adaptation is becoming increasingly urgent; regardless of the speed with which emissions can be reduced, they have already unleashed forces that are leading to increasing temperatures, rising sea levels and more frequent extreme weather events. Adaptation is especially important for poorer countries, where the capacity of governments to act is most limited and threats to livelihoods will be felt most strongly.

**People with more education are more likely to have signed a petition or taken part in a demonstration in relation to the environment**

# SCHOOLING CAN SAVE THE PLANET

Higher levels of education lead to more concern for the environment




*Source:* National Centre for Social Research (2013), based on 2010 International Social Survey Programme data.

Education improves people's understanding of the risks posed by climate change, the need to adapt and measures to reduce the impact of climate change on livelihoods. Farmers in low income countries are most vulnerable to climate change, as they depend heavily on rain-fed agriculture (Below et al., 2010). In Ethiopia, six years of education increase by 20% the chance that a farmer will adapt to climate change through techniques such as soil conservation, varying planting dates and changing crop varieties (Deressa et al., 2009). In Uganda, the likelihood that a family will adopt drought-resistant crop varieties increases when the father has basic education (Hisali et al., 2011). And a survey of farmers in Burkina Faso, Cameroon, Egypt, Ethiopia, Ghana, Kenya, Niger, Senegal, South Africa and Zambia showed that those with education were more likely to make at least one adaptation: a year of education reduced the probability of no adaptation by 1.6% (Maddison, 2007).

## Education empowers women to make life choices

Discrimination prevents some people from getting a fair share of the fruits of overall progress in terms of reducing poverty or improving health. Education can empower these vulnerable people to overcome such barriers, so improving education access and quality will be an essential way of ensuring that efforts to reach post-2015 development goals benefit everyone, especially those who need support the most.

Education is particularly powerful in helping women overcome unequal and oppressive social limits and expectations so they can make choices about their lives. The third MDG, on promoting gender equality and empowering women, acknowledges the central role of education in its inclusion of gender parity in primary and secondary education as a target. After 2015, it will be vital to build on the progress that has been made towards this target, as tackling gender discrimination in school unleashes education's power to help girls and women overcome broader discrimination.

As this chapter has shown, education empowers women to make choices that improve their own and their children's health and chances of survival and boosts women's work prospects. This section looks at further ways in which education empowers women to make choices that improve their welfare, including marrying later and having fewer children. When girls spend more years in school, they tend to marry later and have their first child later, but the effect of education goes beyond this to give girls and young women greater awareness of their rights and improve their confidence in their ability to make decisions that affect their lives (Box 3.4).

Education's influence on empowering women is particularly strong in countries where girls are likely to get married or give birth early and have a large number of children. Such empowerment not only benefits women's own choices, but also improves their health and that of their children, and benefits societies by bringing forward the demographic transition to a stable population with lower fertility and lower mortality.

### *Women's education helps avert child marriage*

Around 2.9 million girls are married by the age of 15 in sub-Saharan Africa and South and West Asia, equivalent to one in eight girls in each region, according to new estimates from Demographic and Health Surveys for this Report based on data for 20- to 24-year-olds. These shocking statistics mean millions of girls are robbed of their childhood and denied an education.

**While just 8% of literate girls are married by age 15 in South and West Asia, more than one in four of those who are not literate are married by this age**

Ensuring that girls stay in school is one of the most effective ways to avert child marriage. If all girls had primary education in sub-Saharan Africa and South and West Asia, child marriage would fall by 14%, from almost 2.9 million to less than 2.5 million, and if they had secondary education it would fall by 64% to just over one million (Infographic: Learning lessens early marriages and births). Education's contribution is evident in the links between literacy and child marriage. While just 4% of literate girls are married by age 15 in sub-Saharan Africa, and 8% in South and West Asia, more than one in five of those who are not literate are married by this age in sub-Saharan Africa, and almost one in four in South and West Asia.

In 13 out of 40 countries, more than 1 girl in 10 marries by the age of 15. And in 28 out of 40 countries this is true for those with less than

### Box 3.4: Education gives women the power to claim their rights

Education gives women more power over their own lives in a variety of ways. As well as widening their choices, it can boost their confidence and perception of their freedom. It can also alter the perceptions of men and social barriers to their autonomy. One key aspect of this transformative power is women's freedom to choose a spouse in countries where arranged marriages are common. According to new analysis for this Report, young women with at least secondary education are 30 percentage points more likely to have a say over their choice of spouse than women with no education in India and 15 percentage points more likely in Pakistan.

Education also influences young women's choice of family size. In Pakistan, while only 30% of women with no education believe they can have a say over the number of their children, the share increases to 52% among women with primary education and to 63% among women with lower secondary education.

The education level of a woman's spouse can also have a key role in her fertility choices. In India, the likelihood that the fertility preferences of a woman with primary education were taken into account rose from 65% for those whose husbands had no education to at least 85% for those whose husbands had at least secondary education. Education also helps prevent the abhorrent practice of infanticide in India, where strong preferences over the sex of the child have been linked to millions of killings of children. While 84% of women with no education would prefer to have a boy if they could only have one child, only 50% of women with at least secondary education would have such a preference.

Further evidence of education's power to change attitudes comes from Sierra Leone, where the expansion of schooling opportunities in the aftermath of the civil war led to a steep increase in the amount of education completed by younger women. An additional year of schooling reduced women's tolerance of domestic violence from 36% to 26%.

Adult education can also make a key difference to women's choices. A literacy programme for women in the Indian state of Uttarakhand that provided continuing education and vocational training, covering issues such as alcoholism, village politics and community conflict resolution, led to a significant increase in the percentages of women who felt able to leave their house without permission (from 58% to 75%) and who participated in village council meetings (from 19% to 41%).

*Sources:* Aslam (2013); Kandpal et al. (2012); Mocan and Cannonier (2012).

**In sub-Saharan Africa and South and West Asia, if all women had secondary education, early births would fall by 59%**

primary education. In Bangladesh, Mali and Sierra Leone, over one in five young women with primary schooling are married by this age. Girls who make it to secondary school are much less likely to be locked into early marriage. In Ethiopia, while almost one in three young women with no education were married by the age of 15 in 2011, among women with secondary education the share was just 9%.

#### *Women's education reduces the chance of early birth*

Empowering women also gives them more control about decisions over when to have their first child. As many as 3.4 million births occur before girls reach age 17 in sub-Saharan Africa and South and West Asia. This corresponds to about one in seven young women in the two regions, according to estimates for this Report using Demographic and Health Survey data for 20- to 24-year-olds. Giving birth so early is a major risk factor for maternal and child deaths.

One reason that girls with more years in school are less likely to give birth early is simply that girls who give birth drop out before they have a chance of more education. But staying in school longer also gives girls more confidence to make choices that prevent them from getting pregnant at a young age. In sub-Saharan Africa and South and West Asia, early births would fall by 10%, from 3.4 to 3.1 million, if all women had primary education. If all women had secondary education, early births would fall by 59%, to 1.4 million (Infographic: Learning lessens early marriages and births).

Average age at first birth differs by more than three years between women without education and women with secondary education. In Nigeria, for example, women with no education gave birth for the first time at age 18, on average, compared with age 25 for those with at least secondary education (ICF International, 2012). Education not only postpones the first birth but also increases the time between births. A period of less than two years between births tends to increase health risks for mothers and their children. In Kenya, the probability that a woman with no education would have another child within two years of the second birth was 27%, compared with 17% among women with secondary education (ICF International, 2012).

# LEARNING LESSENS EARLY MARRIAGES AND BIRTHS

Women with higher levels of education are less likely
to get married or have children at an early age

## Child marriage

14%
fewer marriages if all girls had primary education

64%
fewer marriages if all girls had secondary education

Child marriages for all girls by age 15 in sub-Saharan Africa and South and West Asia

2,867,000 | 2,459,000 | 1,044,000

## Early births

10%
fewer girls would become pregnant if all girls had primary education

59%
fewer girls would become pregnant if all girls had secondary education

Early births for all girls under 17 in sub-Saharan Africa and South and West Asia

3,397,000 | 3,071,000 | 1,393,000

## Fertility rate*

No education | Primary education | Secondary education

Average number of births per woman in sub-Saharan Africa

6.7 | 5.8 | 3.9

*Fertility rate is the average number of children that would be born to a woman over her lifetime

*Sources:* EFA Global Monitoring Report team calculations (2013), based on Demographic and Health Surveys; UNPD (2011).

*Extending girls' education helps bring the demographic transition forward*

Women with more education tend to have fewer children, which benefits them, their families and society more generally. In some parts of the world, education has already been a key factor in bringing forward the transition from high rates of birth and mortality to lower rates. Other parts of the world are lagging, however, particularly sub-Saharan Africa, where women have an average of 5.4 live births, compared with 2.7 in South Asia (UNPD, 2011). Women with no education in sub-Saharan Africa have 6.7 births, on average, while the number falls to 5.8 for those with primary education and to 3.9 for those with secondary education. In Angola, the fertility rate of a woman with no education is 7.8 children, compared with 5.9 children for a woman with primary education and 2.5 children for a woman with secondary education or more.

**If all countries expanded their school systems at the same rate as the Republic of Korea, there would be 843 million fewer people in the world by 2050**

In seven countries, including Niger, Uganda and Zambia, the fertility rate – the number of live births per woman during her childbearing years – is over six children per woman. If all women in sub-Saharan Africa had primary education, the number of births would fall by 7%, from 31 million to 29 million, while if all women had secondary education, the number would fall by 37%, to 19 million.

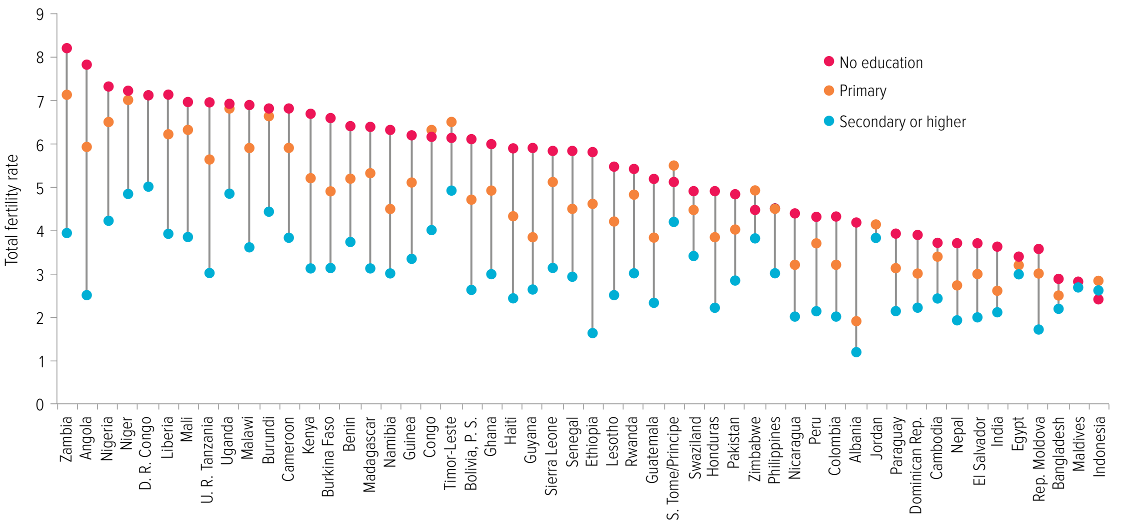
Comparing total fertility rates by education level across countries shows that secondary education is of particular importance. For example, in the United Republic of Tanzania in 2010, the fertility rate of a woman with no education was 7 children, compared with 5.6 children for a woman with primary education and 3 children for a woman with secondary education or higher (Figure 3.7).

The potential contribution of education to stabilizing global population growth should not be underestimated: according to recent projections, if all countries expanded their school systems at the same rate as the Republic of Korea and Singapore, there would be 843 million fewer people in the world by 2050, equivalent to the population of sub-Saharan Africa in 2010, than if enrolment ratios had remained at 2000 levels. In Uganda, for example, the population would increase from 24 million in 2000 to 89 million in 2050 if enrolment grew at the fast rate, but to 105 million if it stayed constant, placing considerable stress on resources (Lutz and KC, 2011).

The results of these projections are supported by recent analysis of the determinants of fertility rates in 49 countries, based on Demographic and Health Surveys for 1986–2008, which identified eight countries where declines in fertility stalled in the 2000s. Of these, seven were in sub-Saharan Africa, and six were in the early phase of the demographic transition: Benin, Guinea, Mozambique, Nigeria, the United Republic of Tanzania and Zambia. Mali and Niger had not started their demographic transition. What these countries have in common is slower progress in female education: the share of women with no schooling fell by only 2.5 percentage points, compared to 6 percentage points in countries with declining fertility rates. Taking into account other factors, such as national income and child mortality, a decline in the share of women of child-bearing age with less than primary education in Guinea from 86% to 24% would reduce the fertility rate by 2 children per woman (Shapiro et al., 2011).

In three countries in eastern Africa, Kenya, Uganda and the United Republic of Tanzania, the decline in overall fertility has slowed, held back by persistently high fertility among women with less education. For example, in Kenya, the percentage of women with at least secondary education stagnated at 30% between 1998 and 2003. Over the same period, the fertility rate of women with secondary education fell by 0.3 children per woman, while the fertility rate of women with less than secondary education increased by 0.6 children per woman. The population of Kenya is expected to reach 59 million in 2025; if the lack of progress in education had not slowed the fertility decline, it could have been at least 15 million lower (Ezeh et al., 2009).

**Figure 3.7: Maternal education greatly reduces fertility rates**
Total fertility rate (number of live births per woman) by level of maternal education, selected countries, 2005–2011




*Source:* ICF International (2012).

In countries that have undergone the demographic transition, education's contribution is clear. In Brazil, around 70% of the fertility decline during the 1960s and 1970s can be explained by improvements in schooling (Lam and Duryea, 1999).

Education played an important role in reducing fertility in Nigeria. An evaluation of the universal primary education programme implemented in the 1970s showed that four years of schooling reduced fertility by one child per woman by the age of 25 (Osili and Long, 2008). However, the percentage of women with at least secondary education was stagnant between 1999 and 2003. Furthermore, the number of girls out of school increased from 4.1 million in 1999 to 5.5 million in 2010. The total fertility rate stagnated at 5.7 children per woman between 2003 and 2008, which risks shifting Nigeria to a pessimistic scenario in which it would take 20 years longer for total fertility to fall below 3 children per woman. This would push the total population 44 million higher by 2050 and almost 300 million higher by 2100, to more than 1 billion, with potentially disastrous consequences for human development (UNPD, 2011).

## Conclusion

The striking evidence laid out in this chapter demonstrates not only education's capacity to accelerate progress towards other development goals, but also how best to tap that potential, most of all by making sure that access to good quality education is available to all, regardless of their circumstances. Education's unique power should secure it a central place in the post-2015 development framework, and in the plans of policy-makers in poor and rich countries alike. This chapter shows why governments and aid donors should renew their political and financial commitment to education not only as a human right and a key goal on its own but also, crucially, as an investment that pays off in every sphere of people's lives and aspirations.

# Part 3 Supporting teachers to end the learning crisis

A lack of attention to education quality and a failure to reach the marginalized have contributed to a learning crisis that needs urgent attention. The importance of education quality was recognized when it was established as the sixth Education for All goal in 2000, and again recently when the UN secretary-general made it one of the three priorities of the Global Education First Initiative. But 250 million children – many of them from disadvantaged backgrounds – are not learning even basic literacy and numeracy skills, let alone the further skills they need to get decent work and lead fulfilling lives.

To solve that crisis, all children must have teachers who are trained, motivated and enjoy teaching, who can identify and support weak learners, and who are backed by well-managed education systems. Good teachers close the gap between poor and good quality education by maximizing the benefits of learning in every classroom for every child. But worldwide, children who already face disadvantage and discrimination – because of factors such as poverty, gender, ethnicity, disability and where they live – are much less likely to be taught by good teachers.

As this Report shows, governments can increase access while also making sure that learning improves for all. Adequately funded national education plans that aim explicitly to meet the needs of the disadvantaged and that make equitable access to well-trained teachers must be a policy priority. But few national plans fulfill these requirements. Instead of getting adequate training and teaching conditions, teachers get the blame for poor learning outcomes. While teacher absenteeism and engagement in private tuition are real problems, policy-makers often ignore underlying reasons such as low pay and a lack of career opportunities.

The key to ending the learning crisis is to recruit the best teacher candidates, give them appropriate training, deploy them where they are needed most and give them incentives to make a long-term commitment to teaching (see illustration).

Education quality is undermined by the need for additional teachers, with 1.6 million required to achieve universal primary education by 2015 and 5.1 million to achieve universal lower secondary education by 2030.

Many teachers are untrained or insufficiently trained, and left to teach in overcrowded classrooms with few resources. On low pay, they soon become discontented, contributing to the lowering of education quality.

To ensure that all children are learning, teachers also need the support of an appropriate curriculum and assessment system that pays particular attention to the needs of children in early grades, when the most vulnerable are in danger of dropping out. Well-designed curricula and assessment practices enrich classroom experiences, help teachers identify and support disadvantaged students and promote the values, attitudes and practical skills needed to face future challenges in their lives. Moreover, children who have had to leave school before achieving the basics need a second chance to acquire these skills.

Beyond teaching the basics, teachers must help children gain important transferable skills, and can also help them become responsible global citizens if issues such as environmental sustainability and peace-building are integrated into a curriculum that focuses on practical action.